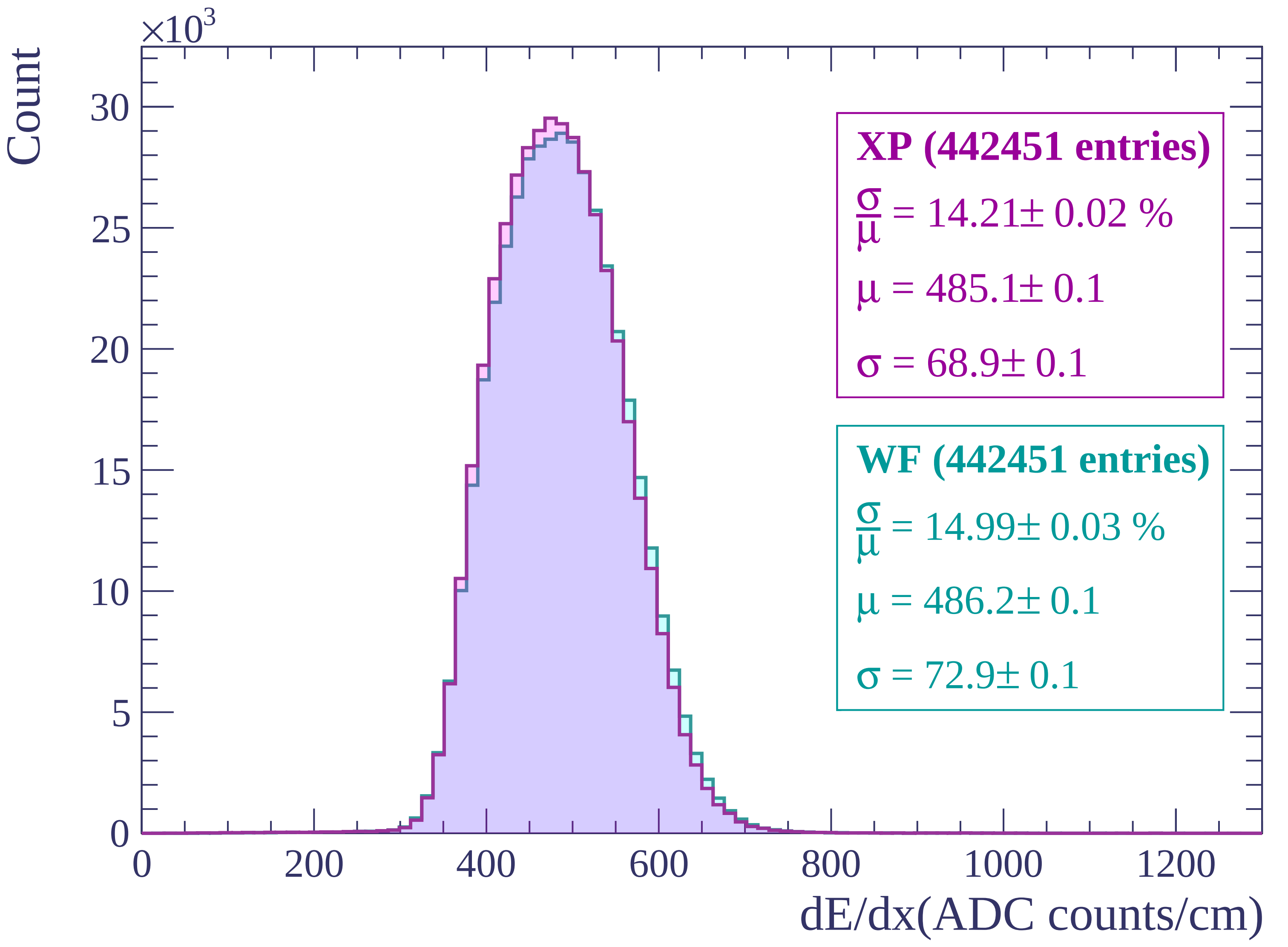
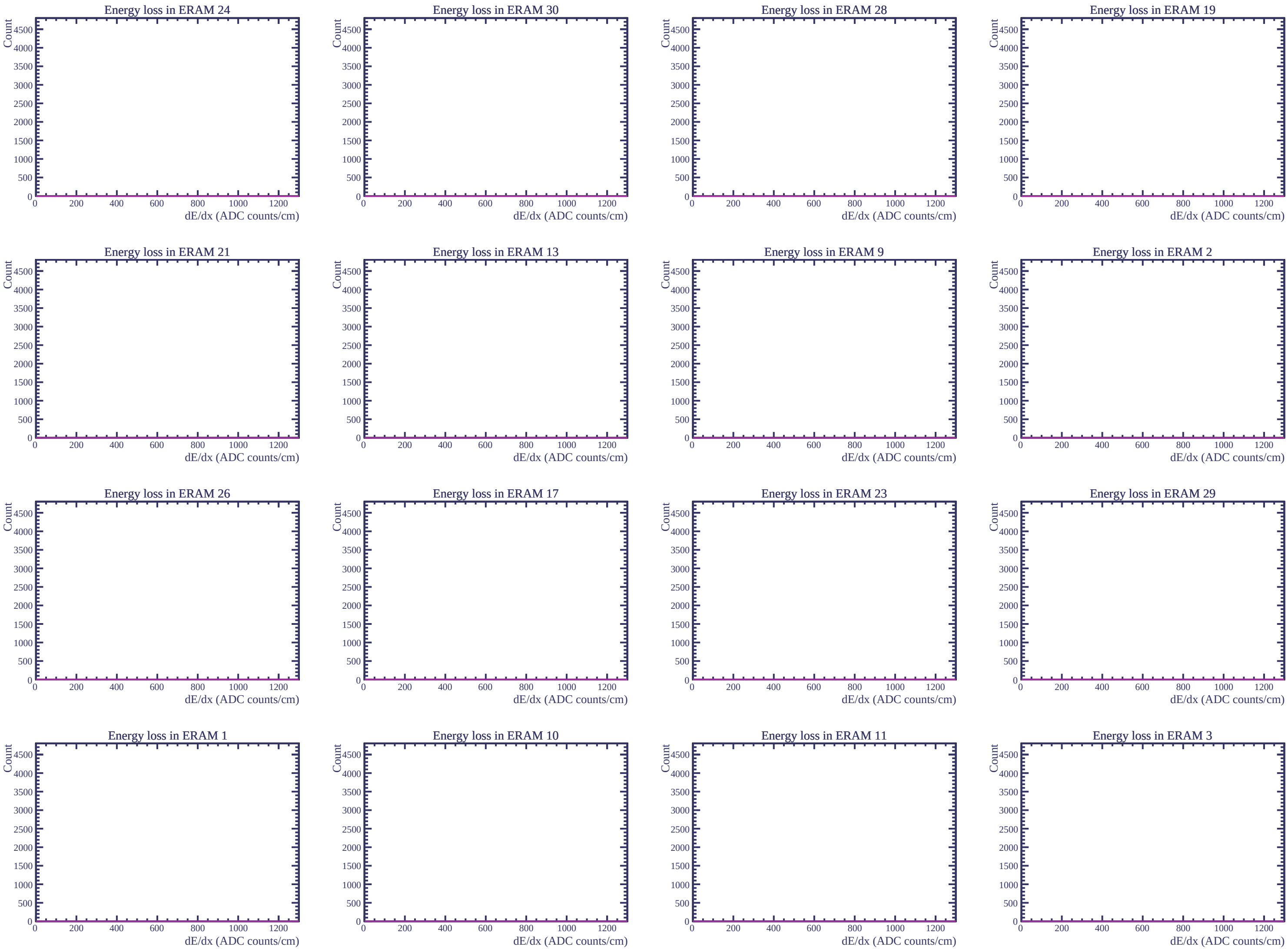
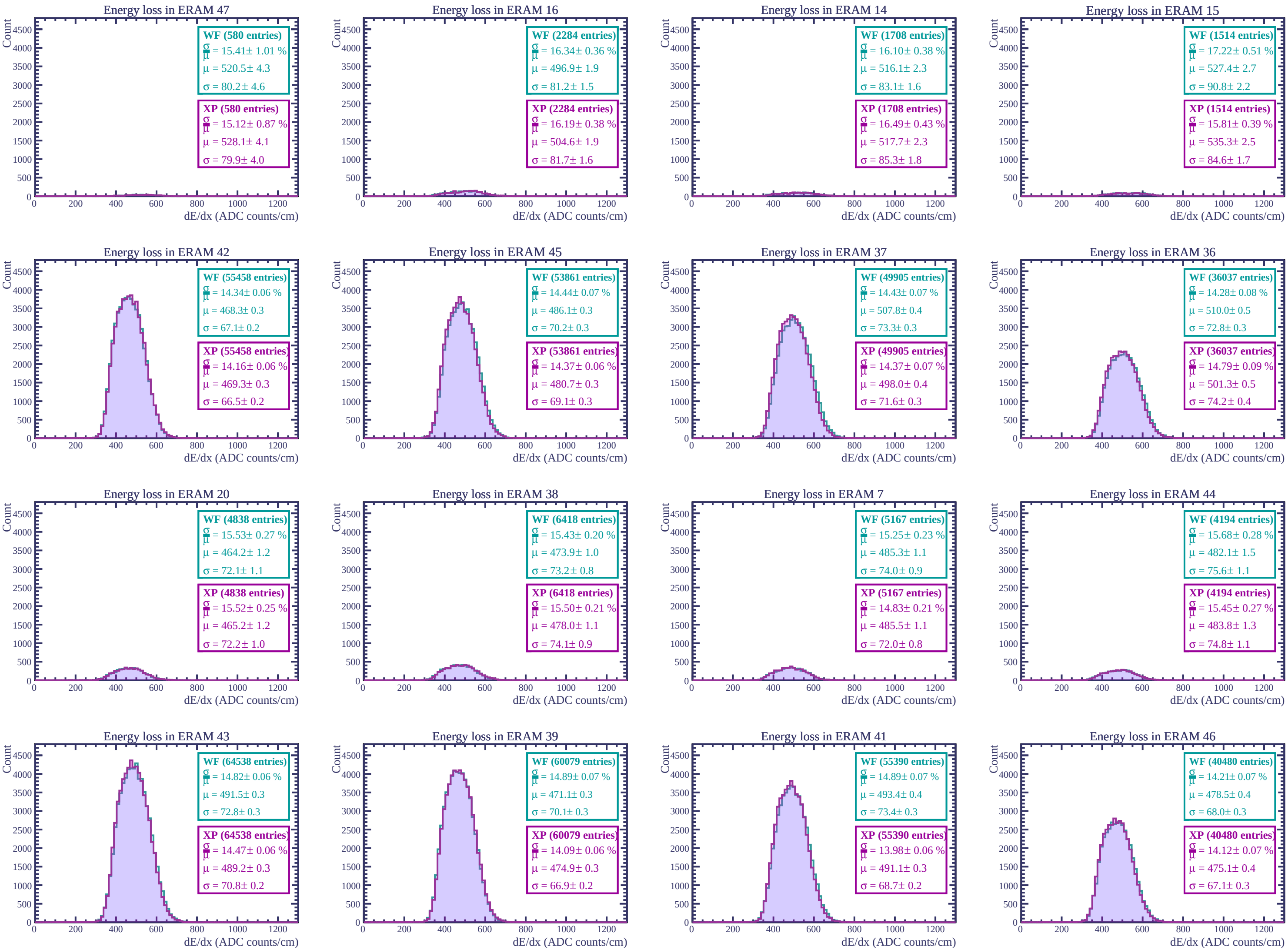
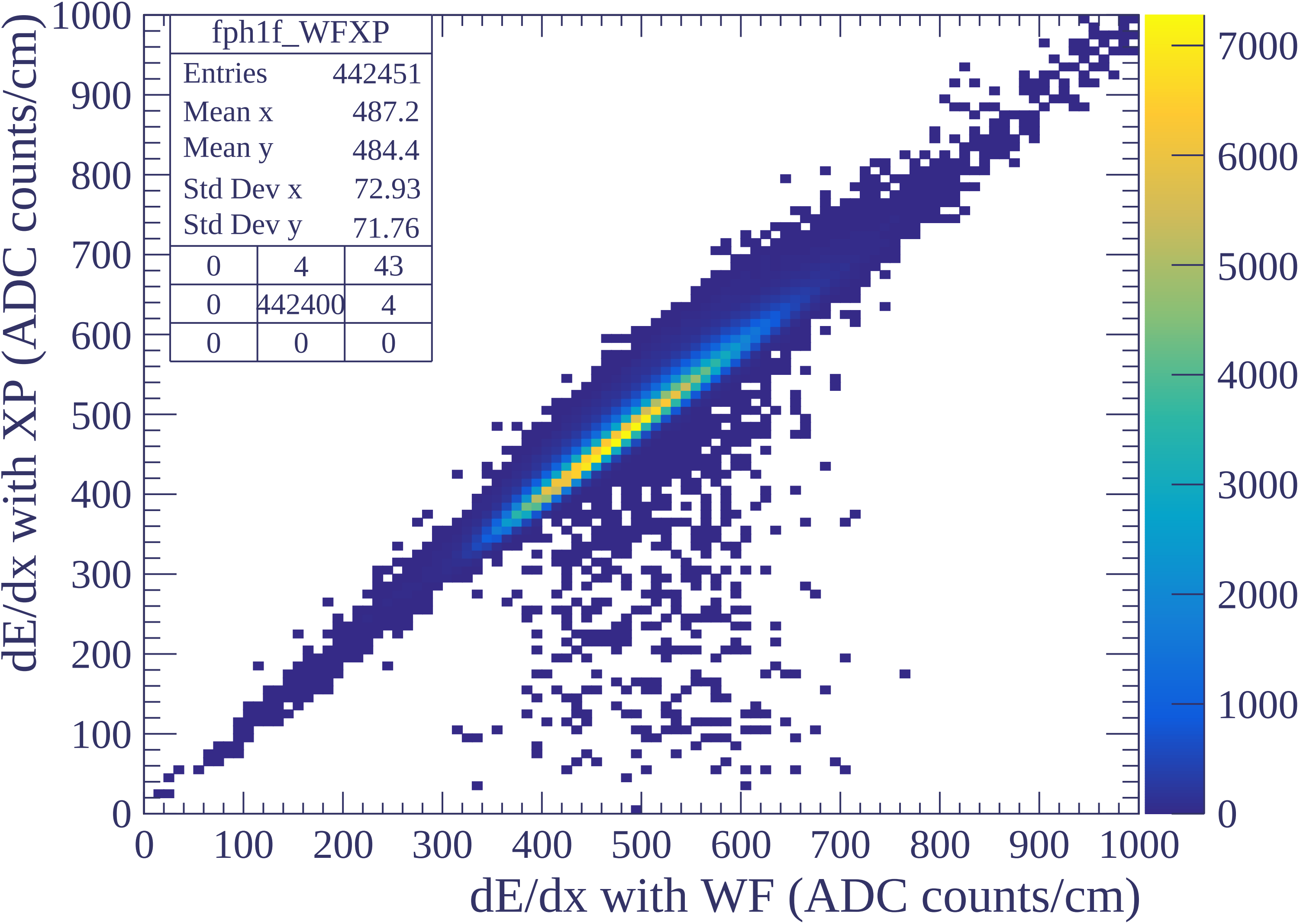


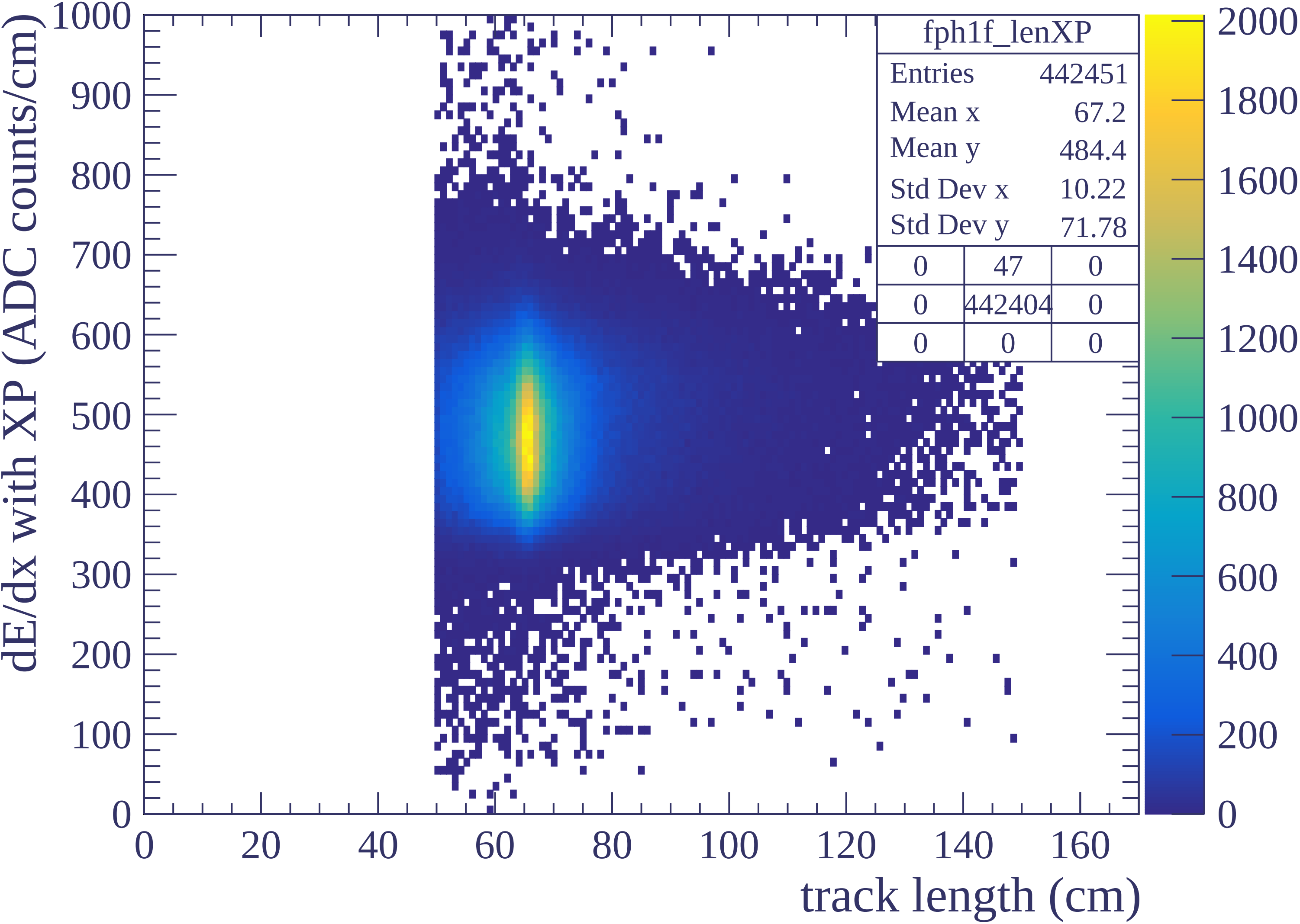
$\times 10^3$
Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (442451 entries)
$\frac{\sigma}{\mu}$ = 14.21± 0.02 %
μ = 485.1± 0.1
σ = 68.9± 0.1
WF (442451 entries)
$\frac{\sigma}{\mu}$ = 14.99± 0.03 %
μ = 486.2± 0.1
σ = 72.9± 0.1


Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
Energy loss in ERAM 13
Energy loss in ERAM 9
Energy loss in ERAM 2
Energy loss in ERAM 26
Energy loss in ERAM 17
Energy loss in ERAM 23
Energy loss in ERAM 29
Energy loss in ERAM 1
Energy loss in ERAM 10
Energy loss in ERAM 11
Energy loss in ERAM 3
Count
dE/dx (ADC counts/cm)

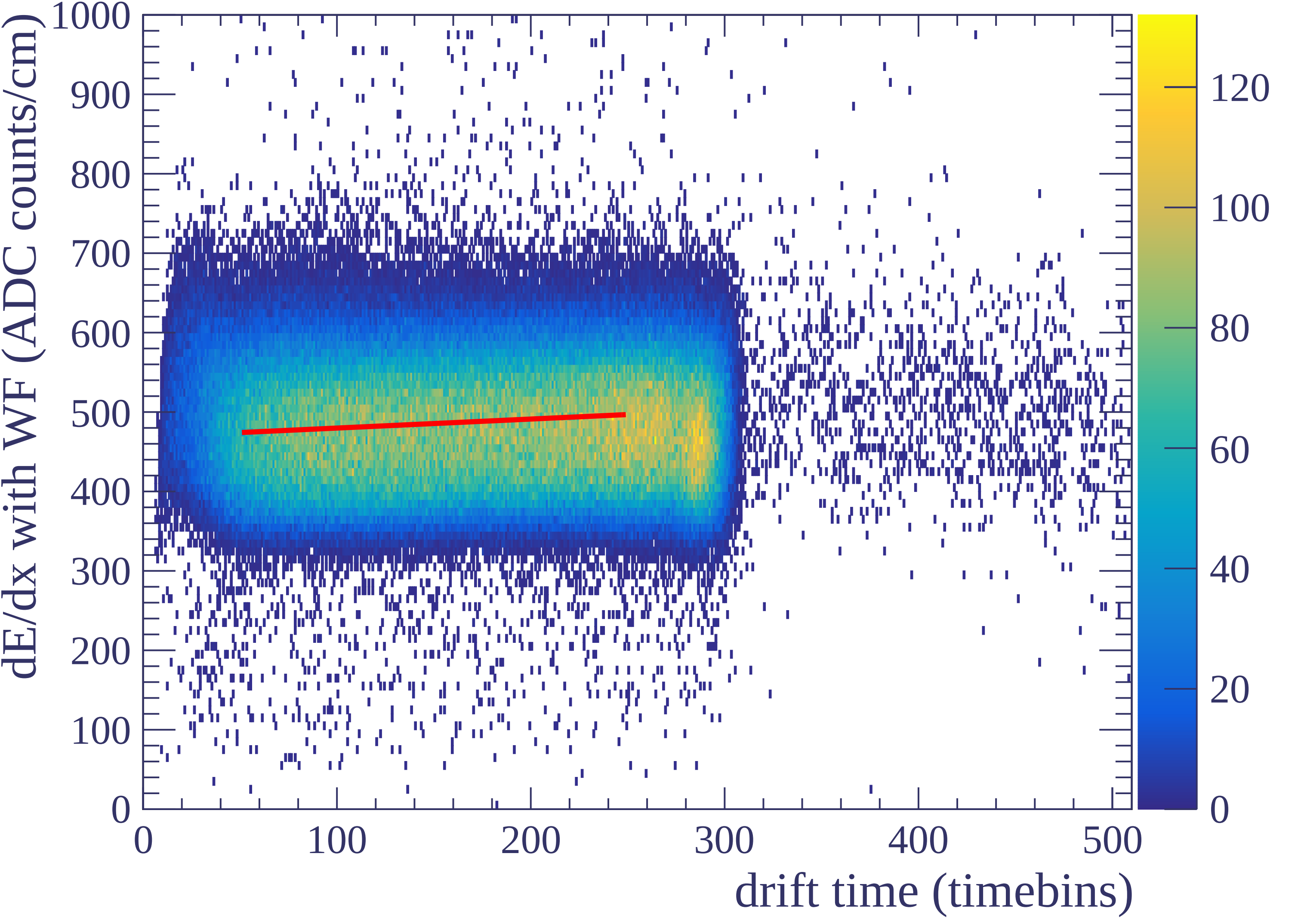


Energy loss in ERAM 47
WF (580 entries)
σ/μ = 15.41± 1.01 %
μ = 520.5± 4.3
σ = 80.2± 4.6
XP (580 entries)
σ/μ = 15.12± 0.87 %
μ = 528.1± 4.1
σ = 79.9± 4.0
Energy loss in ERAM 16
WF (2284 entries)
σ/μ = 16.34± 0.36 %
μ = 496.9± 1.9
σ = 81.2± 1.5
XP (2284 entries)
σ/μ = 16.19± 0.38 %
μ = 504.6± 1.9
σ = 81.7± 1.6
Energy loss in ERAM 14
WF (1708 entries)
σ/μ = 16.10± 0.38 %
μ = 516.1± 2.3
σ = 83.1± 1.6
XP (1708 entries)
σ/μ = 16.49± 0.43 %
μ = 517.7± 2.3
σ = 85.3± 1.8
Energy loss in ERAM 15
WF (1514 entries)
σ/μ = 17.22± 0.51 %
μ = 527.4± 2.7
σ = 90.8± 2.2
XP (1514 entries)
σ/μ = 15.81± 0.39 %
μ = 535.3± 2.5
σ = 84.6± 1.7
Energy loss in ERAM 42
WF (55458 entries)
σ/μ = 14.34± 0.06 %
μ = 468.3± 0.3
σ = 67.1± 0.2
XP (55458 entries)
σ/μ = 14.16± 0.06 %
μ = 469.3± 0.3
σ = 66.5± 0.2
Energy loss in ERAM 45
WF (53861 entries)
σ/μ = 14.44± 0.07 %
μ = 486.1± 0.3
σ = 70.2± 0.3
XP (53861 entries)
σ/μ = 14.37± 0.06 %
μ = 480.7± 0.3
σ = 69.1± 0.3
Energy loss in ERAM 37
WF (49905 entries)
σ/μ = 14.43± 0.07 %
μ = 507.8± 0.4
σ = 73.3± 0.3
XP (49905 entries)
σ/μ = 14.37± 0.07 %
μ = 498.0± 0.4
σ = 71.6± 0.3
Energy loss in ERAM 36
WF (36037 entries)
σ/μ = 14.28± 0.08 %
μ = 510.0± 0.5
σ = 72.8± 0.3
XP (36037 entries)
σ/μ = 14.79± 0.09 %
μ = 501.3± 0.5
σ = 74.2± 0.4
Energy loss in ERAM 20
WF (4838 entries)
σ/μ = 15.53± 0.27 %
μ = 464.2± 1.2
σ = 72.1± 1.1
XP (4838 entries)
σ/μ = 15.52± 0.25 %
μ = 465.2± 1.2
σ = 72.2± 1.0
Energy loss in ERAM 38
WF (6418 entries)
σ/μ = 15.43± 0.20 %
μ = 473.9± 1.0
σ = 73.2± 0.8
XP (6418 entries)
σ/μ = 15.50± 0.21 %
μ = 478.0± 1.1
σ = 74.1± 0.9
Energy loss in ERAM 7
WF (5167 entries)
σ/μ = 15.25± 0.23 %
μ = 485.3± 1.1
σ = 74.0± 0.9
XP (5167 entries)
σ/μ = 14.83± 0.21 %
μ = 485.5± 1.1
σ = 72.0± 0.8
Energy loss in ERAM 44
WF (4194 entries)
σ/μ = 15.68± 0.28 %
μ = 482.1± 1.5
σ = 75.6± 1.1
XP (4194 entries)
σ/μ = 15.45± 0.27 %
μ = 483.8± 1.3
σ = 74.8± 1.1
Energy loss in ERAM 43
WF (64538 entries)
σ/μ = 14.82± 0.06 %
μ = 491.5± 0.3
σ = 72.8± 0.3
XP (64538 entries)
σ/μ = 14.47± 0.06 %
μ = 489.2± 0.3
σ = 70.8± 0.2
Energy loss in ERAM 39
WF (60079 entries)
σ/μ = 14.89± 0.07 %
μ = 471.1± 0.3
σ = 70.1± 0.3
XP (60079 entries)
σ/μ = 14.09± 0.06 %
μ = 474.9± 0.3
σ = 66.9± 0.2
Energy loss in ERAM 41
WF (55390 entries)
σ/μ = 14.89± 0.07 %
μ = 493.4± 0.4
σ = 73.4± 0.3
XP (55390 entries)
σ/μ = 13.98± 0.06 %
μ = 491.1± 0.3
σ = 68.7± 0.2
Energy loss in ERAM 46
WF (40480 entries)
σ/μ = 14.21± 0.07 %
μ = 478.5± 0.4
σ = 68.0± 0.3
XP (40480 entries)
σ/μ = 14.12± 0.07 %
μ = 475.1± 0.4
σ = 67.1± 0.3
Count
dE/dx (ADC counts/cm)

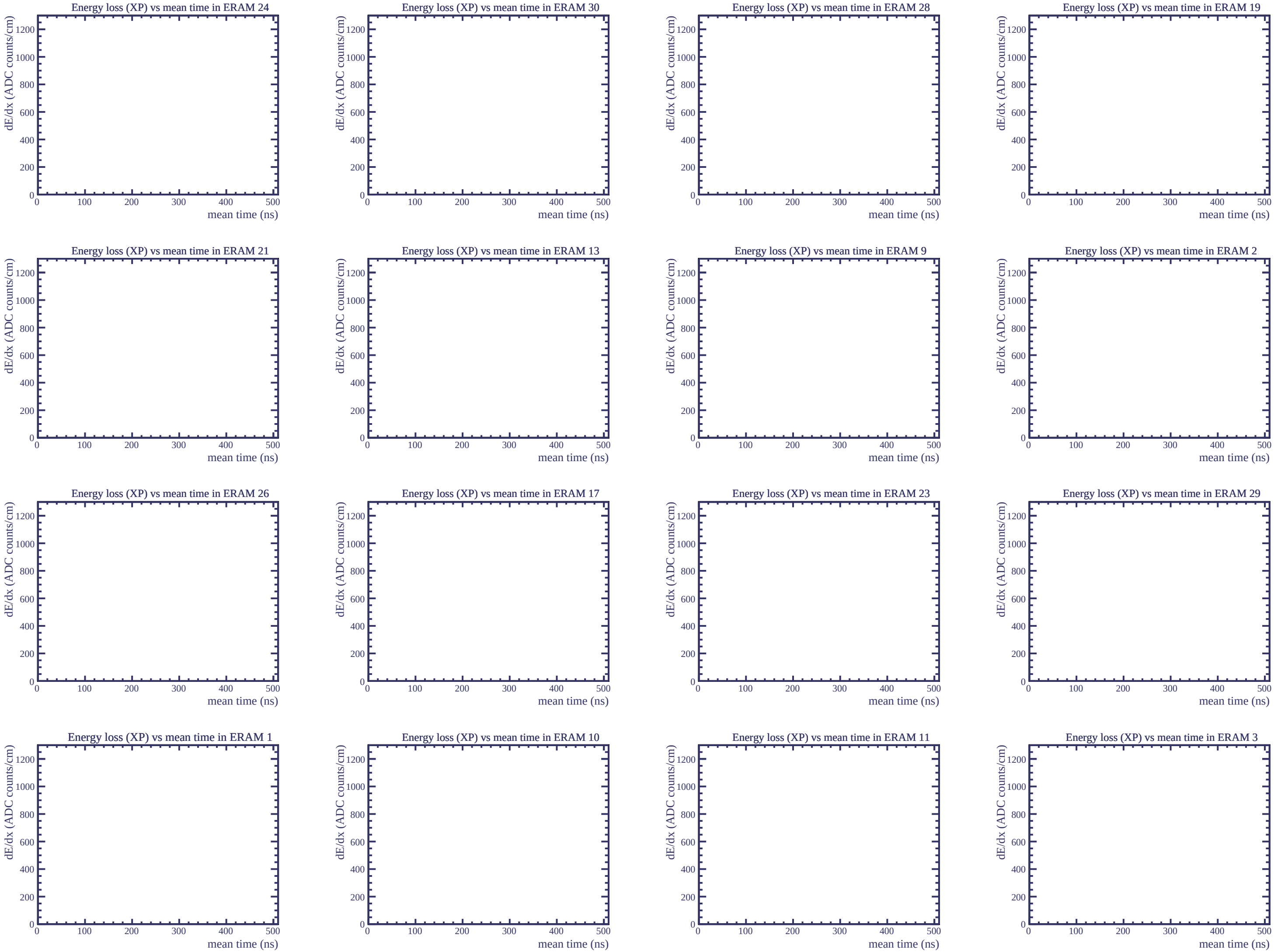

fph1f_WFXP
Entries 442451
Mean x 487.2
Mean y 484.4
Std Dev x 72.93
Std Dev y 71.76
0 4 43
0 442400 4
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
1000
2000
3000
4000
5000
6000
7000

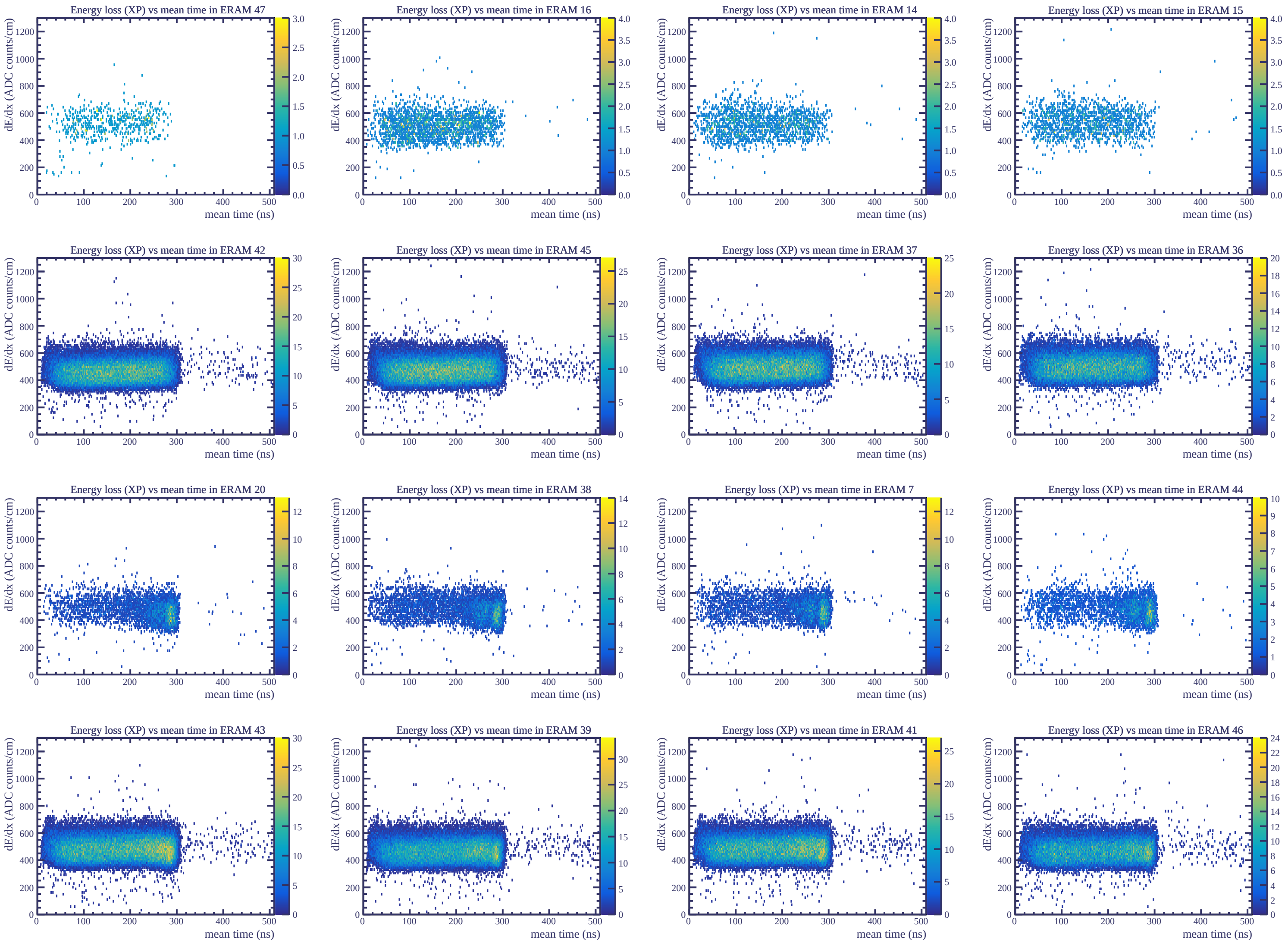


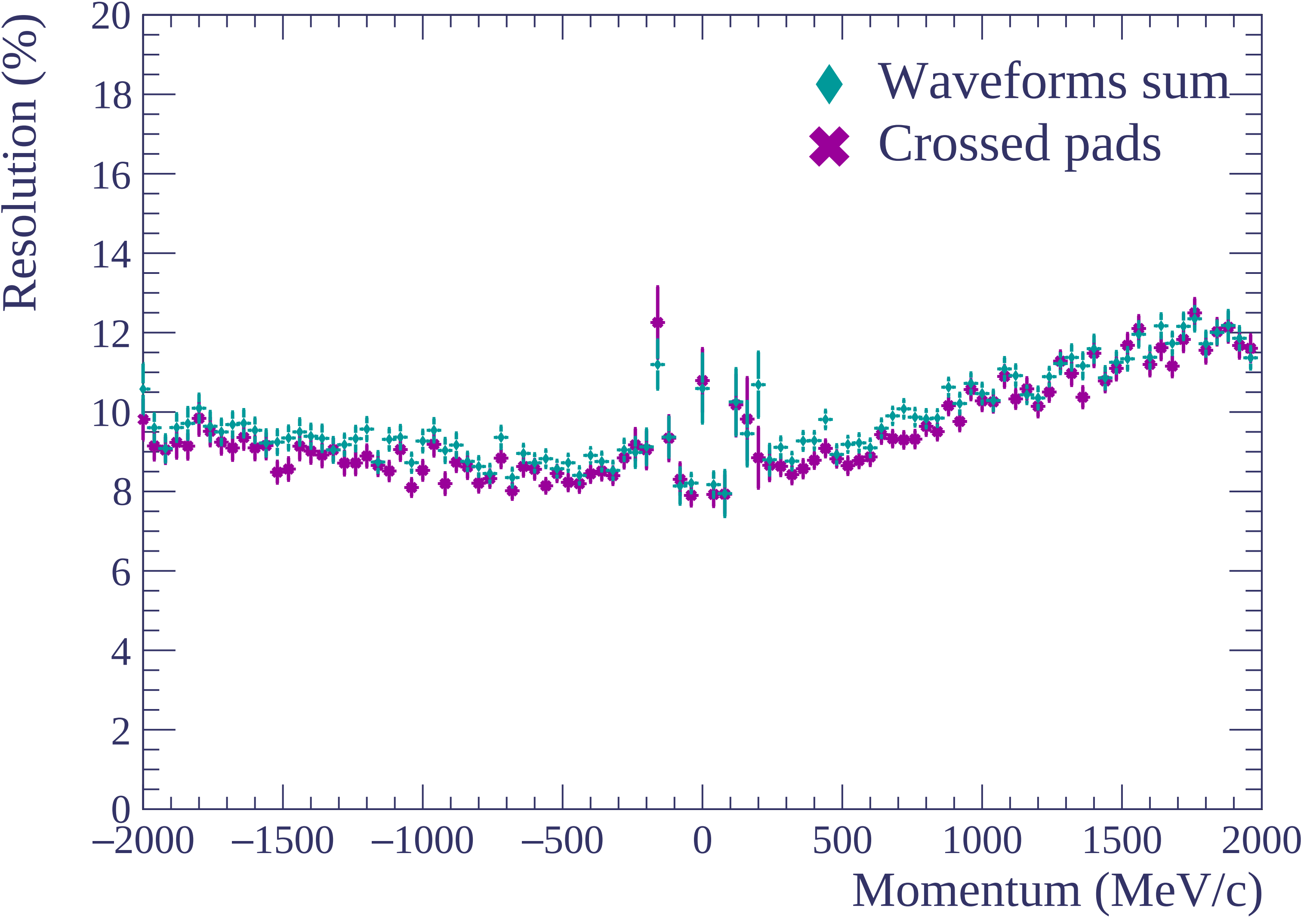
fph1f_lenXP
Entries 442451
Mean x 67.2
Mean y 484.4
Std Dev x 10.22
Std Dev y 71.78
0 47 0
0 442404 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000

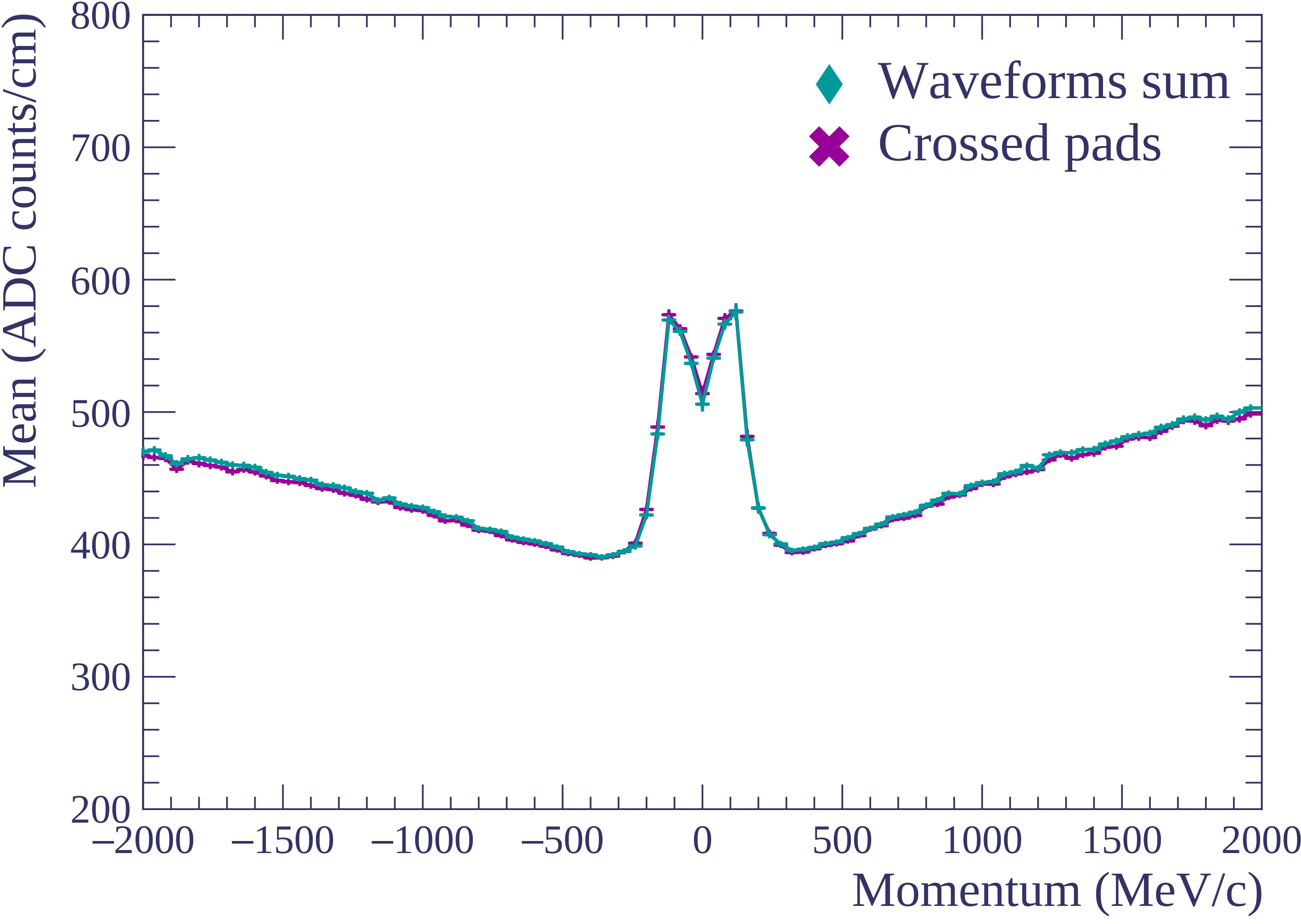


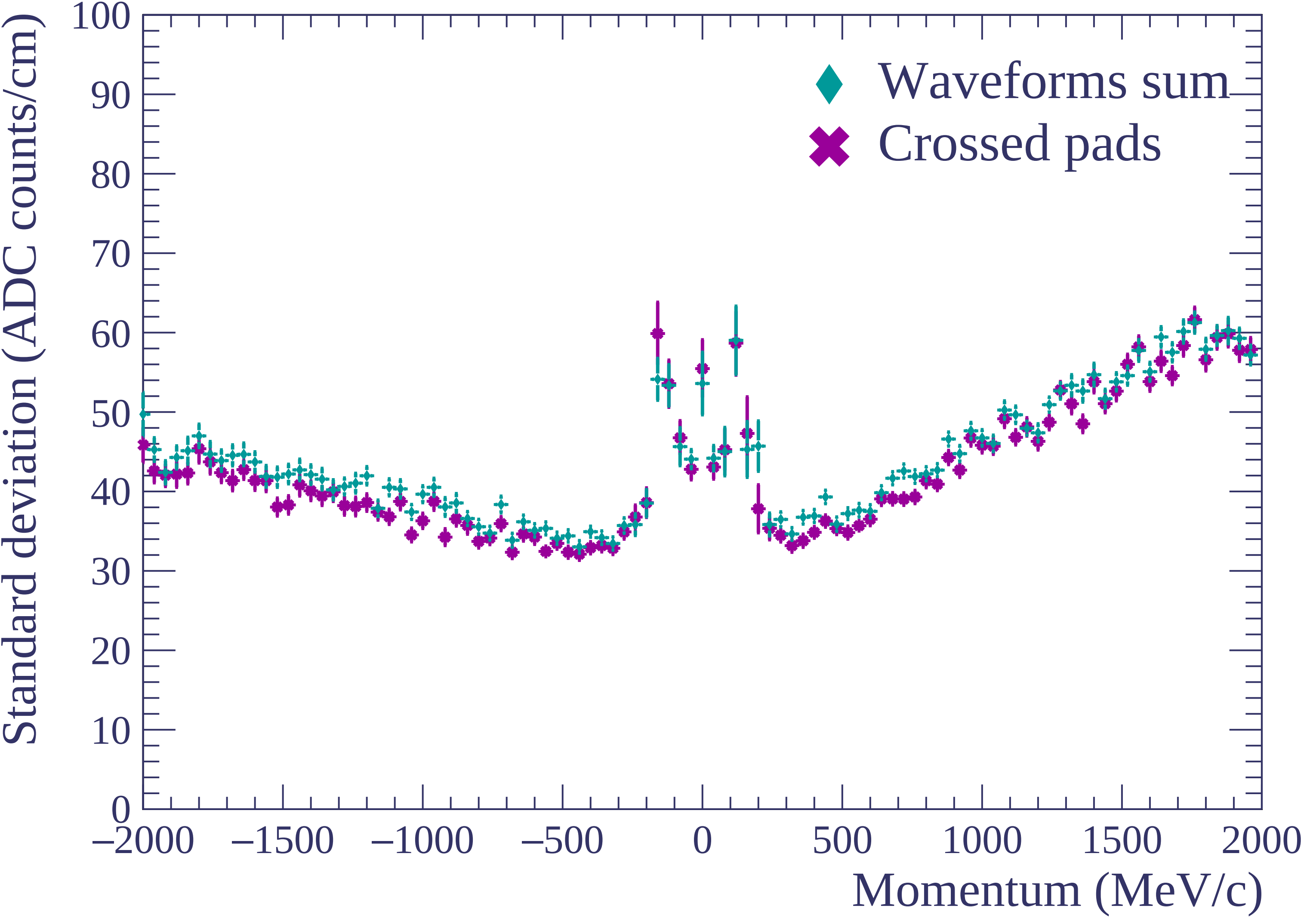
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
20
40
60
80
100
120


Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

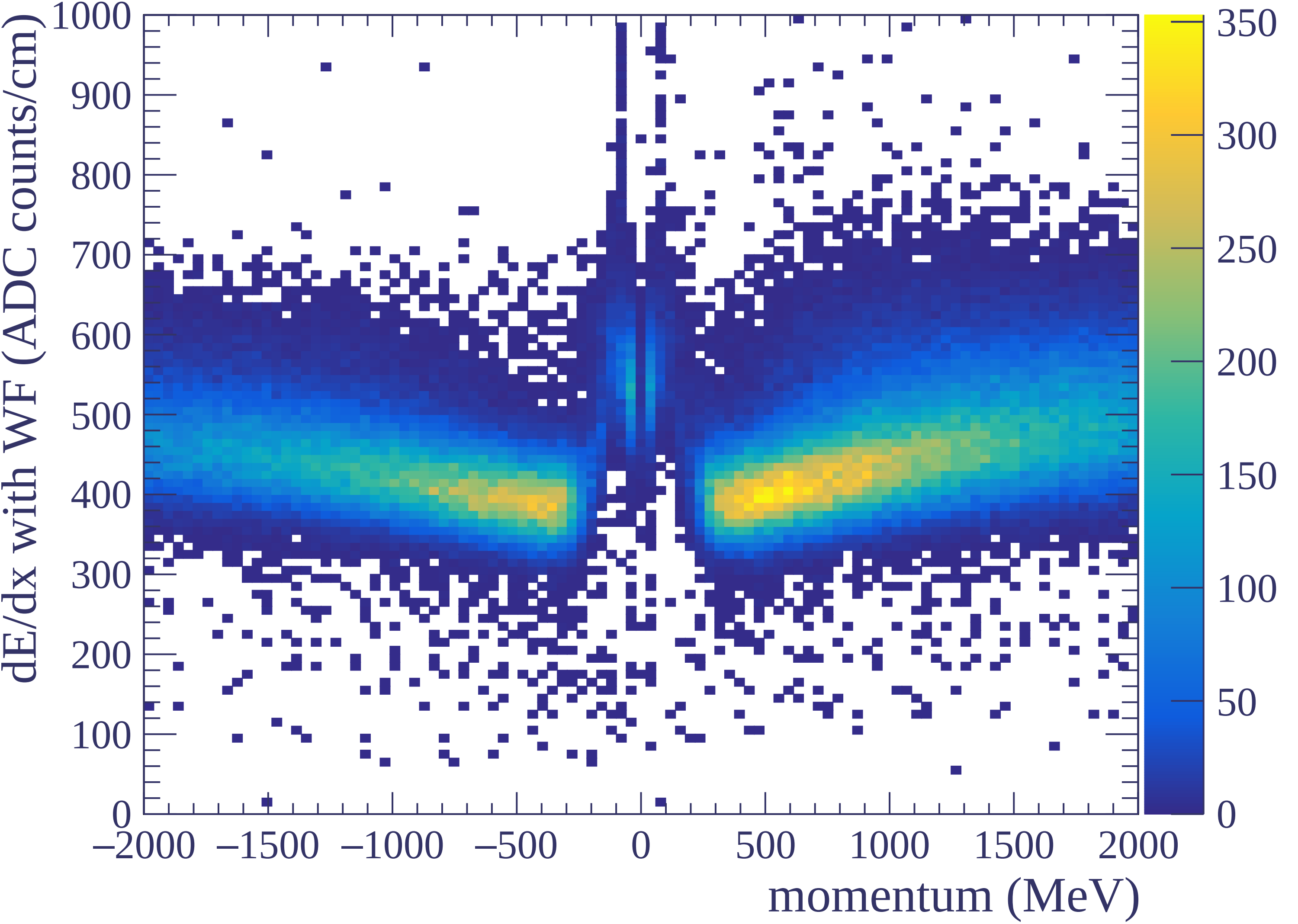

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)


Resolution (%)
Momentum (MeV/c)
Waveforms sum
Crossed pads
0
2
4
6
8
10
12
14
16
18
20
−2000
−1500
−1000
−500
0
500
1000
1500
2000

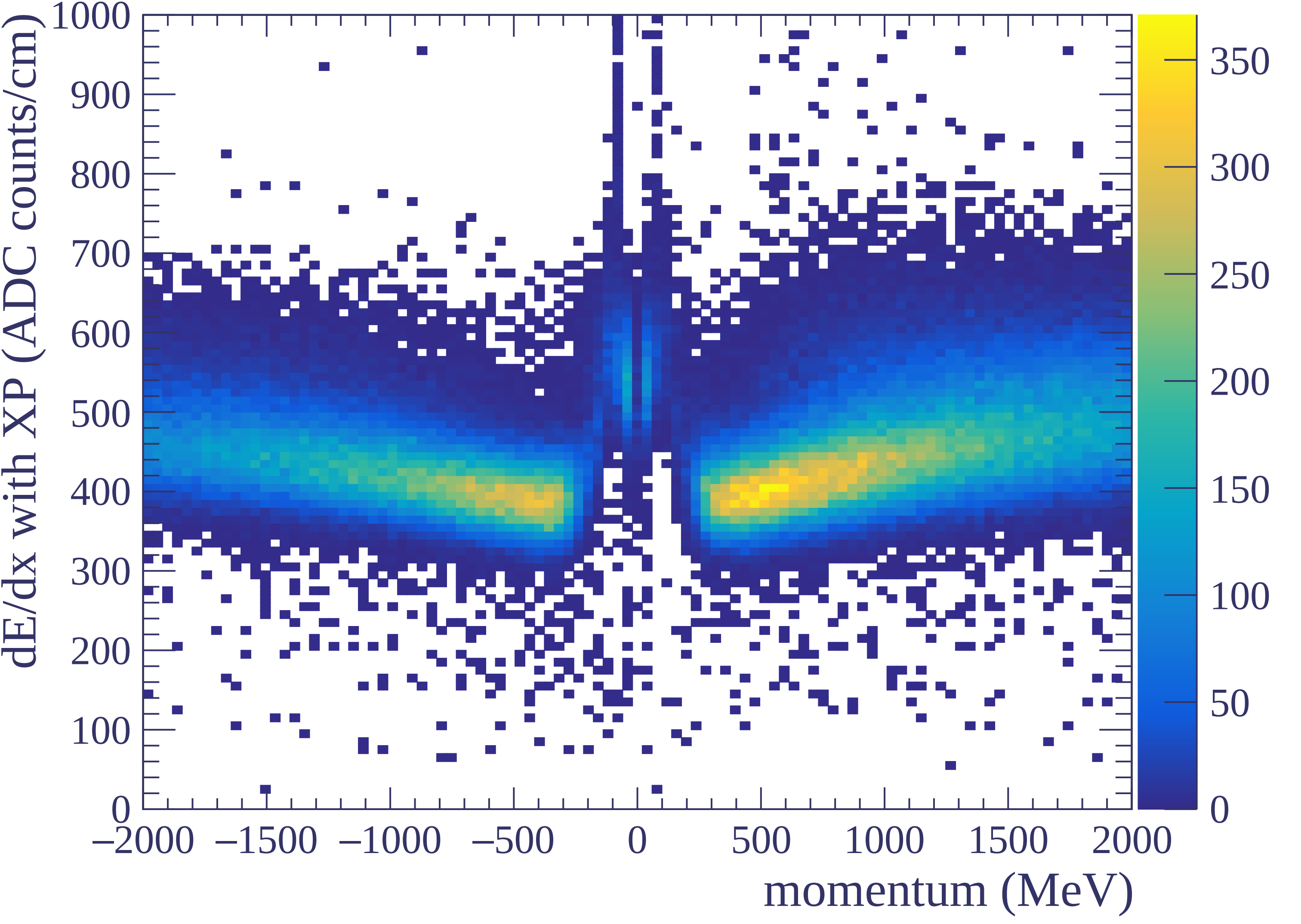

Waveforms sum
Crossed pads
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)

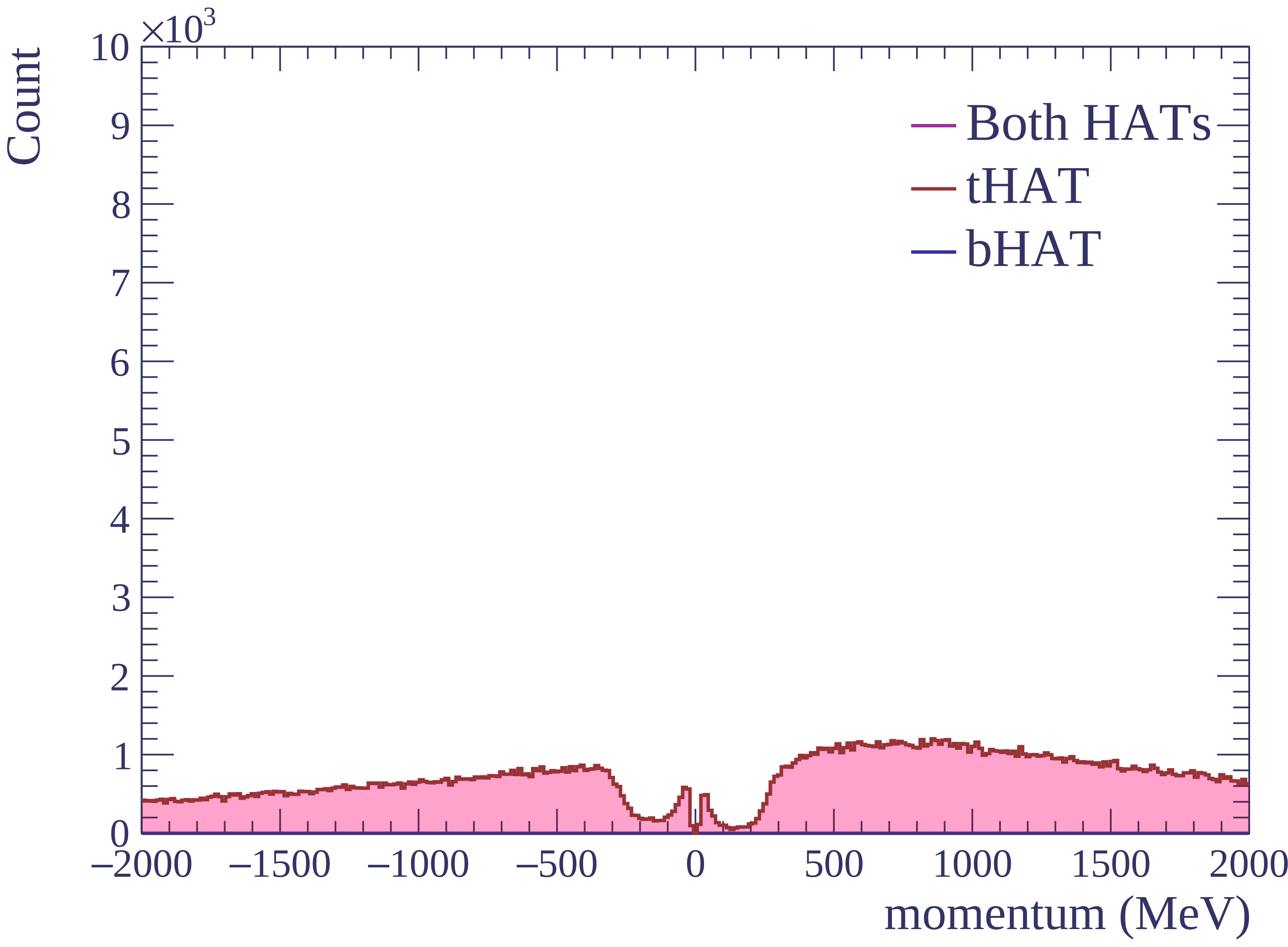


Standard deviation (ADC counts/cm)
Momentum (MeV/c)
Waveforms sum
Crossed pads
0
10
20
30
40
50
60
70
80
90
100
−2000
−1500
−1000
−500
0
500
1000
1500
2000

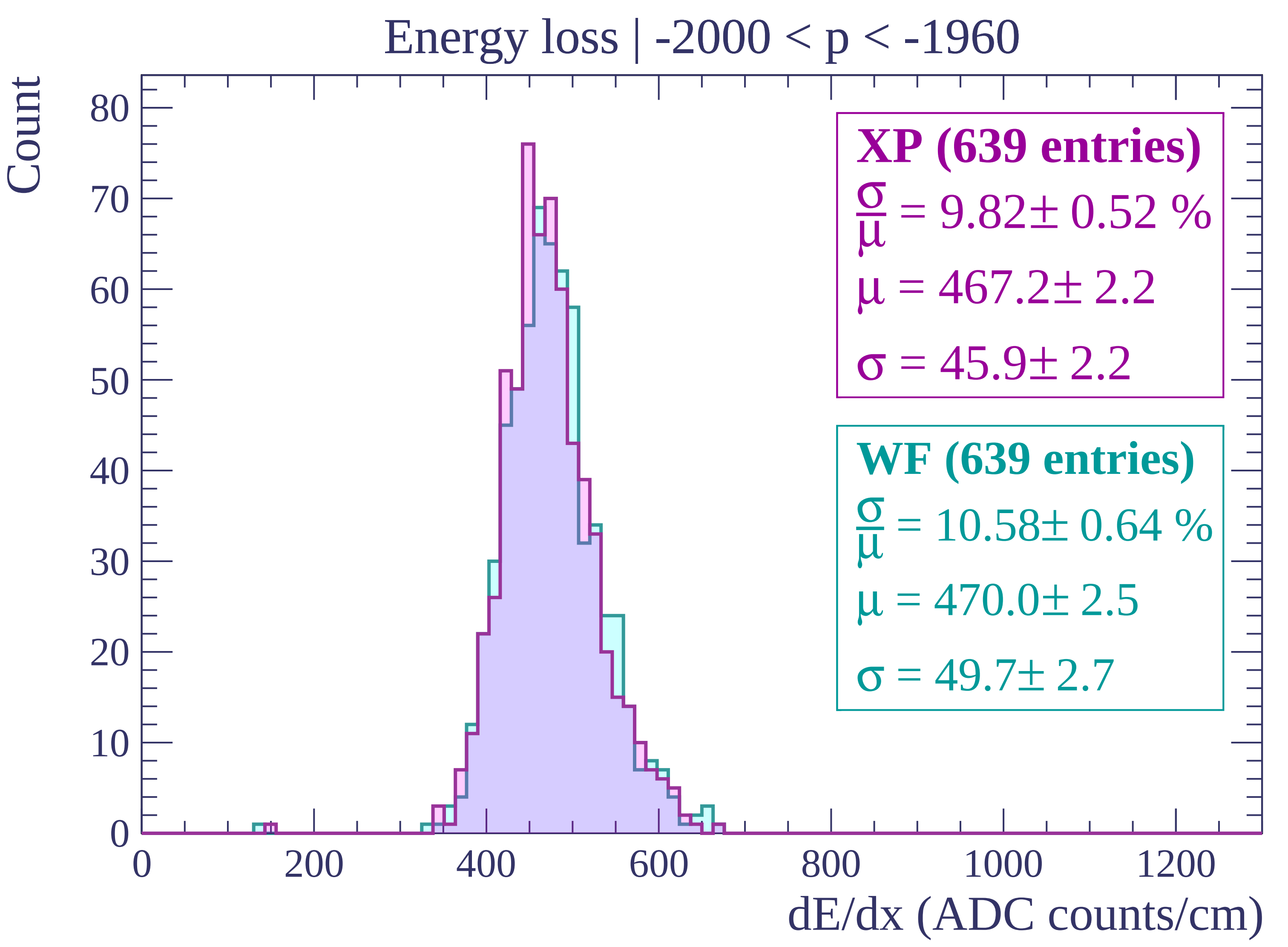


dE/dx with WF (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
50
100
150
200
250
300
350

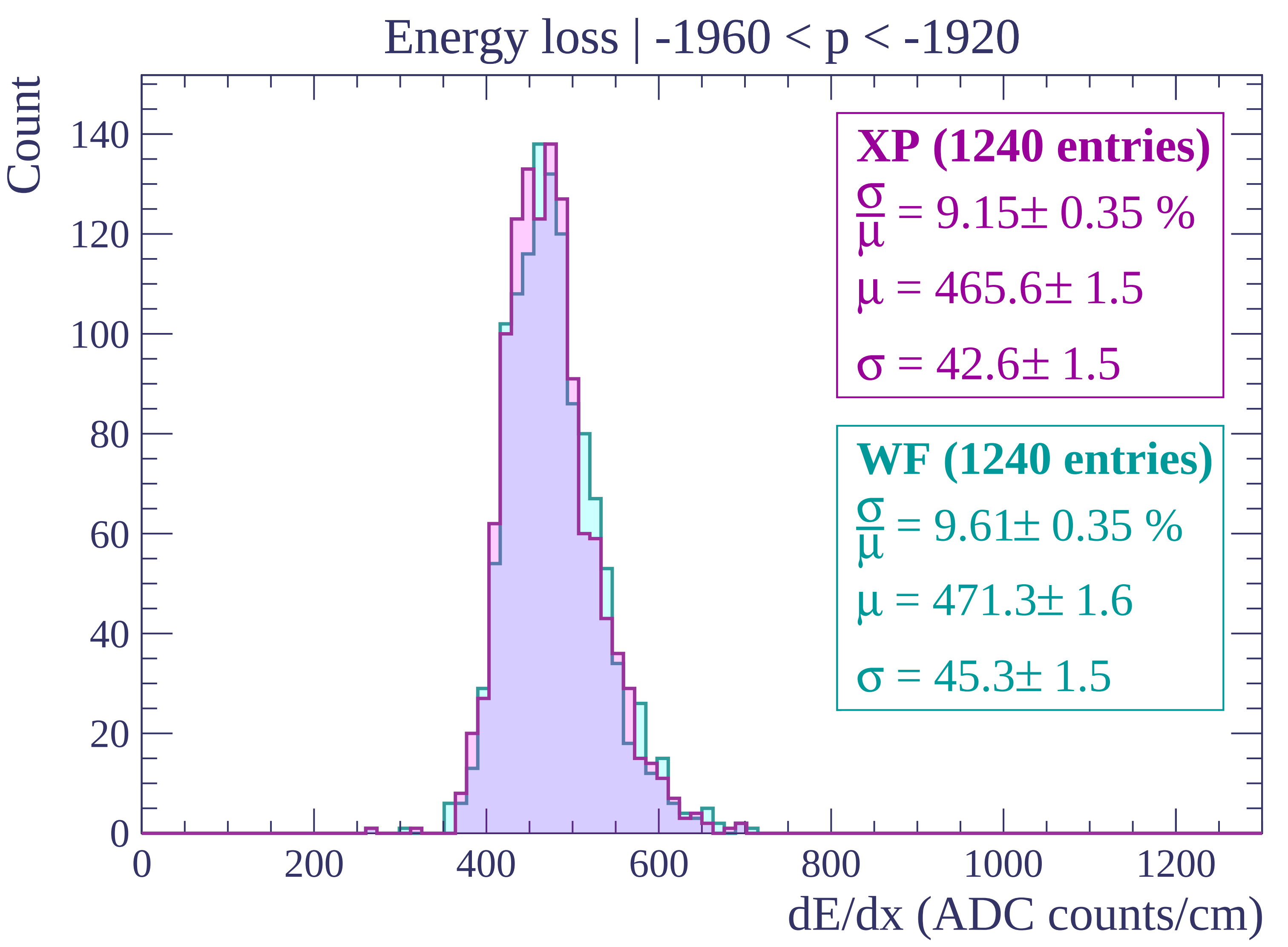

dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
350
300
250
200
150
100
50
0

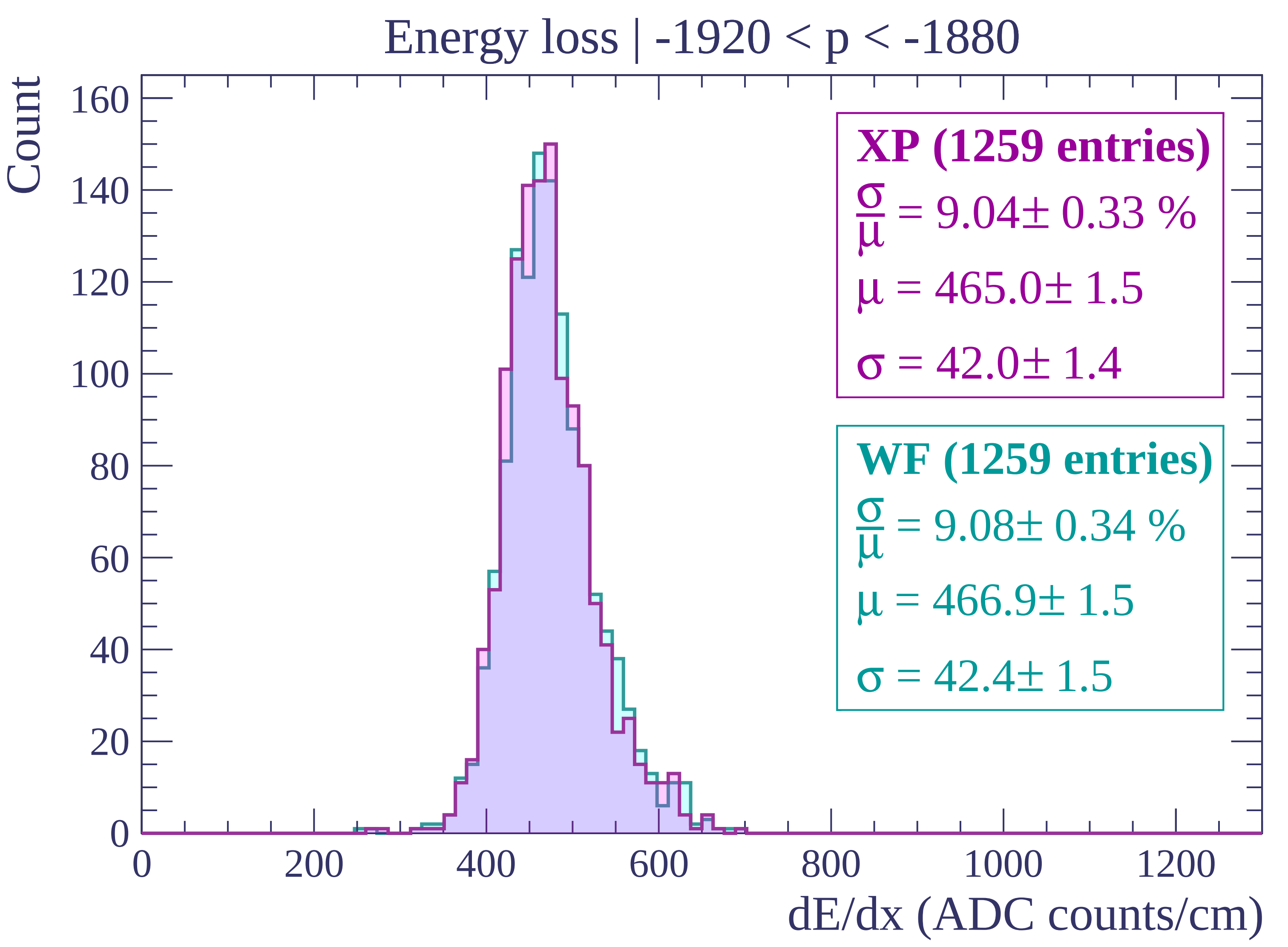


Count
×10^3
10
9
8
7
6
5
4
3
2
1
0
Both HATs
tHAT
bHAT
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)

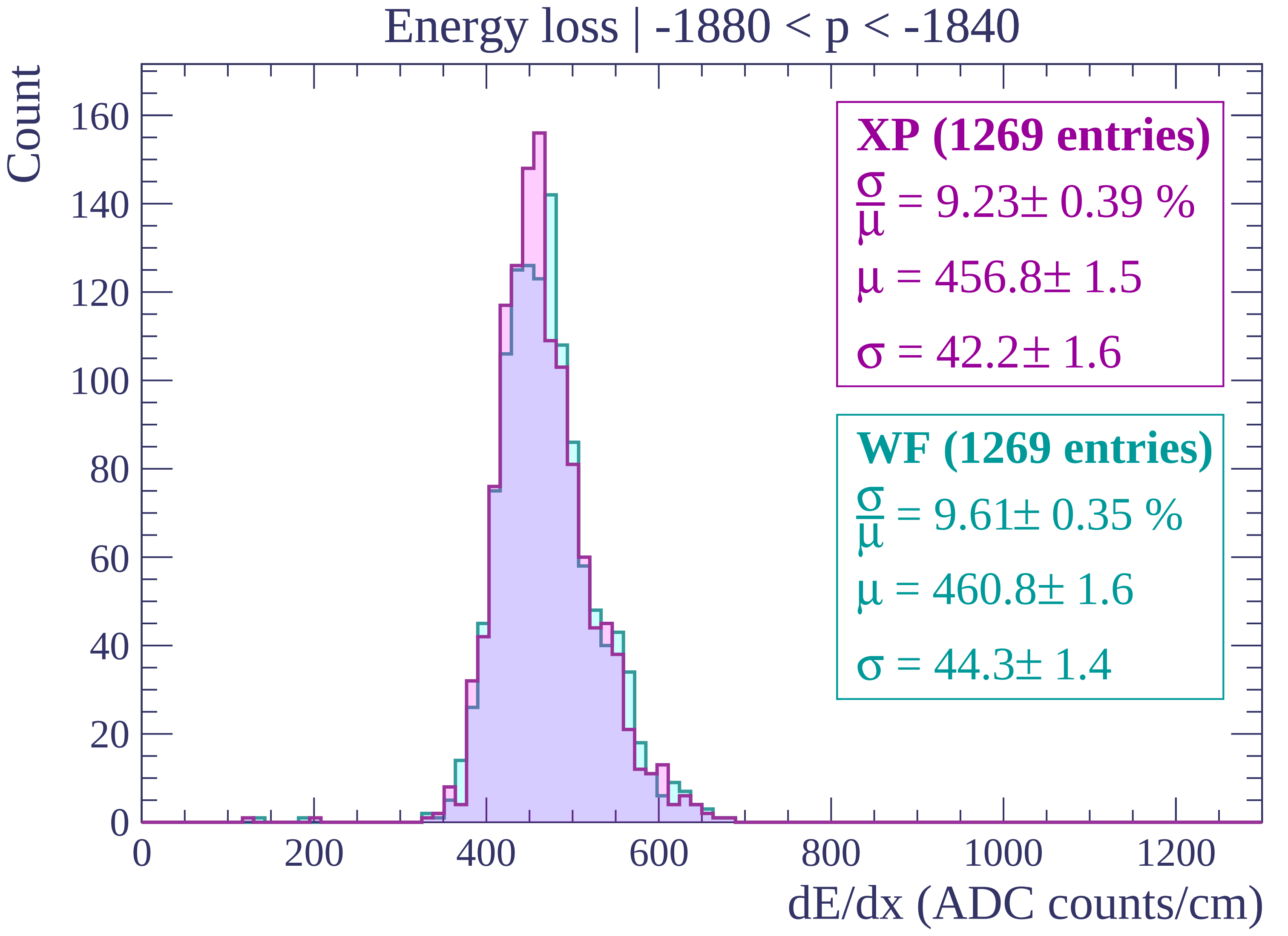


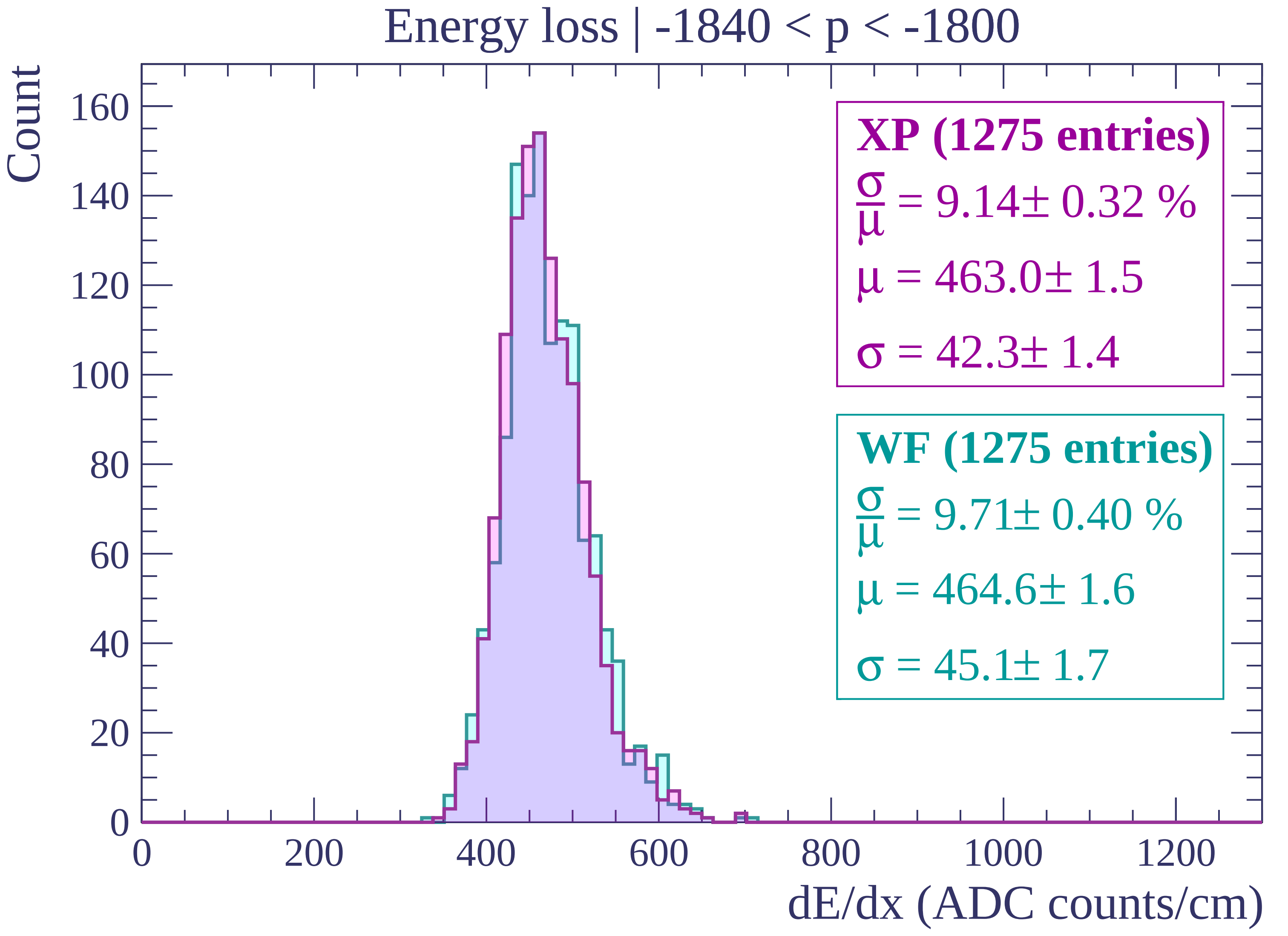
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (639 entries)
σ/μ = 9.82± 0.52 %
μ = 467.2± 2.2
σ = 45.9± 2.2
WF (639 entries)
σ/μ = 10.58± 0.64 %
μ = 470.0± 2.5
σ = 49.7± 2.7

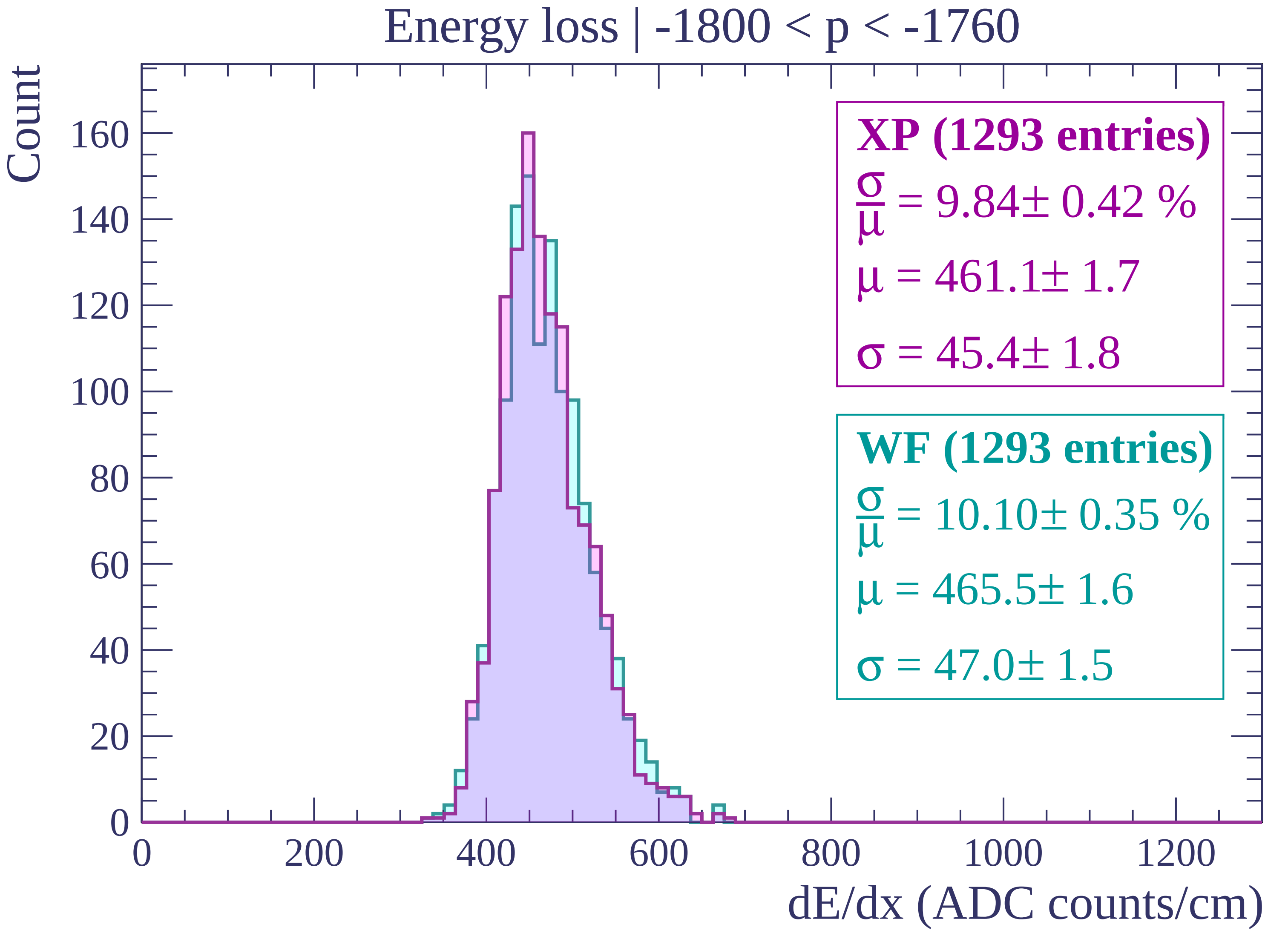

Energy loss | -1960 < p < -1920
Count
XP (1240 entries)
σ/μ = 9.15± 0.35 %
μ = 465.6± 1.5
σ = 42.6± 1.5
WF (1240 entries)
σ/μ = 9.61± 0.35 %
μ = 471.3± 1.6
σ = 45.3± 1.5
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

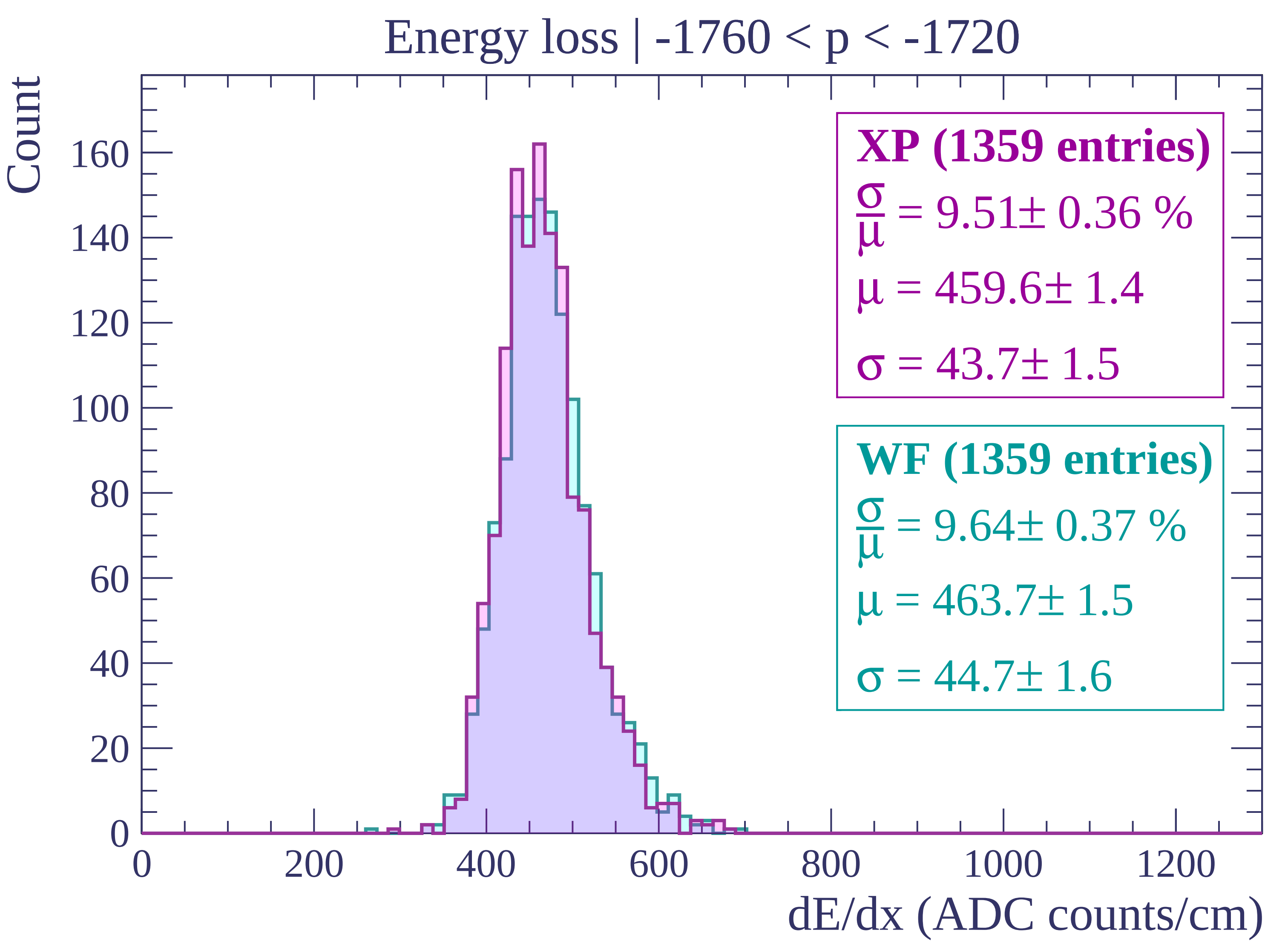


Energy loss | -1920 < p < -1880
Count
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1259 entries)
σ/μ = 9.04± 0.33 %
μ = 465.0± 1.5
σ = 42.0± 1.4
WF (1259 entries)
σ/μ = 9.08± 0.34 %
μ = 466.9± 1.5
σ = 42.4± 1.5

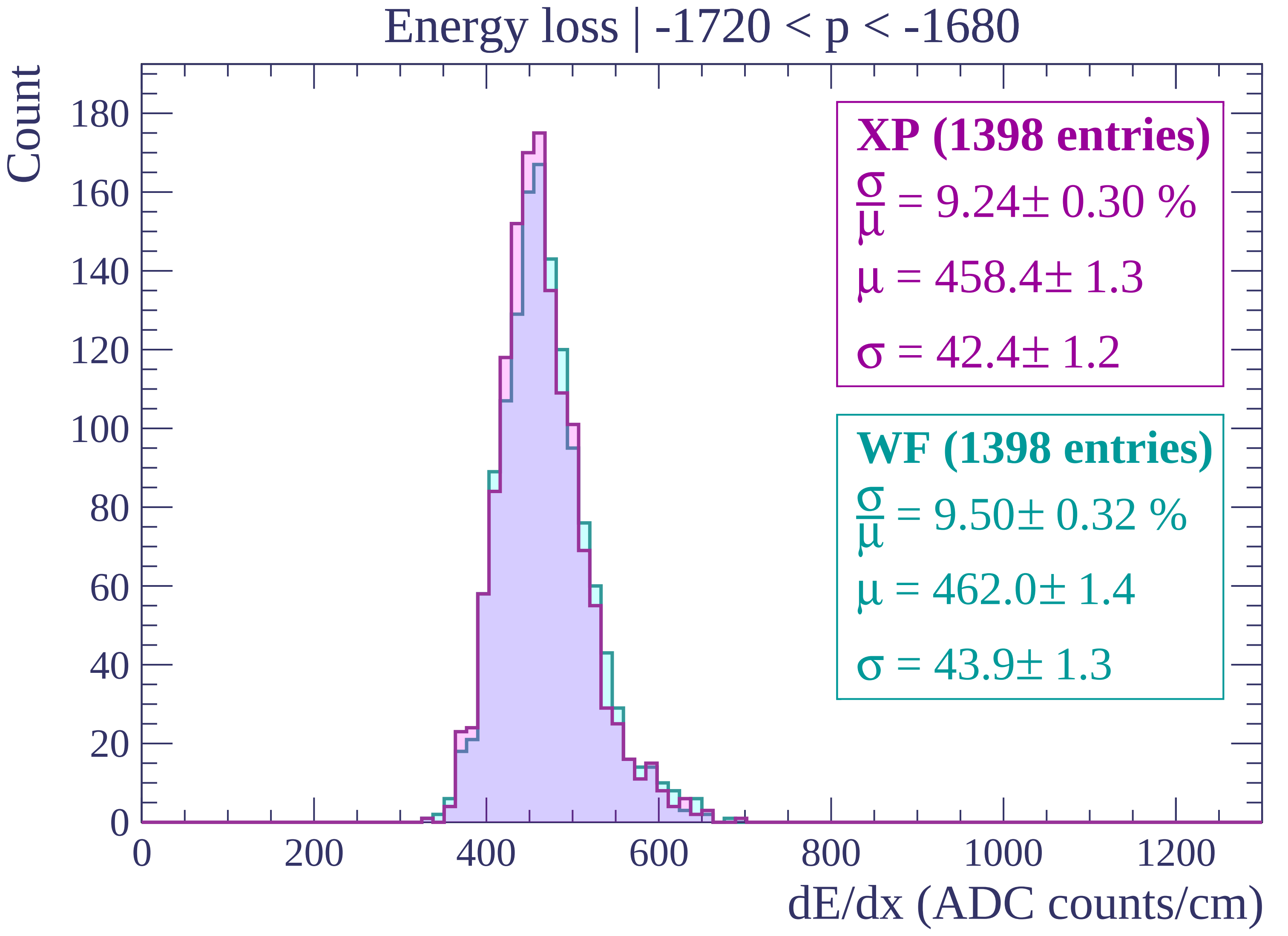

Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1269 entries)
σ/μ = 9.23± 0.39 %
μ = 456.8± 1.5
σ = 42.2± 1.6
WF (1269 entries)
σ/μ = 9.61± 0.35 %
μ = 460.8± 1.6
σ = 44.3± 1.4


Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
XP (1275 entries)
σ/μ = 9.14± 0.32 %
μ = 463.0± 1.5
σ = 42.3± 1.4
WF (1275 entries)
σ/μ = 9.71± 0.40 %
μ = 464.6± 1.6
σ = 45.1± 1.7
0
20
40
60
80
100
120
140
160
200
400
600
800
1000
1200

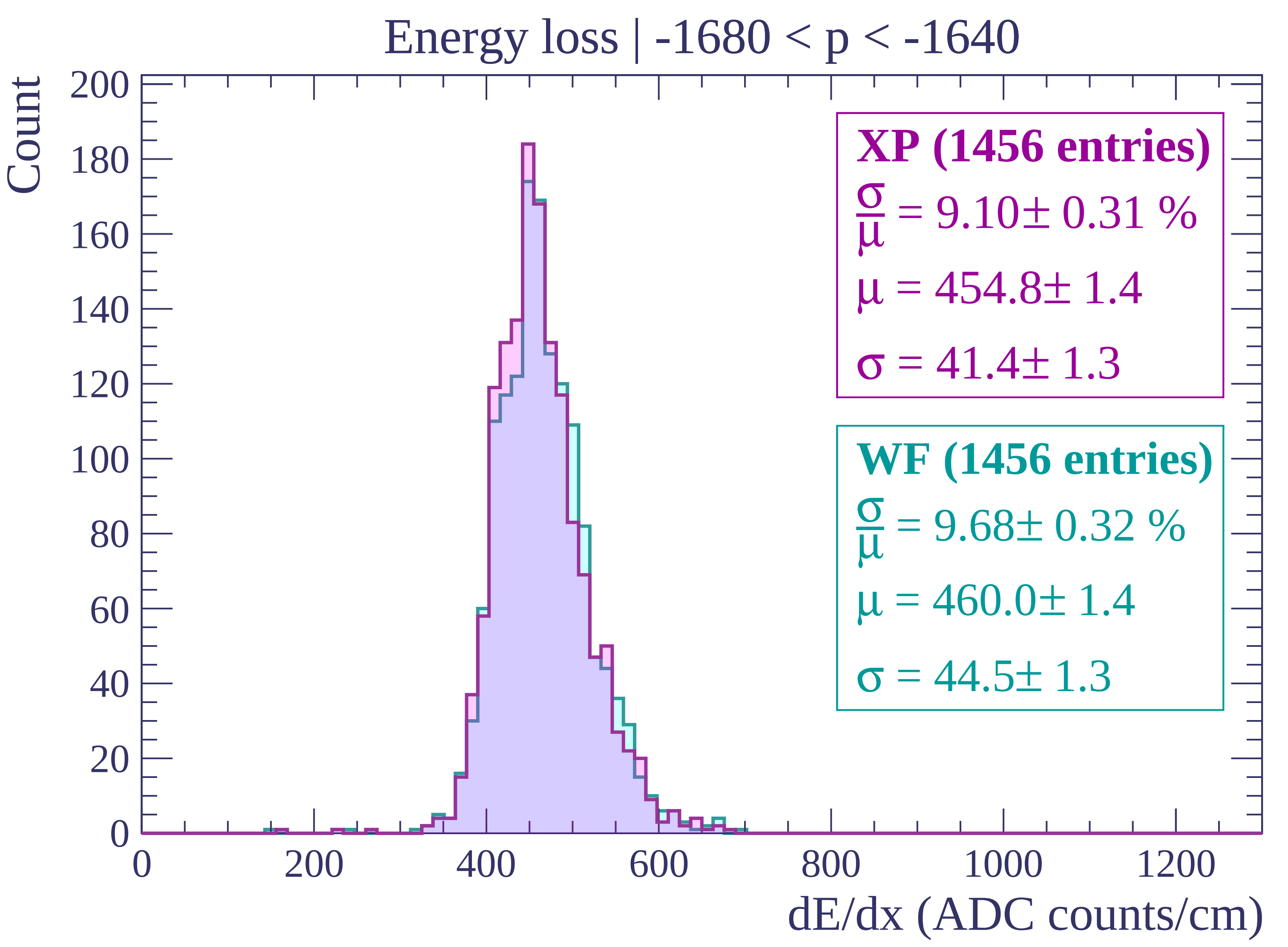

Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1293 entries)
σ/μ = 9.84± 0.42 %
μ = 461.1± 1.7
σ = 45.4± 1.8
WF (1293 entries)
σ/μ = 10.10± 0.35 %
μ = 465.5± 1.6
σ = 47.0± 1.5

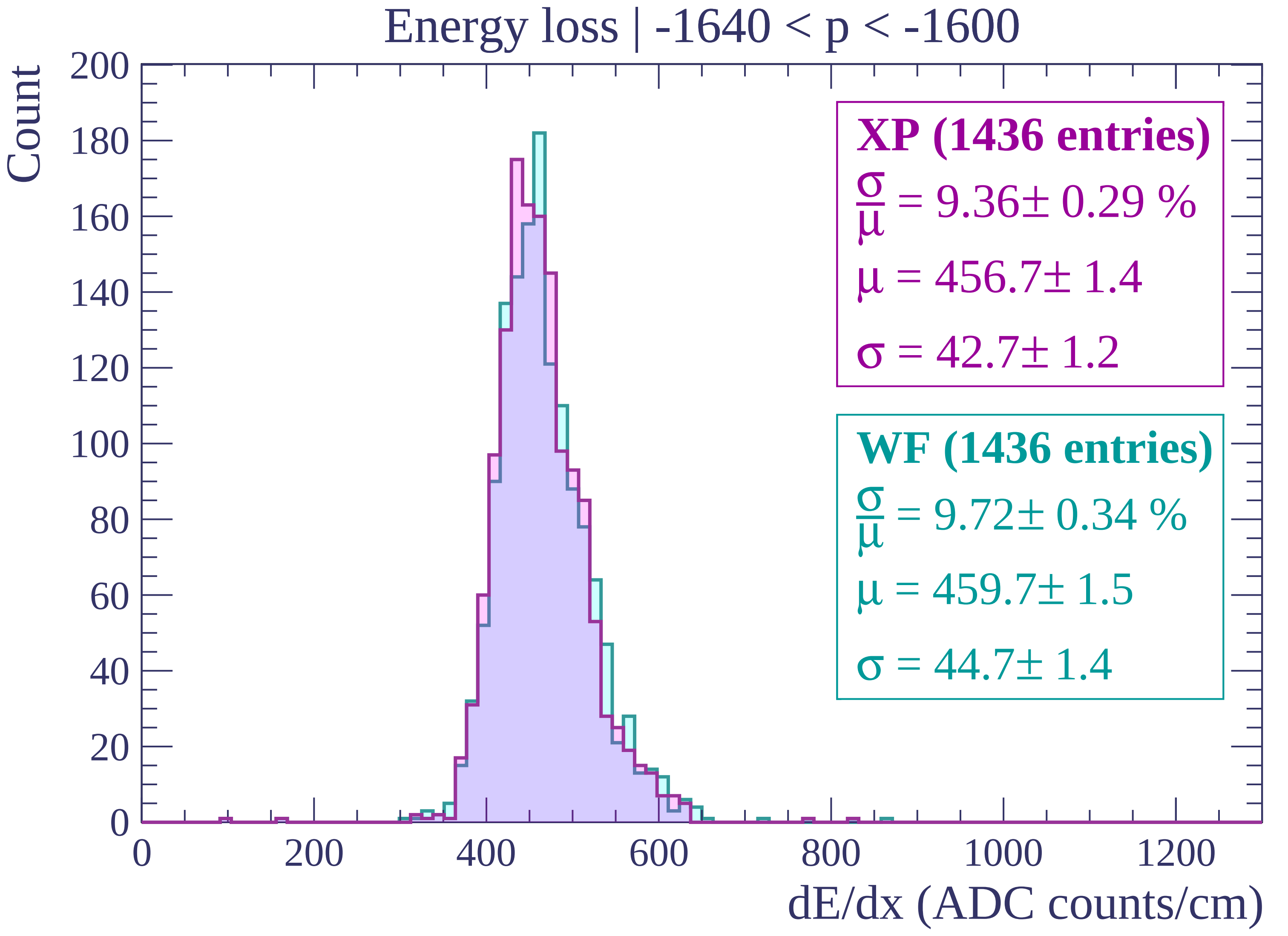


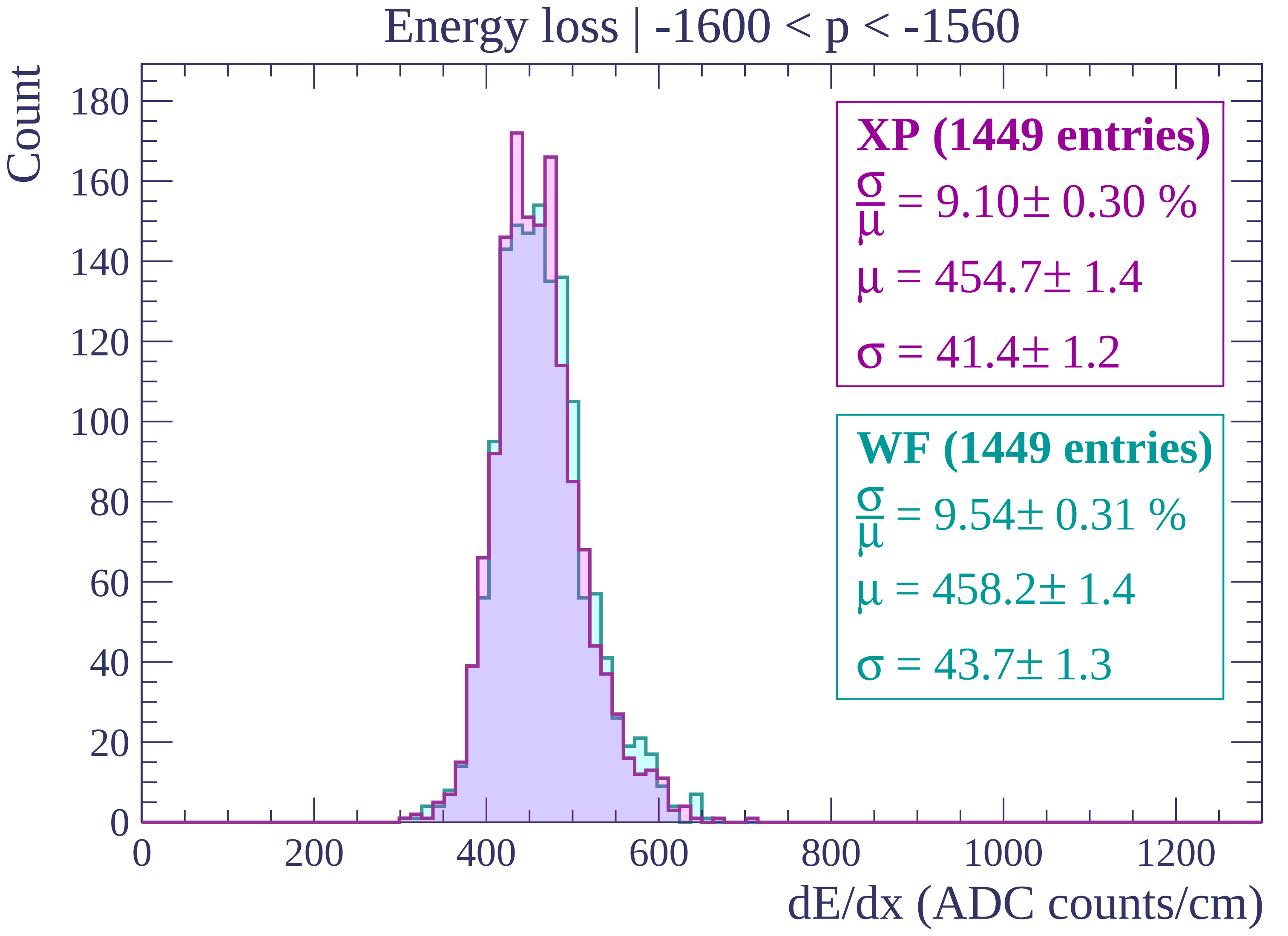
Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
XP (1359 entries)
σ/μ = 9.51± 0.36 %
μ = 459.6± 1.4
σ = 43.7± 1.5
WF (1359 entries)
σ/μ = 9.64± 0.37 %
μ = 463.7± 1.5
σ = 44.7± 1.6



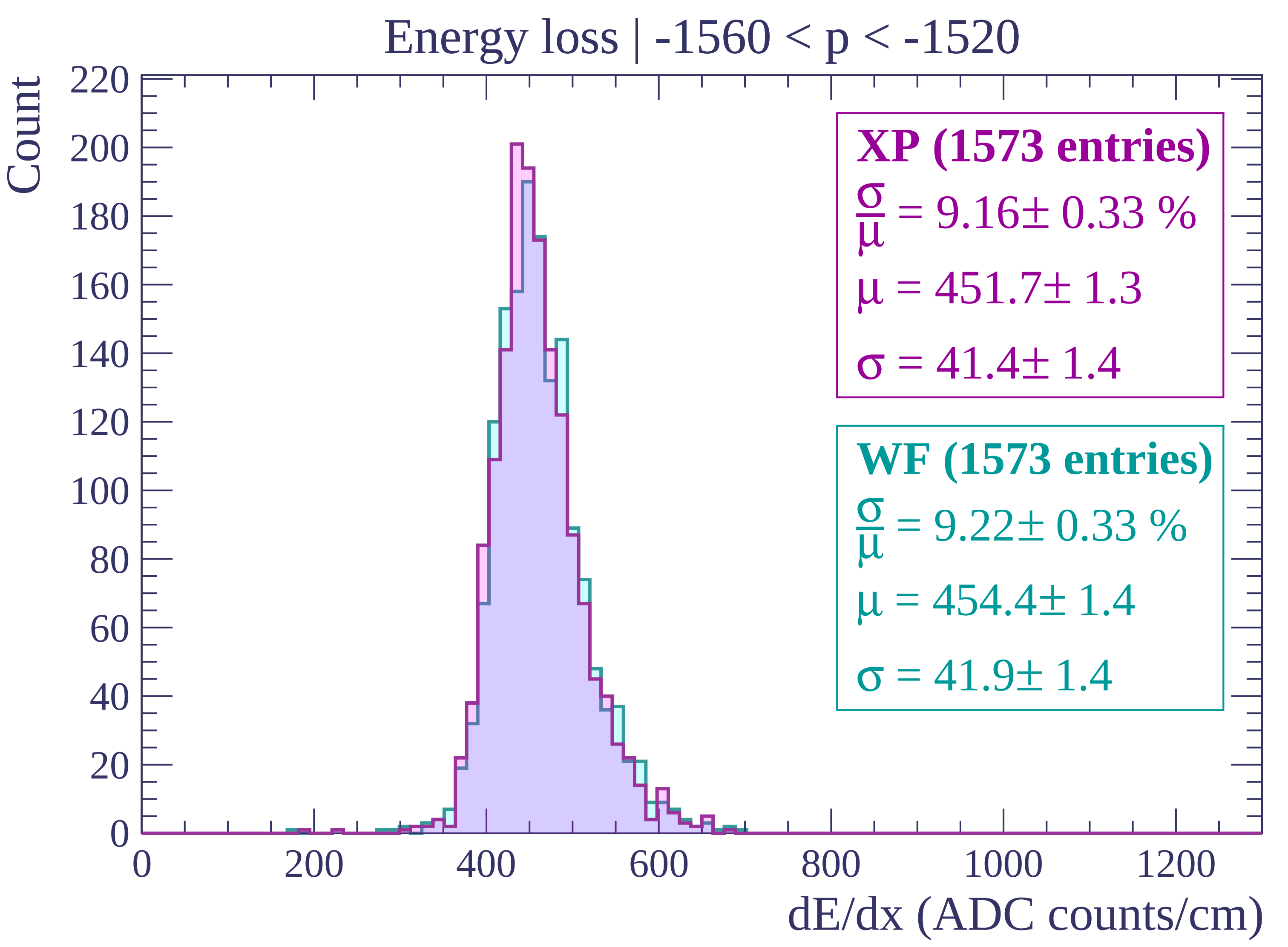
Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP (1398 entries)
σ/μ = 9.24± 0.30 %
μ = 458.4± 1.3
σ = 42.4± 1.2
WF (1398 entries)
σ/μ = 9.50± 0.32 %
μ = 462.0± 1.4
σ = 43.9± 1.3

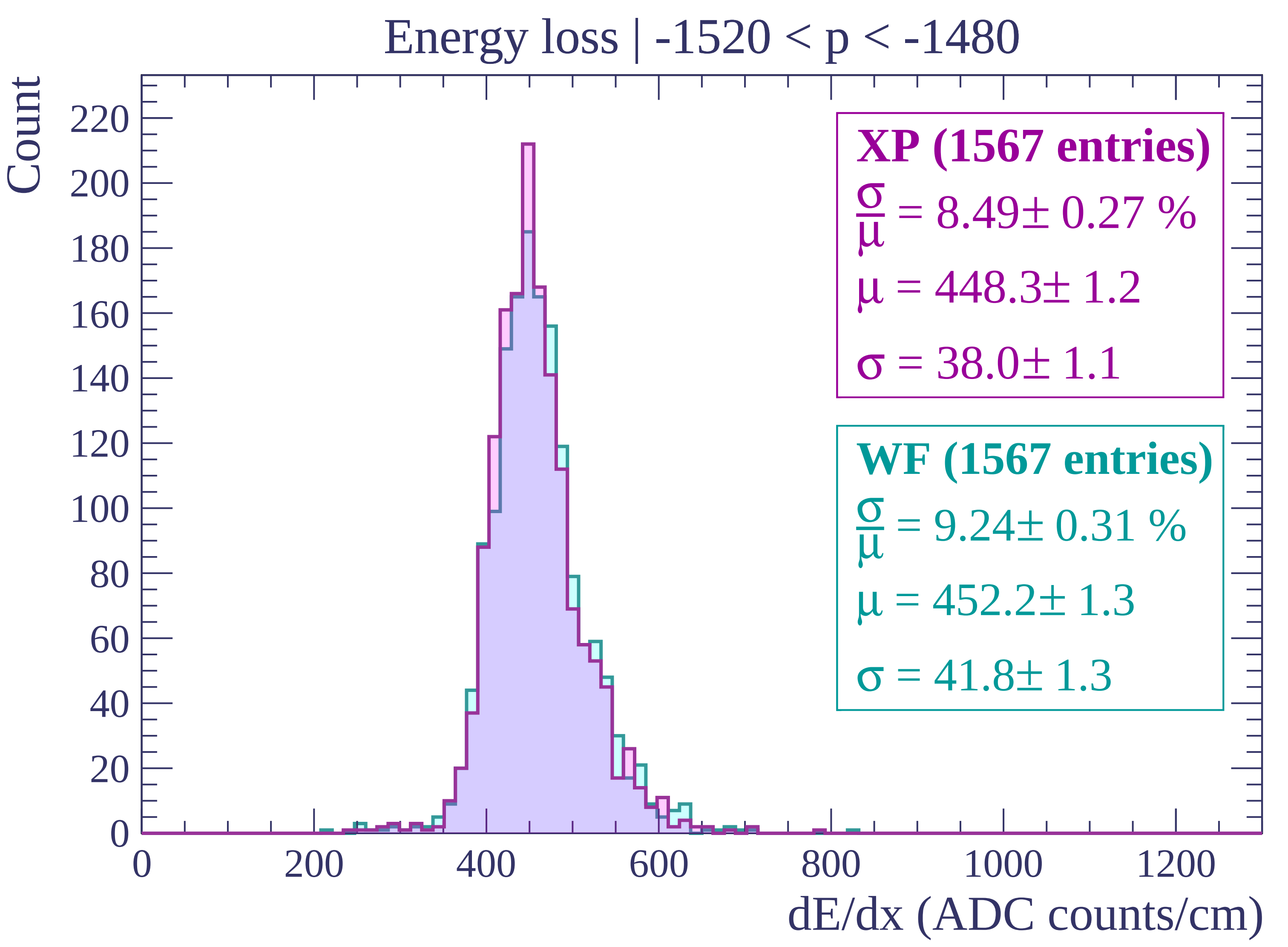

Energy loss | -1680 < p < -1640
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (1456 entries)
$\frac{\sigma}{\mu}$ = 9.10± 0.31 %
μ = 454.8± 1.4
σ = 41.4± 1.3
WF (1456 entries)
$\frac{\sigma}{\mu}$ = 9.68± 0.32 %
μ = 460.0± 1.4
σ = 44.5± 1.3

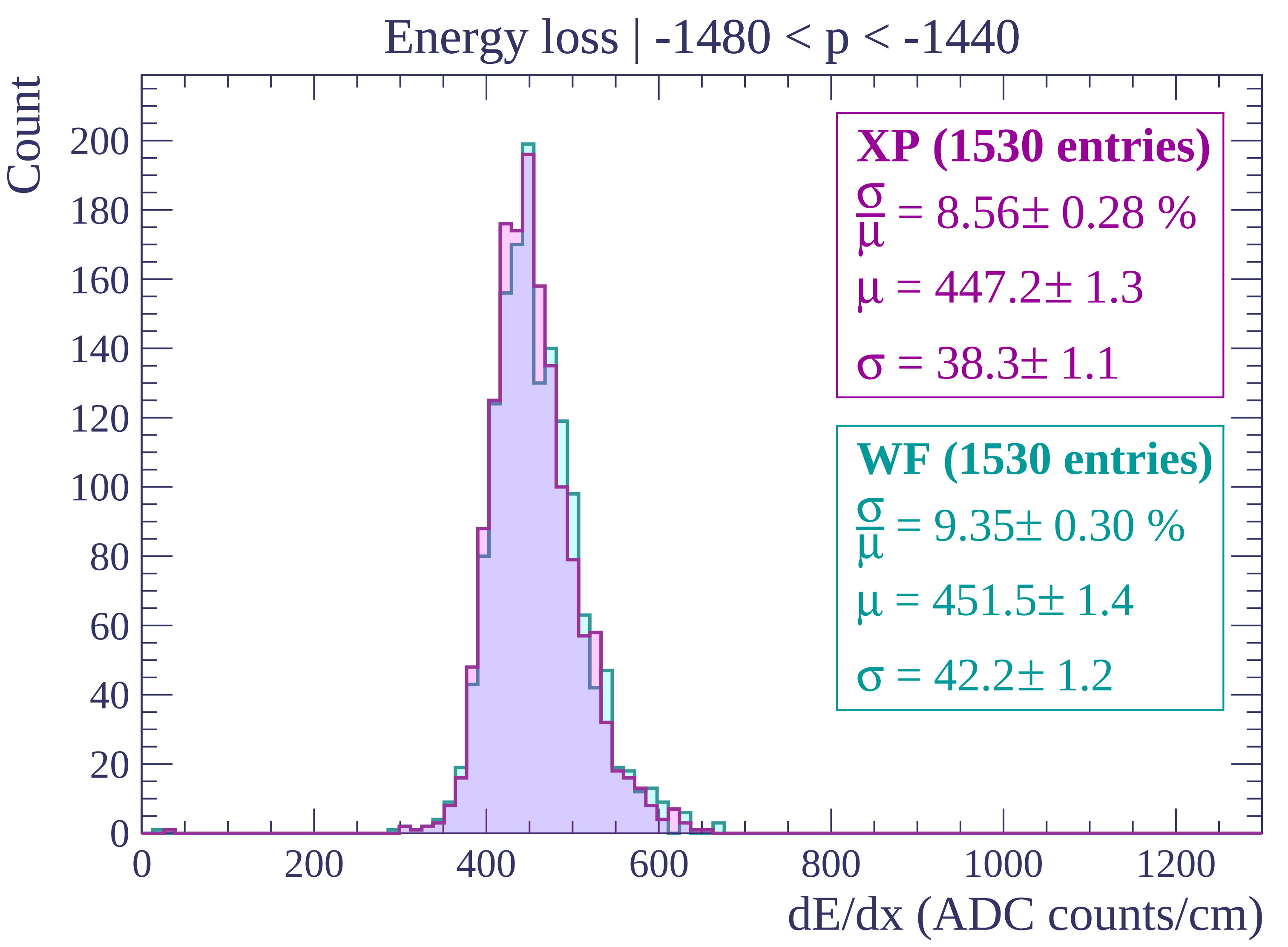

Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP (1436 entries)
σ/μ = 9.36± 0.29 %
μ = 456.7± 1.4
σ = 42.7± 1.2
WF (1436 entries)
σ/μ = 9.72± 0.34 %
μ = 459.7± 1.5
σ = 44.7± 1.4



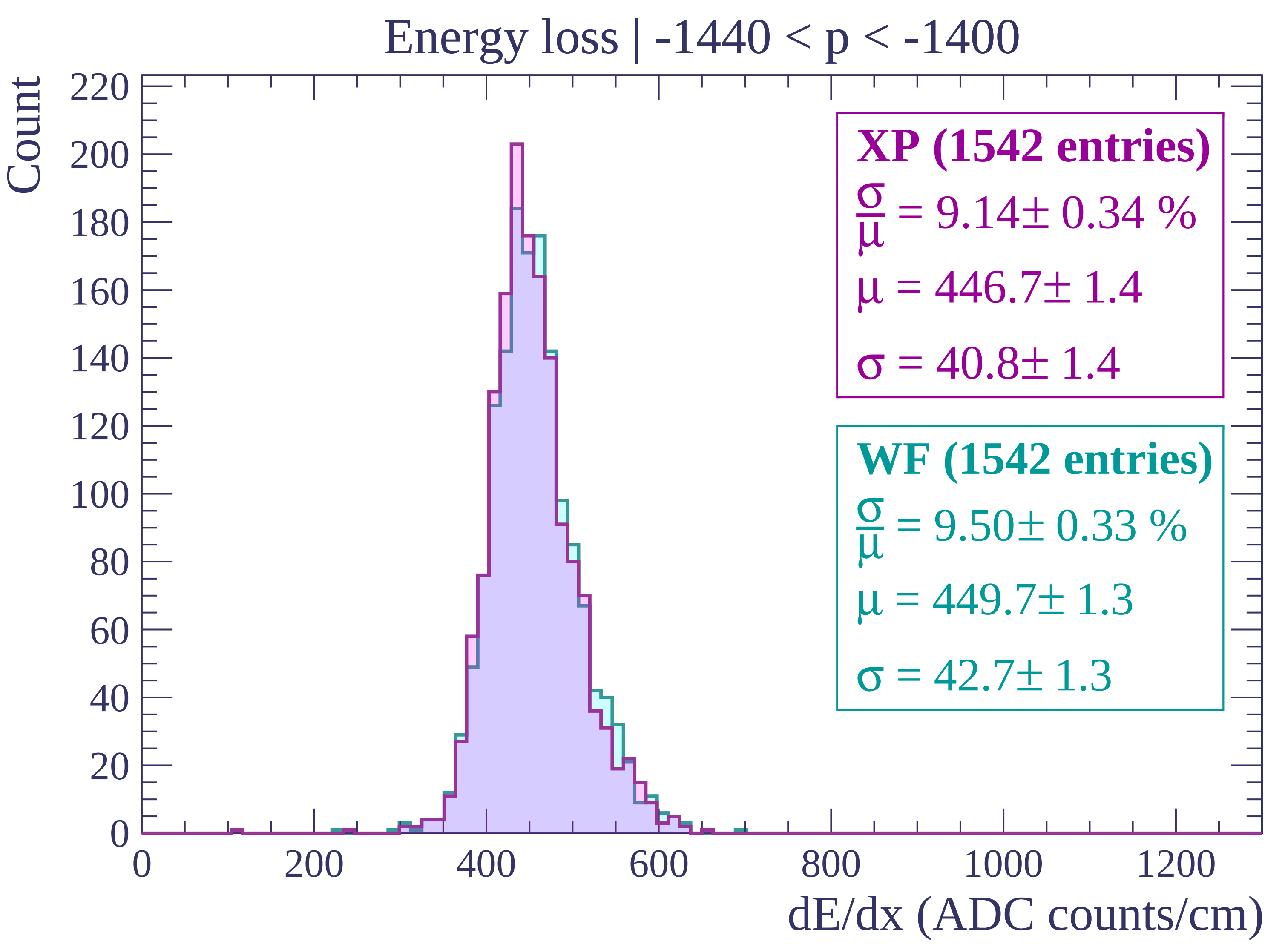
Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP (1449 entries)
σ/μ = 9.10± 0.30 %
μ = 454.7± 1.4
σ = 41.4± 1.2
WF (1449 entries)
σ/μ = 9.54± 0.31 %
μ = 458.2± 1.4
σ = 43.7± 1.3


Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
XP (1573 entries)
σ/μ = 9.16± 0.33 %
μ = 451.7± 1.3
σ = 41.4± 1.4
WF (1573 entries)
σ/μ = 9.22± 0.33 %
μ = 454.4± 1.4
σ = 41.9± 1.4
0
20
40
60
80
100
120
140
160
180
200
220
200
400
600
800
1000
1200

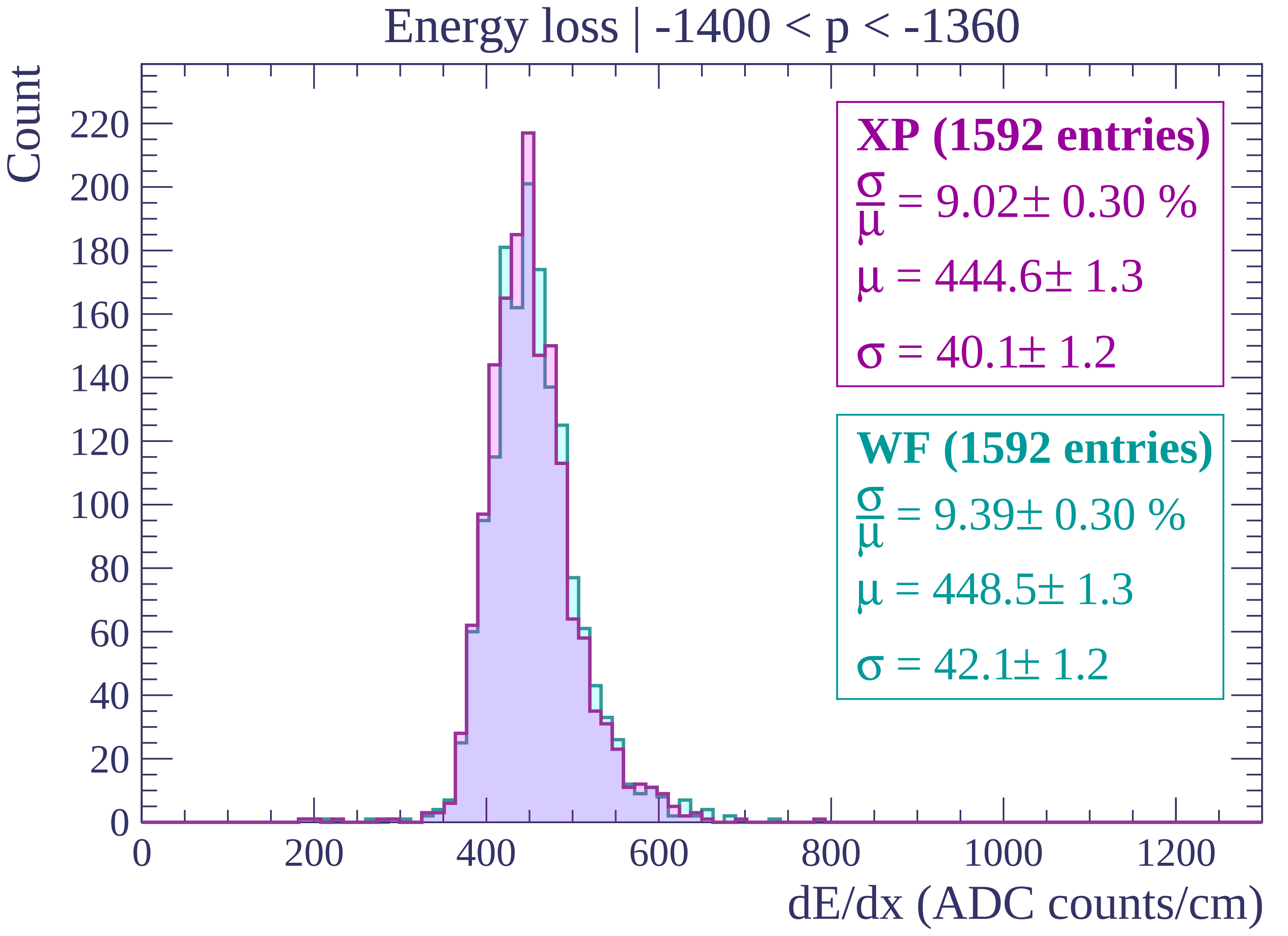

Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP (1567 entries)
σ/μ = 8.49± 0.27 %
μ = 448.3± 1.2
σ = 38.0± 1.1
WF (1567 entries)
σ/μ = 9.24± 0.31 %
μ = 452.2± 1.3
σ = 41.8± 1.3



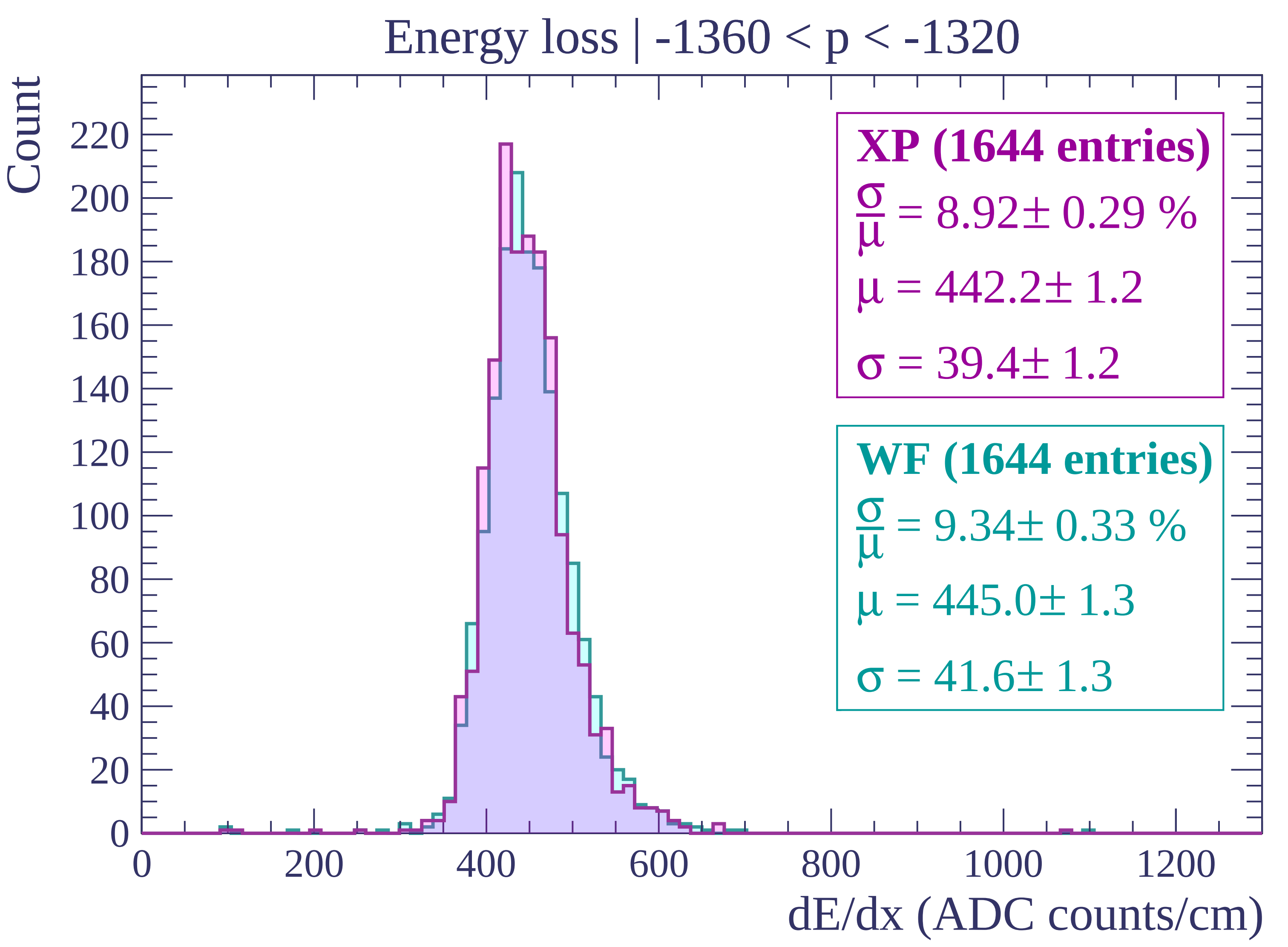
Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (1530 entries)
$\frac{\sigma}{\mu}$ = 8.56± 0.28 %
μ = 447.2± 1.3
σ = 38.3± 1.1
WF (1530 entries)
$\frac{\sigma}{\mu}$ = 9.35± 0.30 %
μ = 451.5± 1.4
σ = 42.2± 1.2



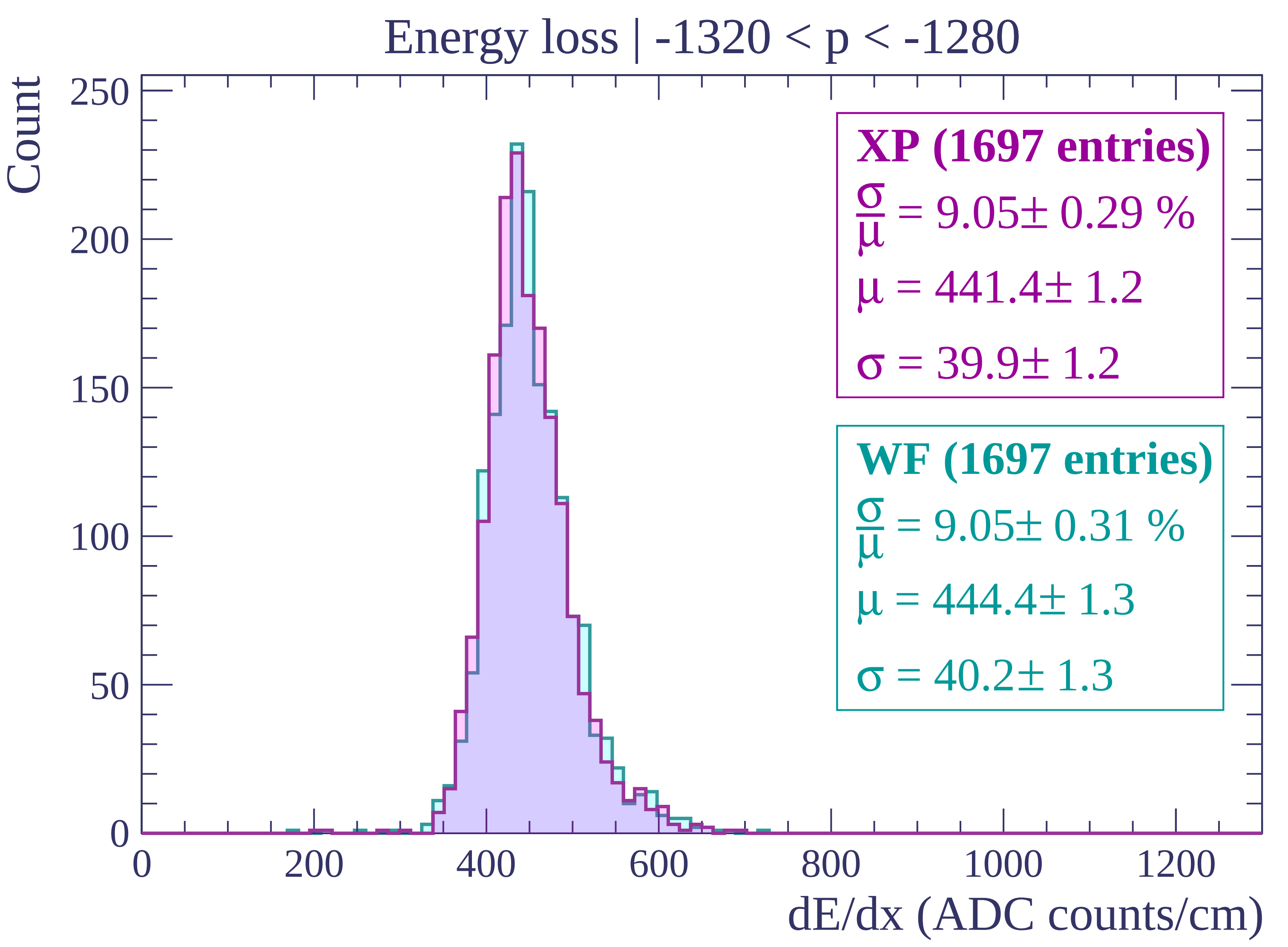
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1542 entries)
$\frac{\sigma}{\mu}$ = 9.14± 0.34 %
μ = 446.7± 1.4
σ = 40.8± 1.4
WF (1542 entries)
$\frac{\sigma}{\mu}$ = 9.50± 0.33 %
μ = 449.7± 1.3
σ = 42.7± 1.3

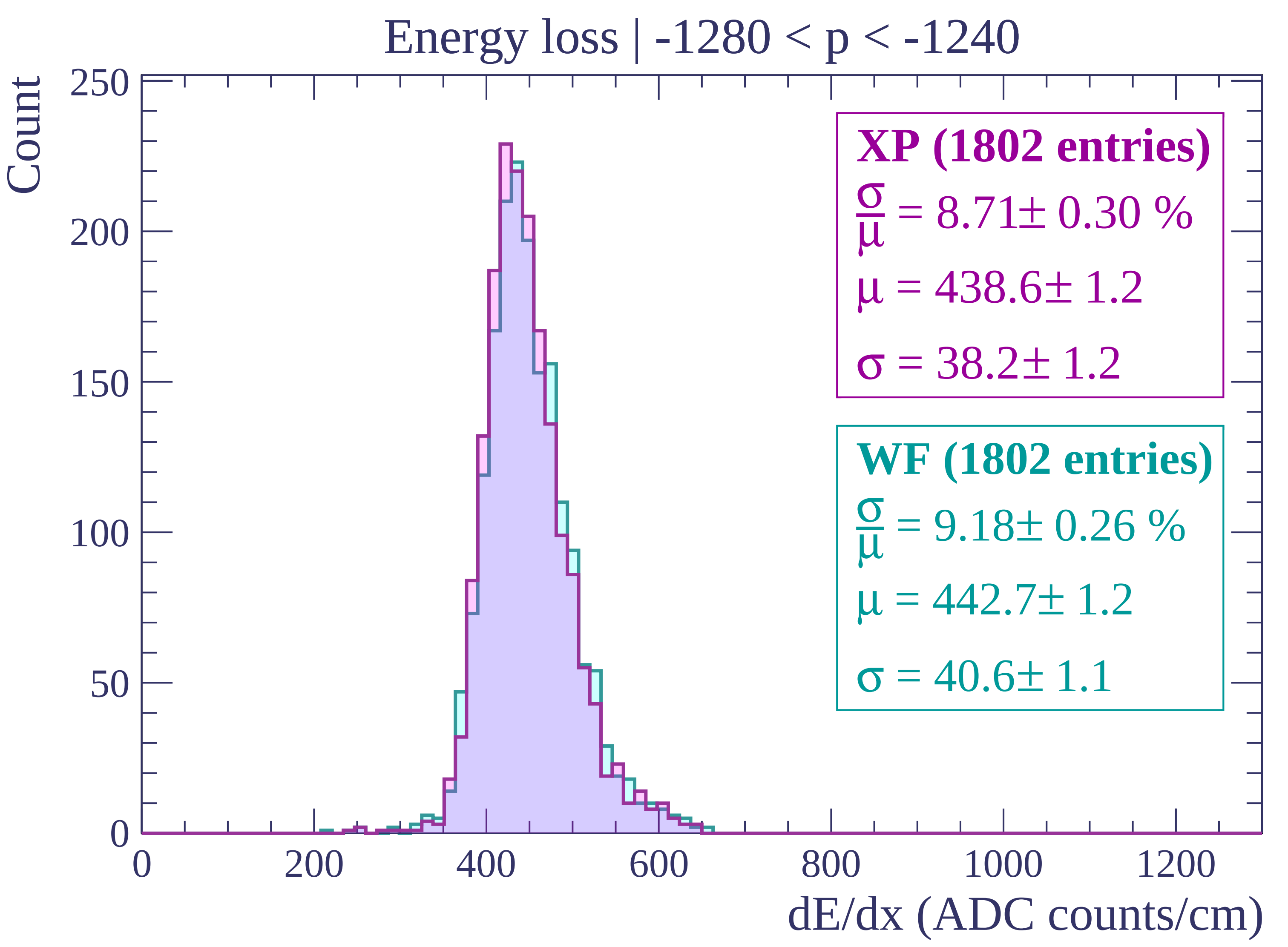


Energy loss | -1400 < p < -1360
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1592 entries)
$\frac{\sigma}{\mu}$ = 9.02± 0.30 %
μ = 444.6± 1.3
σ = 40.1± 1.2
WF (1592 entries)
$\frac{\sigma}{\mu}$ = 9.39± 0.30 %
μ = 448.5± 1.3
σ = 42.1± 1.2

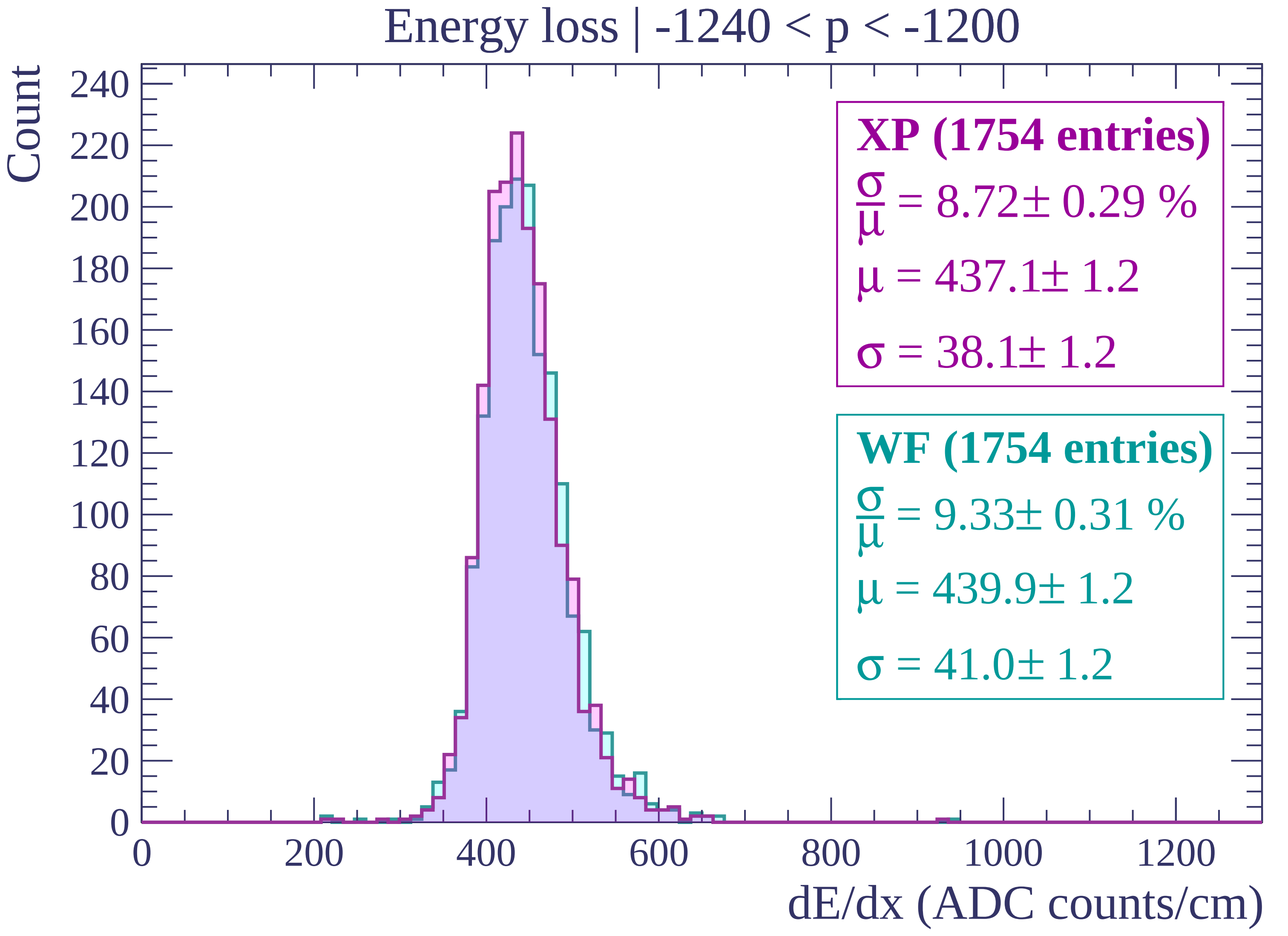


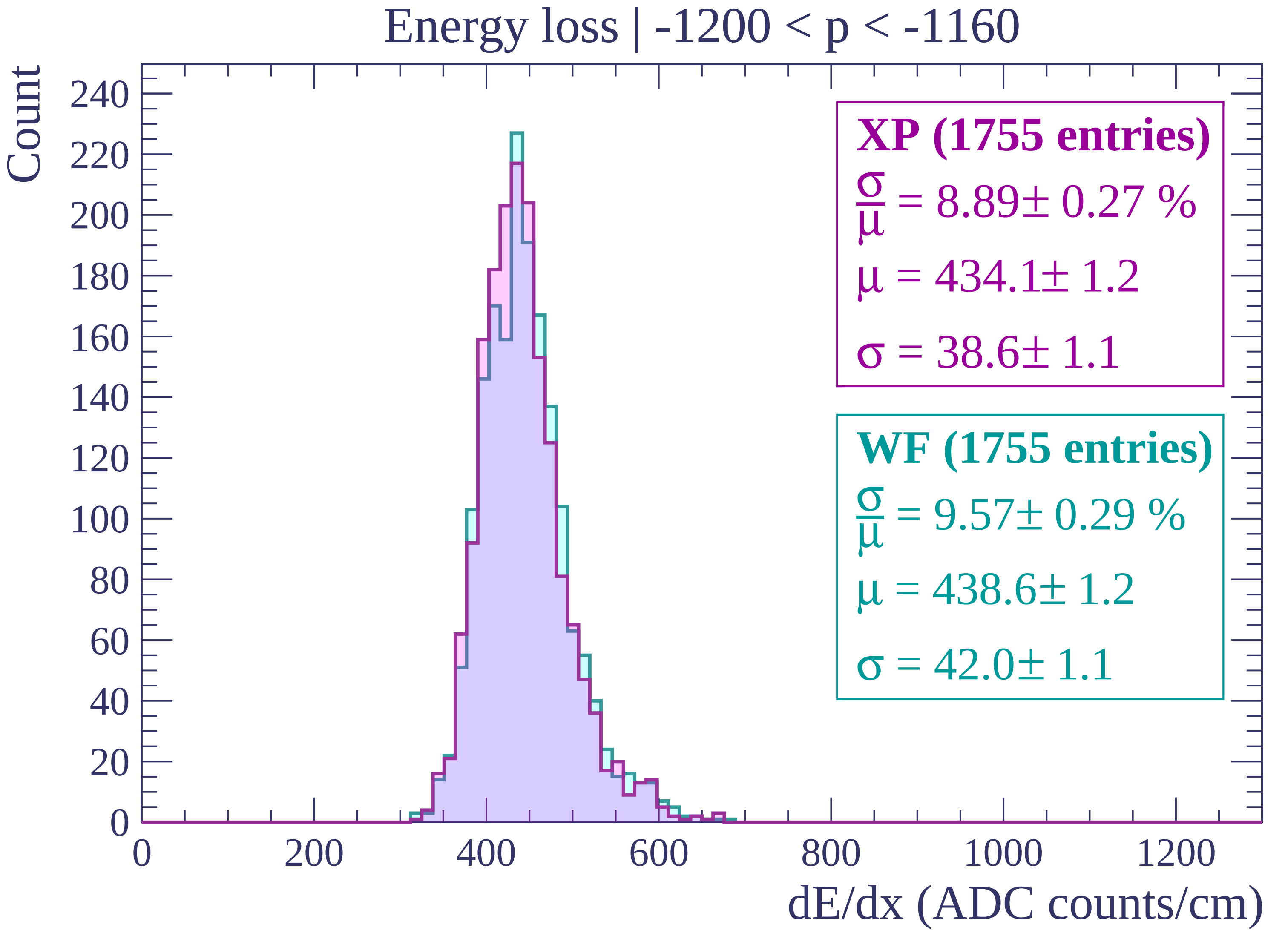
Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
XP (1644 entries)
σ/μ = 8.92± 0.29 %
μ = 442.2± 1.2
σ = 39.4± 1.2
WF (1644 entries)
σ/μ = 9.34± 0.33 %
μ = 445.0± 1.3
σ = 41.6± 1.3


Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
XP (1697 entries)
σ/μ = 9.05± 0.29 %
μ = 441.4± 1.2
σ = 39.9± 1.2
WF (1697 entries)
σ/μ = 9.05± 0.31 %
μ = 444.4± 1.3
σ = 40.2± 1.3
0
50
100
150
200
250
0
200
400
600
800
1000
1200

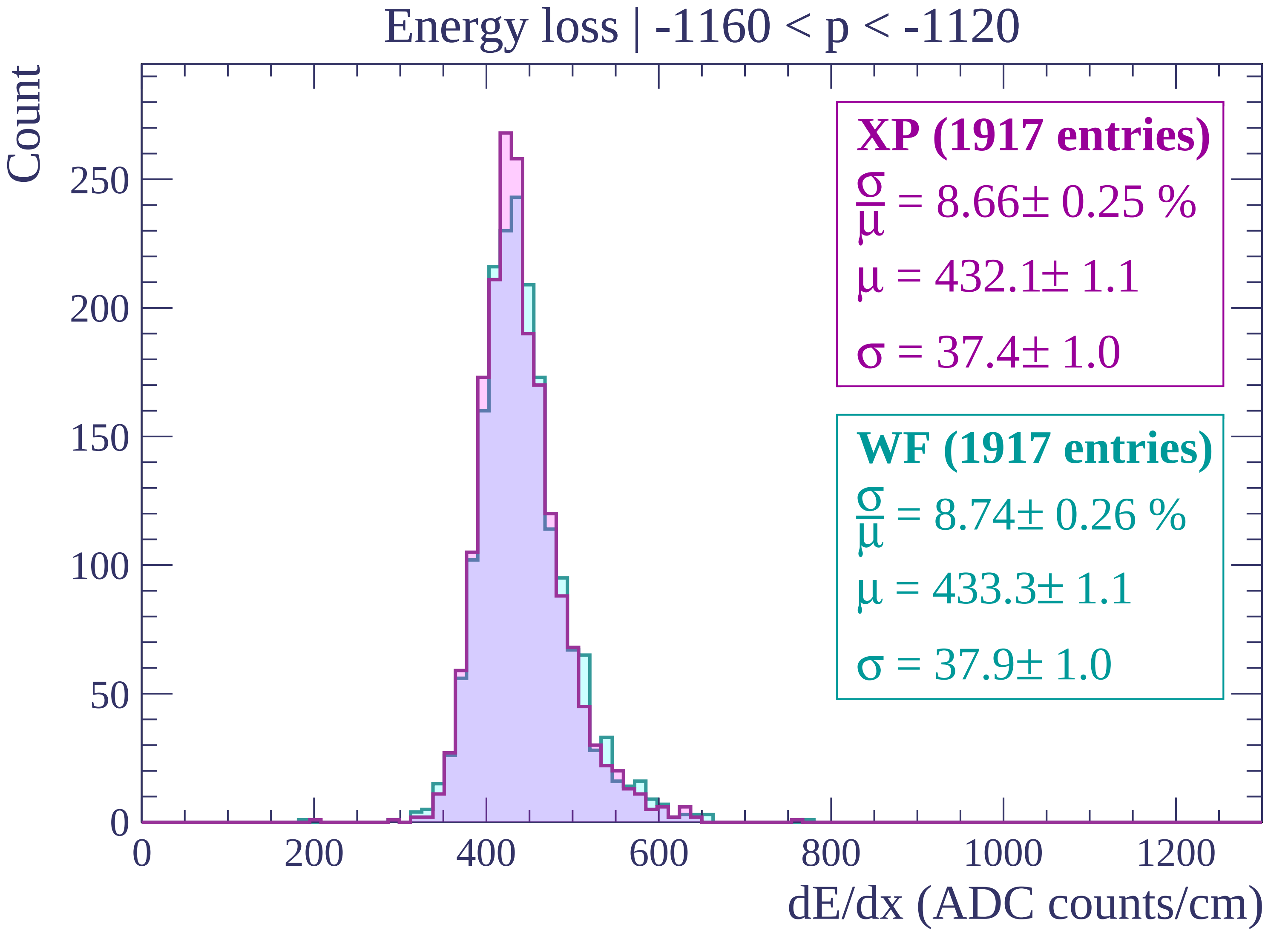


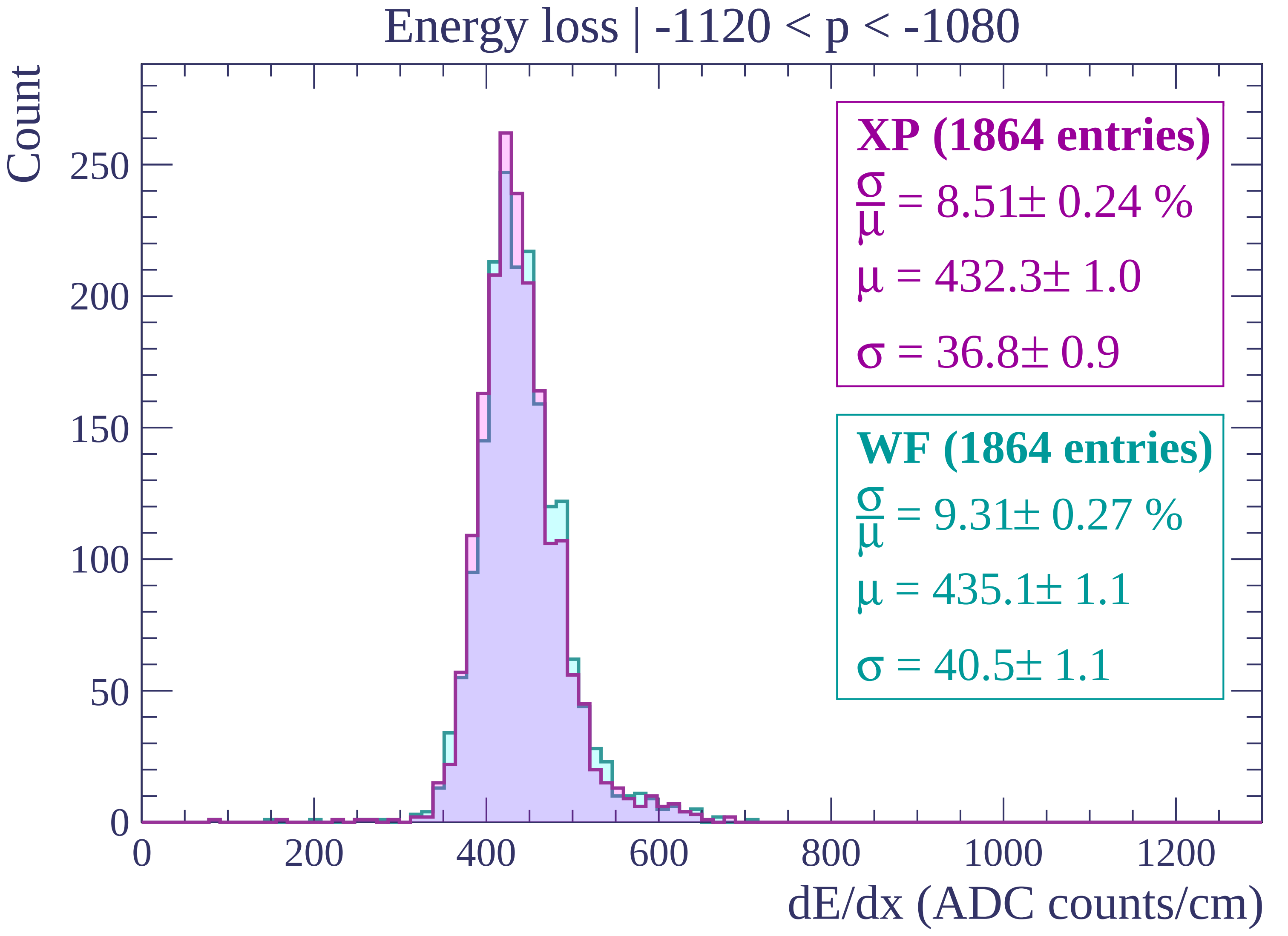
Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP (1802 entries)
σ/μ = 8.71± 0.30 %
μ = 438.6± 1.2
σ = 38.2± 1.2
WF (1802 entries)
σ/μ = 9.18± 0.26 %
μ = 442.7± 1.2
σ = 40.6± 1.1


Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP (1754 entries)
σ/μ = 8.72± 0.29 %
μ = 437.1± 1.2
σ = 38.1± 1.2
WF (1754 entries)
σ/μ = 9.33± 0.31 %
μ = 439.9± 1.2
σ = 41.0± 1.2

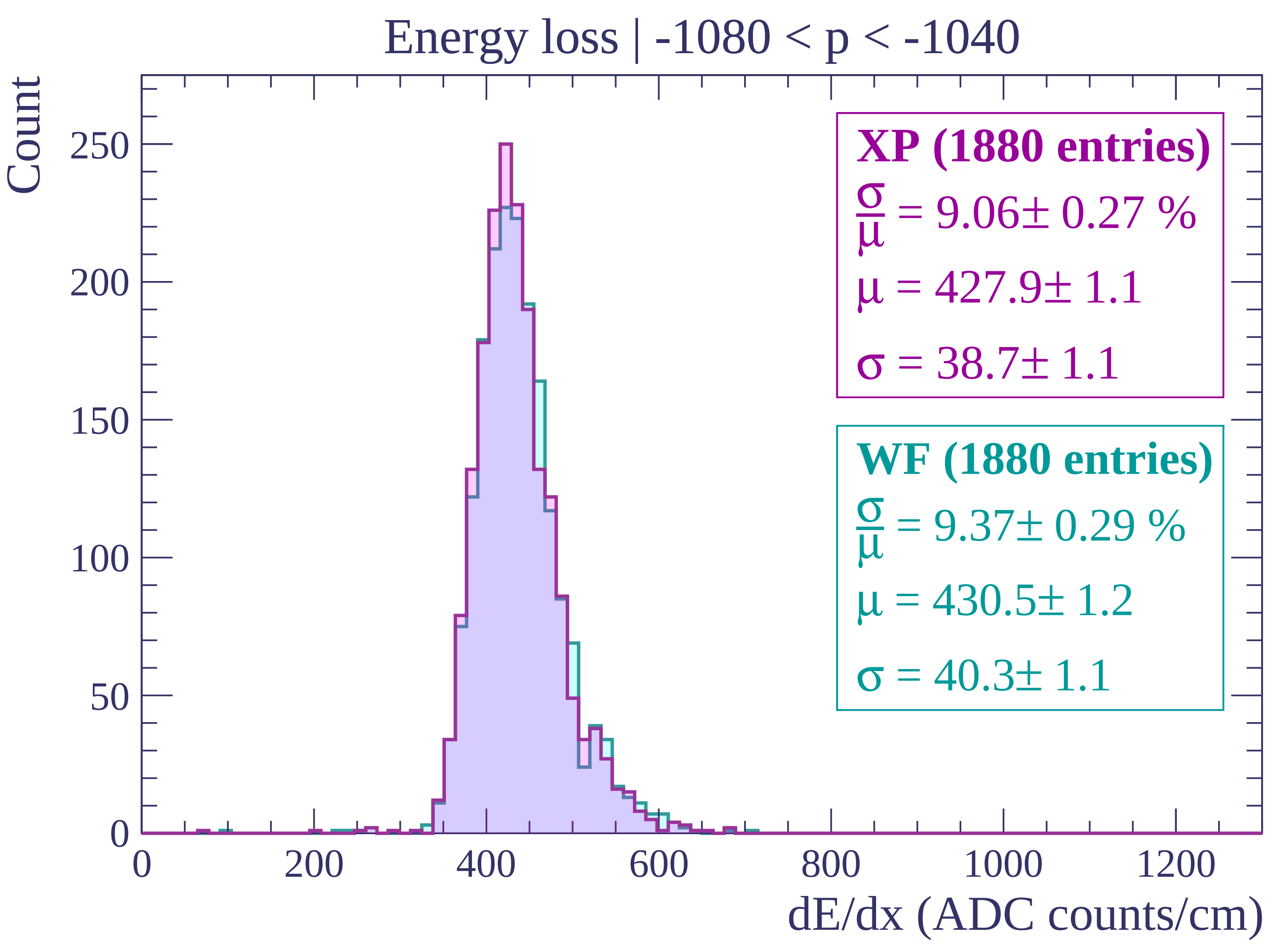


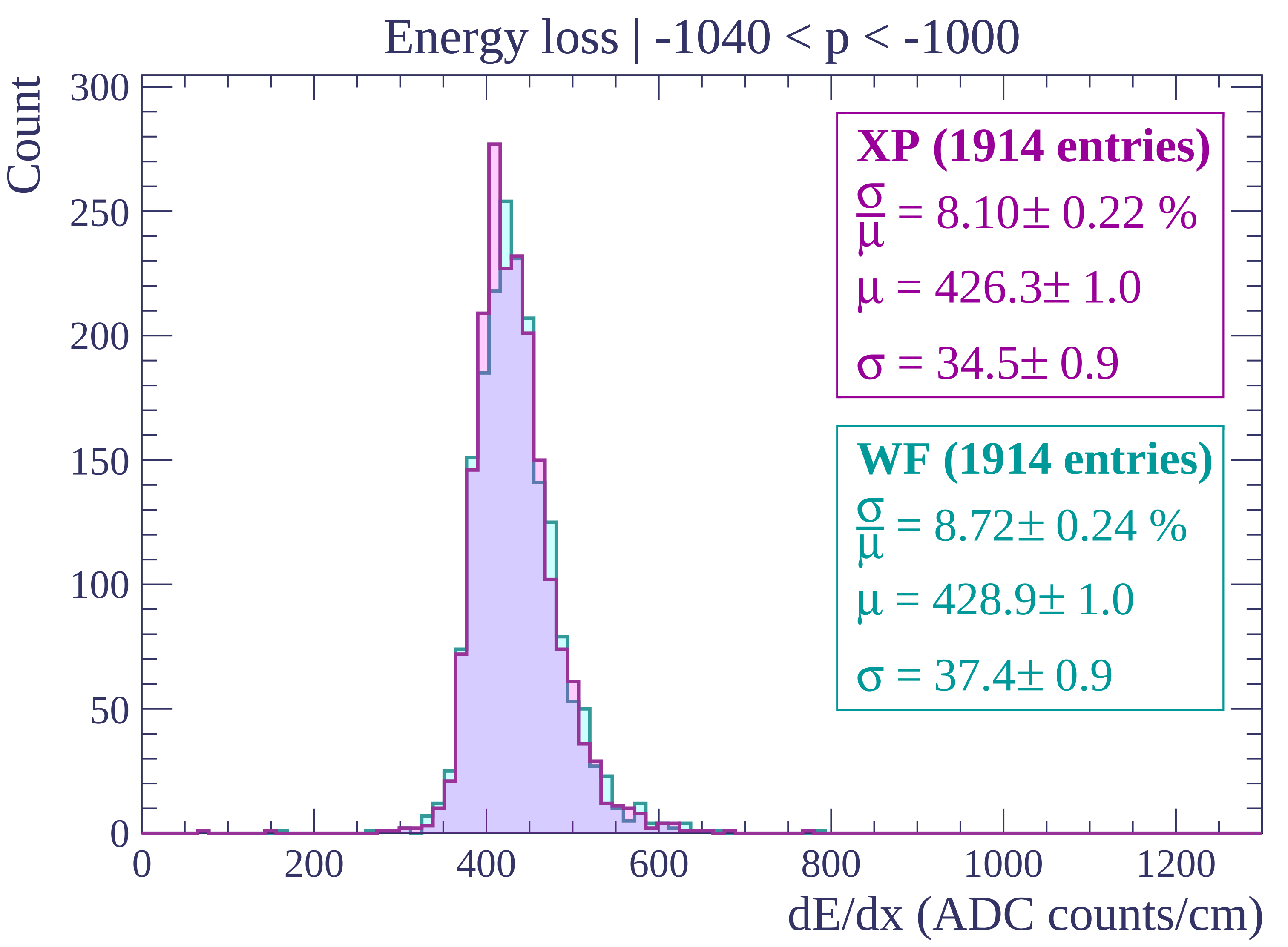
Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
XP (1755 entries)
σ/μ = 8.89± 0.27 %
μ = 434.1± 1.2
σ = 38.6± 1.1
WF (1755 entries)
σ/μ = 9.57± 0.29 %
μ = 438.6± 1.2
σ = 42.0± 1.1
0
20
40
60
80
100
120
140
160
180
200
220
240
200
400
600
800
1000
1200

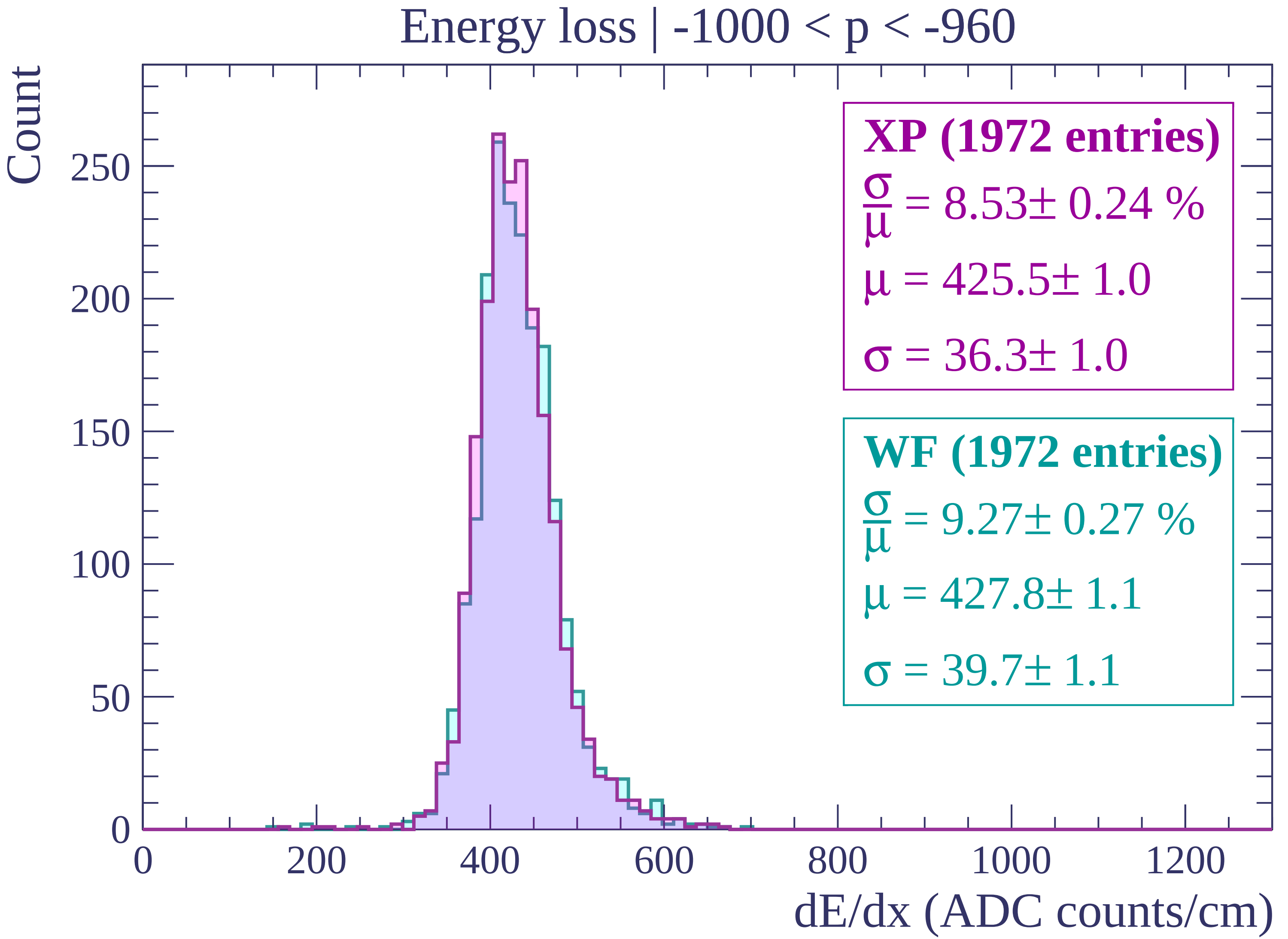

Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP (1917 entries)
$\frac{\sigma}{\mu}$ = 8.66± 0.25 %
μ = 432.1± 1.1
σ = 37.4± 1.0
WF (1917 entries)
$\frac{\sigma}{\mu}$ = 8.74± 0.26 %
μ = 433.3± 1.1
σ = 37.9± 1.0



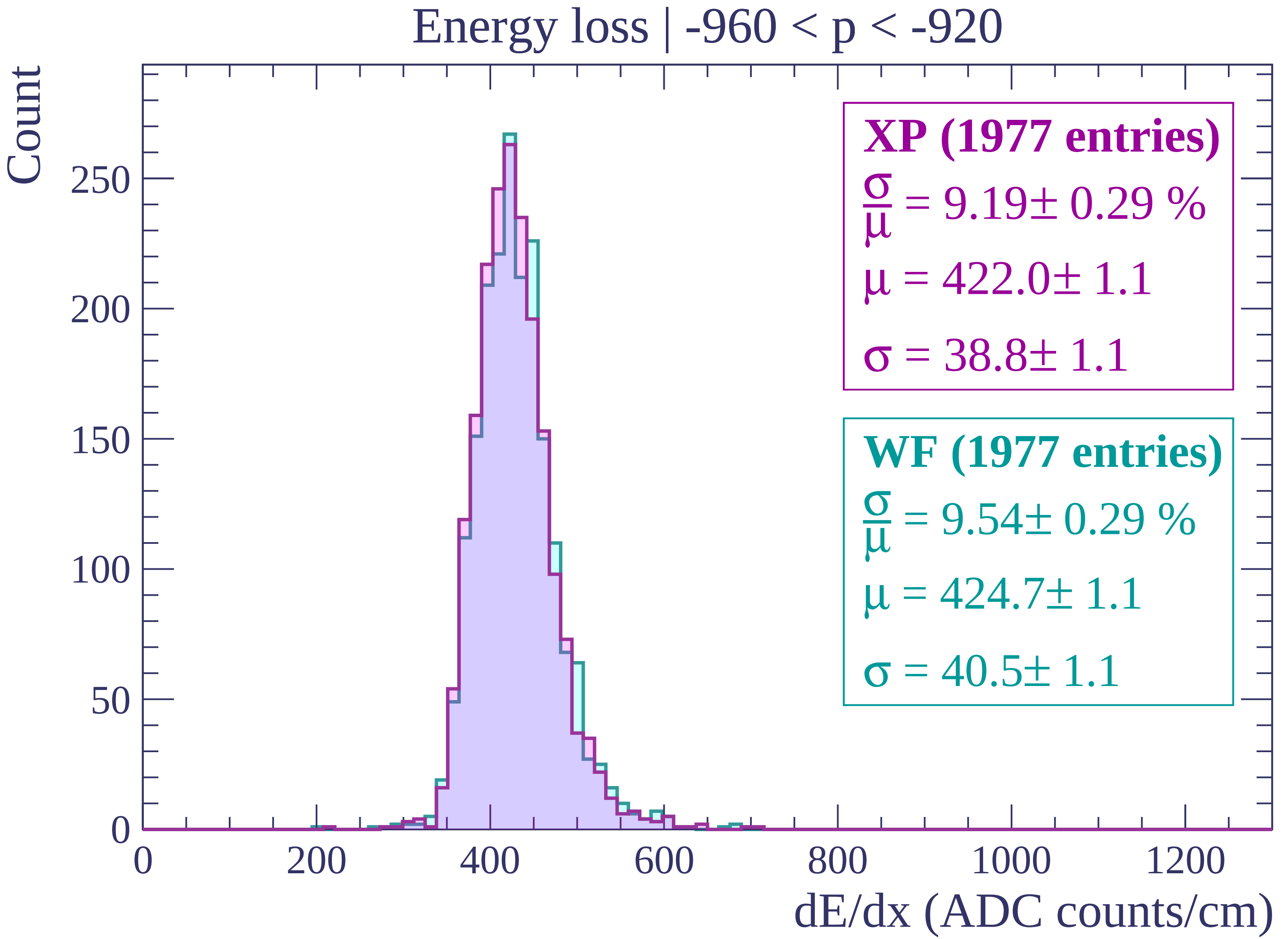
Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP (1864 entries)
σ/μ = 8.51± 0.24 %
μ = 432.3± 1.0
σ = 36.8± 0.9
WF (1864 entries)
σ/μ = 9.31± 0.27 %
μ = 435.1± 1.1
σ = 40.5± 1.1

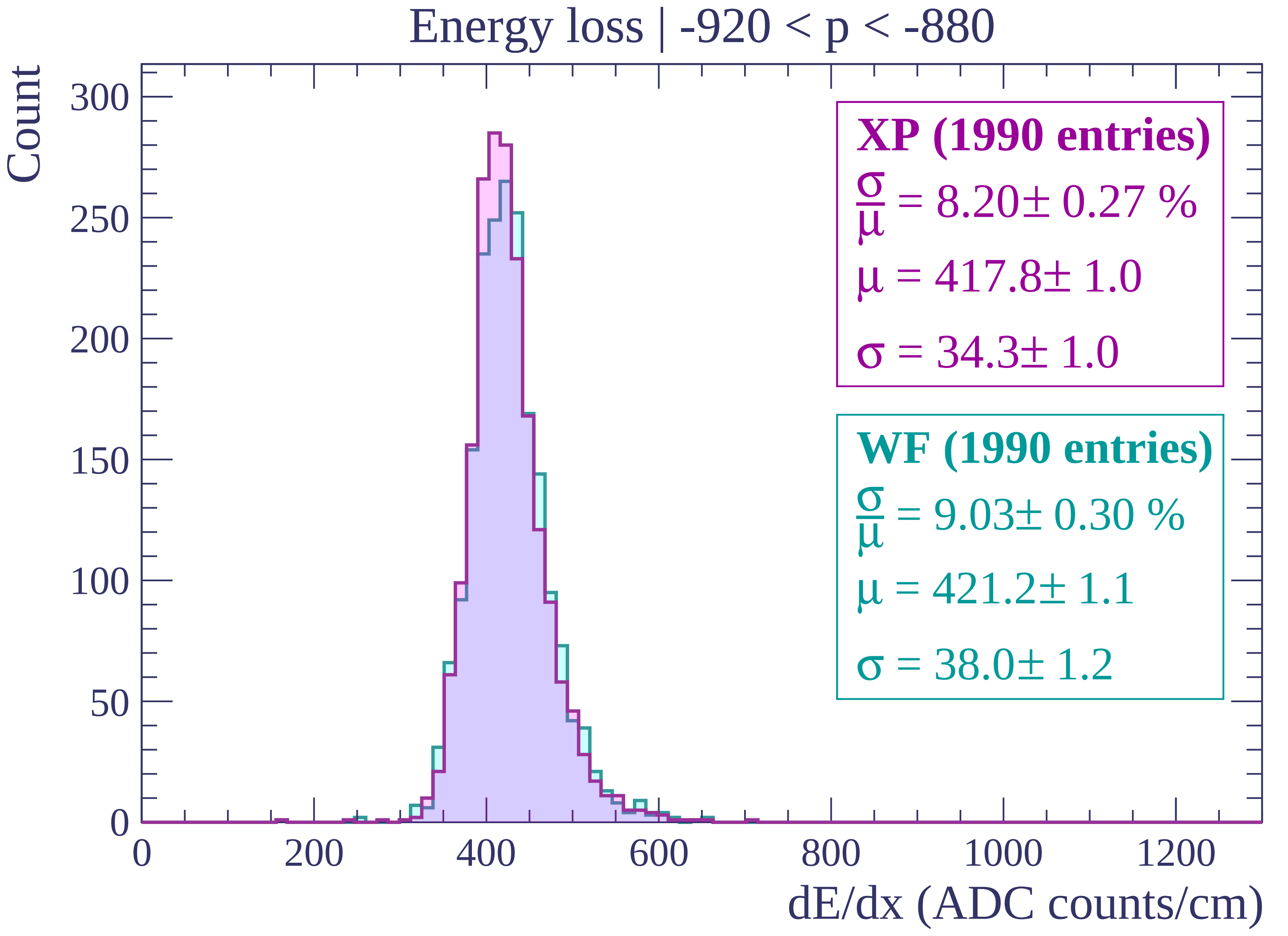

Energy loss | -1080 < p < -1040
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1880 entries)
$\frac{\sigma}{\mu}$ = 9.06± 0.27 %
μ = 427.9± 1.1
σ = 38.7± 1.1
WF (1880 entries)
$\frac{\sigma}{\mu}$ = 9.37± 0.29 %
μ = 430.5± 1.2
σ = 40.3± 1.1


Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (1914 entries)
$\frac{\sigma}{\mu}$ = 8.10± 0.22 %
μ = 426.3± 1.0
σ = 34.5± 0.9
WF (1914 entries)
$\frac{\sigma}{\mu}$ = 8.72± 0.24 %
μ = 428.9± 1.0
σ = 37.4± 0.9

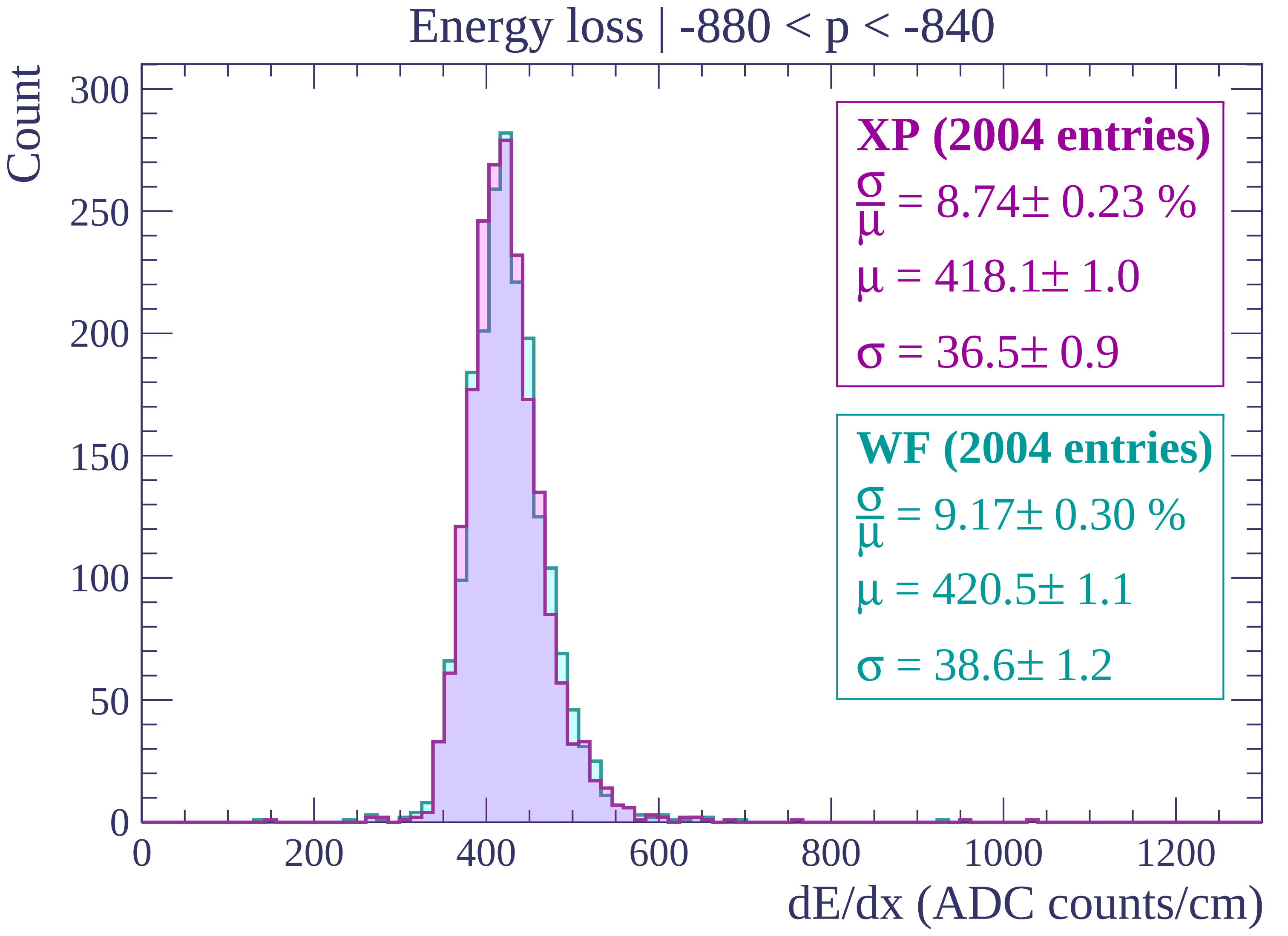


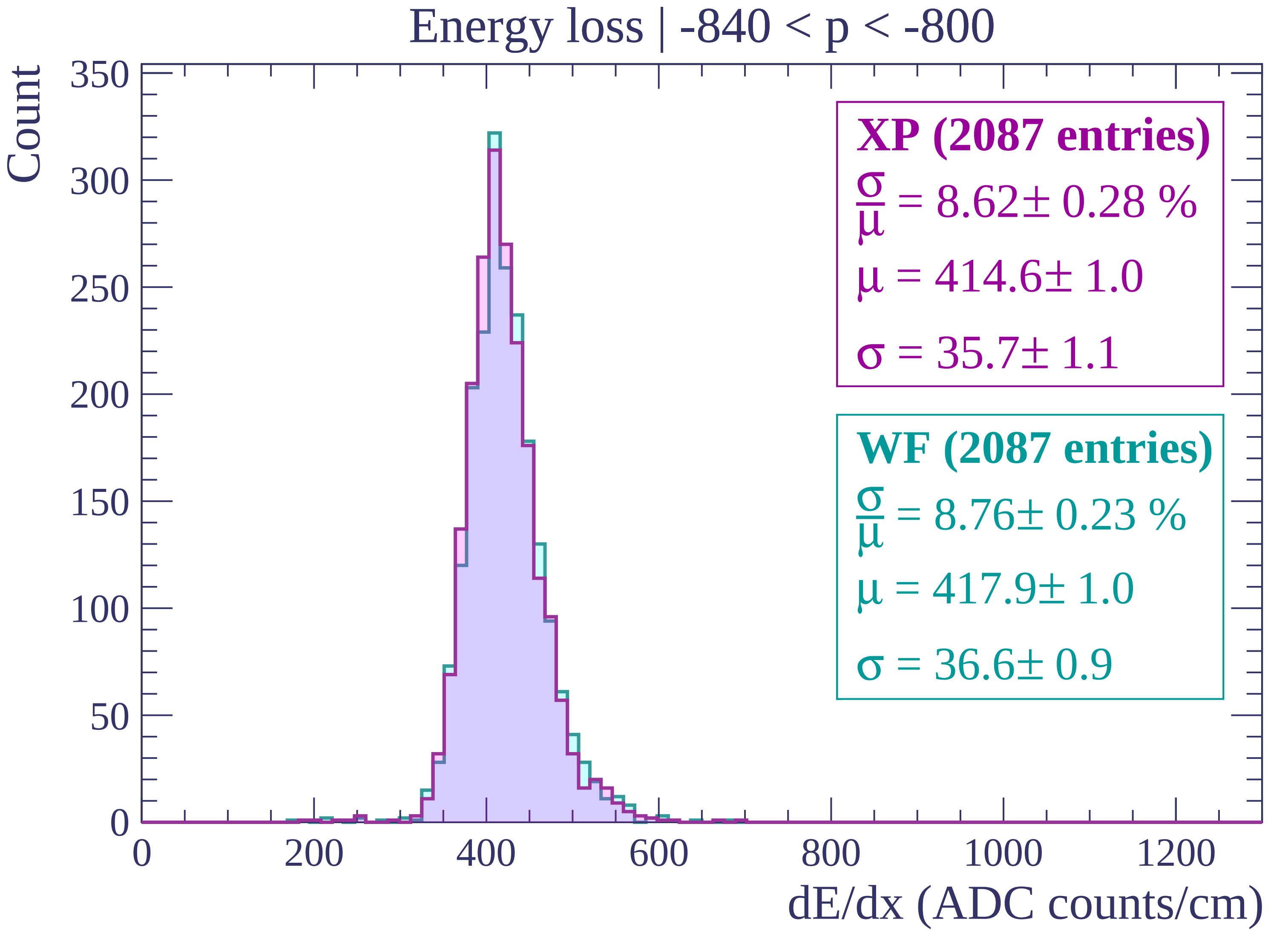
Energy loss | -1000 < p < -960
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1972 entries)
σ/μ = 8.53± 0.24 %
μ = 425.5± 1.0
σ = 36.3± 1.0
WF (1972 entries)
σ/μ = 9.27± 0.27 %
μ = 427.8± 1.1
σ = 39.7± 1.1

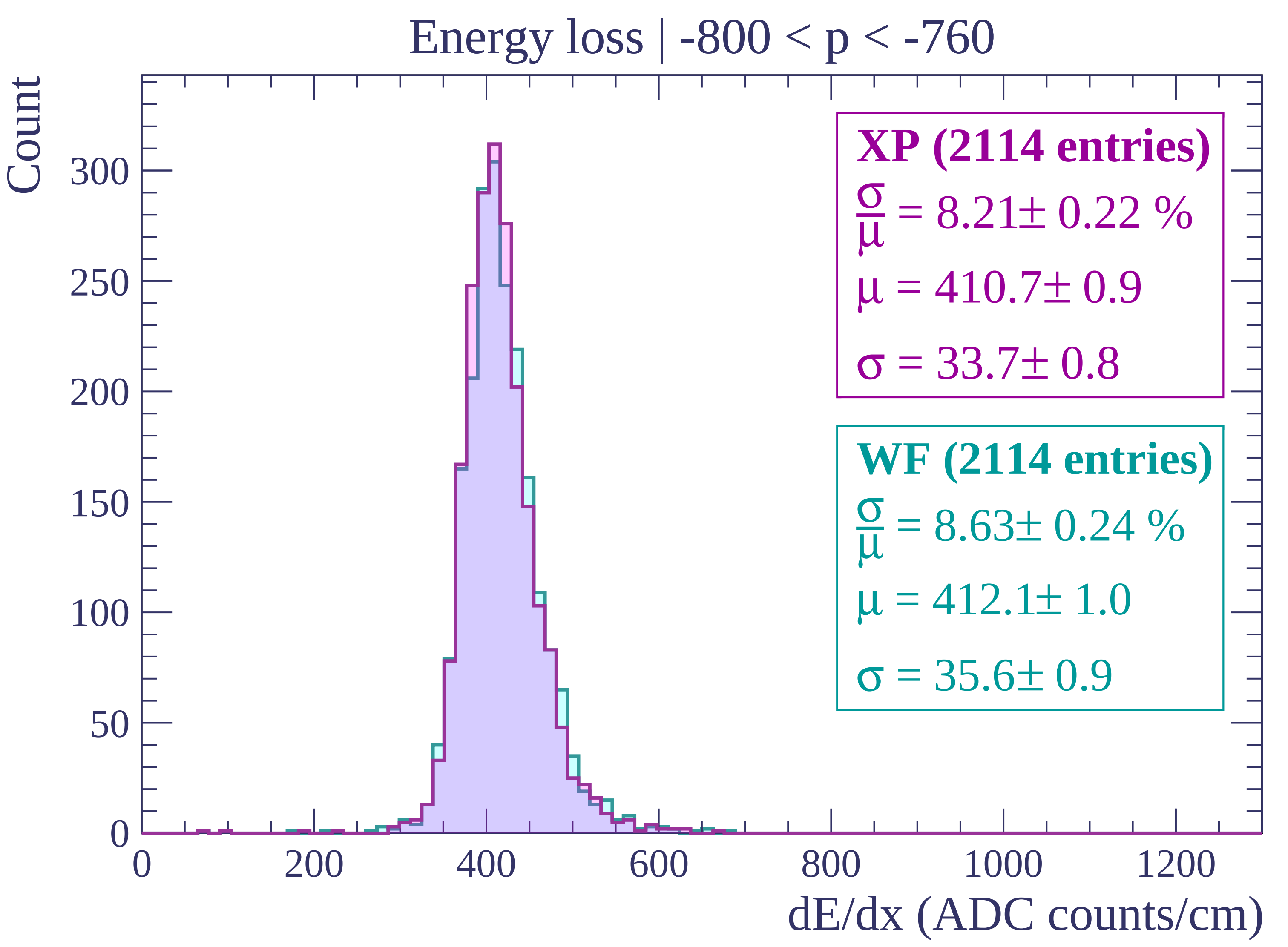

Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP (1977 entries)
$\frac{\sigma}{\mu}$ = 9.19± 0.29 %
μ = 422.0± 1.1
σ = 38.8± 1.1
WF (1977 entries)
$\frac{\sigma}{\mu}$ = 9.54± 0.29 %
μ = 424.7± 1.1
σ = 40.5± 1.1



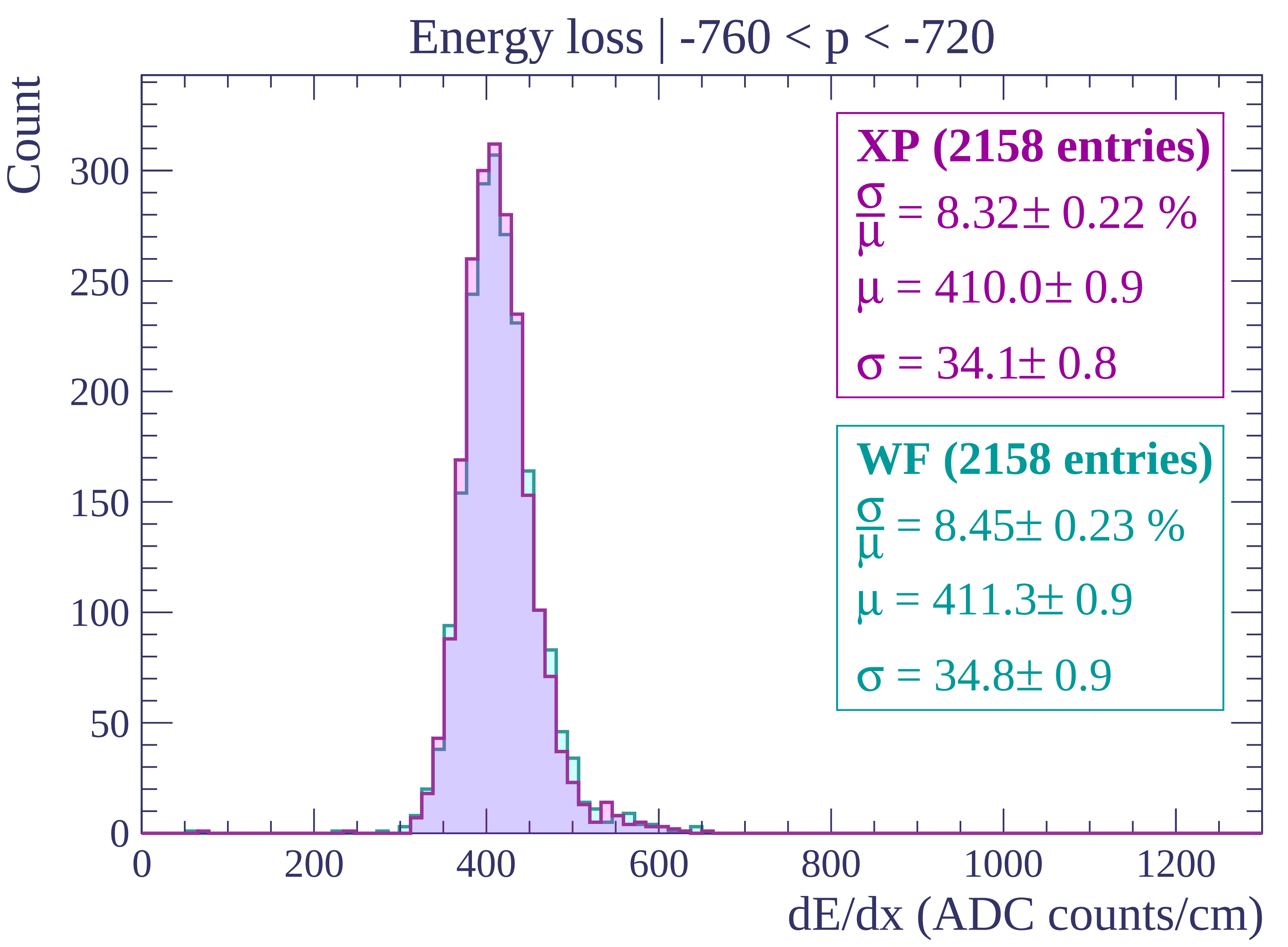
Energy loss | -920 < p < -880
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1990 entries)
σ/μ = 8.20± 0.27 %
μ = 417.8± 1.0
σ = 34.3± 1.0
WF (1990 entries)
σ/μ = 9.03± 0.30 %
μ = 421.2± 1.1
σ = 38.0± 1.2

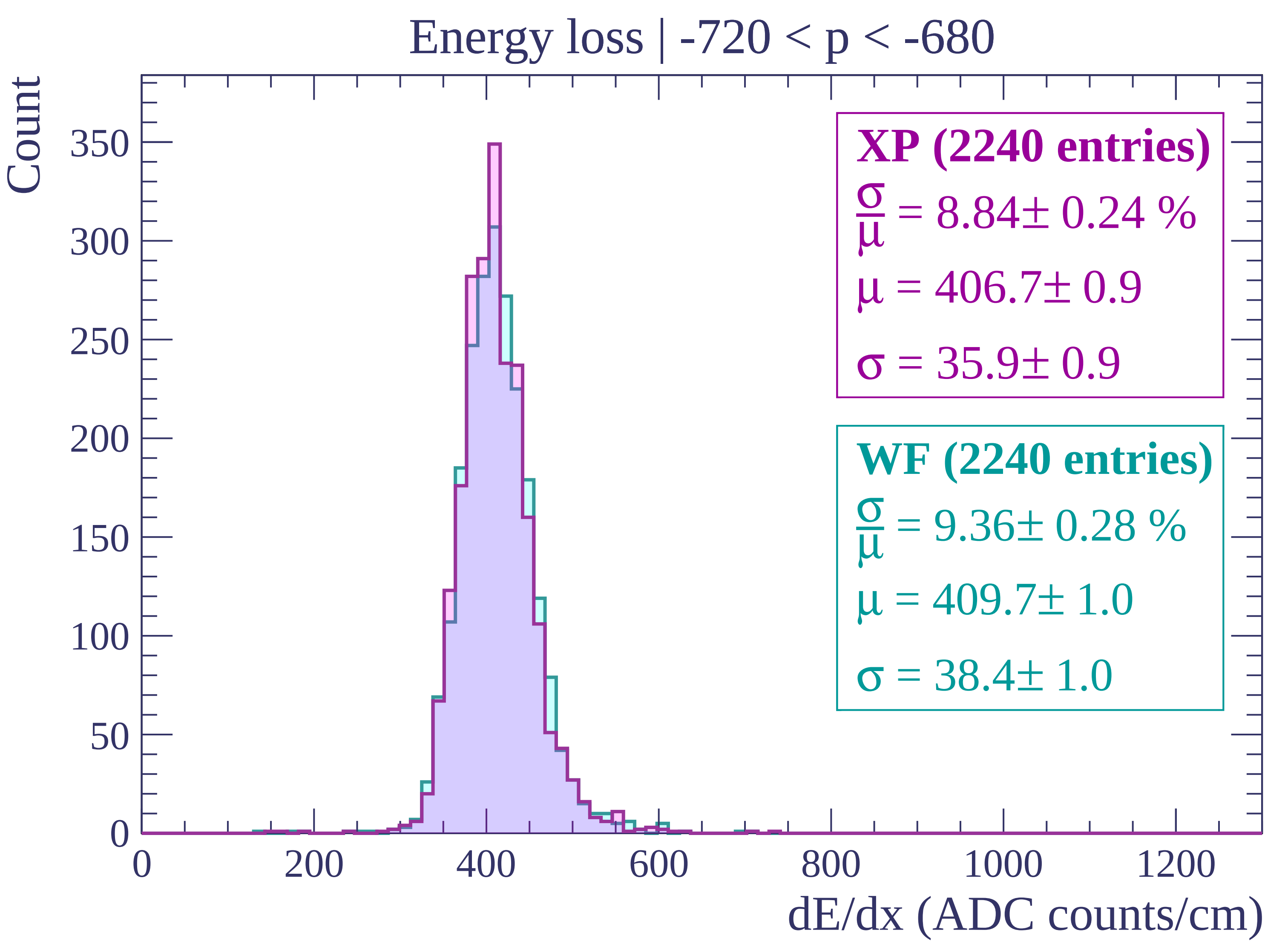

Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
XP (2004 entries)
σ/μ = 8.74± 0.23 %
μ = 418.1± 1.0
σ = 36.5± 0.9
WF (2004 entries)
σ/μ = 9.17± 0.30 %
μ = 420.5± 1.1
σ = 38.6± 1.2
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200


Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2087 entries)
σ/μ = 8.62± 0.28 %
μ = 414.6± 1.0
σ = 35.7± 1.1
WF (2087 entries)
σ/μ = 8.76± 0.23 %
μ = 417.9± 1.0
σ = 36.6± 0.9

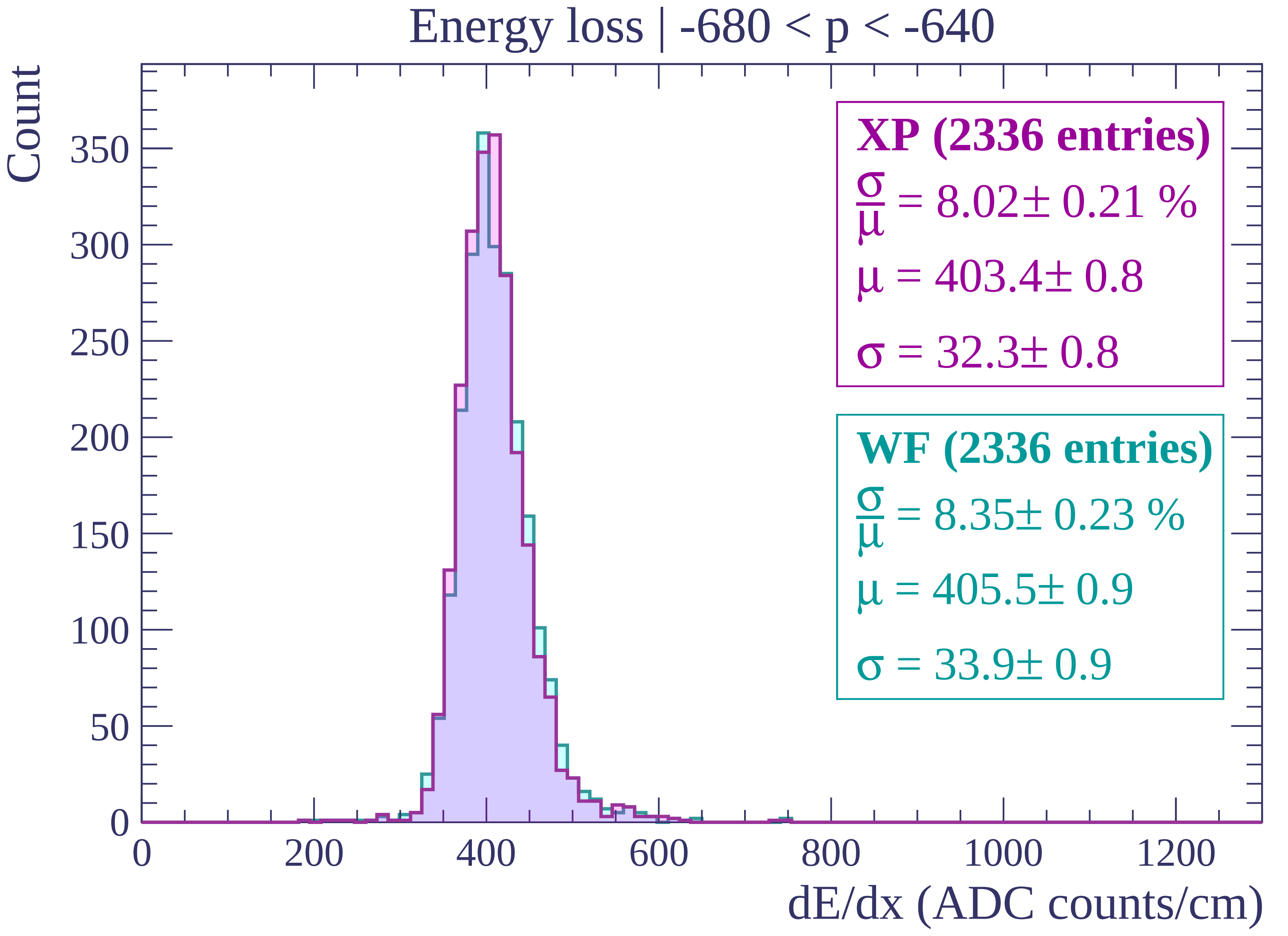


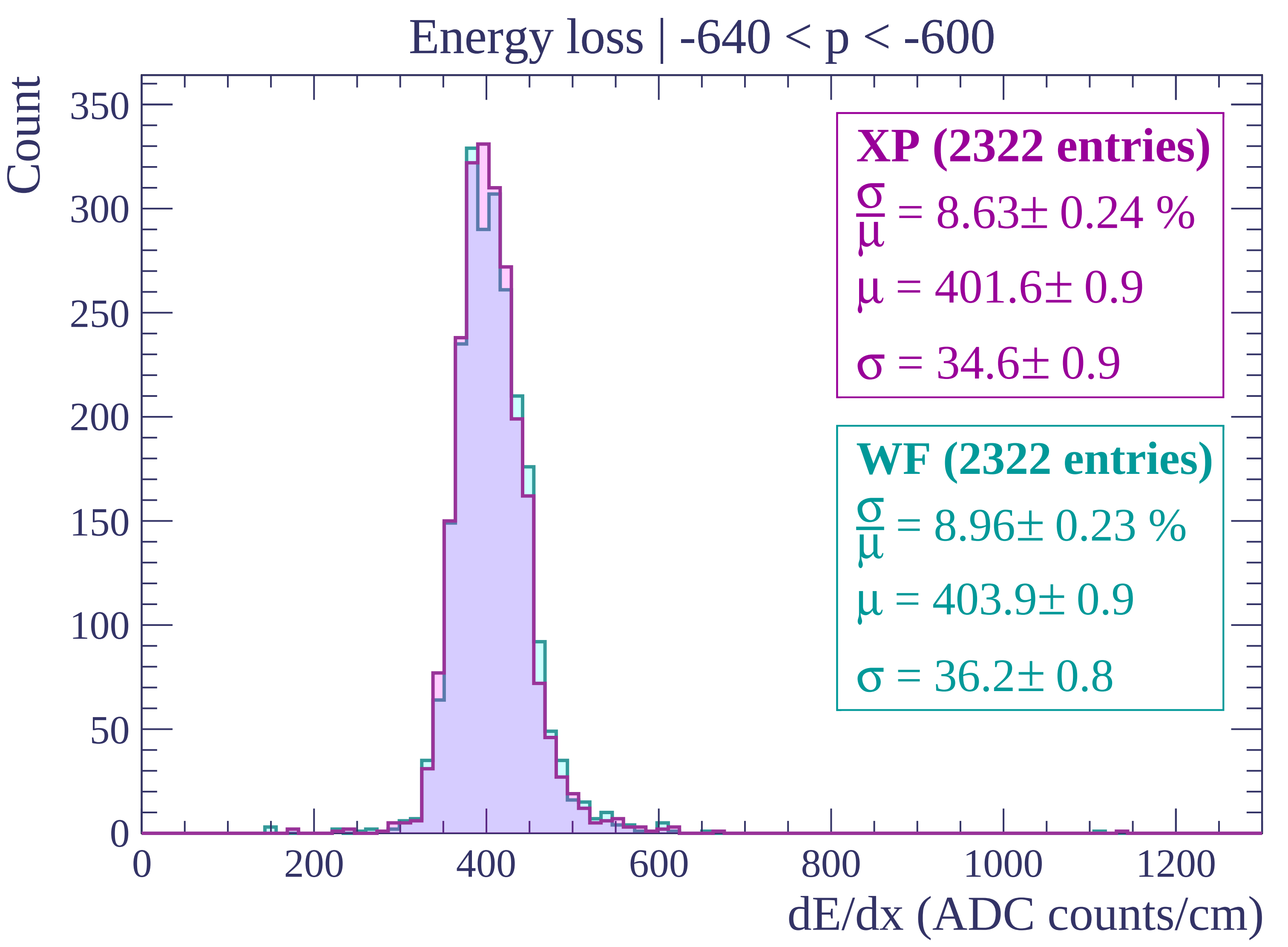
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2114 entries)
σ/μ = 8.21± 0.22 %
μ = 410.7± 0.9
σ = 33.7± 0.8
WF (2114 entries)
σ/μ = 8.63± 0.24 %
μ = 412.1± 1.0
σ = 35.6± 0.9

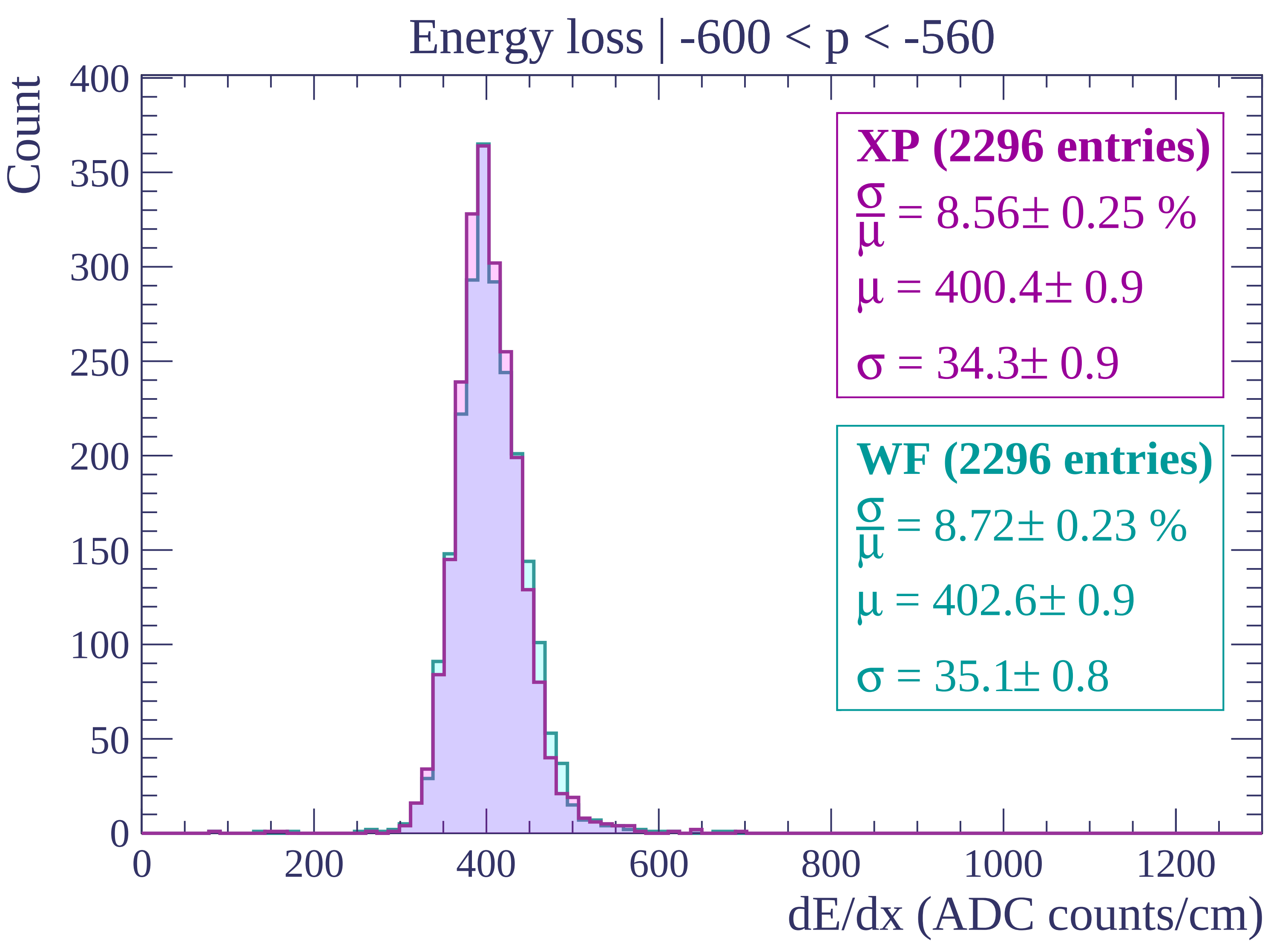

Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2158 entries)
σ/μ = 8.32± 0.22 %
μ = 410.0± 0.9
σ = 34.1± 0.8
WF (2158 entries)
σ/μ = 8.45± 0.23 %
μ = 411.3± 0.9
σ = 34.8± 0.9



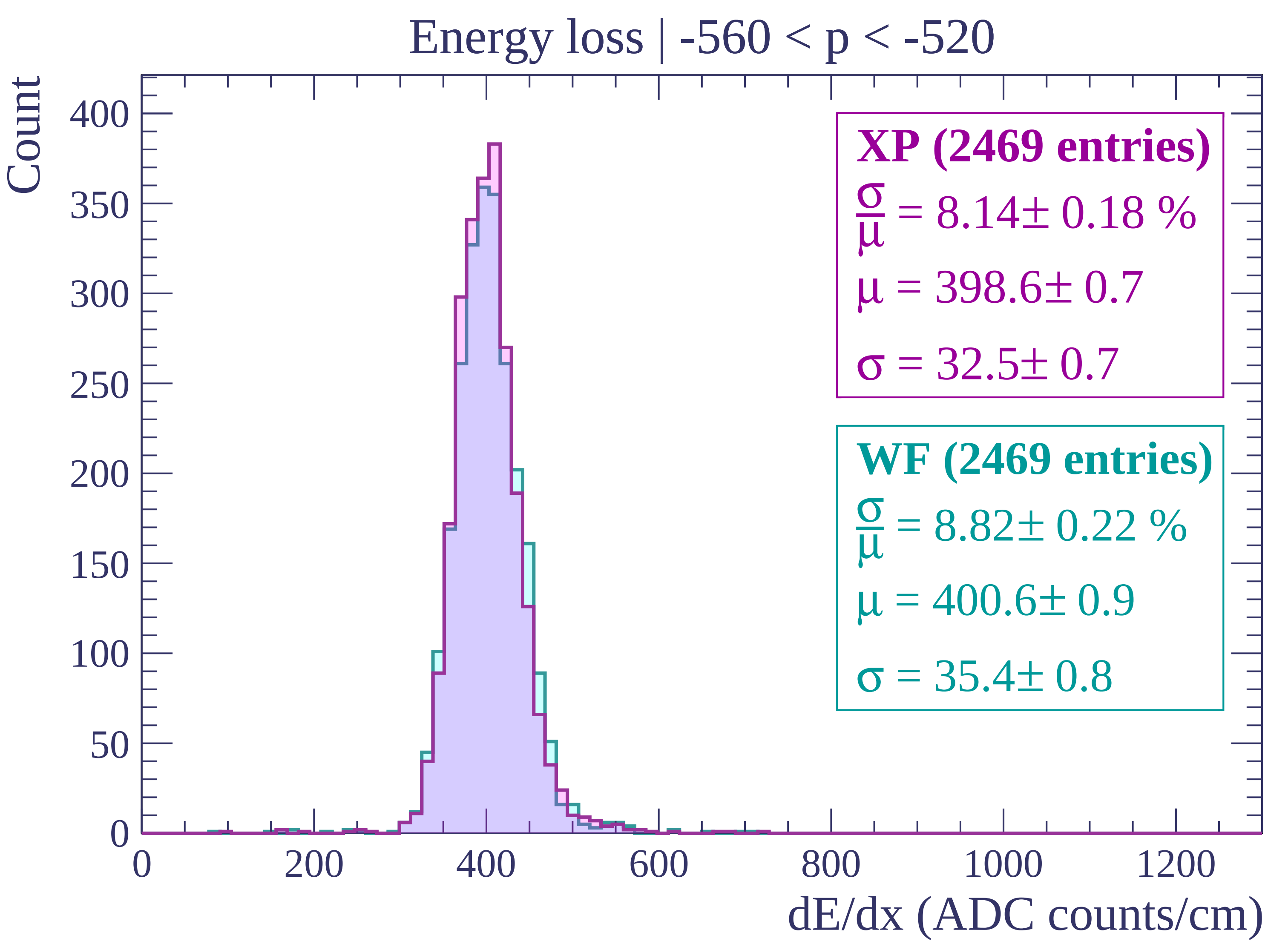
Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
XP (2240 entries)
σ/μ = 8.84± 0.24 %
μ = 406.7± 0.9
σ = 35.9± 0.9
WF (2240 entries)
σ/μ = 9.36± 0.28 %
μ = 409.7± 1.0
σ = 38.4± 1.0


Energy loss | -680 < p < -640
Count
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2336 entries)
$\frac{\sigma}{\mu}$ = 8.02± 0.21 %
μ = 403.4± 0.8
σ = 32.3± 0.8
WF (2336 entries)
$\frac{\sigma}{\mu}$ = 8.35± 0.23 %
μ = 405.5± 0.9
σ = 33.9± 0.9

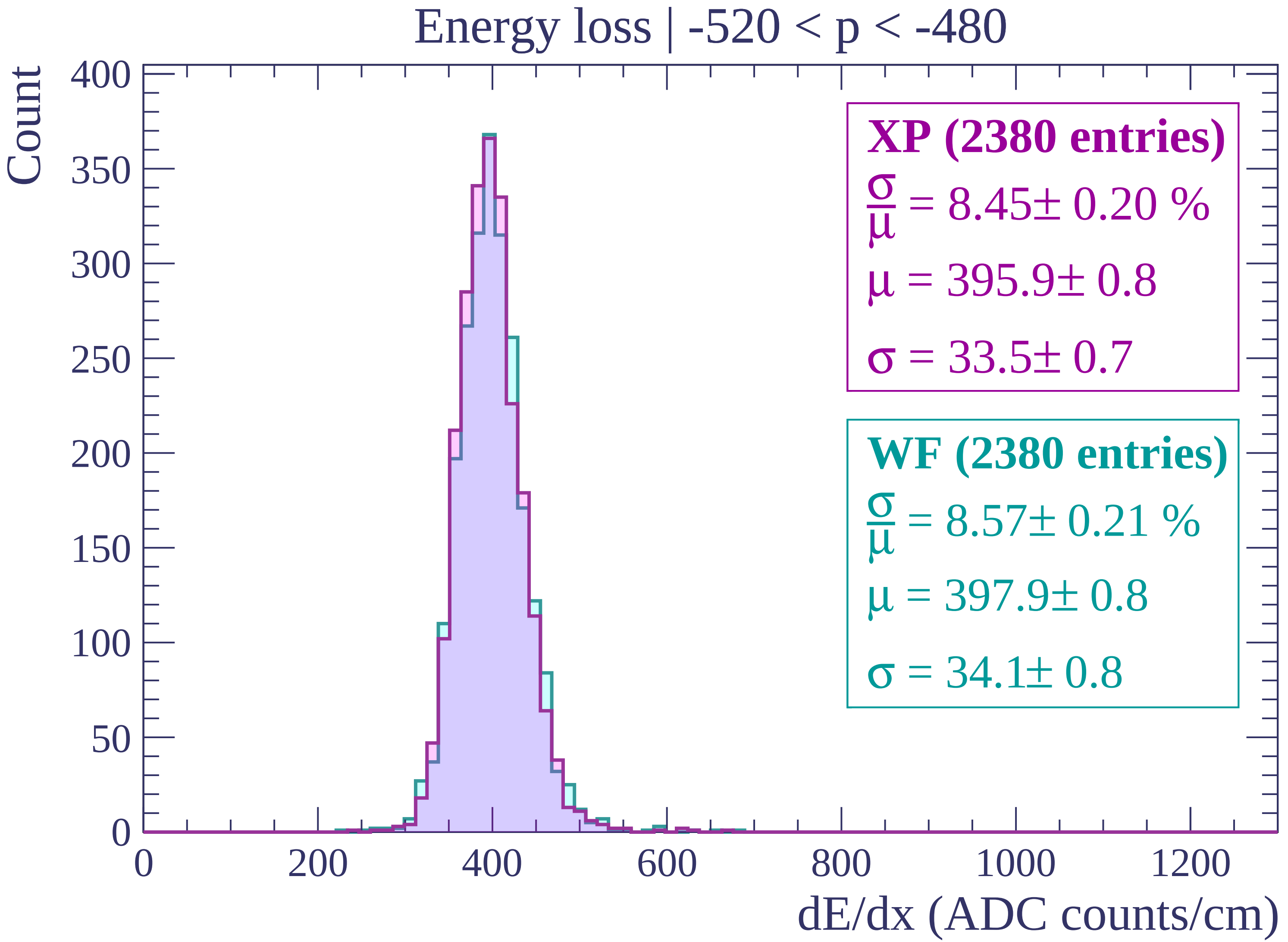

Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
XP (2322 entries)
σ/μ = 8.63± 0.24 %
μ = 401.6± 0.9
σ = 34.6± 0.9
WF (2322 entries)
σ/μ = 8.96± 0.23 %
μ = 403.9± 0.9
σ = 36.2± 0.8
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200



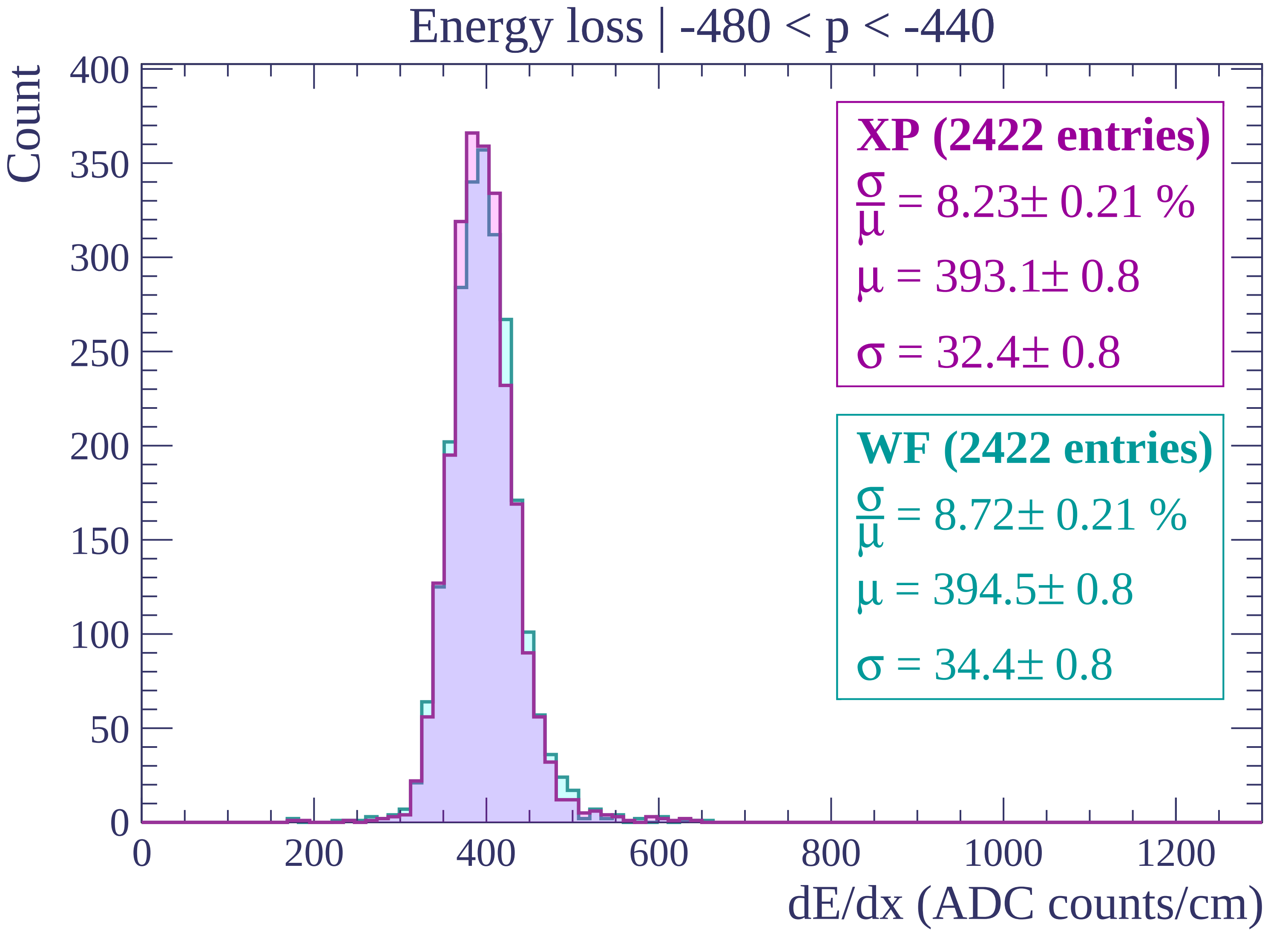
Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (2296 entries)
σ/μ = 8.56± 0.25 %
μ = 400.4± 0.9
σ = 34.3± 0.9
WF (2296 entries)
σ/μ = 8.72± 0.23 %
μ = 402.6± 0.9
σ = 35.1± 0.8



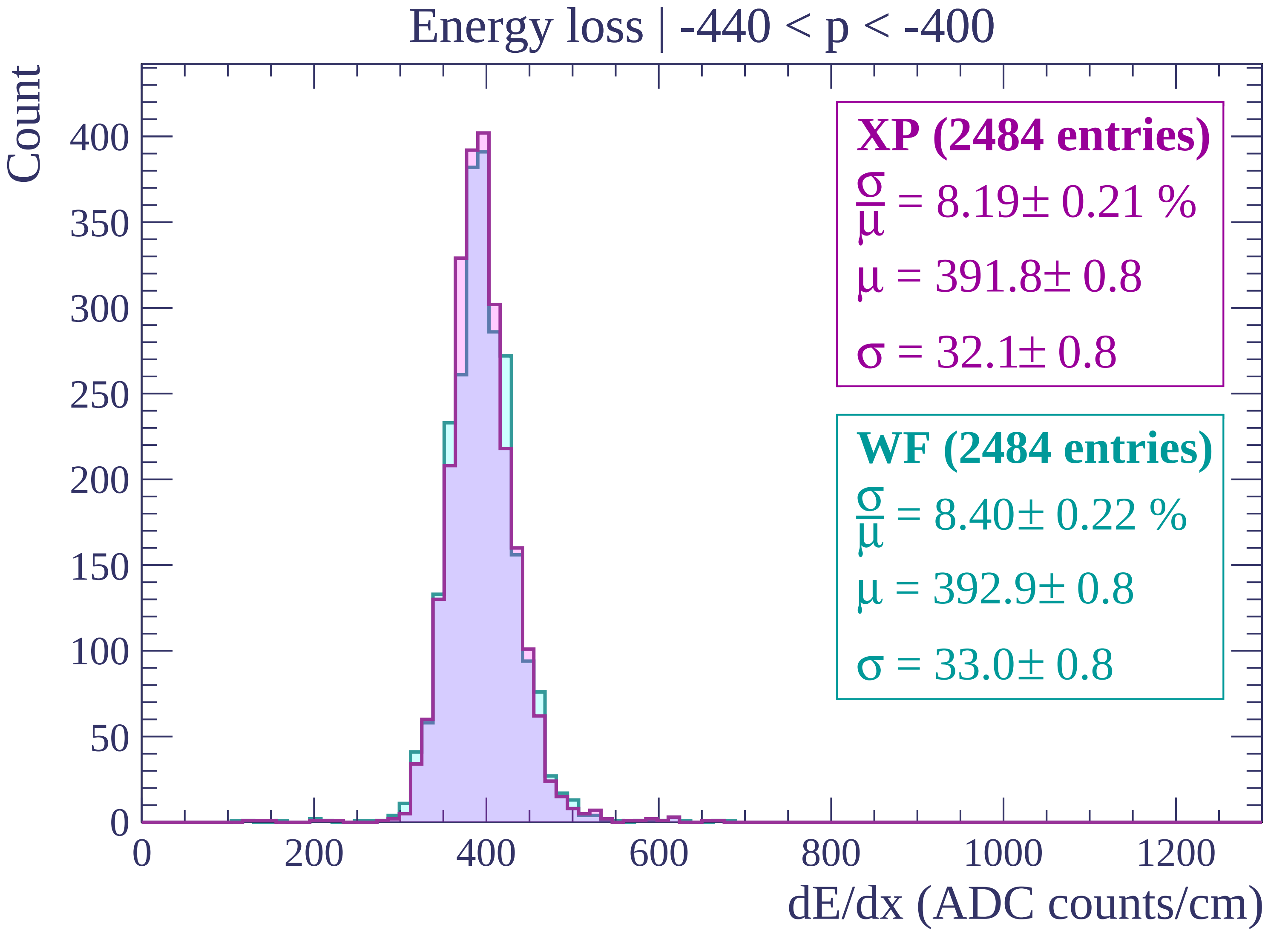
Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (2469 entries)
$\frac{\sigma}{\mu}$ = 8.14± 0.18 %
μ = 398.6± 0.7
σ = 32.5± 0.7
WF (2469 entries)
$\frac{\sigma}{\mu}$ = 8.82± 0.22 %
μ = 400.6± 0.9
σ = 35.4± 0.8

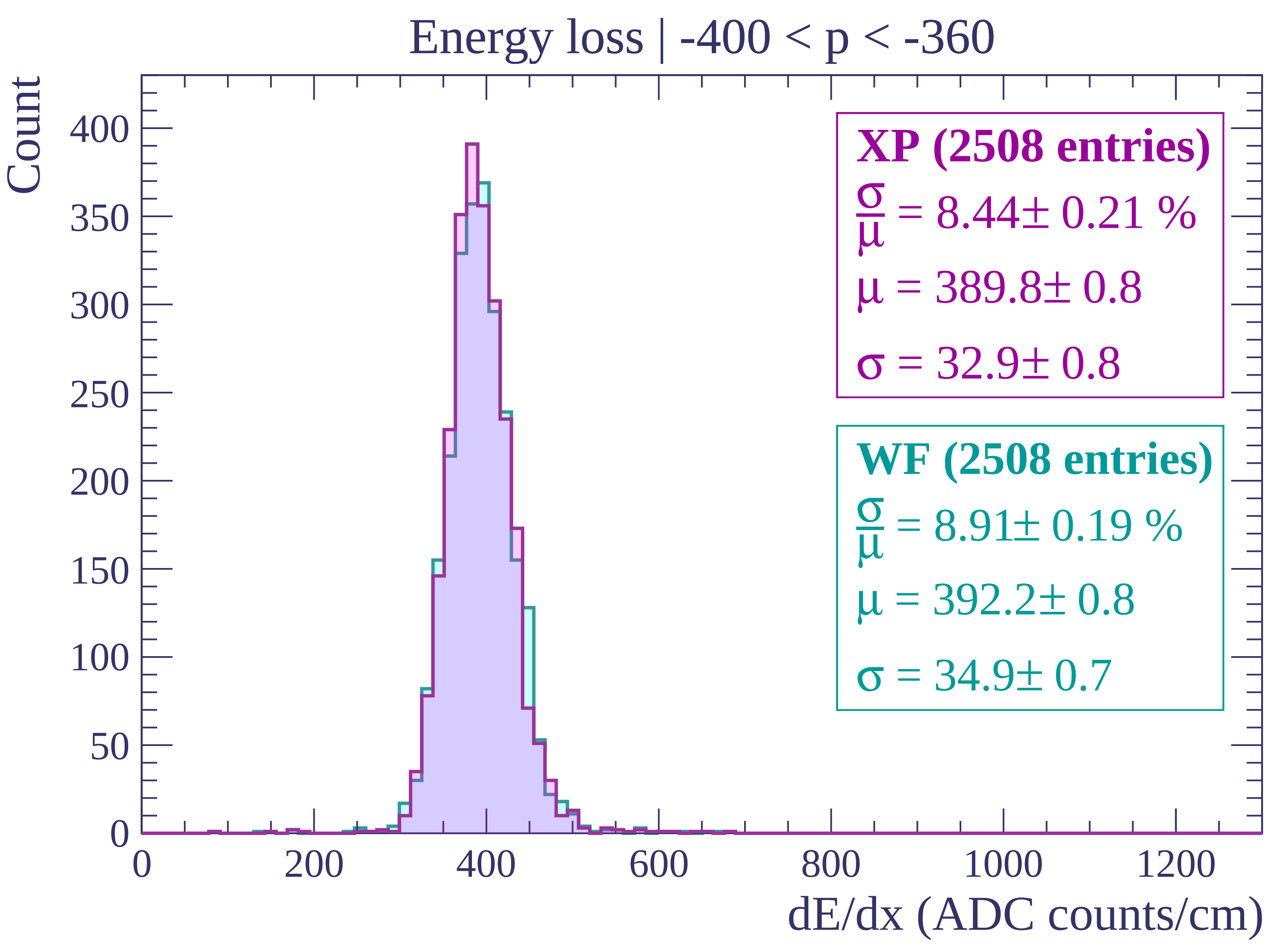


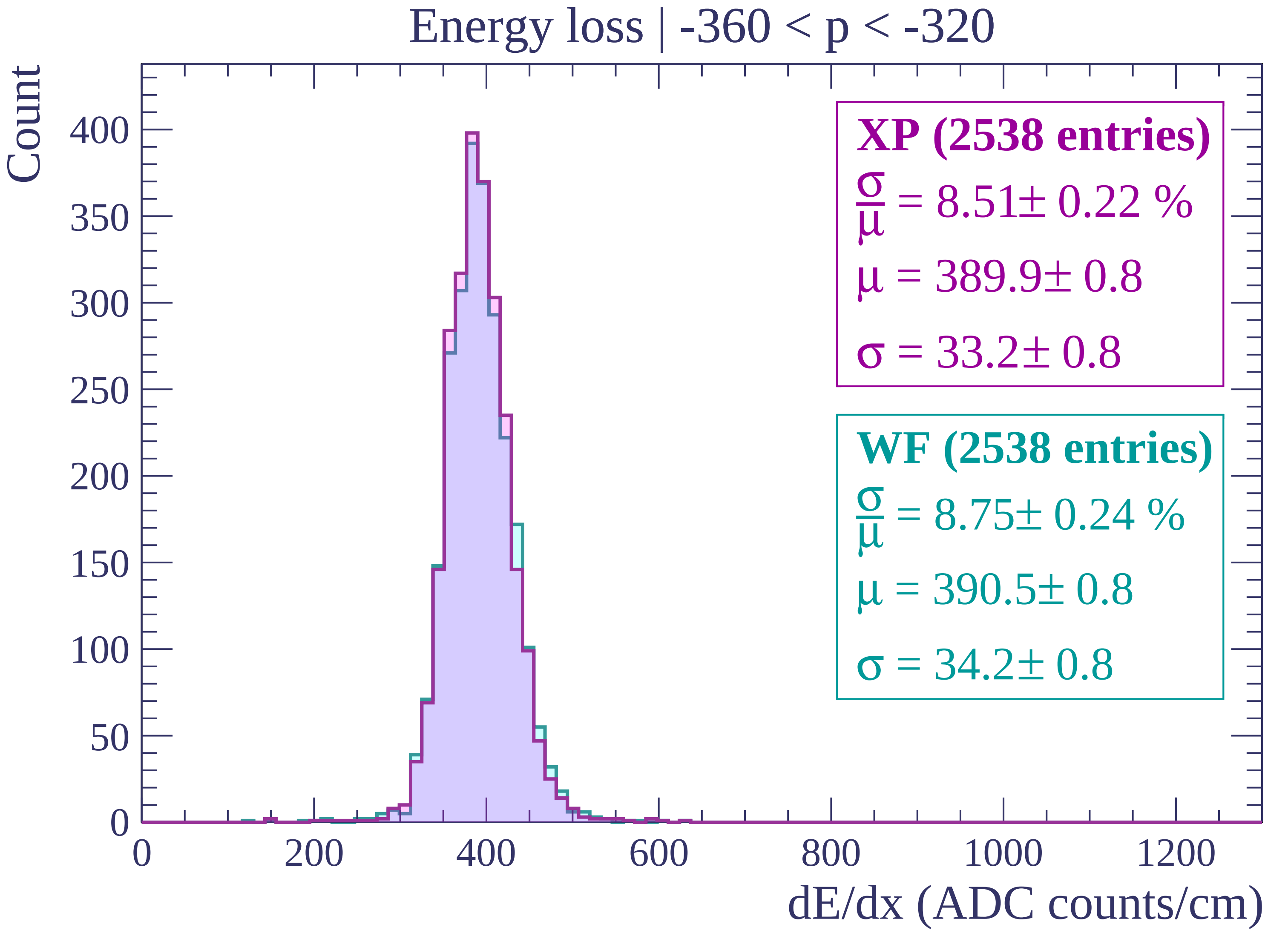
Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
XP (2380 entries)
$\frac{\sigma}{\mu}$ = 8.45± 0.20 %
μ = 395.9± 0.8
σ = 33.5± 0.7
WF (2380 entries)
$\frac{\sigma}{\mu}$ = 8.57± 0.21 %
μ = 397.9± 0.8
σ = 34.1± 0.8
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200



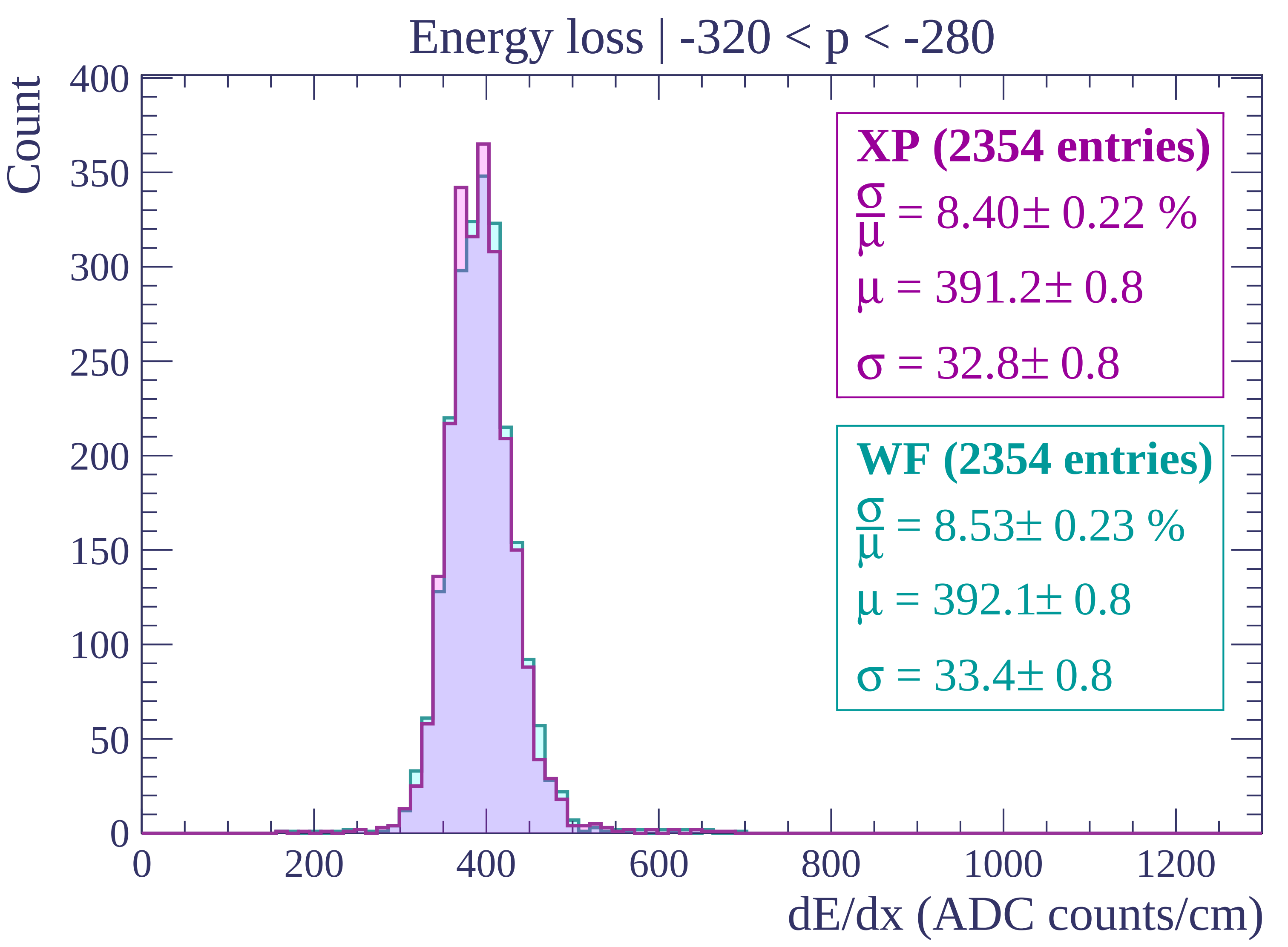
Energy loss | -480 < p < -440
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2422 entries)
σ/μ = 8.23± 0.21 %
μ = 393.1± 0.8
σ = 32.4± 0.8
WF (2422 entries)
σ/μ = 8.72± 0.21 %
μ = 394.5± 0.8
σ = 34.4± 0.8


Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
XP (2484 entries)
σ/μ = 8.19± 0.21 %
μ = 391.8± 0.8
σ = 32.1± 0.8
WF (2484 entries)
σ/μ = 8.40± 0.22 %
μ = 392.9± 0.8
σ = 33.0± 0.8
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

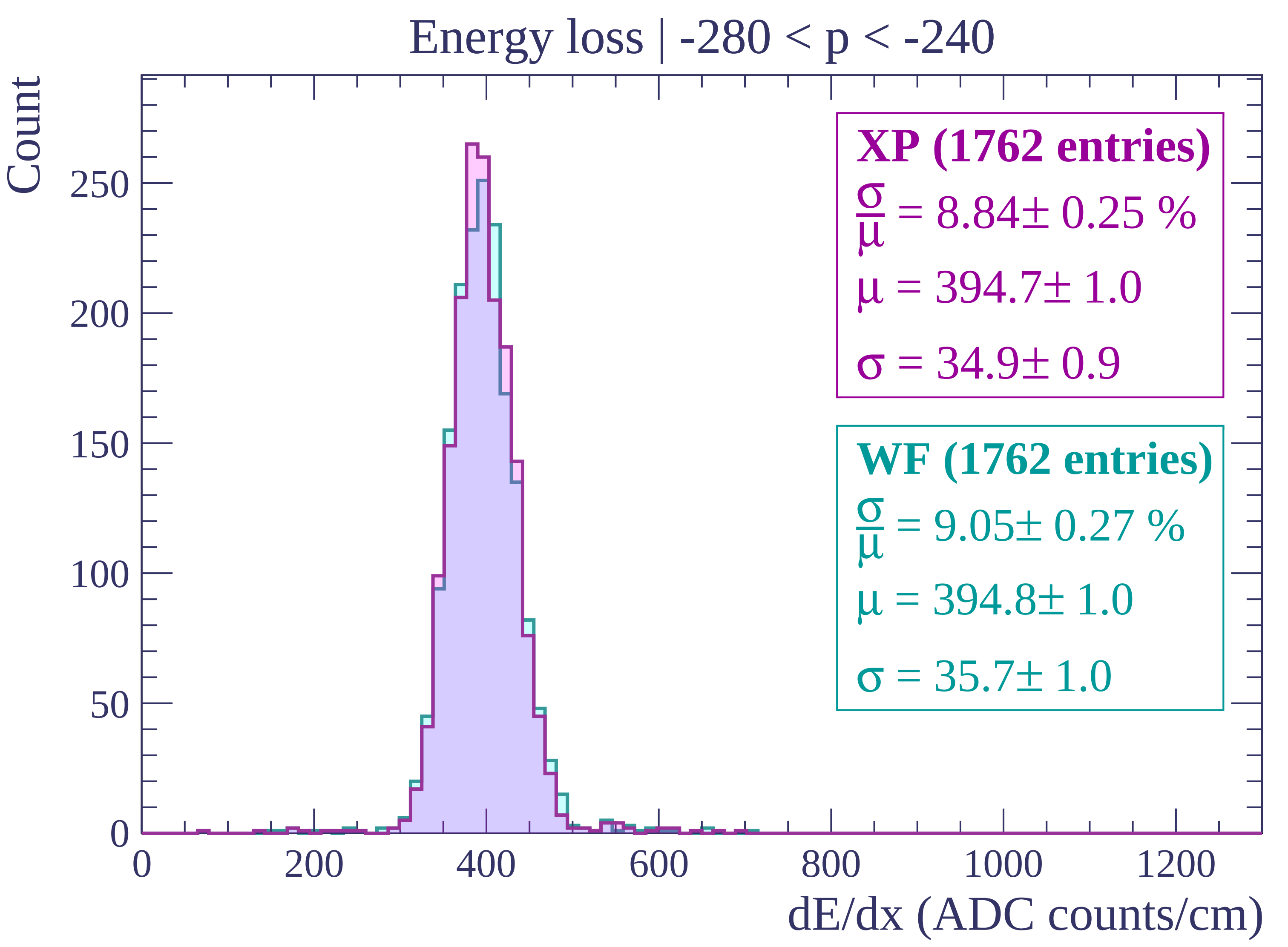

Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
XP (2508 entries)
σ/μ = 8.44± 0.21 %
μ = 389.8± 0.8
σ = 32.9± 0.8
WF (2508 entries)
σ/μ = 8.91± 0.19 %
μ = 392.2± 0.8
σ = 34.9± 0.7
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200



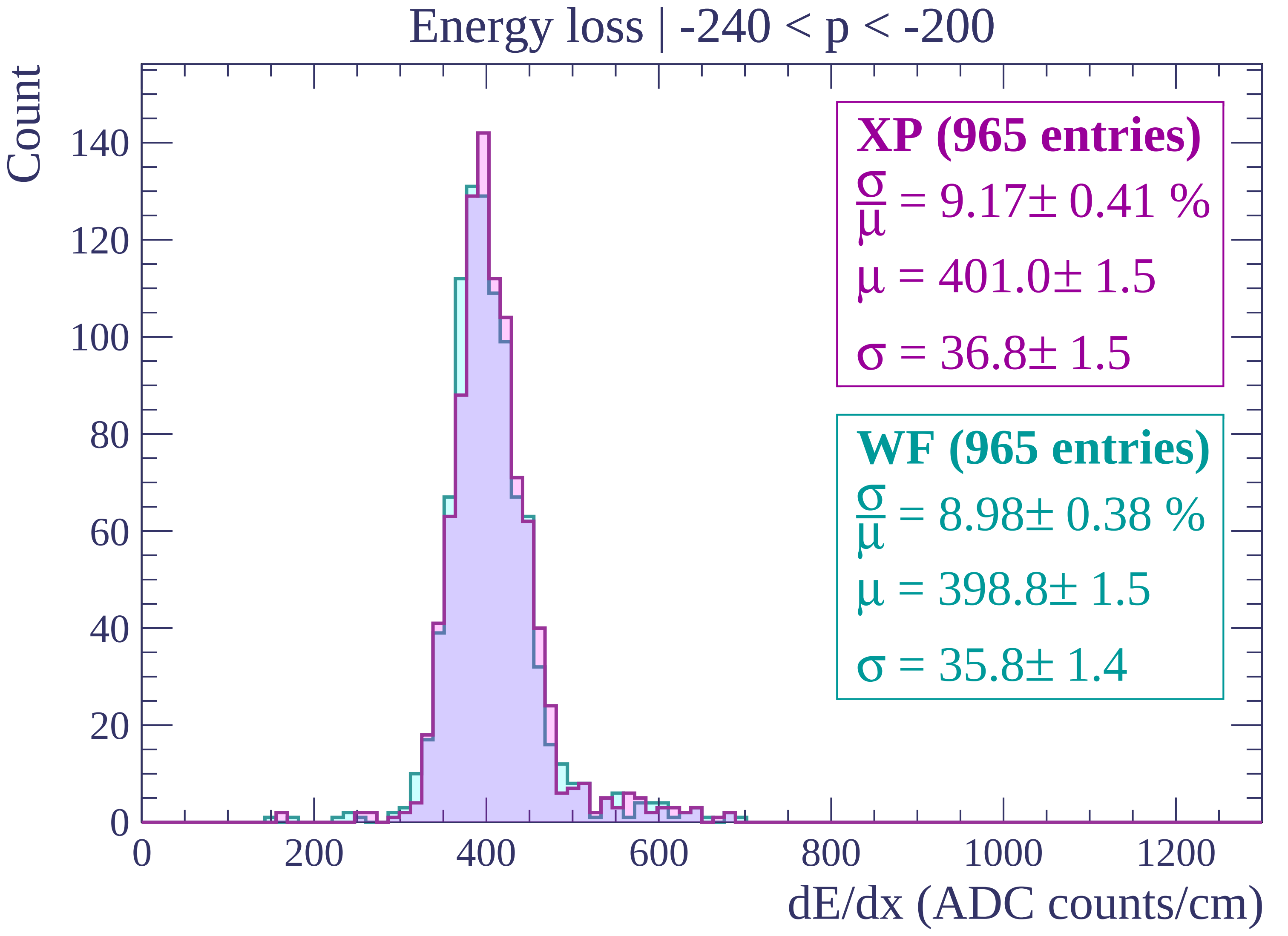
Energy loss | -360 < p < -320
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2538 entries)
σ/μ = 8.51± 0.22 %
μ = 389.9± 0.8
σ = 33.2± 0.8
WF (2538 entries)
σ/μ = 8.75± 0.24 %
μ = 390.5± 0.8
σ = 34.2± 0.8



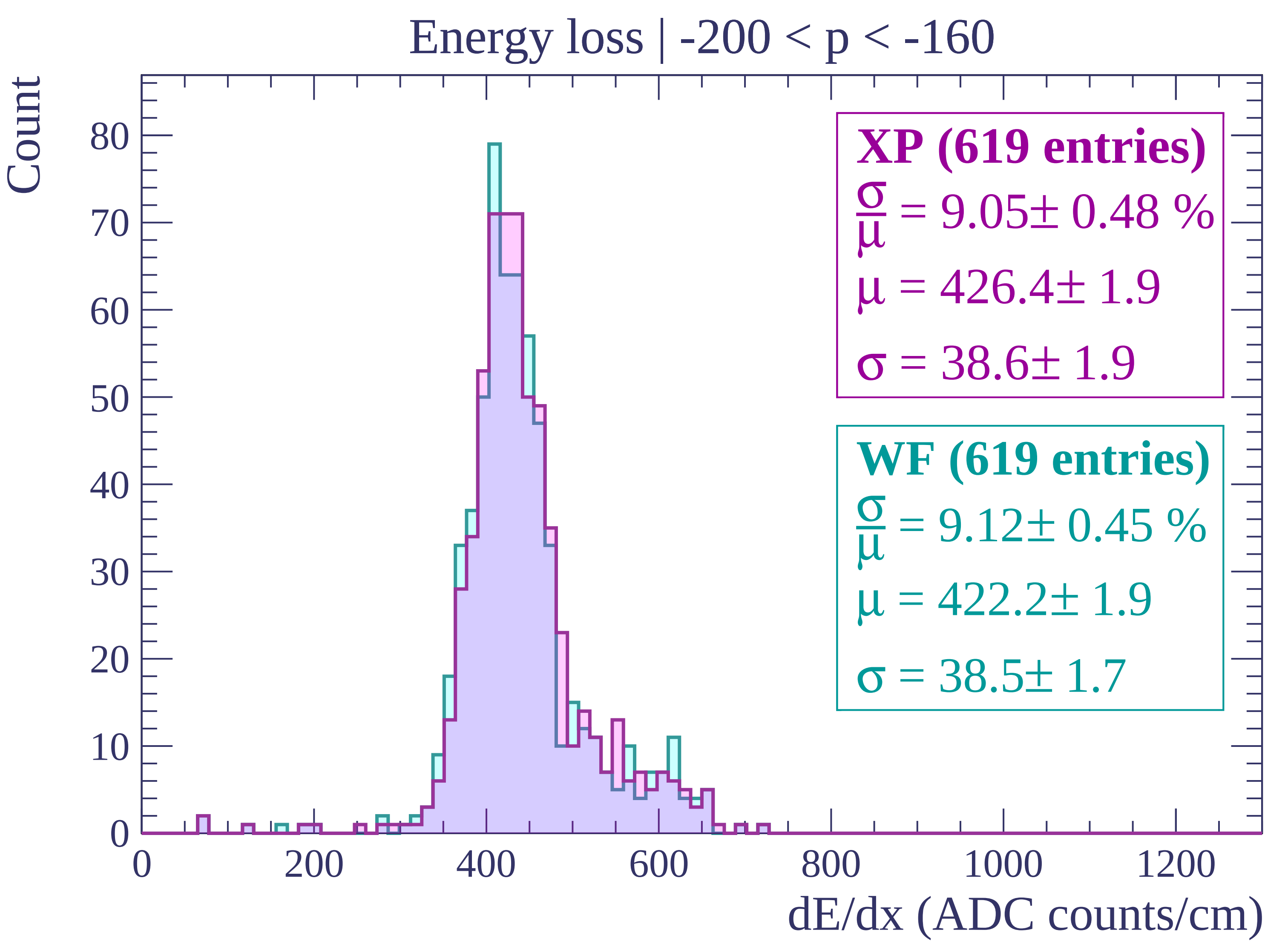
Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (2354 entries)
σ/μ = 8.40± 0.22 %
μ = 391.2± 0.8
σ = 32.8± 0.8
WF (2354 entries)
σ/μ = 8.53± 0.23 %
μ = 392.1± 0.8
σ = 33.4± 0.8

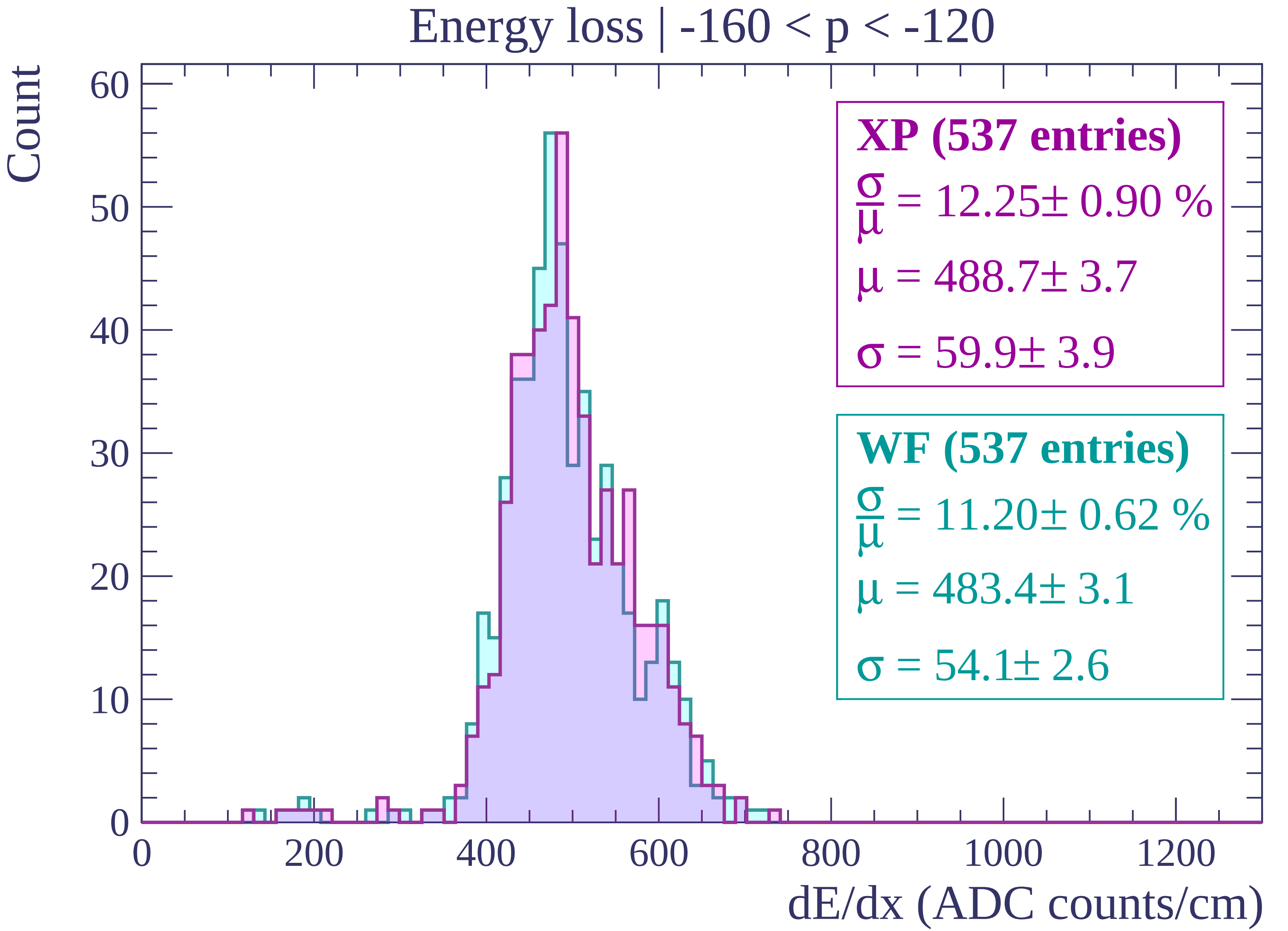


Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (1762 entries)
σ/μ = 8.84± 0.25 %
μ = 394.7± 1.0
σ = 34.9± 0.9
WF (1762 entries)
σ/μ = 9.05± 0.27 %
μ = 394.8± 1.0
σ = 35.7± 1.0

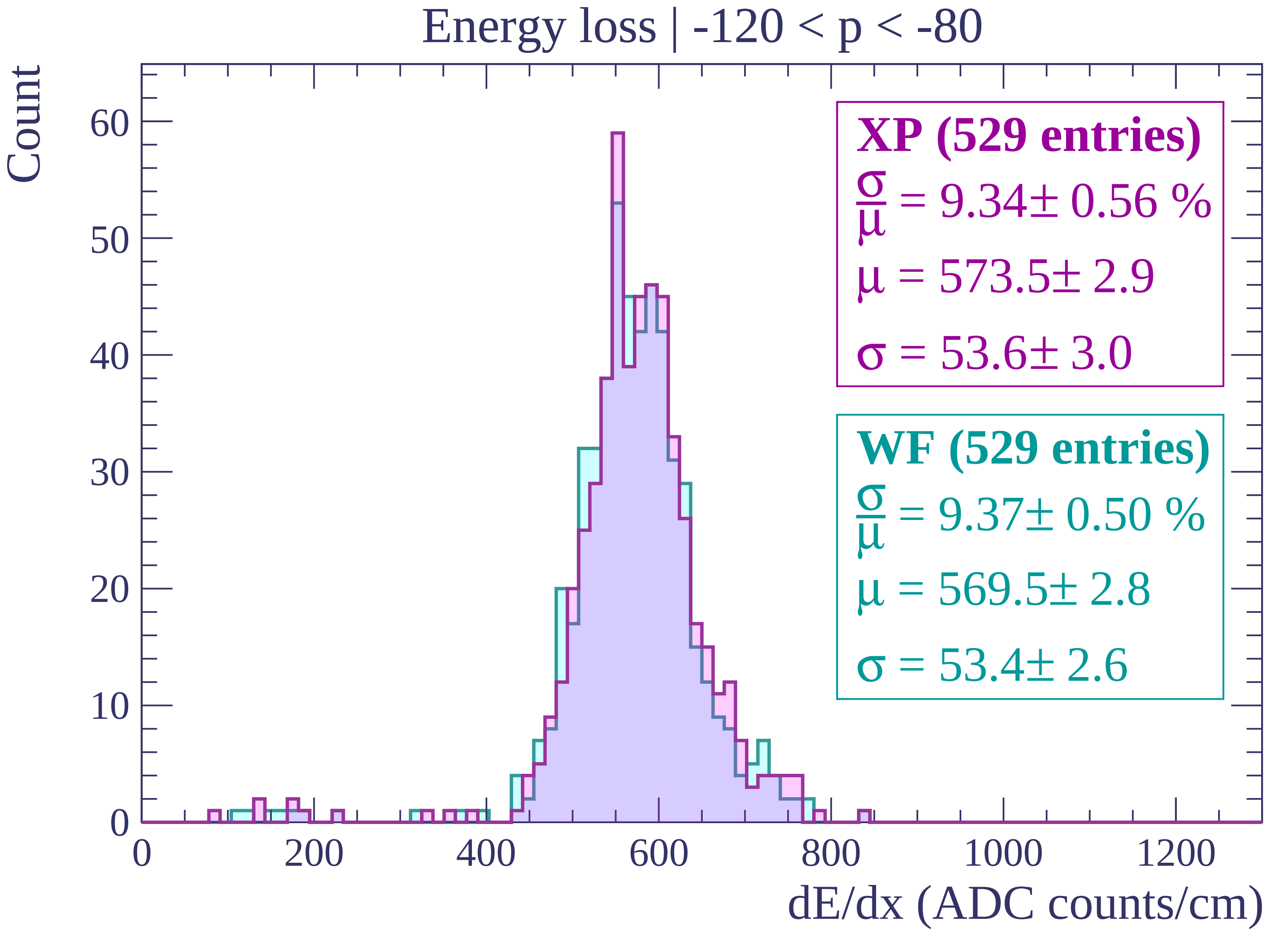


Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP (965 entries)
$\frac{\sigma}{\mu}$ = 9.17± 0.41 %
μ = 401.0± 1.5
σ = 36.8± 1.5
WF (965 entries)
$\frac{\sigma}{\mu}$ = 8.98± 0.38 %
μ = 398.8± 1.5
σ = 35.8± 1.4

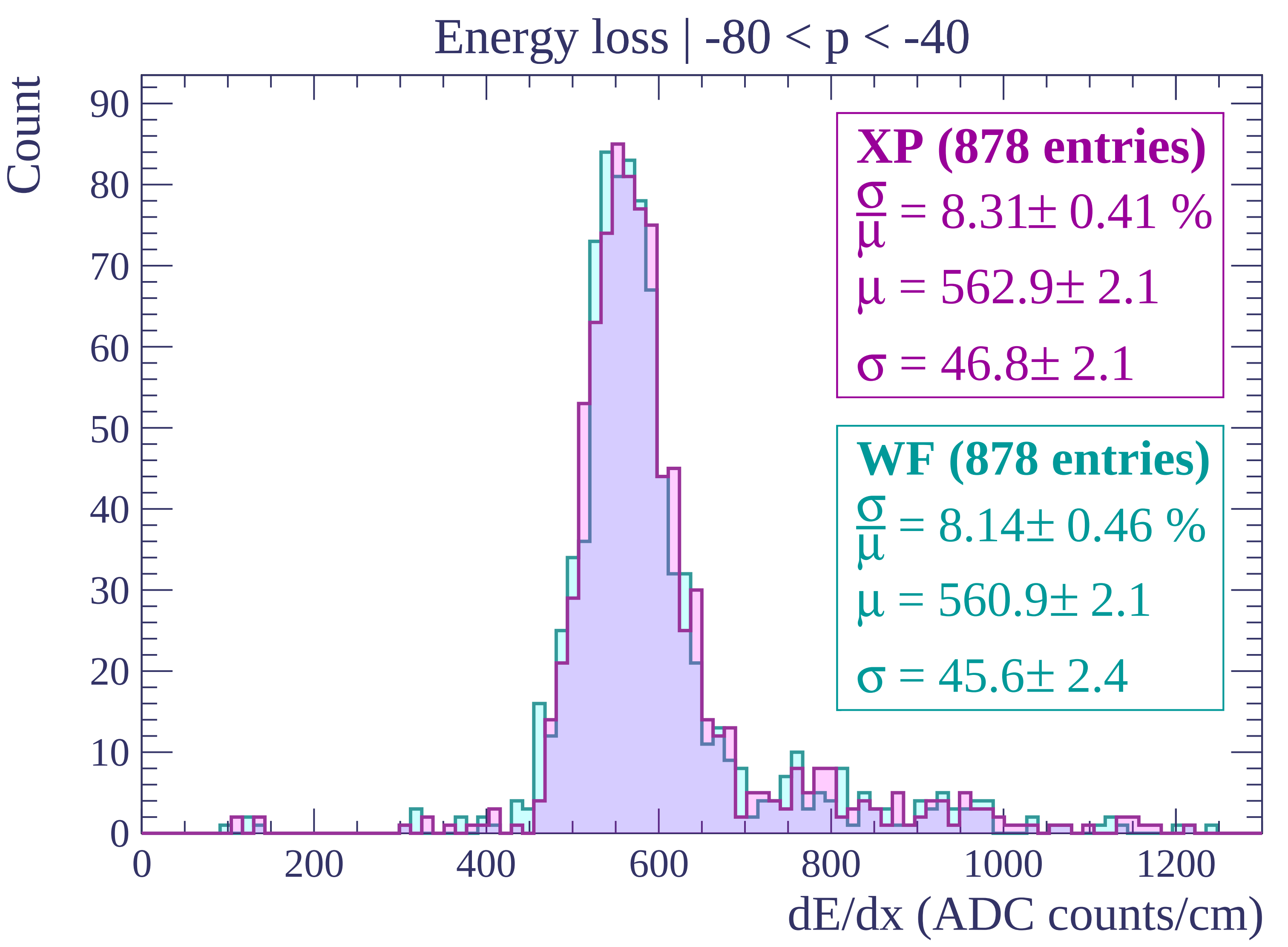

Energy loss | -200 < p < -160
Count
dE/dx (ADC counts/cm)
XP (619 entries)
σ/μ = 9.05± 0.48 %
μ = 426.4± 1.9
σ = 38.6± 1.9
WF (619 entries)
σ/μ = 9.12± 0.45 %
μ = 422.2± 1.9
σ = 38.5± 1.7
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200

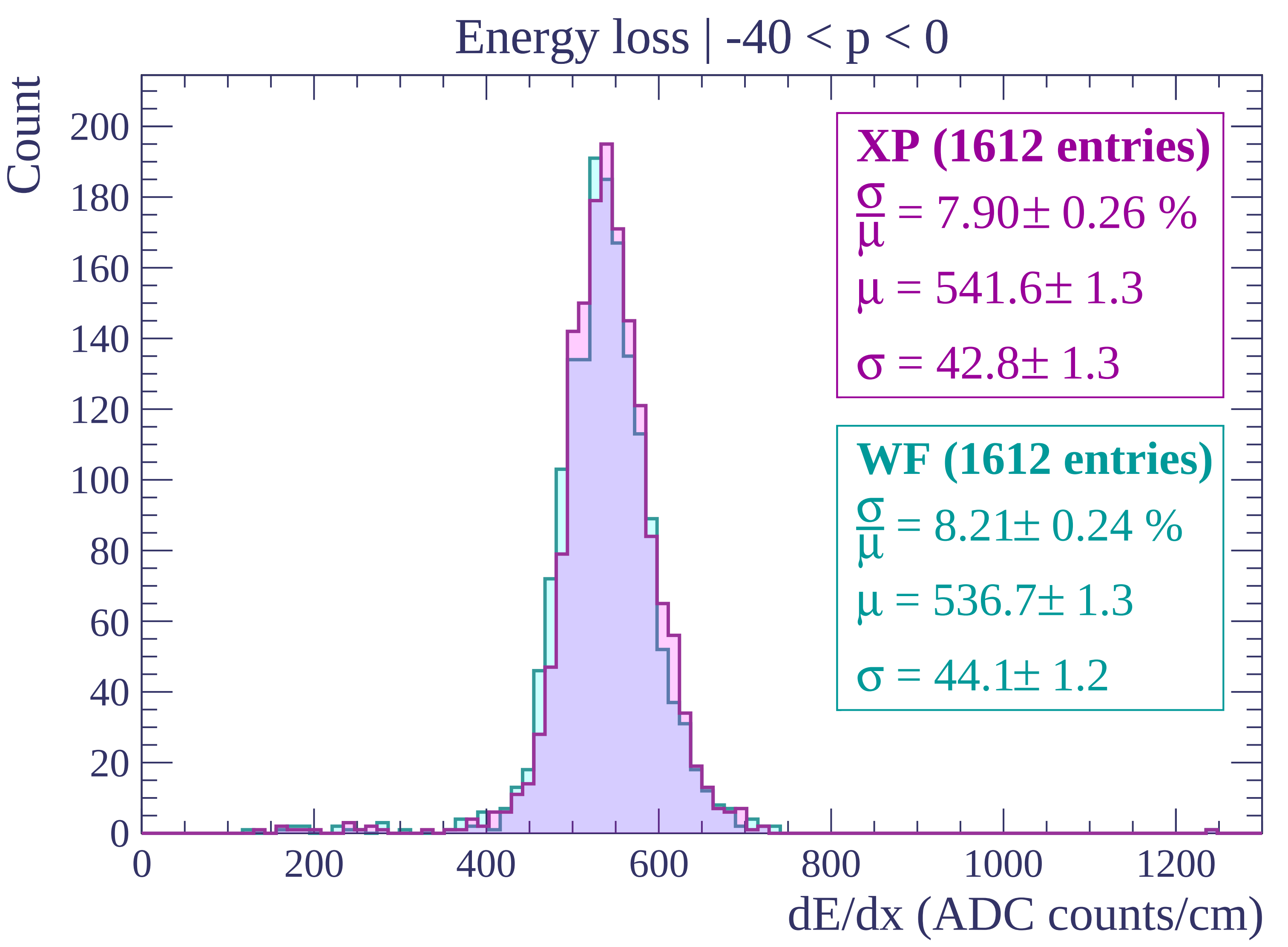


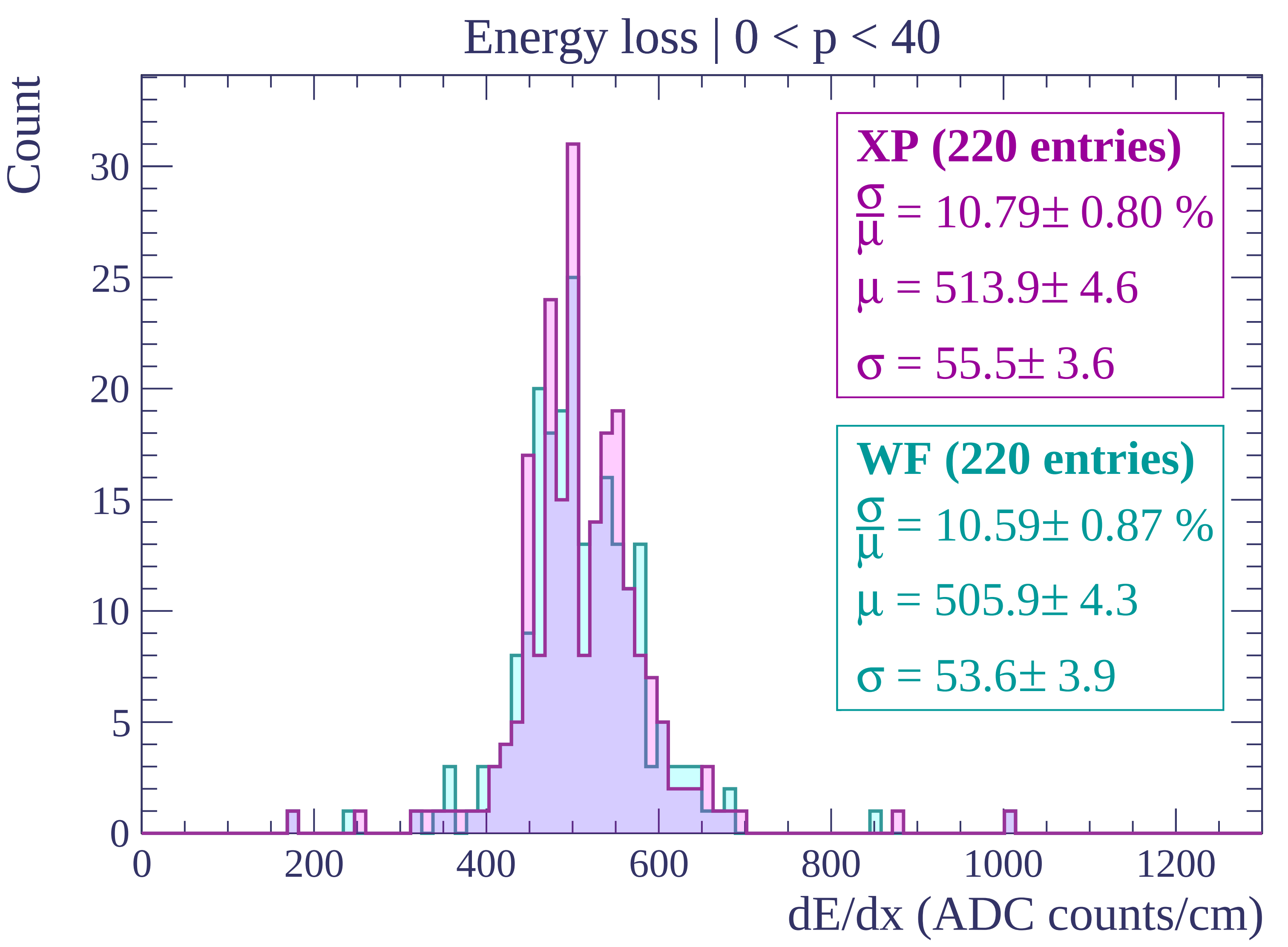
Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP (537 entries)
σ/μ = 12.25± 0.90 %
μ = 488.7± 3.7
σ = 59.9± 3.9
WF (537 entries)
σ/μ = 11.20± 0.62 %
μ = 483.4± 3.1
σ = 54.1± 2.6

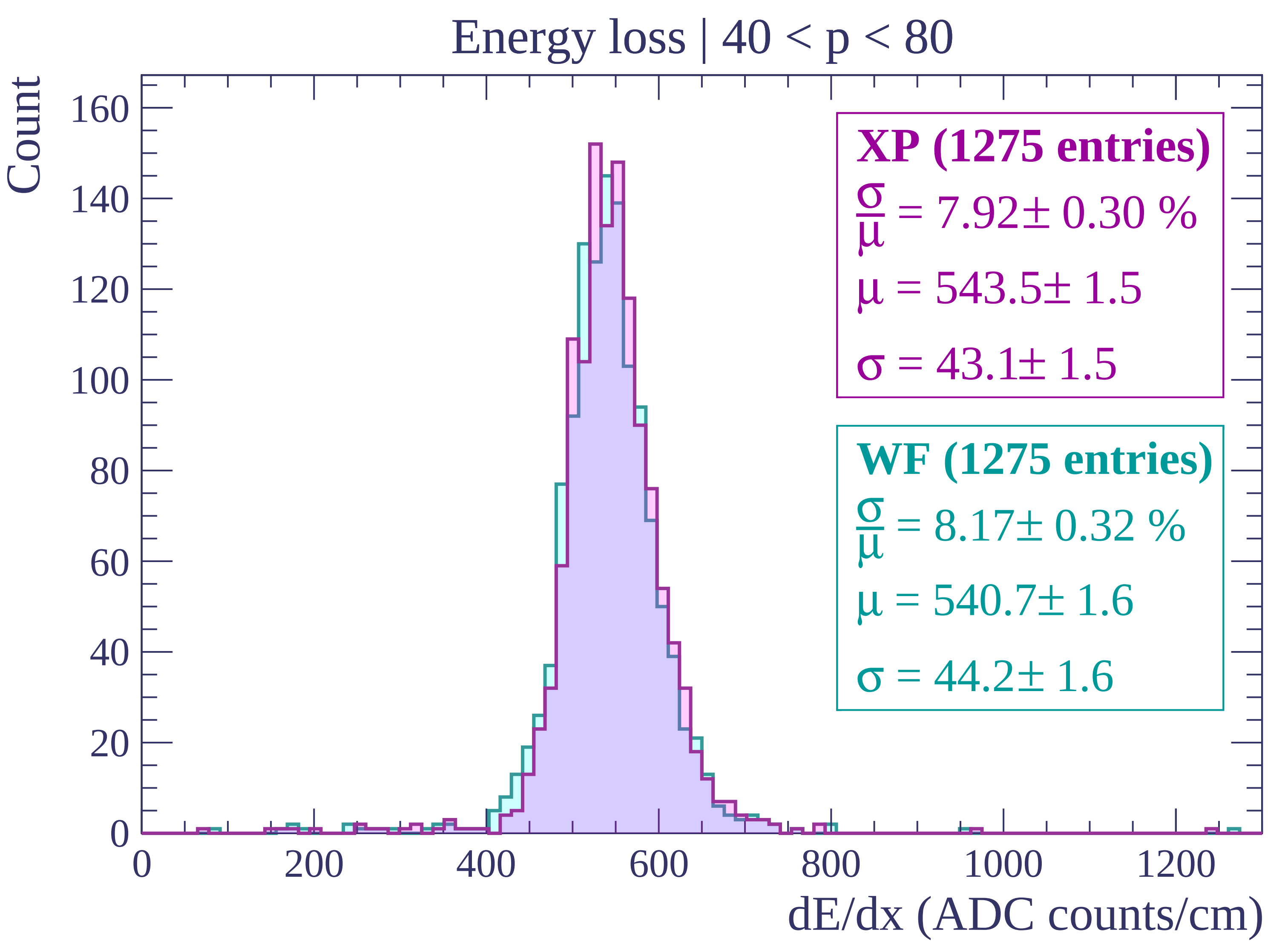


Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
XP (529 entries)
$\frac{\sigma}{\mu}$ = 9.34± 0.56 %
μ = 573.5± 2.9
σ = 53.6± 3.0
WF (529 entries)
$\frac{\sigma}{\mu}$ = 9.37± 0.50 %
μ = 569.5± 2.8
σ = 53.4± 2.6
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200

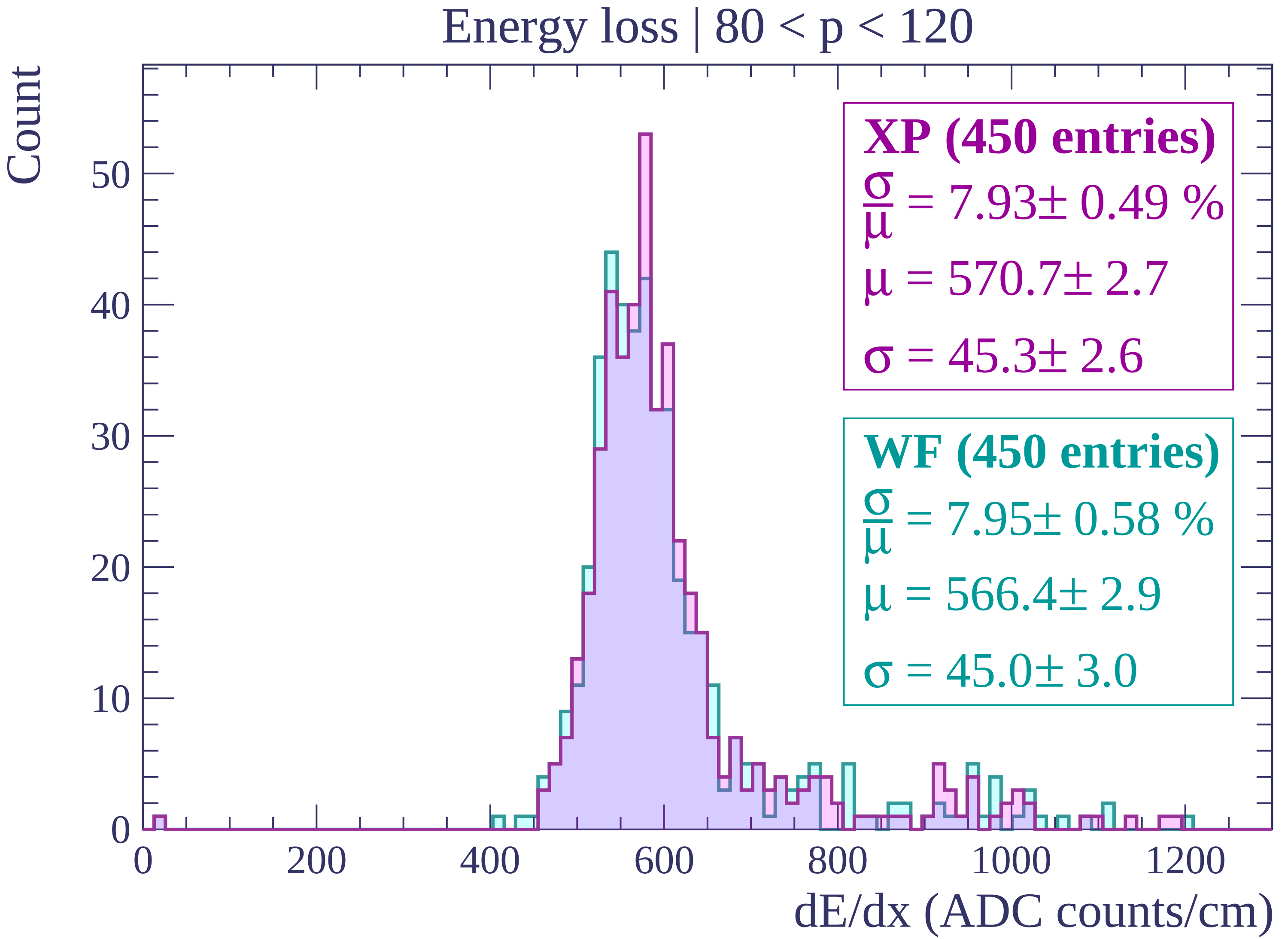

Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
XP (878 entries)
σ/μ = 8.31± 0.41 %
μ = 562.9± 2.1
σ = 46.8± 2.1
WF (878 entries)
σ/μ = 8.14± 0.46 %
μ = 560.9± 2.1
σ = 45.6± 2.4

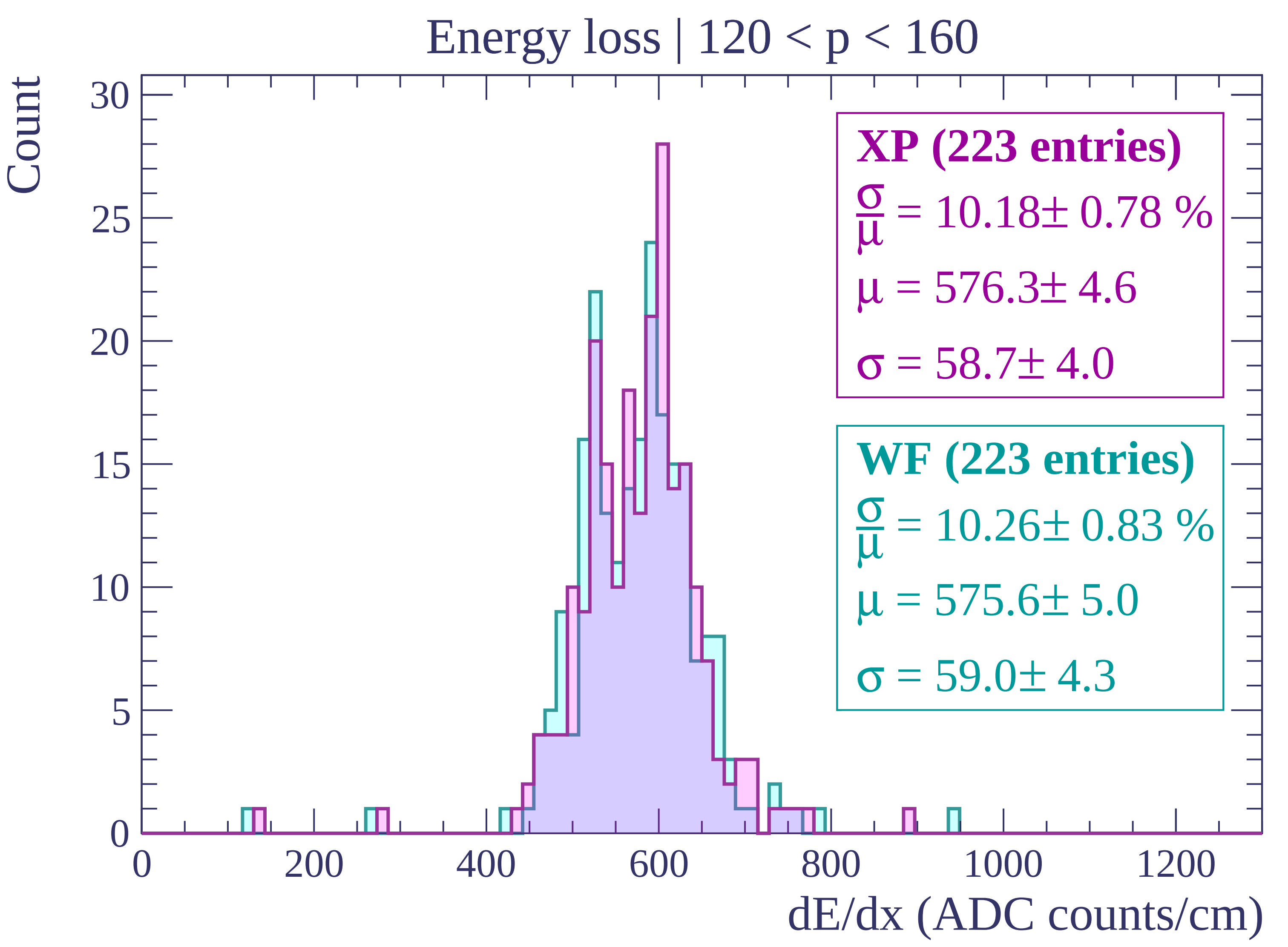

Energy loss | -40 < p < 0
Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1612 entries)
σ/μ = 7.90± 0.26 %
μ = 541.6± 1.3
σ = 42.8± 1.3
WF (1612 entries)
σ/μ = 8.21± 0.24 %
μ = 536.7± 1.3
σ = 44.1± 1.2


Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP (220 entries)
σ/μ = 10.79± 0.80 %
μ = 513.9± 4.6
σ = 55.5± 3.6
WF (220 entries)
σ/μ = 10.59± 0.87 %
μ = 505.9± 4.3
σ = 53.6± 3.9

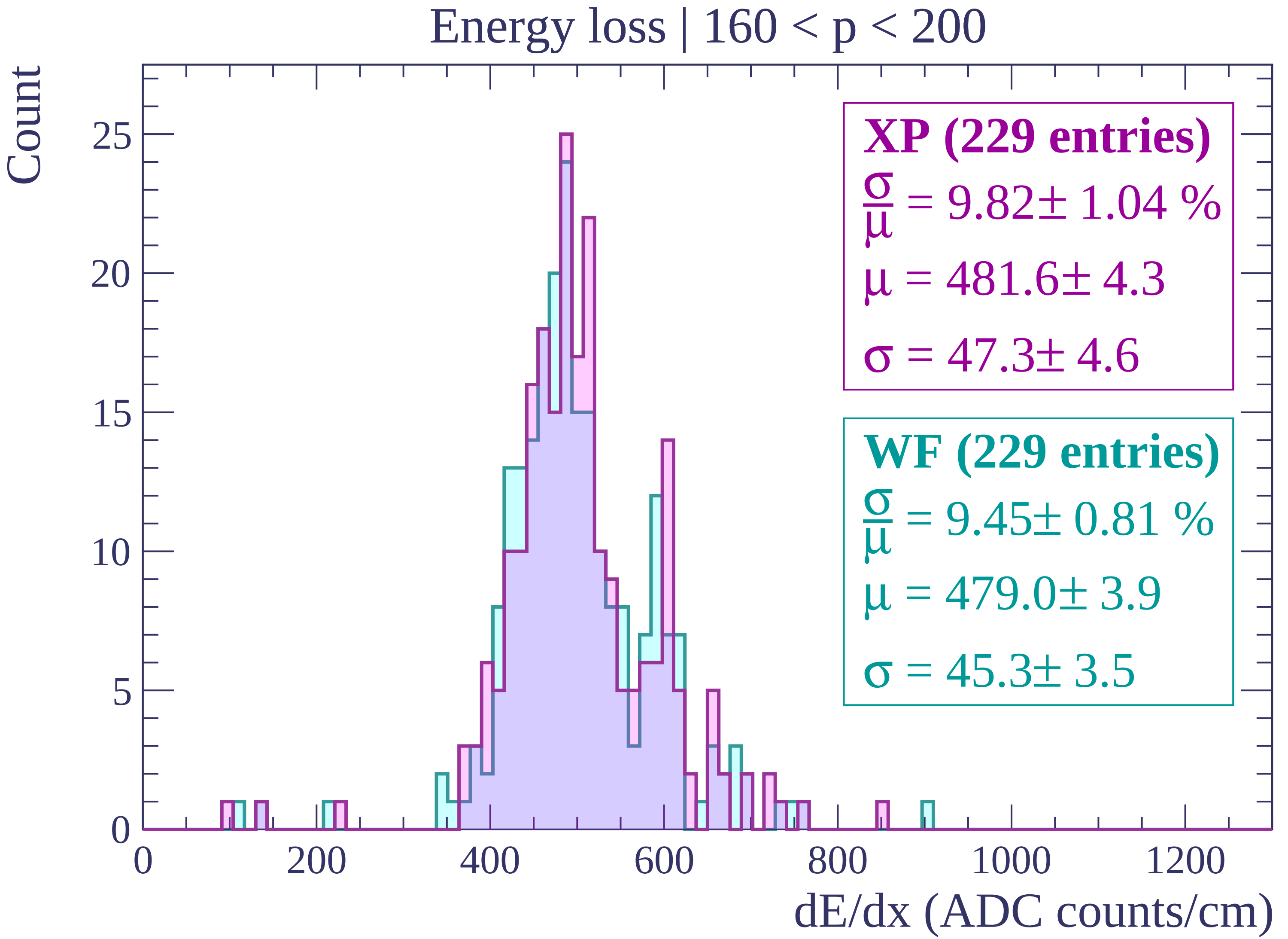


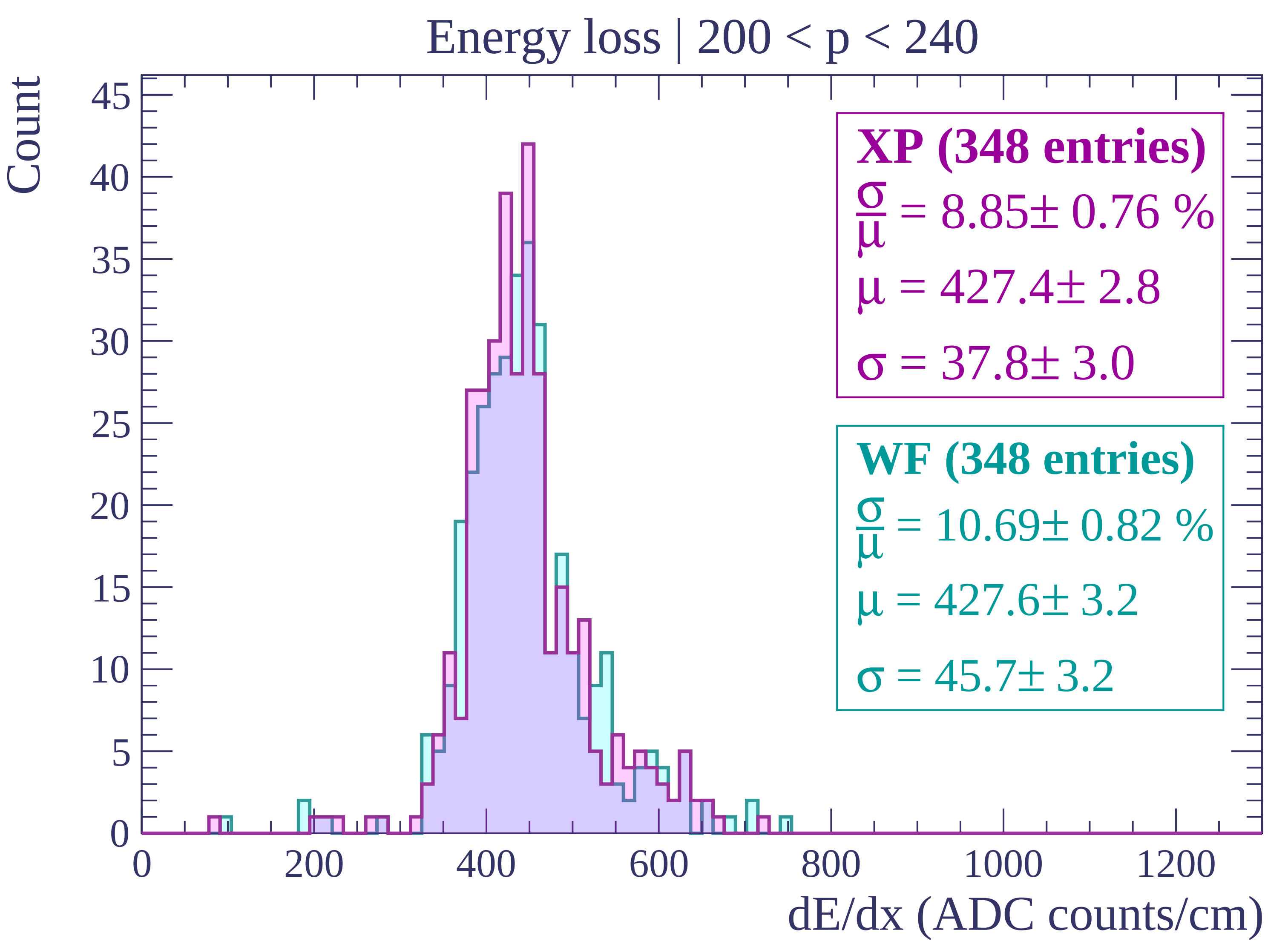
Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1275 entries)
σ/μ = 7.92± 0.30 %
μ = 543.5± 1.5
σ = 43.1± 1.5
WF (1275 entries)
σ/μ = 8.17± 0.32 %
μ = 540.7± 1.6
σ = 44.2± 1.6

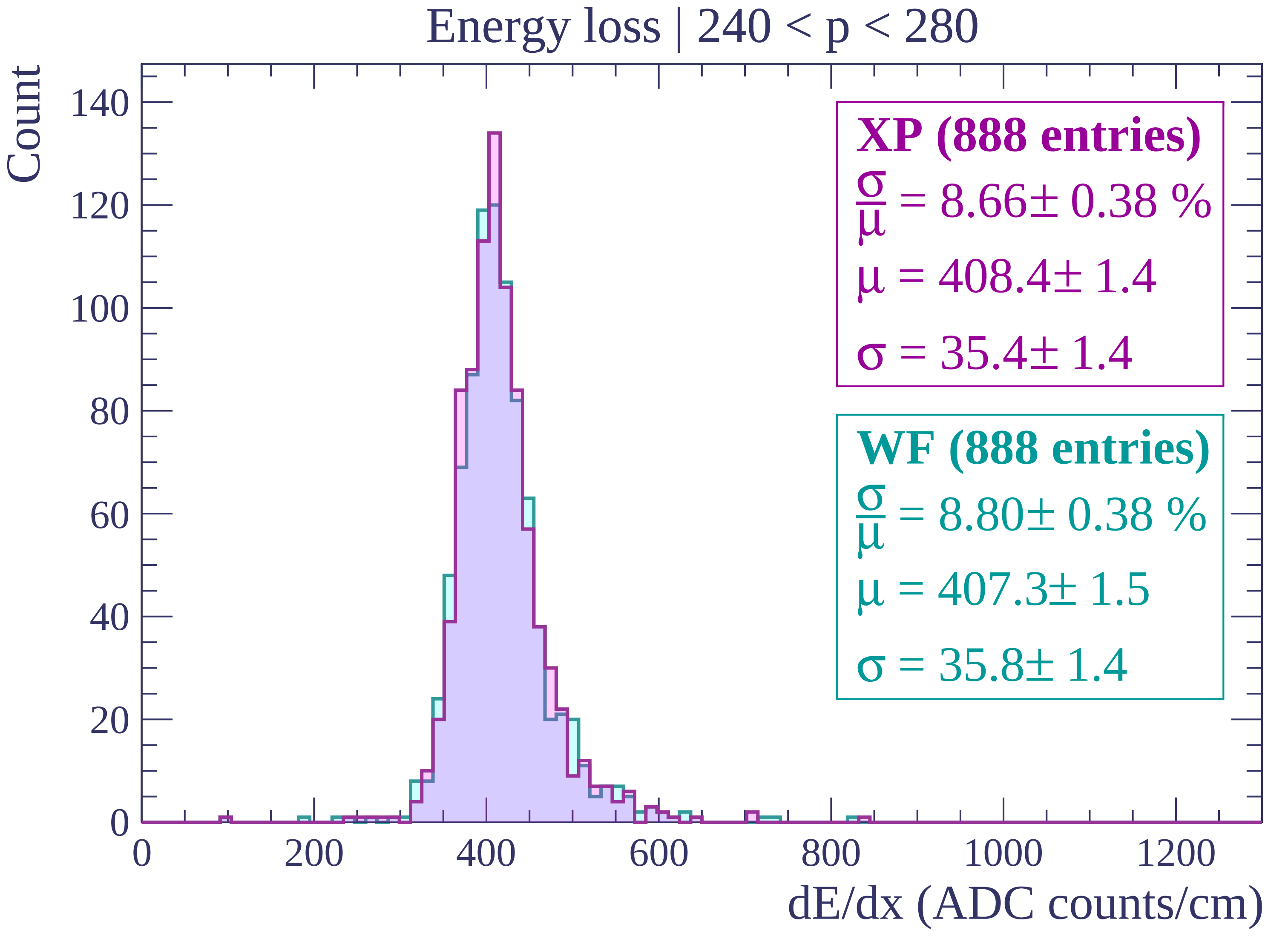

Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP (450 entries)
$\frac{\sigma}{\mu}$ = 7.93± 0.49 %
μ = 570.7± 2.7
σ = 45.3± 2.6
WF (450 entries)
$\frac{\sigma}{\mu}$ = 7.95± 0.58 %
μ = 566.4± 2.9
σ = 45.0± 3.0



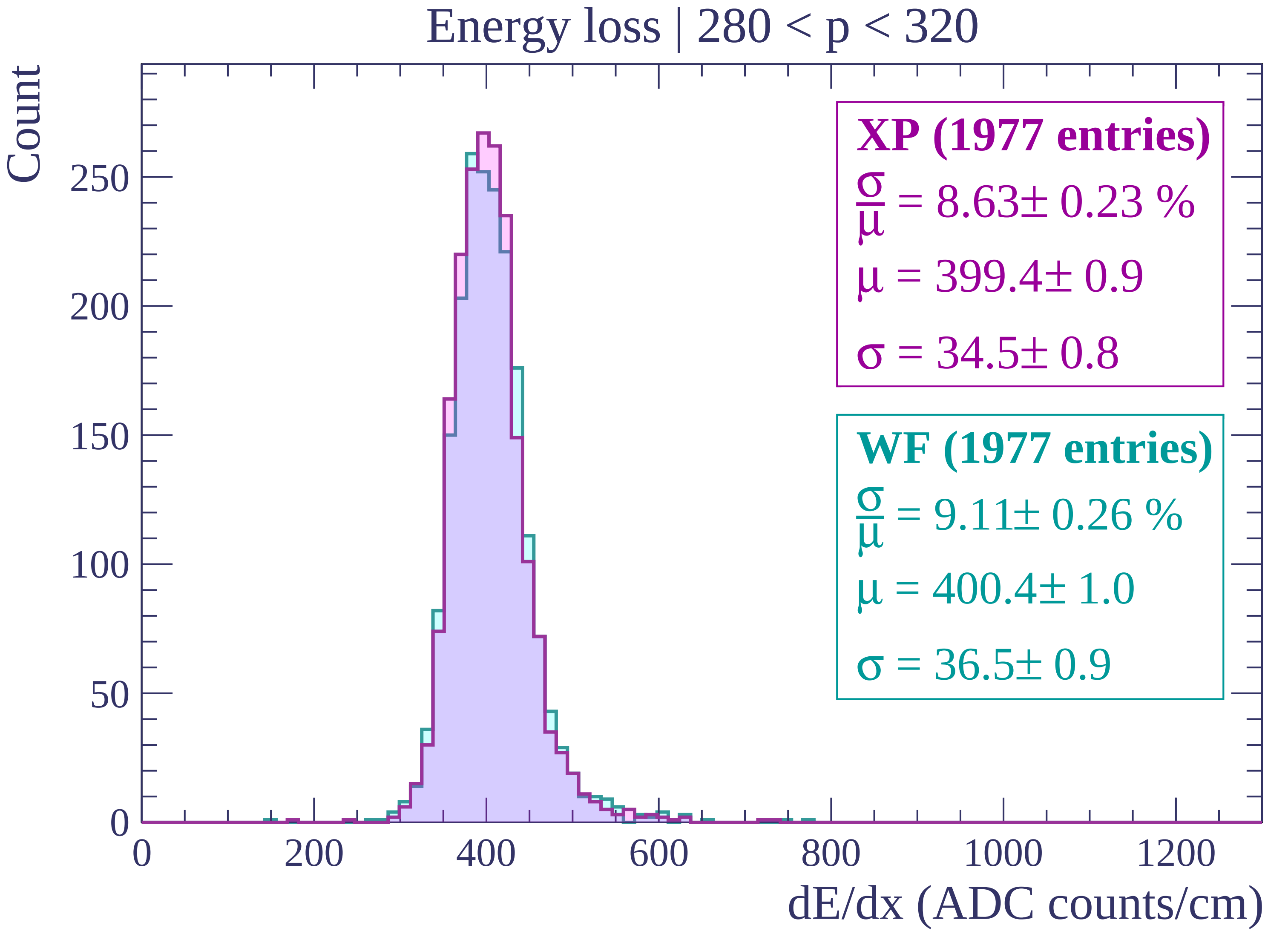
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP (223 entries)
σ/μ = 10.18± 0.78 %
μ = 576.3± 4.6
σ = 58.7± 4.0
WF (223 entries)
σ/μ = 10.26± 0.83 %
μ = 575.6± 5.0
σ = 59.0± 4.3


Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP (229 entries)
σ/μ = 9.82± 1.04 %
μ = 481.6± 4.3
σ = 47.3± 4.6
WF (229 entries)
σ/μ = 9.45± 0.81 %
μ = 479.0± 3.9
σ = 45.3± 3.5

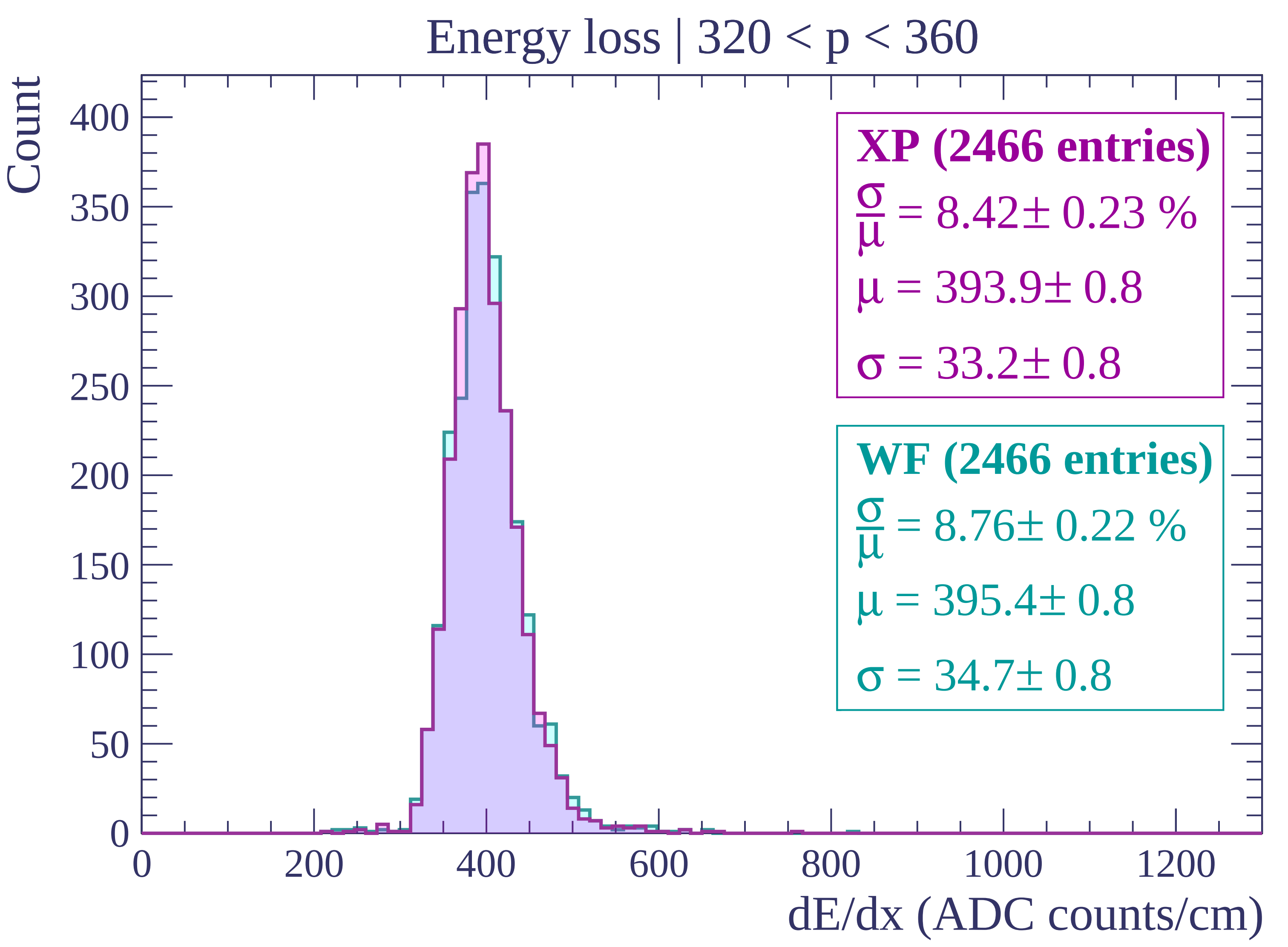

Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP (348 entries)
σ/μ = 8.85± 0.76 %
μ = 427.4± 2.8
σ = 37.8± 3.0
WF (348 entries)
σ/μ = 10.69± 0.82 %
μ = 427.6± 3.2
σ = 45.7± 3.2

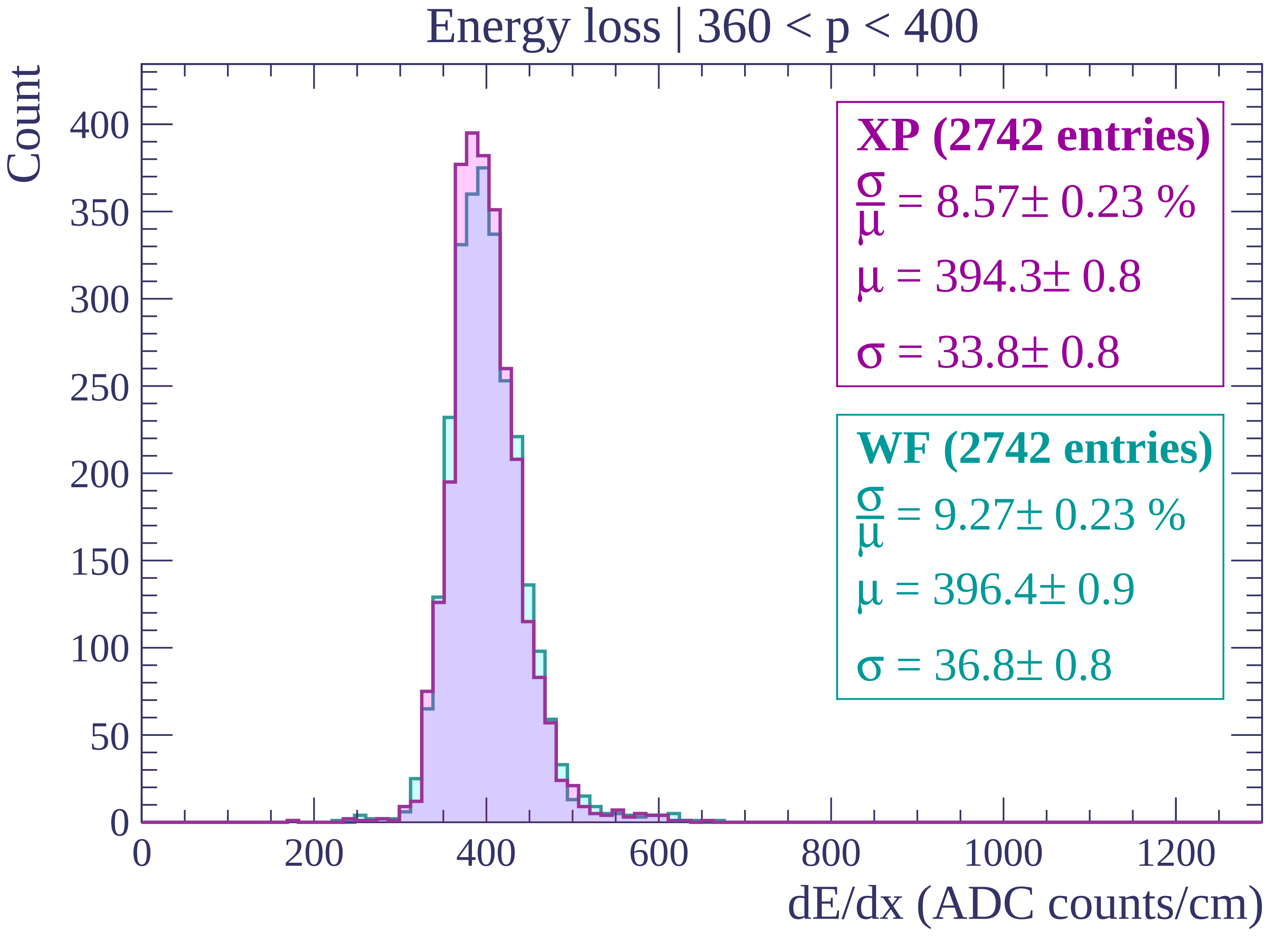


Energy loss | 240 < p < 280
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (888 entries)
$\frac{\sigma}{\mu}$ = 8.66± 0.38 %
μ = 408.4± 1.4
σ = 35.4± 1.4
WF (888 entries)
$\frac{\sigma}{\mu}$ = 8.80± 0.38 %
μ = 407.3± 1.5
σ = 35.8± 1.4

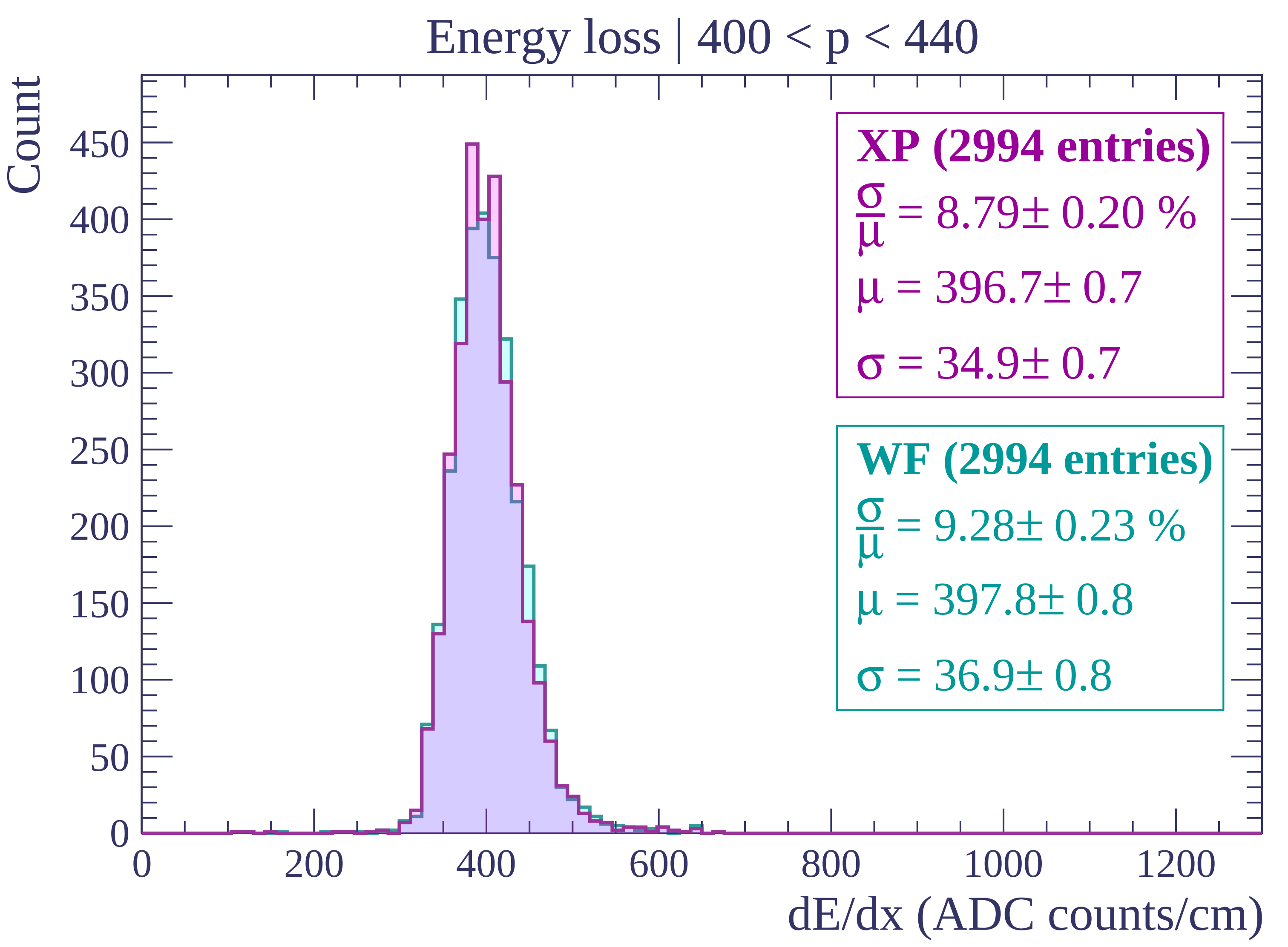


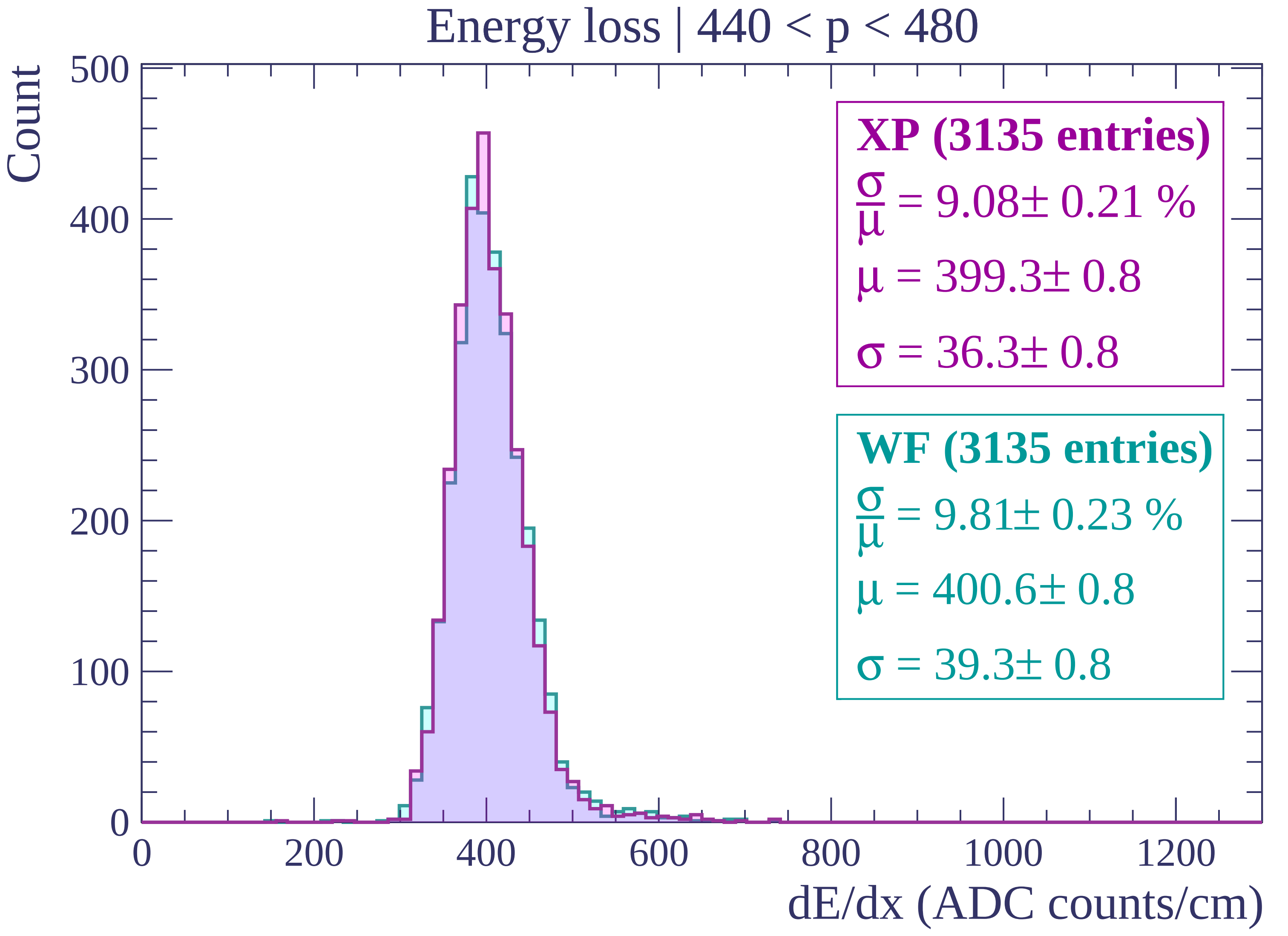
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (1977 entries)
$\frac{\sigma}{\mu}$ = 8.63± 0.23 %
μ = 399.4± 0.9
σ = 34.5± 0.8
WF (1977 entries)
$\frac{\sigma}{\mu}$ = 9.11± 0.26 %
μ = 400.4± 1.0
σ = 36.5± 0.9


Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (2466 entries)
σ/μ = 8.42± 0.23 %
μ = 393.9± 0.8
σ = 33.2± 0.8
WF (2466 entries)
σ/μ = 8.76± 0.22 %
μ = 395.4± 0.8
σ = 34.7± 0.8

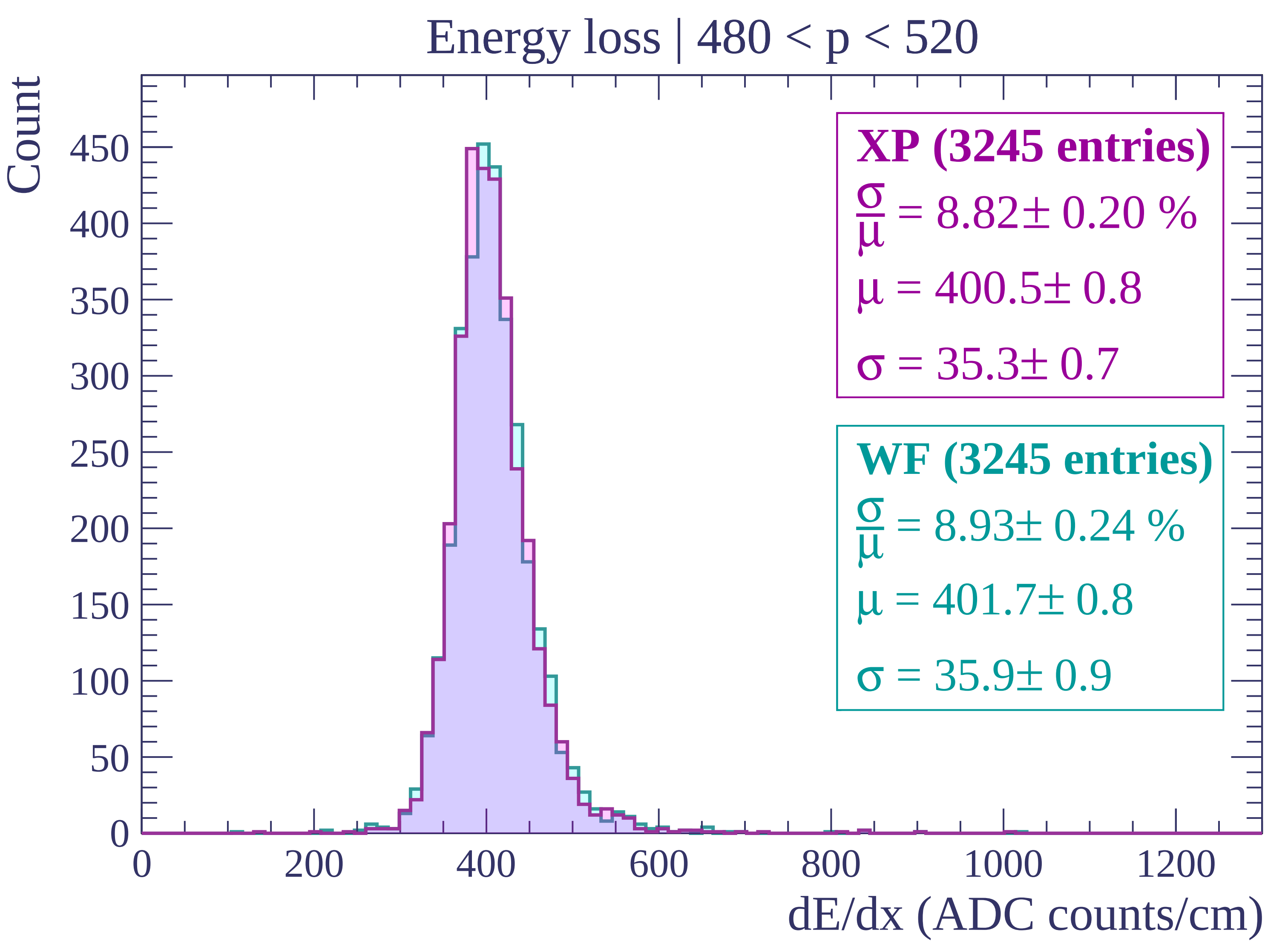


Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (2742 entries)
σ/μ = 8.57± 0.23 %
μ = 394.3± 0.8
σ = 33.8± 0.8
WF (2742 entries)
σ/μ = 9.27± 0.23 %
μ = 396.4± 0.9
σ = 36.8± 0.8

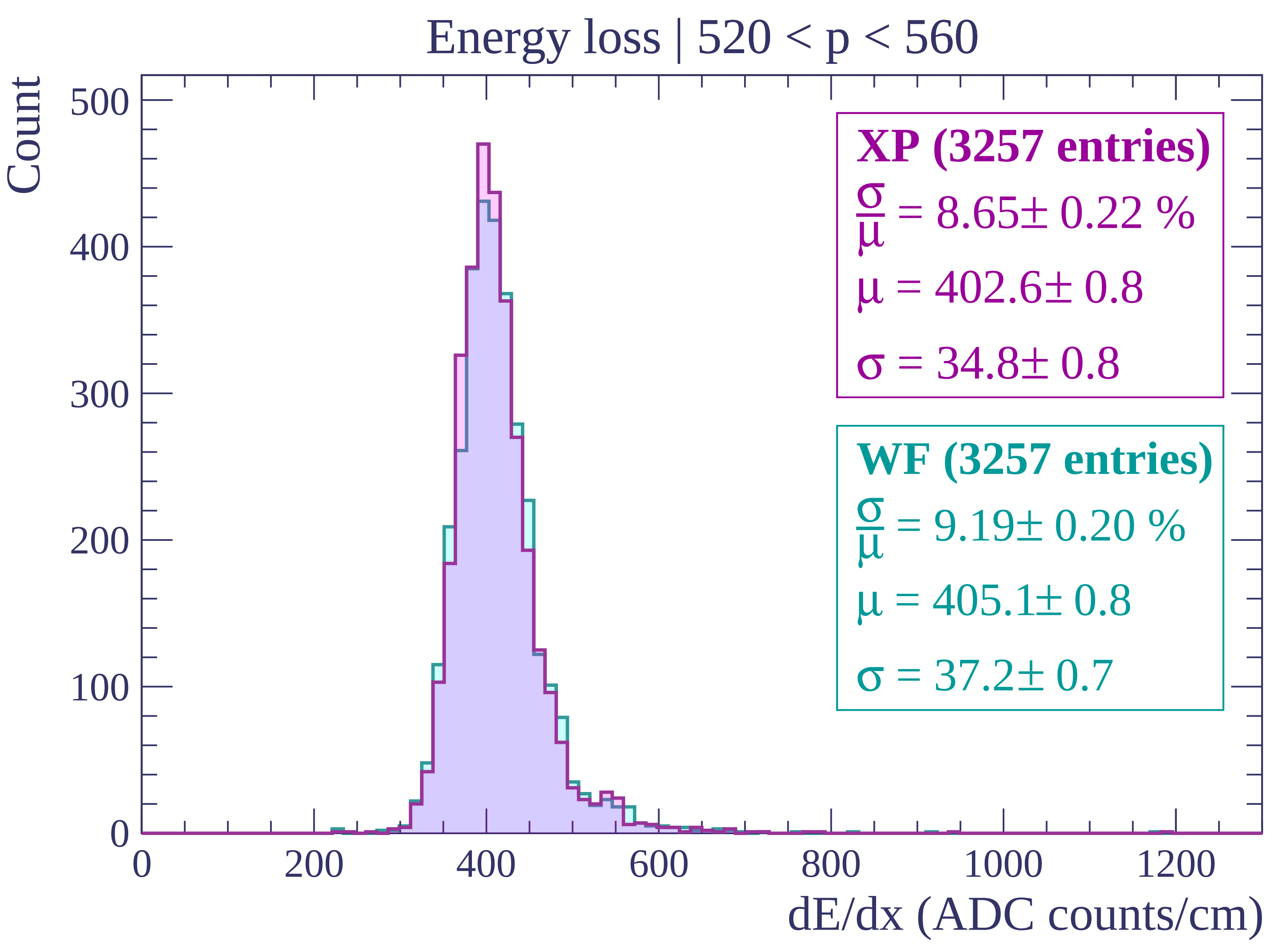

Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (2994 entries)
σ/μ = 8.79± 0.20 %
μ = 396.7± 0.7
σ = 34.9± 0.7
WF (2994 entries)
σ/μ = 9.28± 0.23 %
μ = 397.8± 0.8
σ = 36.9± 0.8



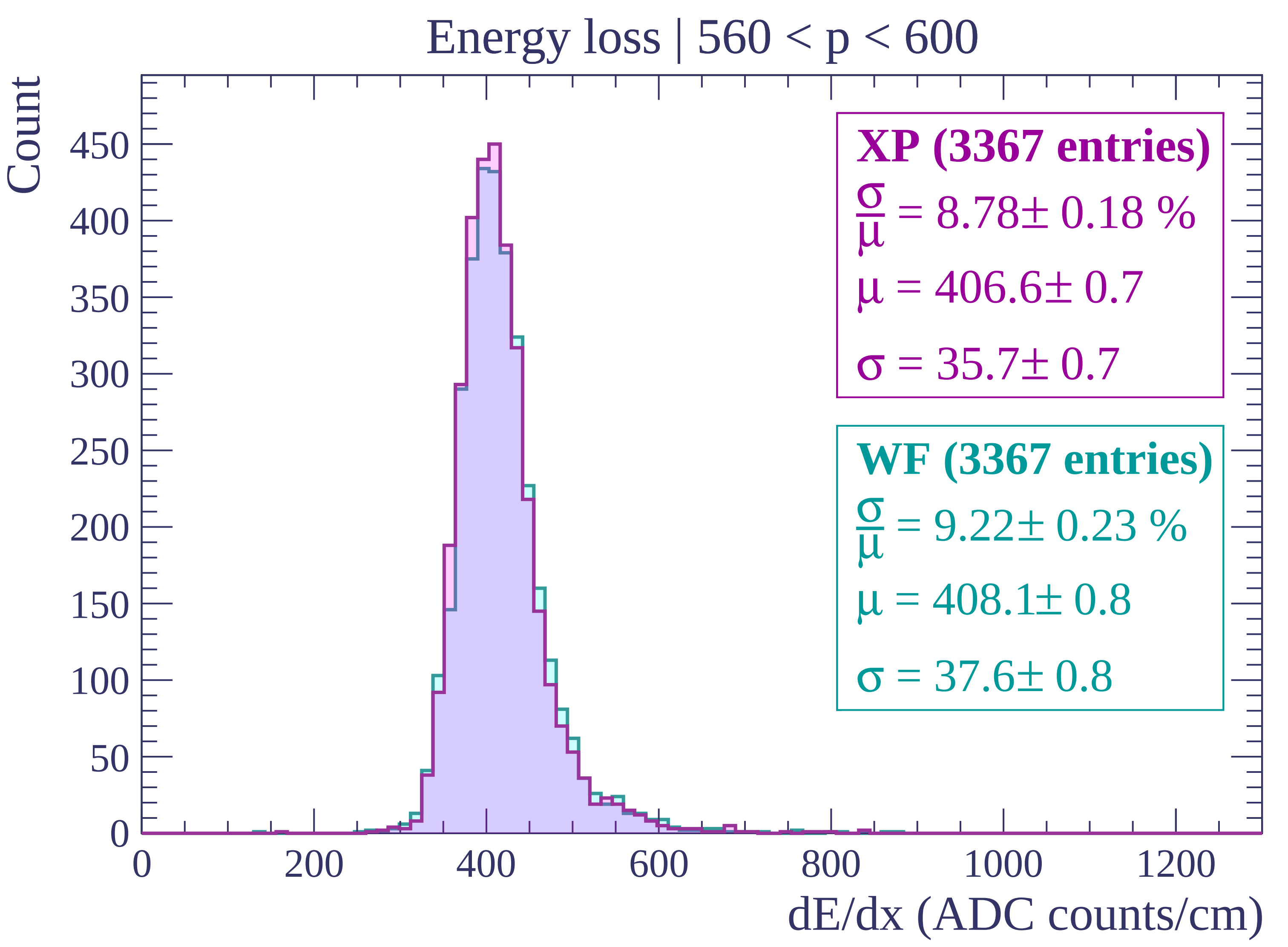
Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3135 entries)
σ/μ = 9.08± 0.21 %
μ = 399.3± 0.8
σ = 36.3± 0.8
WF (3135 entries)
σ/μ = 9.81± 0.23 %
μ = 400.6± 0.8
σ = 39.3± 0.8



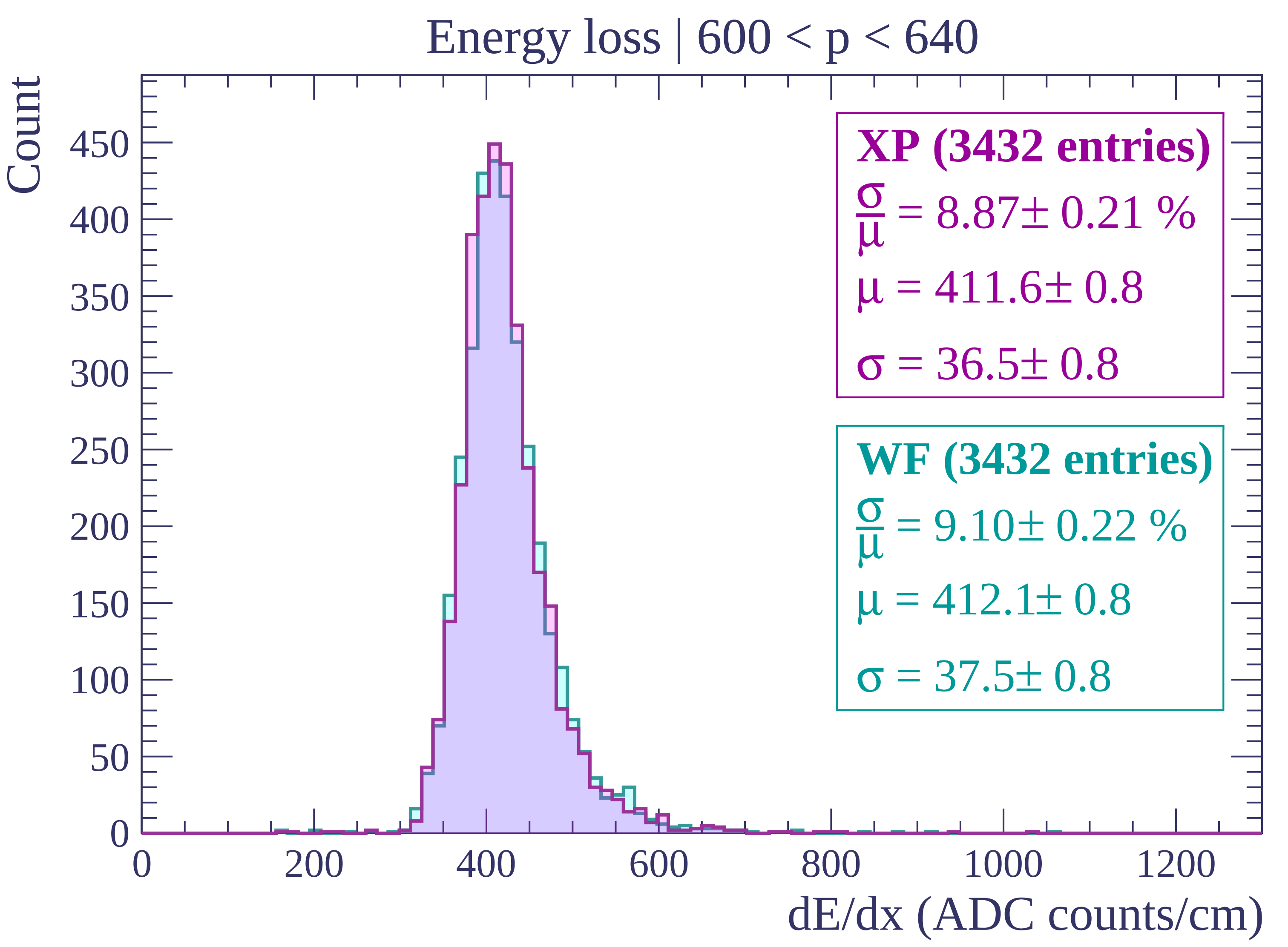
Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3245 entries)
$\frac{\sigma}{\mu}$ = 8.82± 0.20 %
μ = 400.5± 0.8
σ = 35.3± 0.7
WF (3245 entries)
$\frac{\sigma}{\mu}$ = 8.93± 0.24 %
μ = 401.7± 0.8
σ = 35.9± 0.9



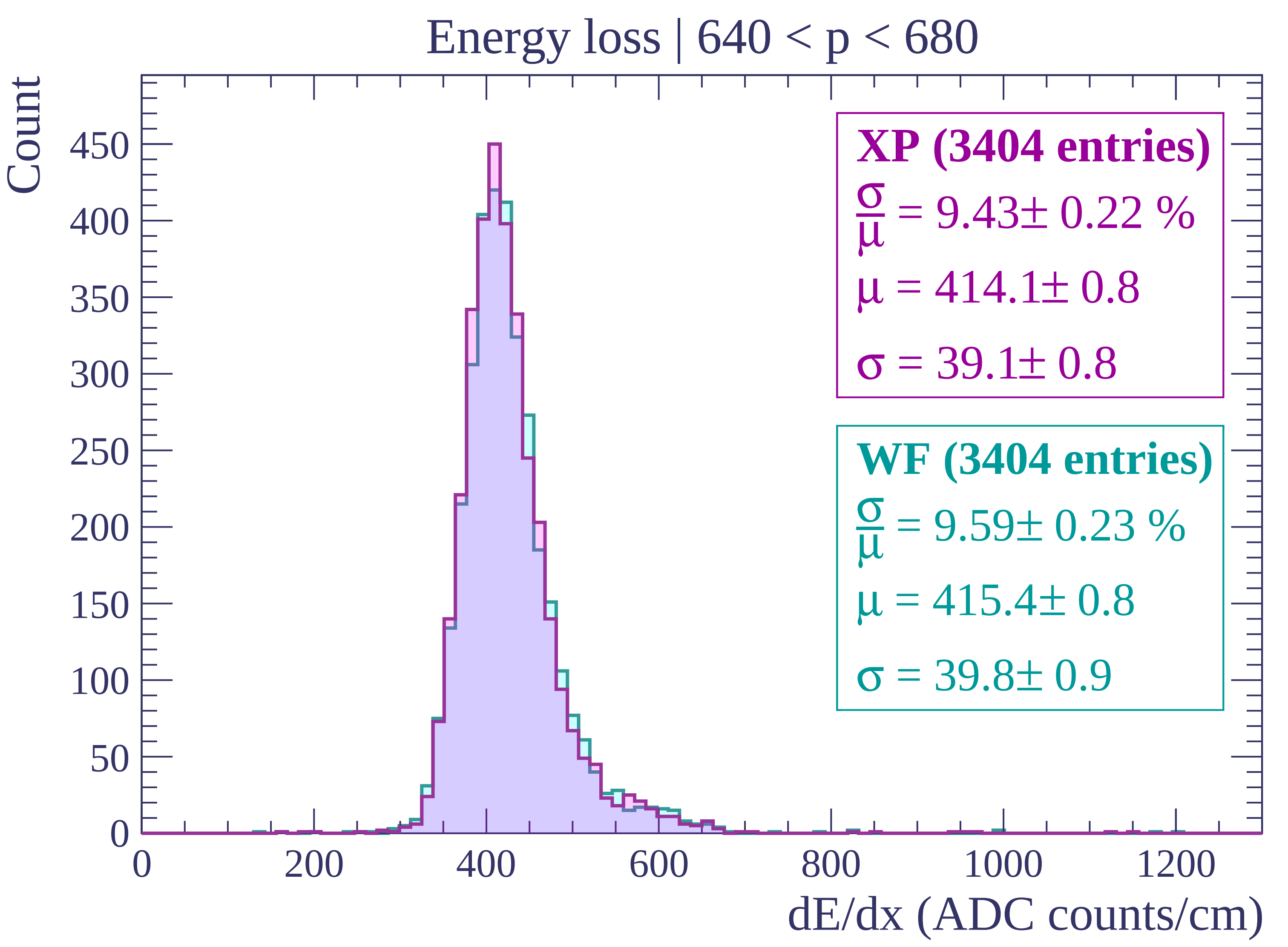
Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3257 entries)
σ/μ = 8.65± 0.22 %
μ = 402.6± 0.8
σ = 34.8± 0.8
WF (3257 entries)
σ/μ = 9.19± 0.20 %
μ = 405.1± 0.8
σ = 37.2± 0.7

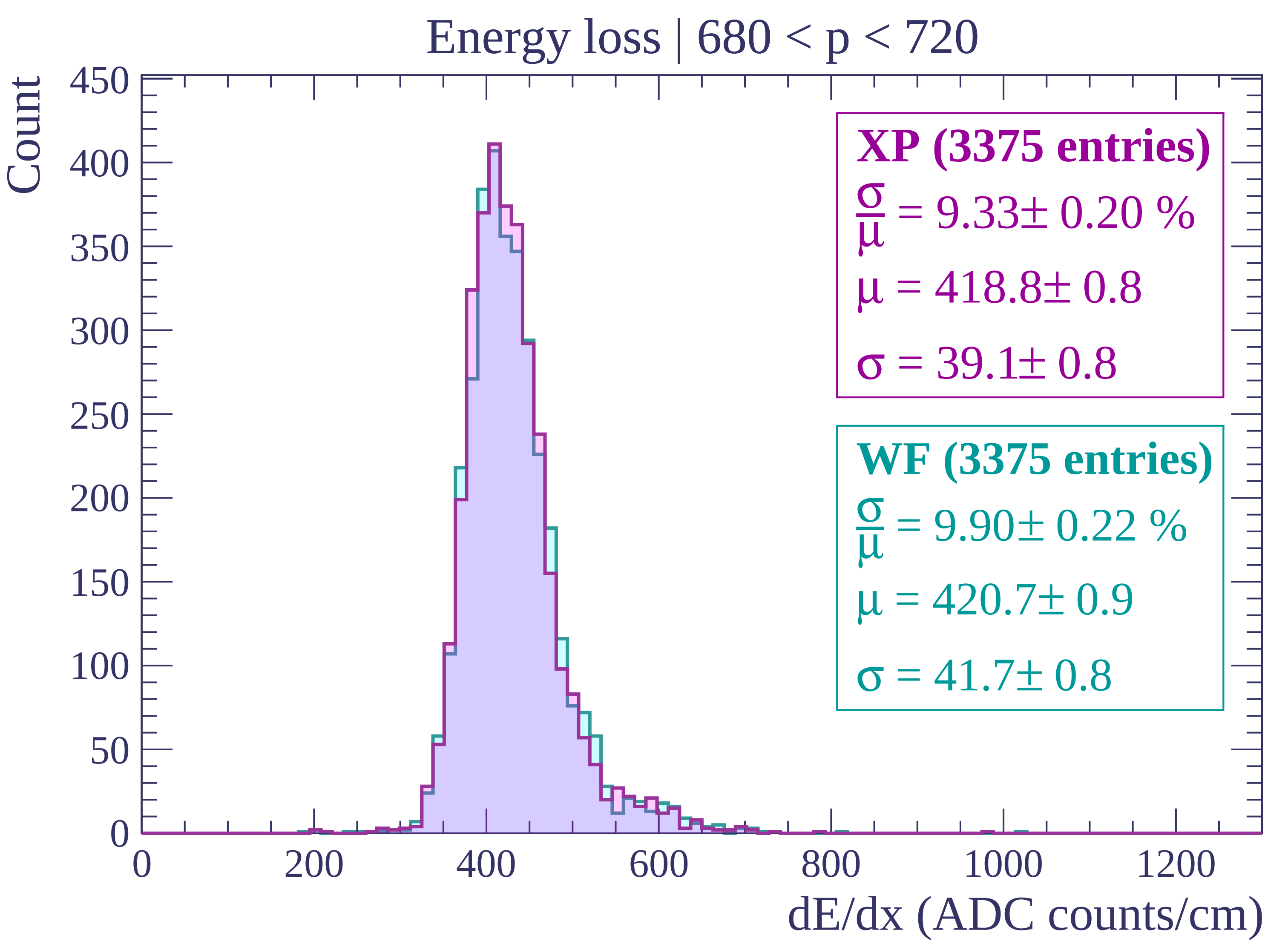


Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3367 entries)
σ/μ = 8.78± 0.18 %
μ = 406.6± 0.7
σ = 35.7± 0.7
WF (3367 entries)
σ/μ = 9.22± 0.23 %
μ = 408.1± 0.8
σ = 37.6± 0.8

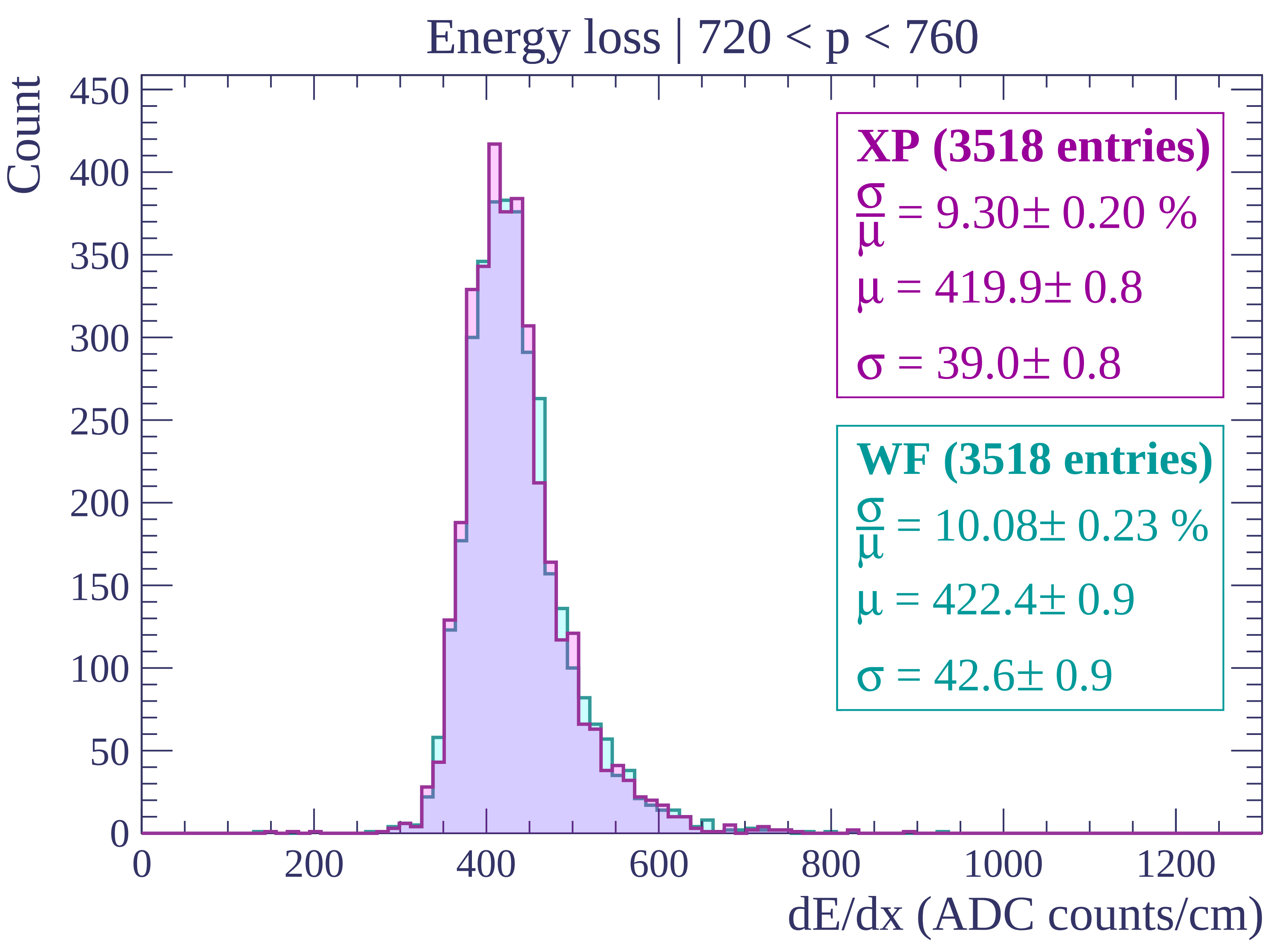


Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3432 entries)
σ/μ = 8.87± 0.21 %
μ = 411.6± 0.8
σ = 36.5± 0.8
WF (3432 entries)
σ/μ = 9.10± 0.22 %
μ = 412.1± 0.8
σ = 37.5± 0.8

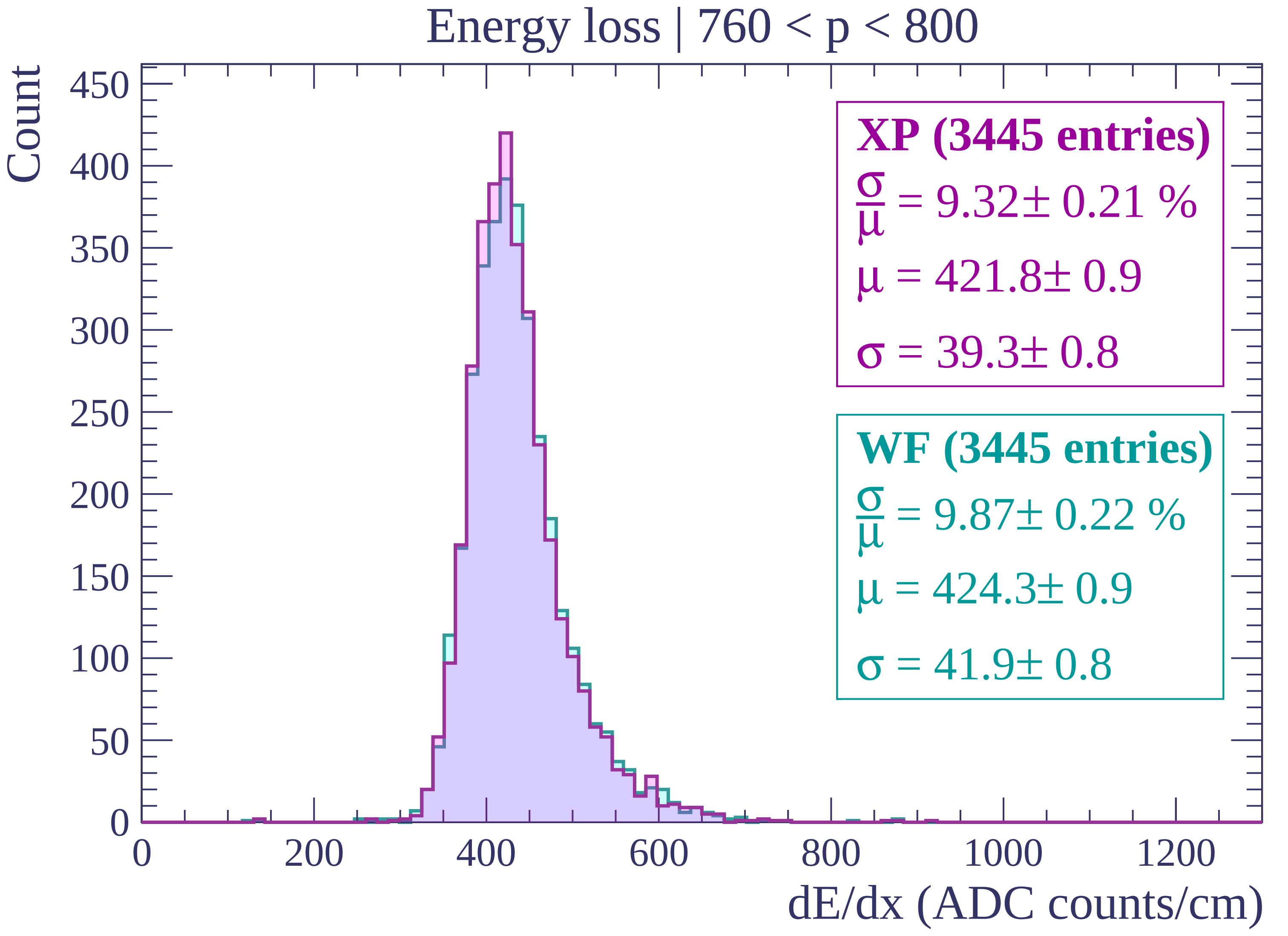

Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3404 entries)
σ/μ = 9.43± 0.22 %
μ = 414.1± 0.8
σ = 39.1± 0.8
WF (3404 entries)
σ/μ = 9.59± 0.23 %
μ = 415.4± 0.8
σ = 39.8± 0.9

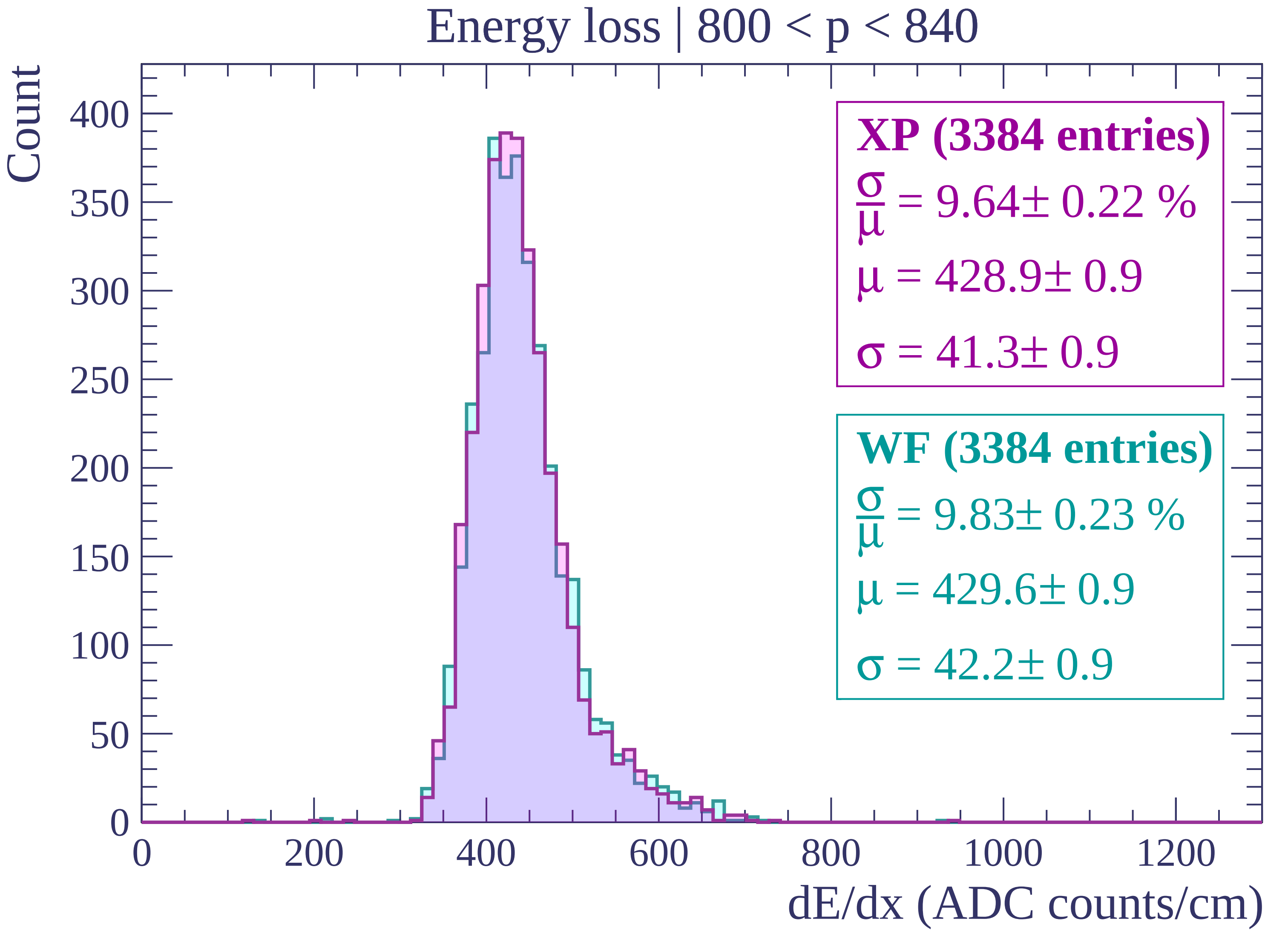


Energy loss | 680 < p < 720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3375 entries)
σ/μ = 9.33± 0.20 %
μ = 418.8± 0.8
σ = 39.1± 0.8
WF (3375 entries)
σ/μ = 9.90± 0.22 %
μ = 420.7± 0.9
σ = 41.7± 0.8

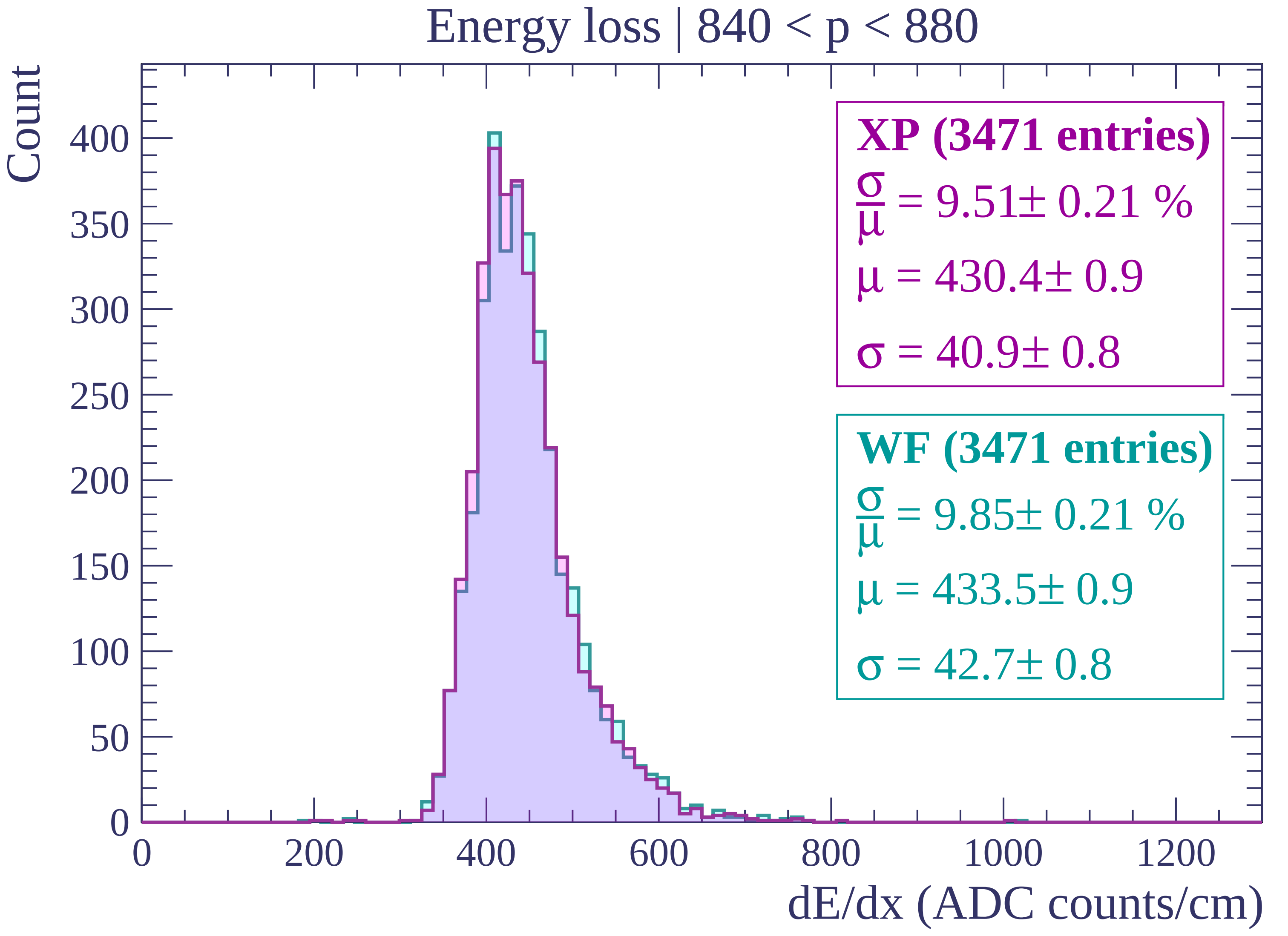


Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
XP (3518 entries)
$\frac{\sigma}{\mu}$ = 9.30± 0.20 %
μ = 419.9± 0.8
σ = 39.0± 0.8
WF (3518 entries)
$\frac{\sigma}{\mu}$ = 10.08± 0.23 %
μ = 422.4± 0.9
σ = 42.6± 0.9
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

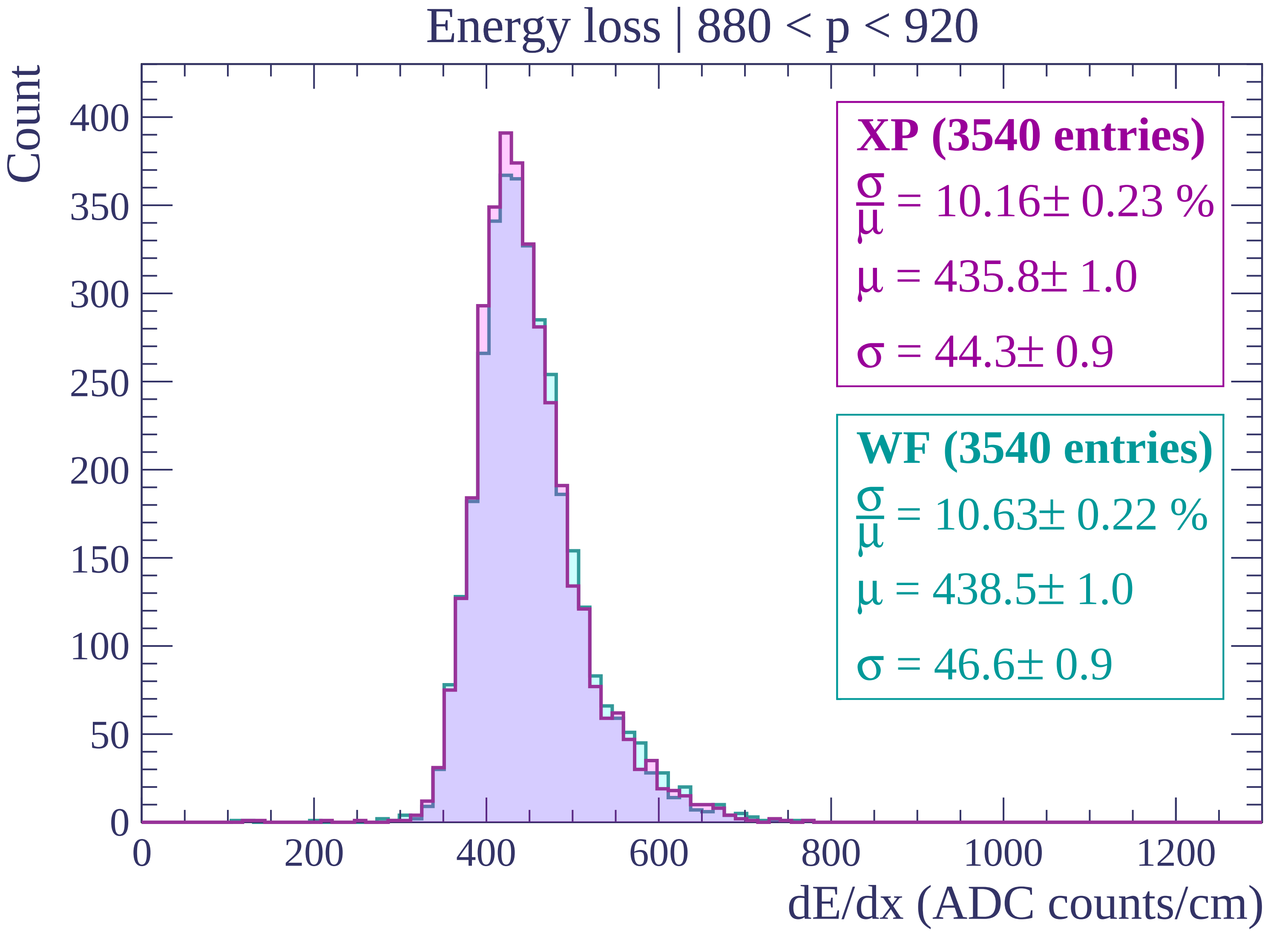

Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
XP (3445 entries)
$\frac{\sigma}{\mu}$ = 9.32± 0.21 %
μ = 421.8± 0.9
σ = 39.3± 0.8
WF (3445 entries)
$\frac{\sigma}{\mu}$ = 9.87± 0.22 %
μ = 424.3± 0.9
σ = 41.9± 0.8
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



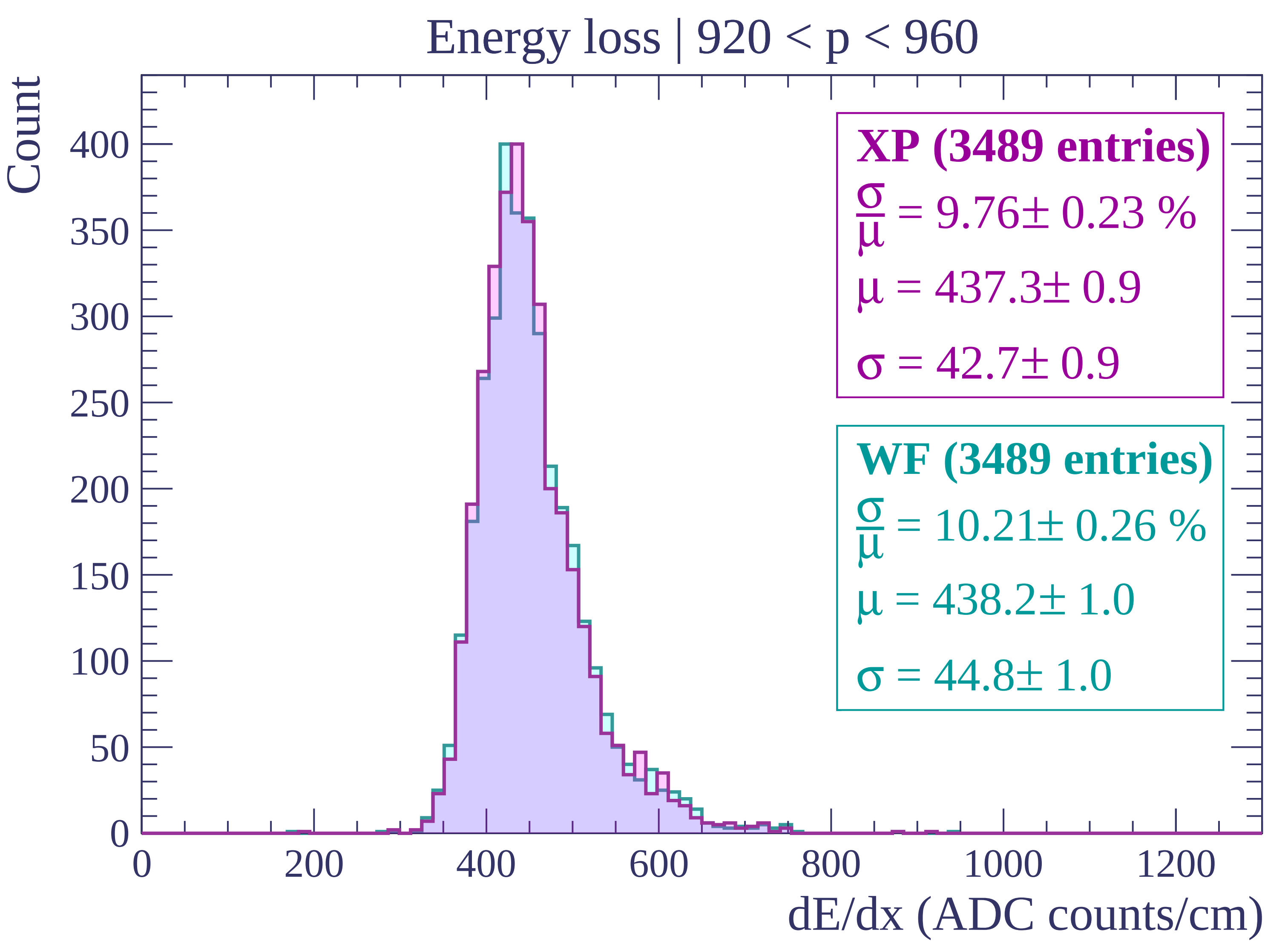
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP (3384 entries)
$\frac{\sigma}{\mu}$ = 9.64± 0.22 %
μ = 428.9± 0.9
σ = 41.3± 0.9
WF (3384 entries)
$\frac{\sigma}{\mu}$ = 9.83± 0.23 %
μ = 429.6± 0.9
σ = 42.2± 0.9



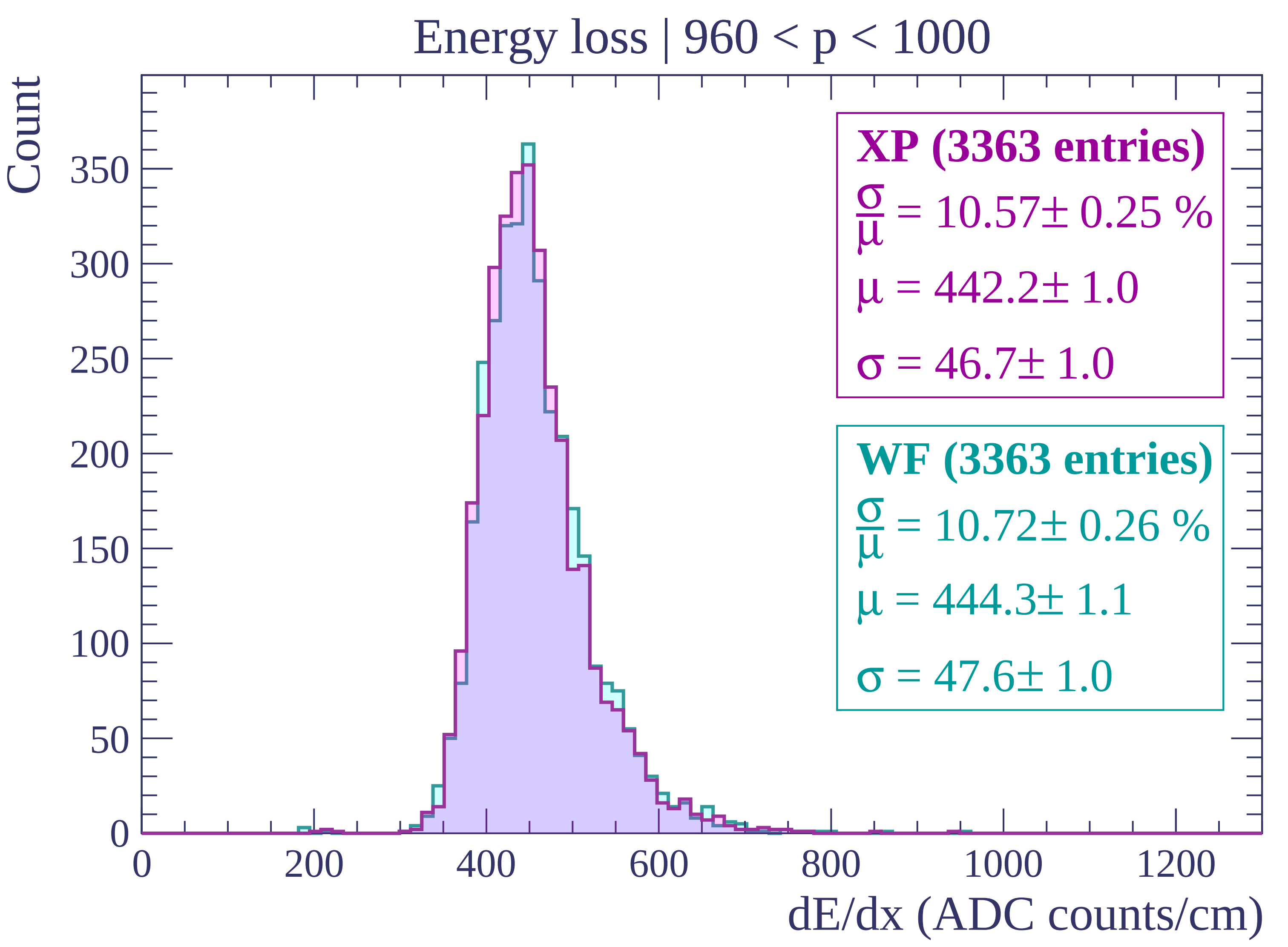
Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3471 entries)
σ/μ = 9.51± 0.21 %
μ = 430.4± 0.9
σ = 40.9± 0.8
WF (3471 entries)
σ/μ = 9.85± 0.21 %
μ = 433.5± 0.9
σ = 42.7± 0.8



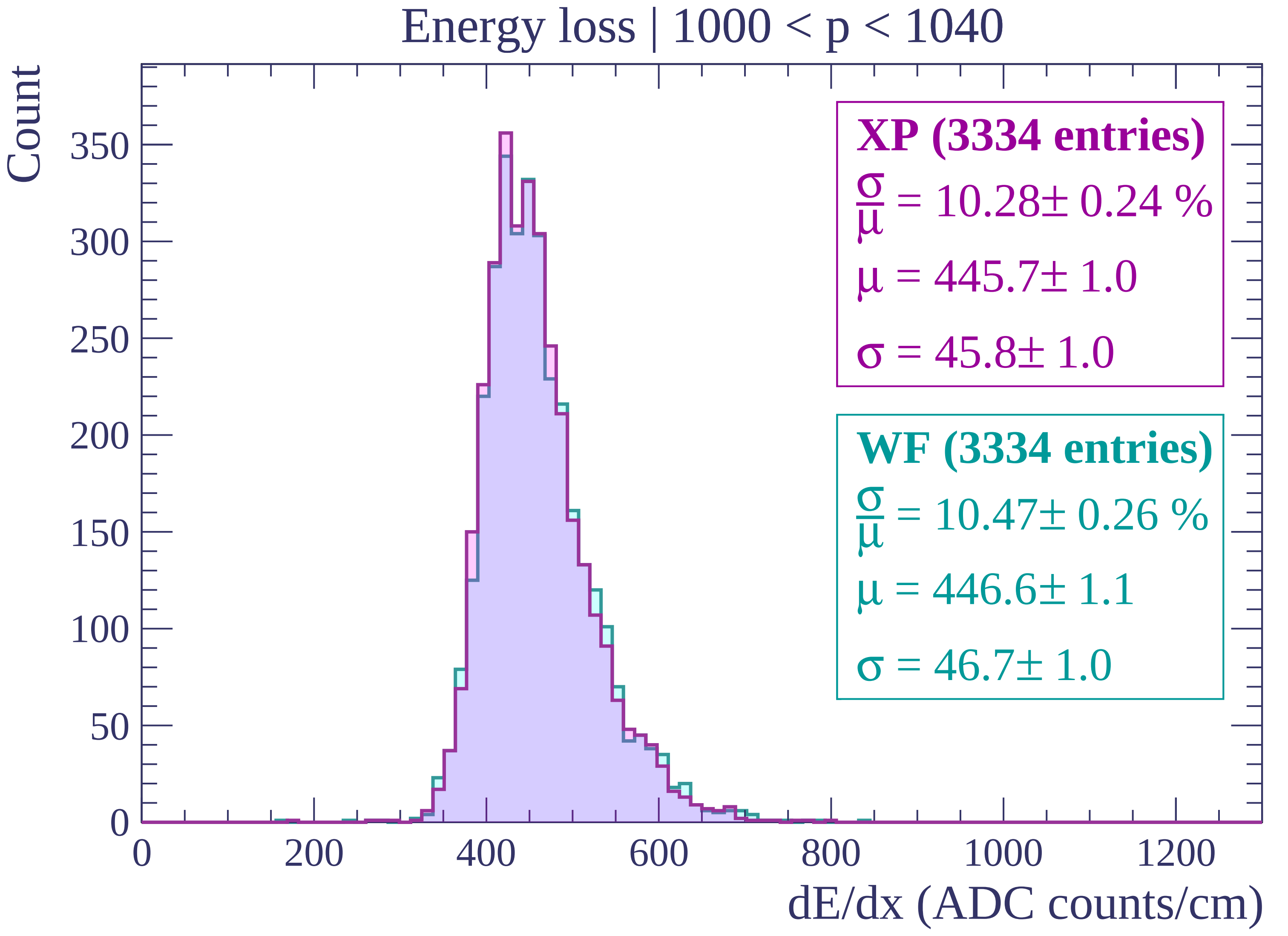
Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3540 entries)
σ/μ = 10.16± 0.23 %
μ = 435.8± 1.0
σ = 44.3± 0.9
WF (3540 entries)
σ/μ = 10.63± 0.22 %
μ = 438.5± 1.0
σ = 46.6± 0.9

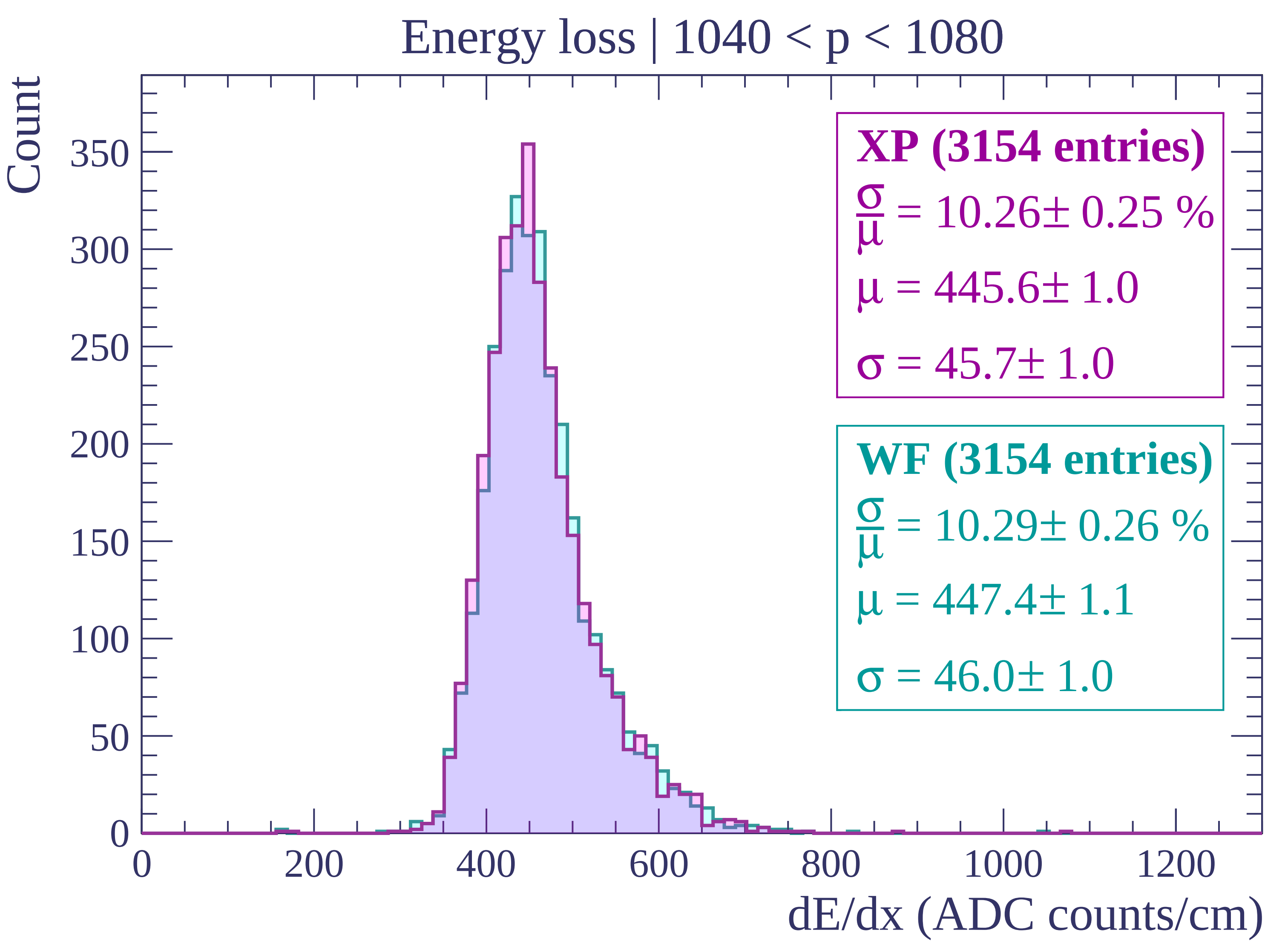


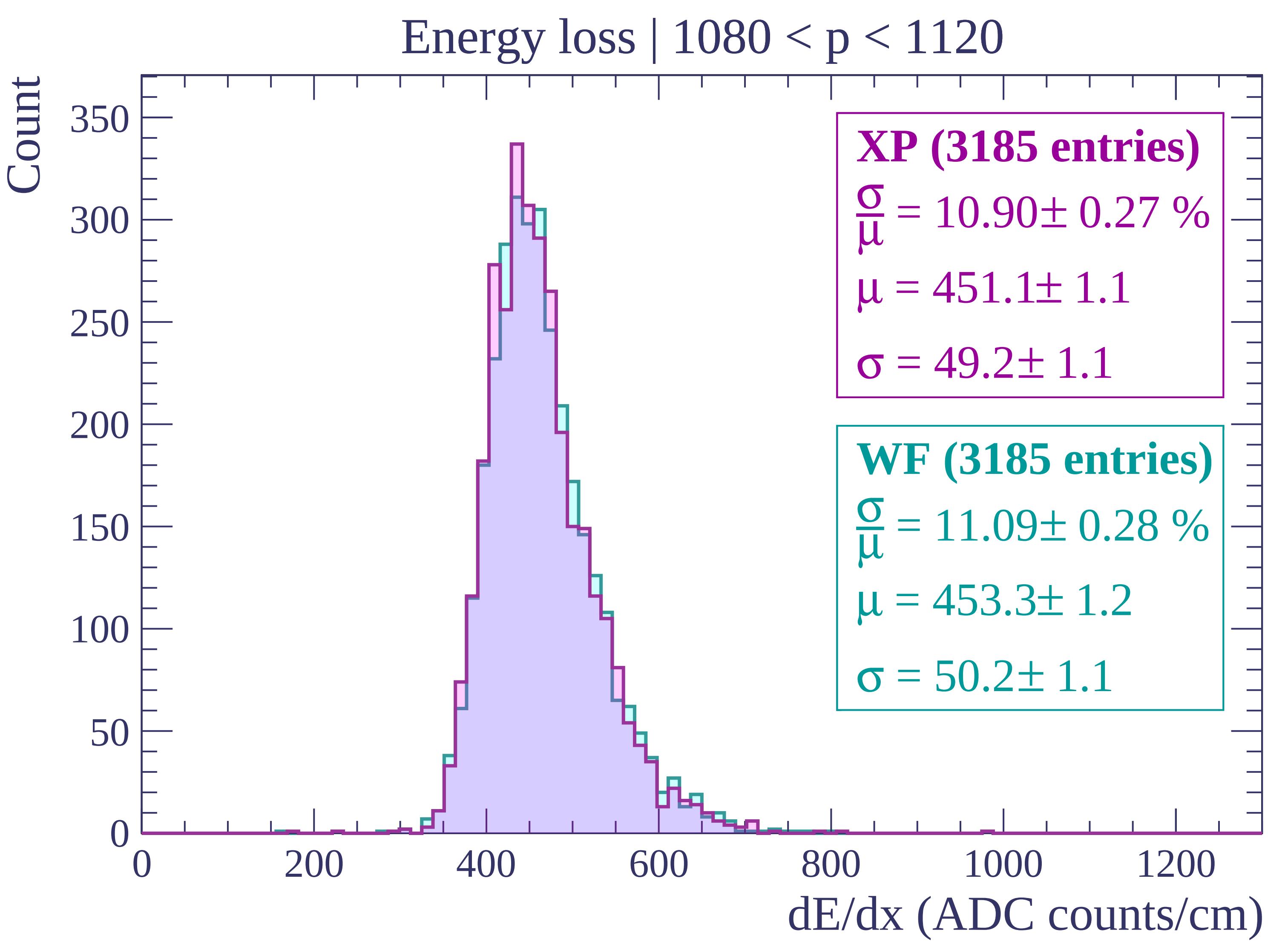
Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3489 entries)
$\frac{\sigma}{\mu}$ = 9.76± 0.23 %
μ = 437.3± 0.9
σ = 42.7± 0.9
WF (3489 entries)
$\frac{\sigma}{\mu}$ = 10.21± 0.26 %
μ = 438.2± 1.0
σ = 44.8± 1.0

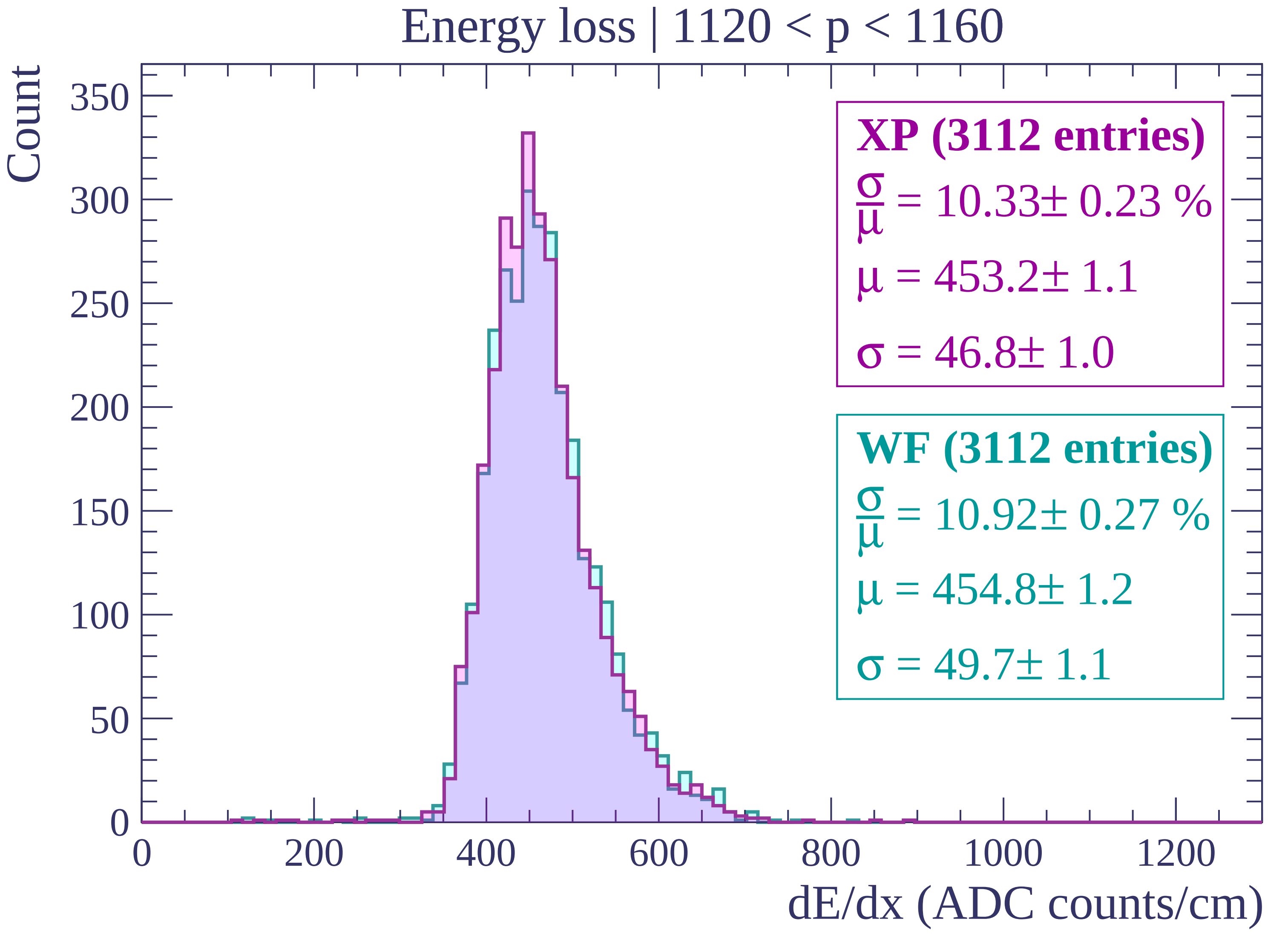


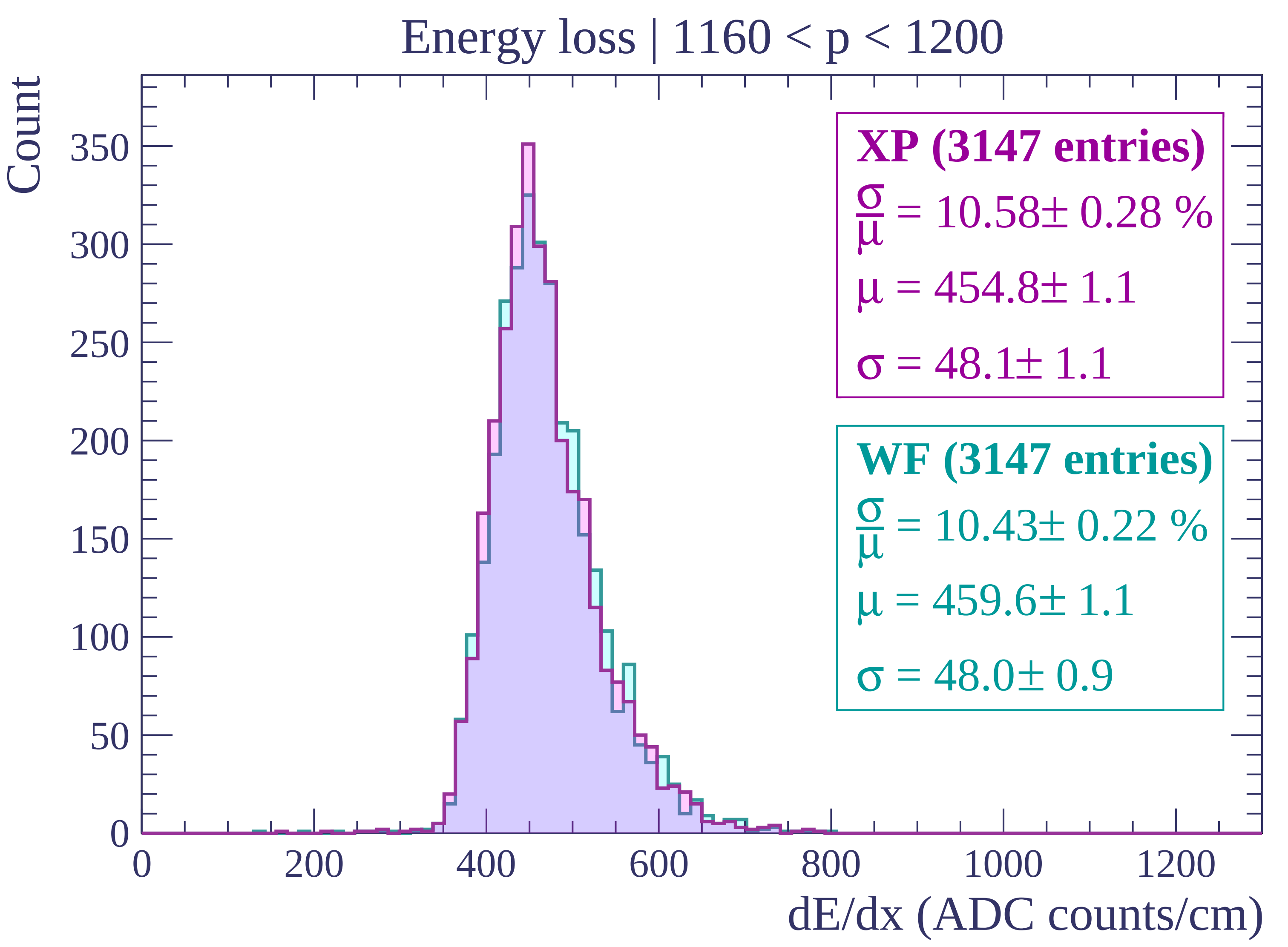
Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP (3363 entries)
$\frac{\sigma}{\mu}$ = 10.57± 0.25 %
μ = 442.2± 1.0
σ = 46.7± 1.0
WF (3363 entries)
$\frac{\sigma}{\mu}$ = 10.72± 0.26 %
μ = 444.3± 1.1
σ = 47.6± 1.0

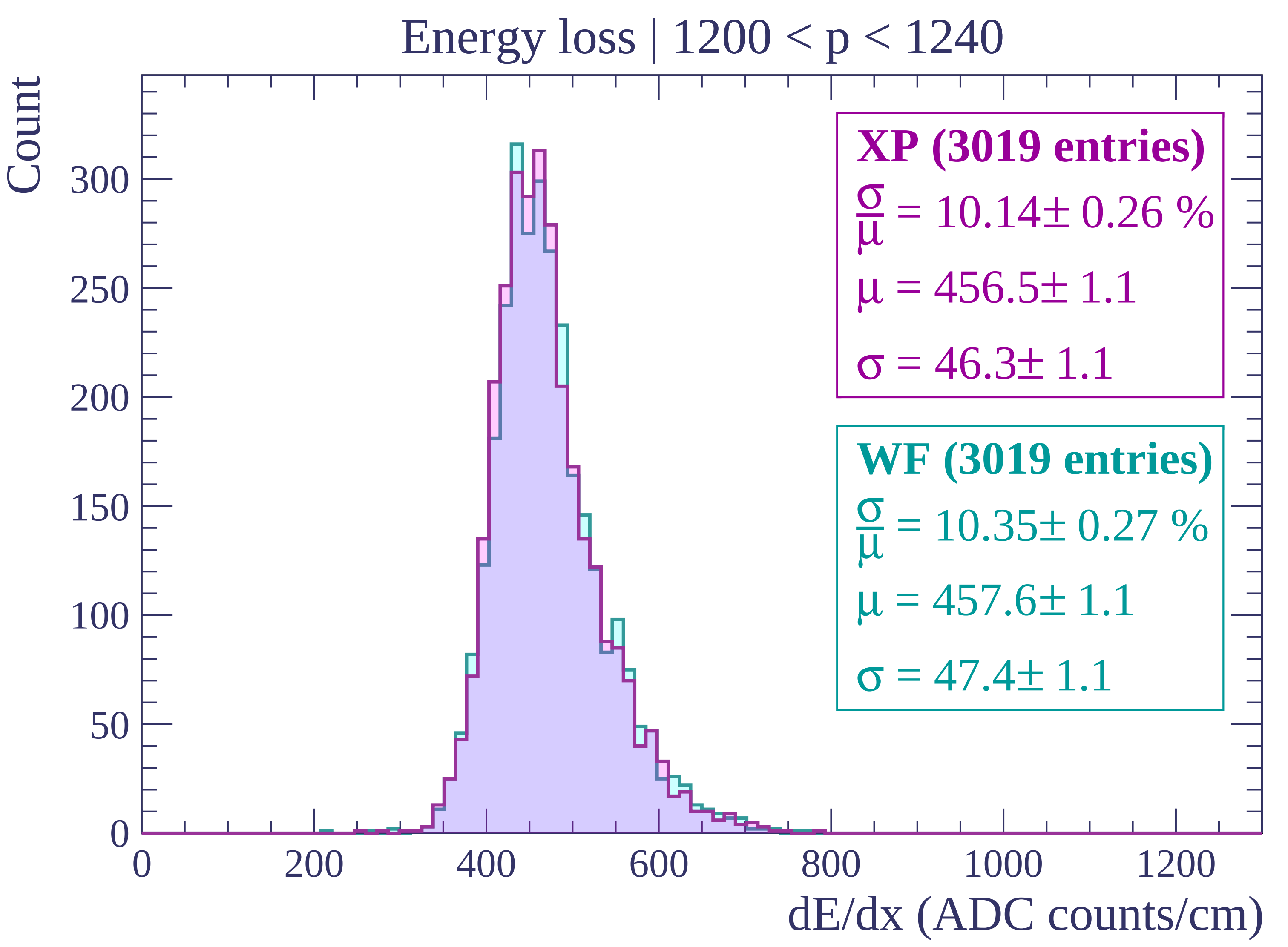

Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3334 entries)
σ/μ = 10.28± 0.24 %
μ = 445.7± 1.0
σ = 45.8± 1.0
WF (3334 entries)
σ/μ = 10.47± 0.26 %
μ = 446.6± 1.1
σ = 46.7± 1.0


Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3154 entries)
$\frac{\sigma}{\mu}$ = 10.26± 0.25 %
μ = 445.6± 1.0
σ = 45.7± 1.0
WF (3154 entries)
$\frac{\sigma}{\mu}$ = 10.29± 0.26 %
μ = 447.4± 1.1
σ = 46.0± 1.0

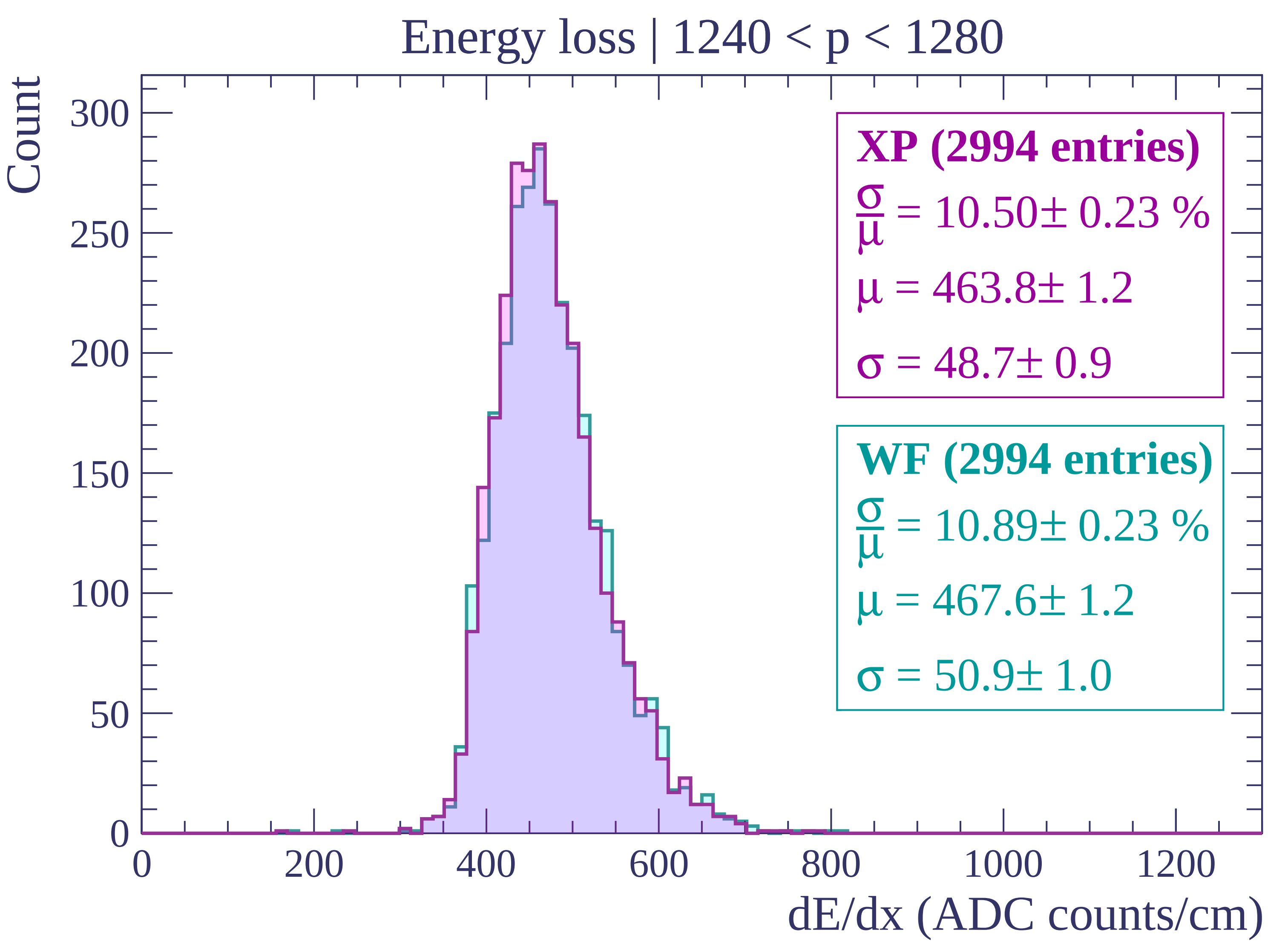

Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3185 entries)
σ/μ = 10.90± 0.27 %
μ = 451.1± 1.1
σ = 49.2± 1.1
WF (3185 entries)
σ/μ = 11.09± 0.28 %
μ = 453.3± 1.2
σ = 50.2± 1.1


Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP (3112 entries)
σ/μ = 10.33± 0.23 %
μ = 453.2± 1.1
σ = 46.8± 1.0
WF (3112 entries)
σ/μ = 10.92± 0.27 %
μ = 454.8± 1.2
σ = 49.7± 1.1

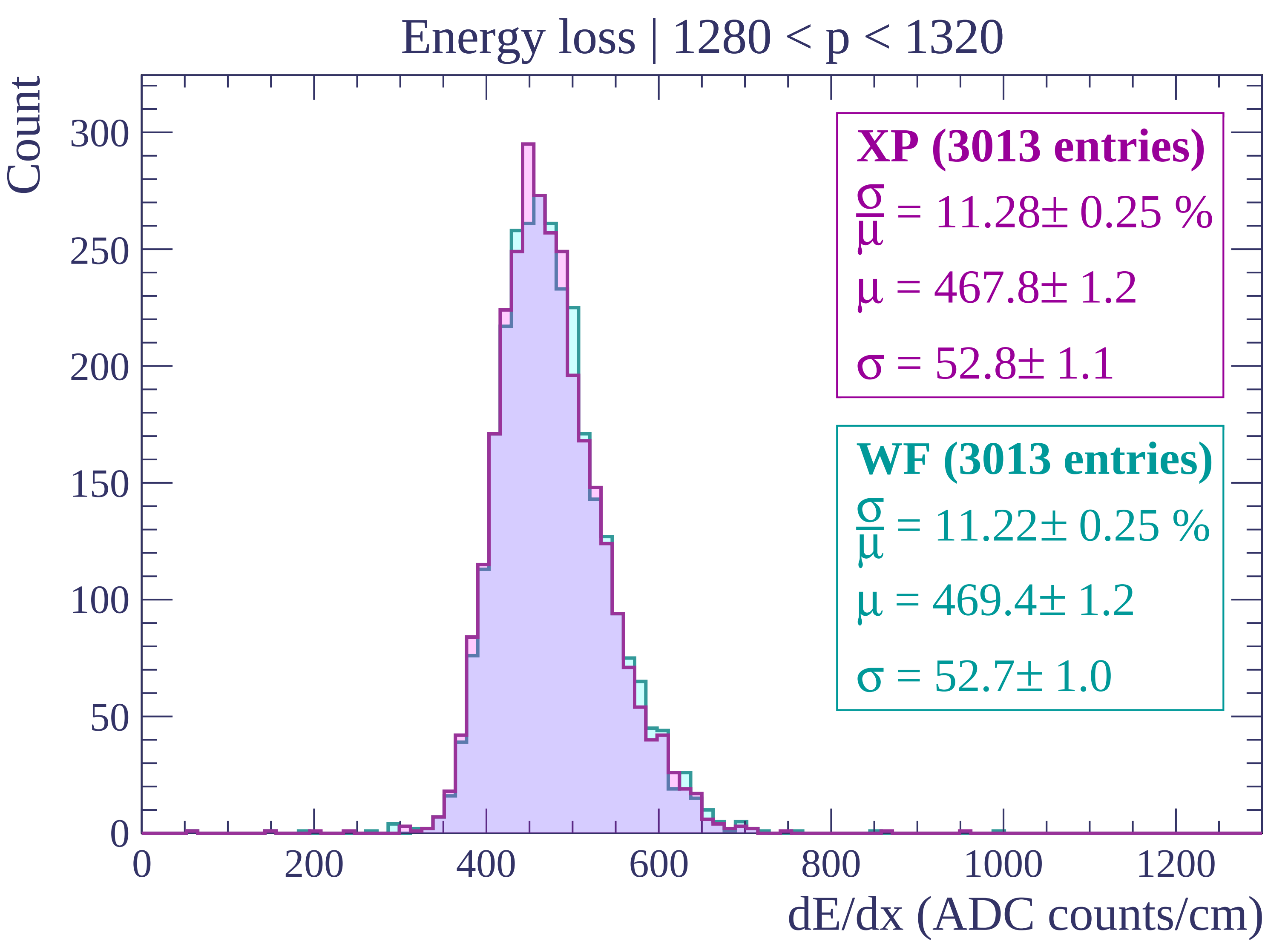

Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3147 entries)
σ/μ = 10.58± 0.28 %
μ = 454.8± 1.1
σ = 48.1± 1.1
WF (3147 entries)
σ/μ = 10.43± 0.22 %
μ = 459.6± 1.1
σ = 48.0± 0.9



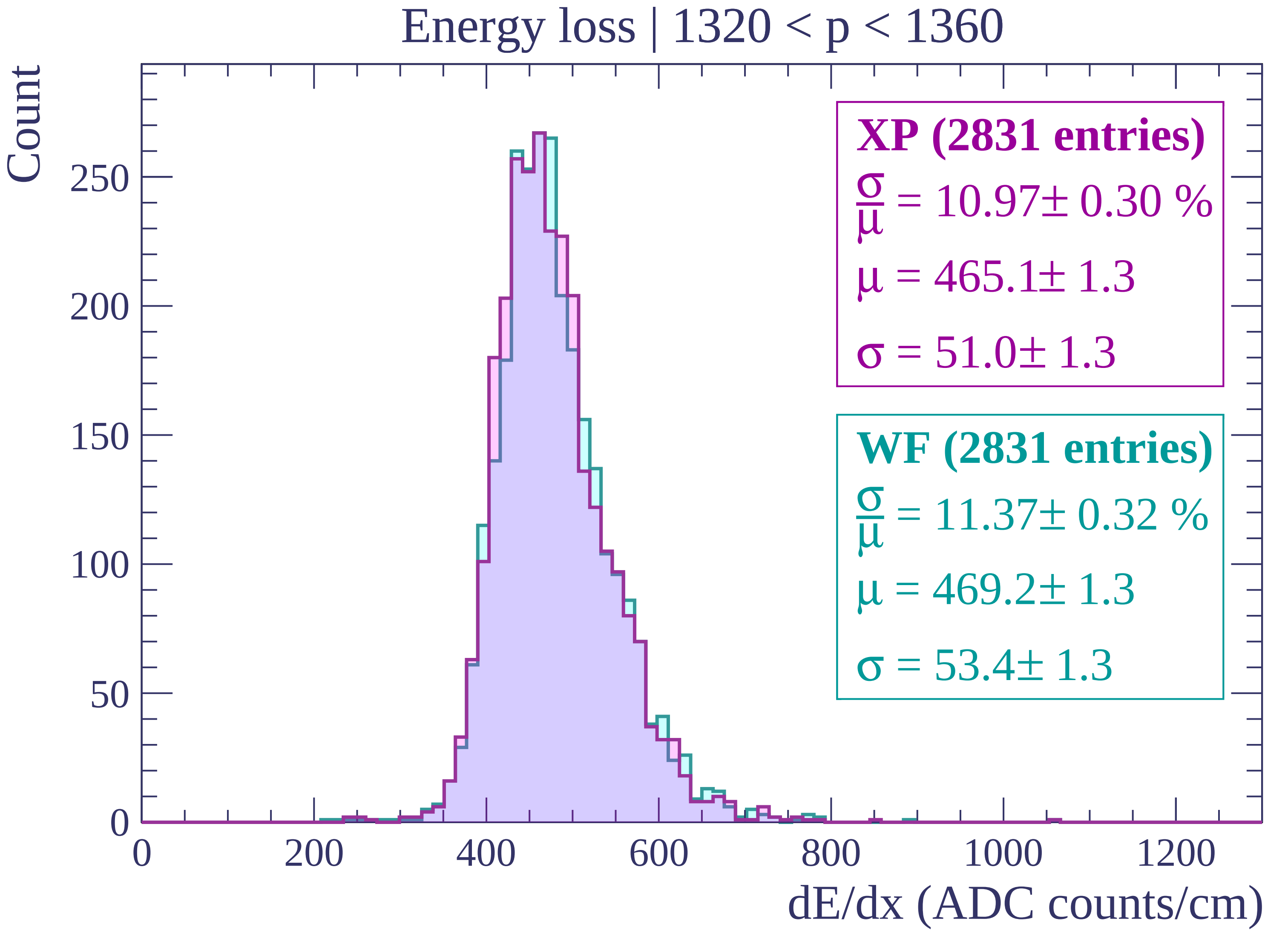
Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3019 entries)
$\frac{\sigma}{\mu}$ = 10.14± 0.26 %
μ = 456.5± 1.1
σ = 46.3± 1.1
WF (3019 entries)
$\frac{\sigma}{\mu}$ = 10.35± 0.27 %
μ = 457.6± 1.1
σ = 47.4± 1.1



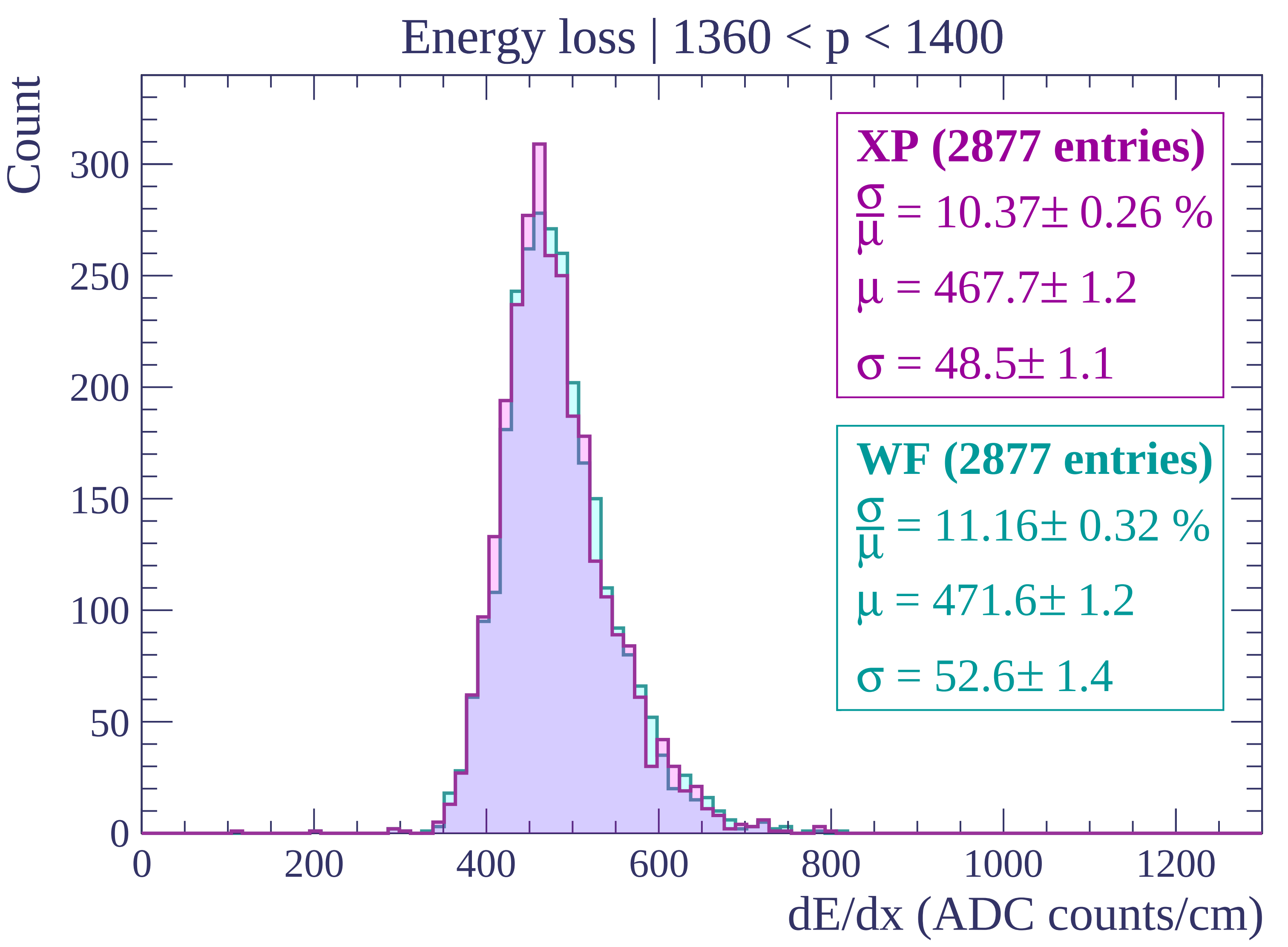
Energy loss | 1240 < p < 1280
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2994 entries)
σ/μ = 10.50± 0.23 %
μ = 463.8± 1.2
σ = 48.7± 0.9
WF (2994 entries)
σ/μ = 10.89± 0.23 %
μ = 467.6± 1.2
σ = 50.9± 1.0

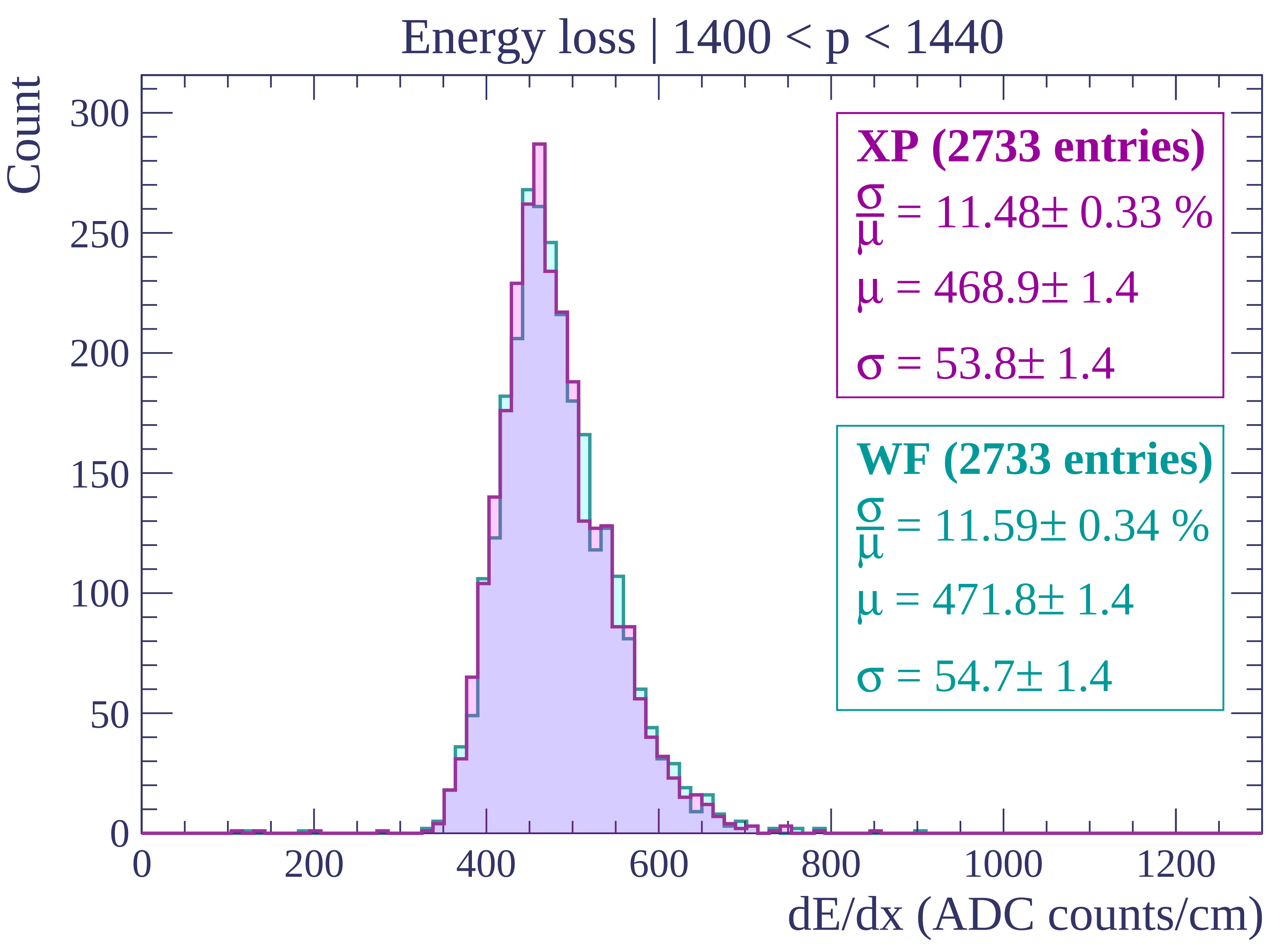


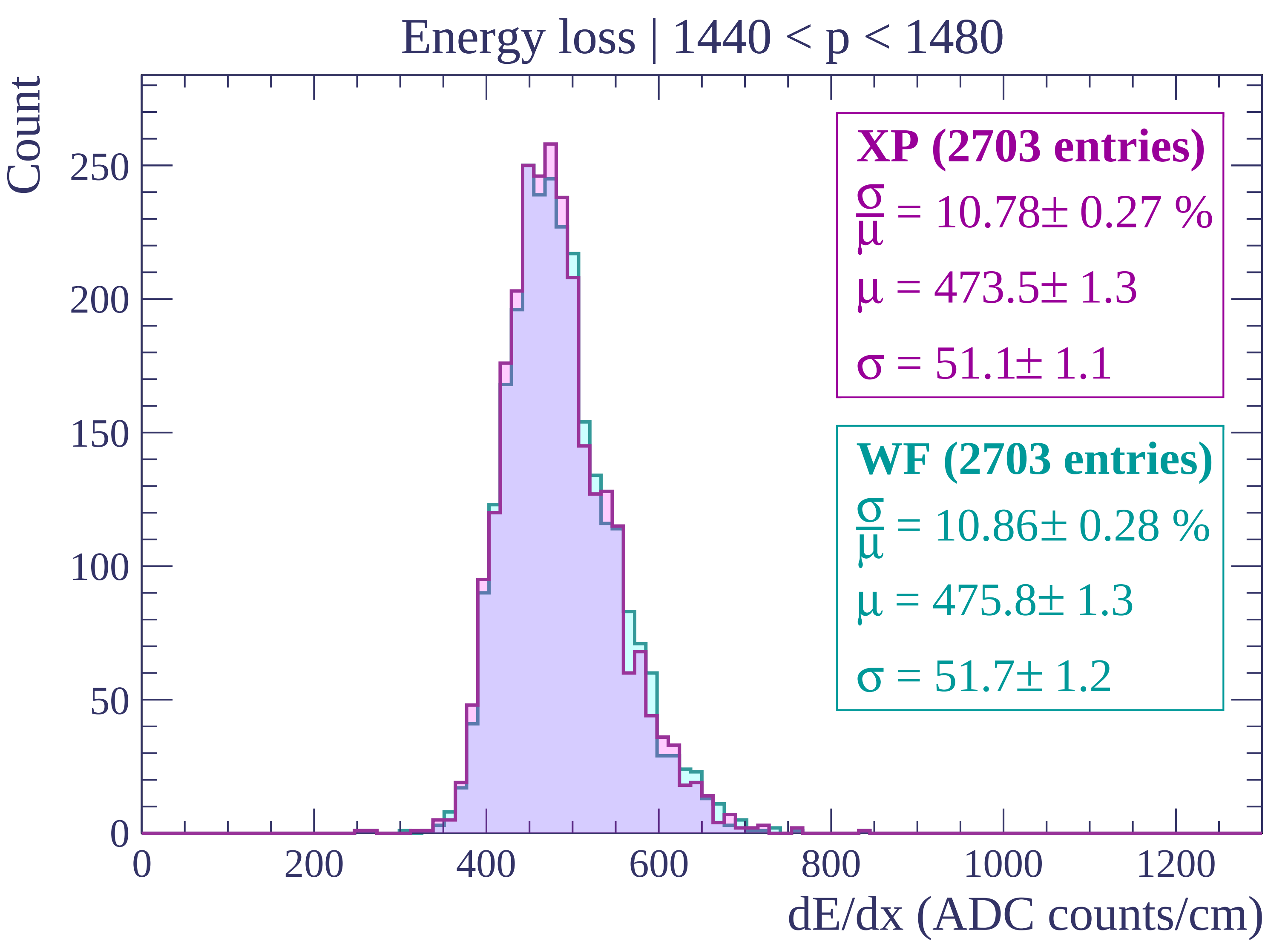
Energy loss | 1280 < p < 1320
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3013 entries)
σ/μ = 11.28± 0.25 %
μ = 467.8± 1.2
σ = 52.8± 1.1
WF (3013 entries)
σ/μ = 11.22± 0.25 %
μ = 469.4± 1.2
σ = 52.7± 1.0

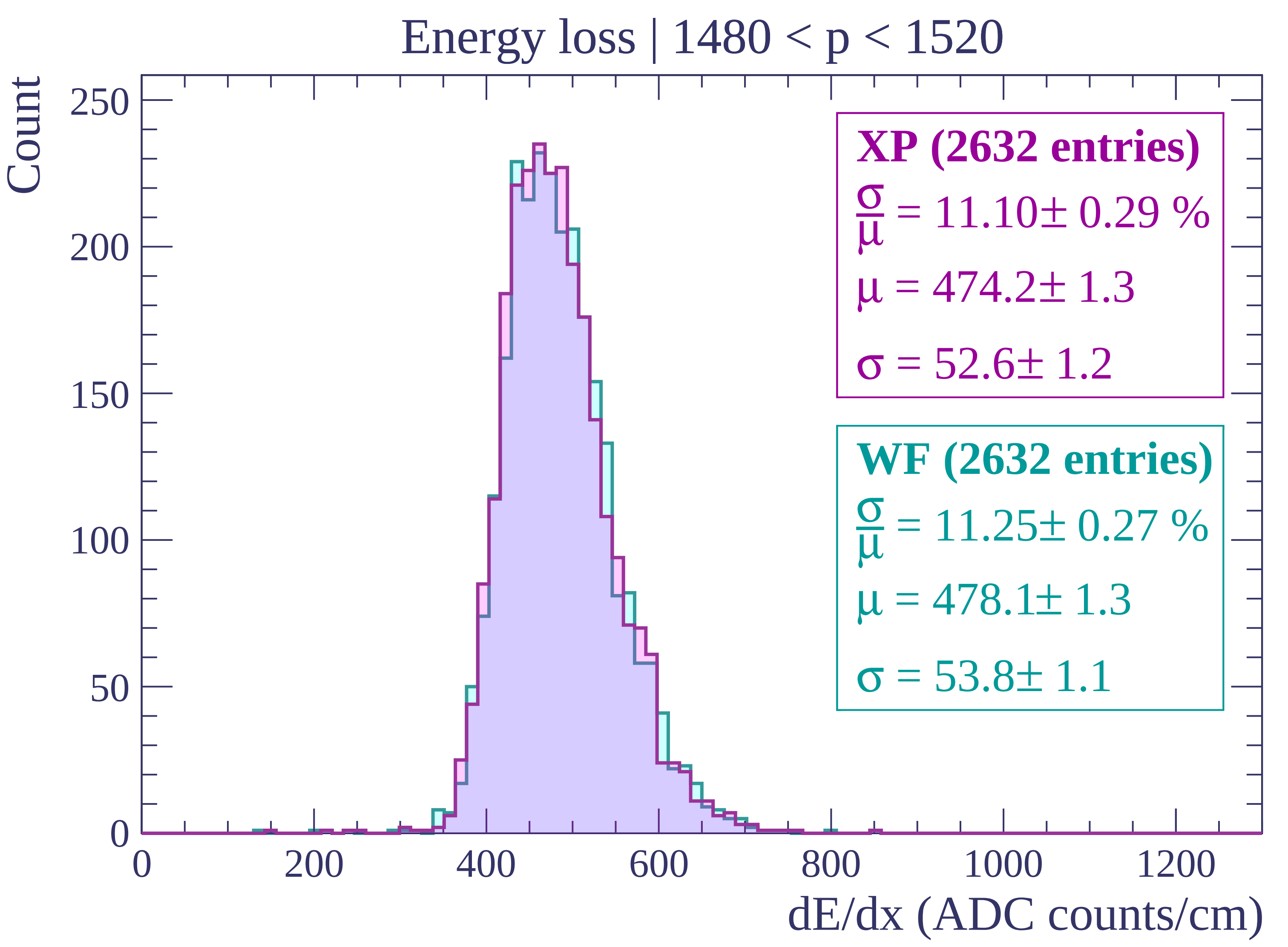


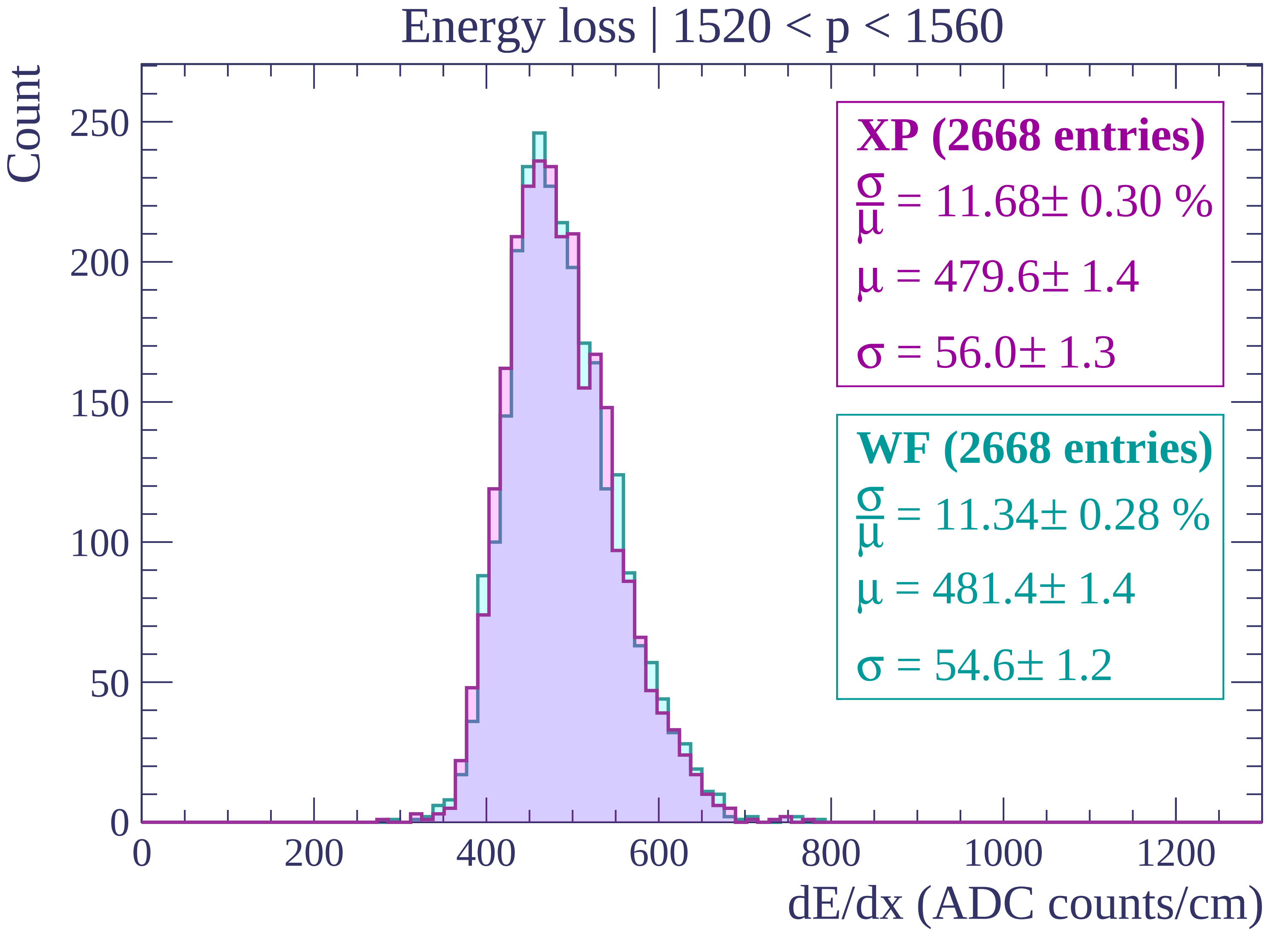
Energy loss | 1320 < p < 1360
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2831 entries)
σ/μ = 10.97± 0.30 %
μ = 465.1± 1.3
σ = 51.0± 1.3
WF (2831 entries)
σ/μ = 11.37± 0.32 %
μ = 469.2± 1.3
σ = 53.4± 1.3

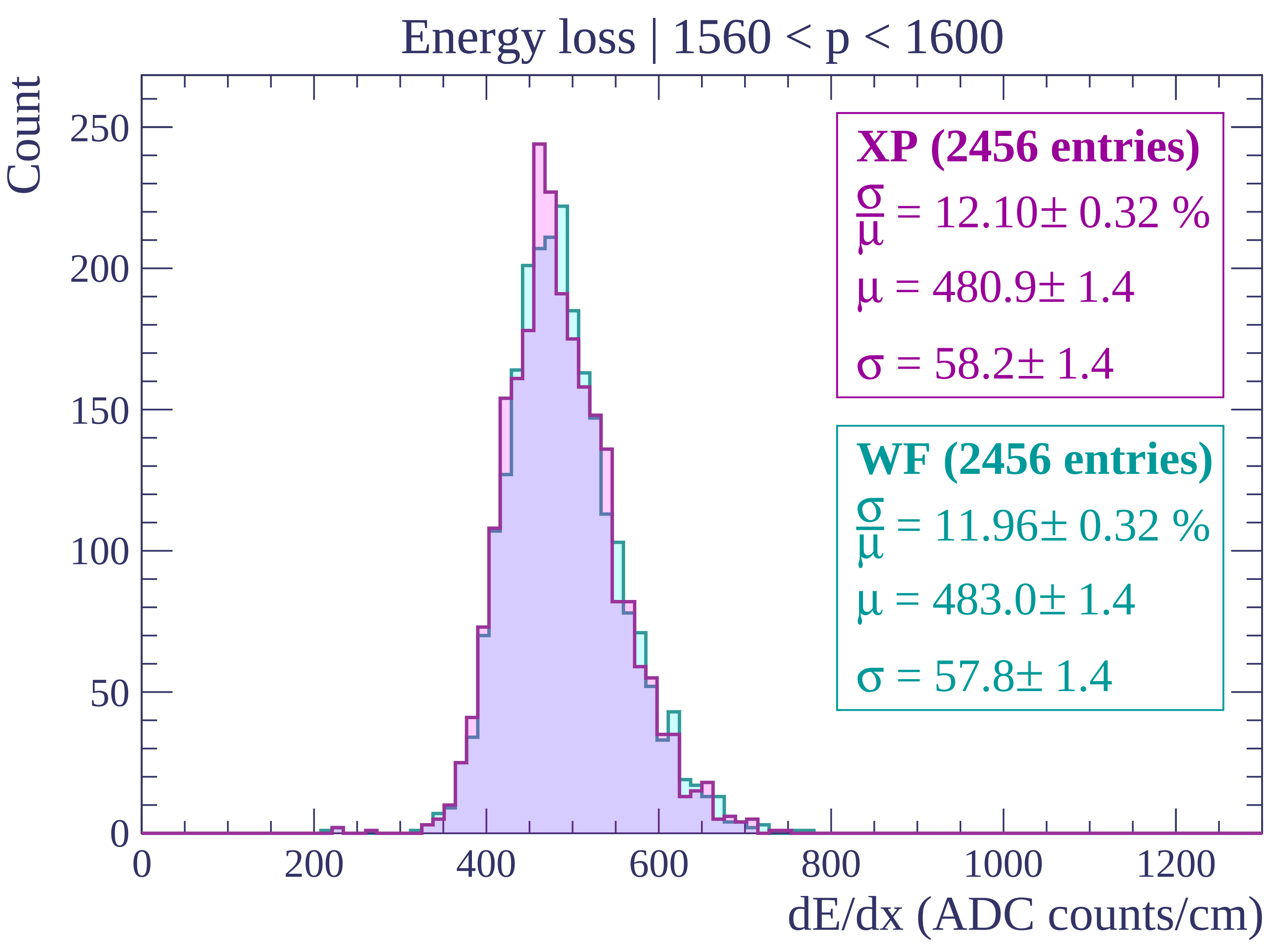

Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
XP (2877 entries)
σ/μ = 10.37± 0.26 %
μ = 467.7± 1.2
σ = 48.5± 1.1
WF (2877 entries)
σ/μ = 11.16± 0.32 %
μ = 471.6± 1.2
σ = 52.6± 1.4
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

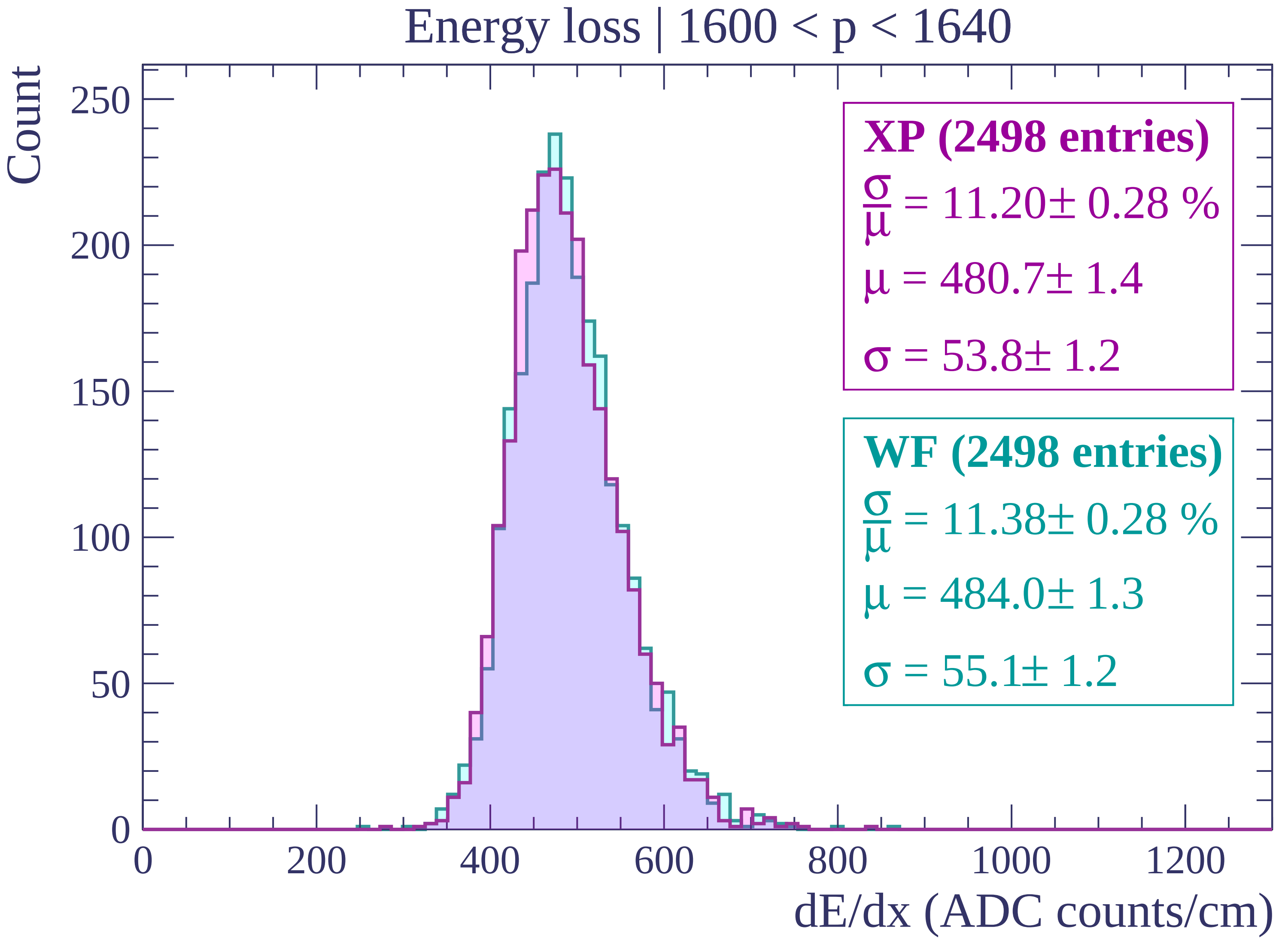

Energy loss | 1400 < p < 1440
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2733 entries)
σ/μ = 11.48± 0.33 %
μ = 468.9± 1.4
σ = 53.8± 1.4
WF (2733 entries)
σ/μ = 11.59± 0.34 %
μ = 471.8± 1.4
σ = 54.7± 1.4


Energy loss | 1440 < p < 1480
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2703 entries)
σ/μ = 10.78± 0.27 %
μ = 473.5± 1.3
σ = 51.1± 1.1
WF (2703 entries)
σ/μ = 10.86± 0.28 %
μ = 475.8± 1.3
σ = 51.7± 1.2

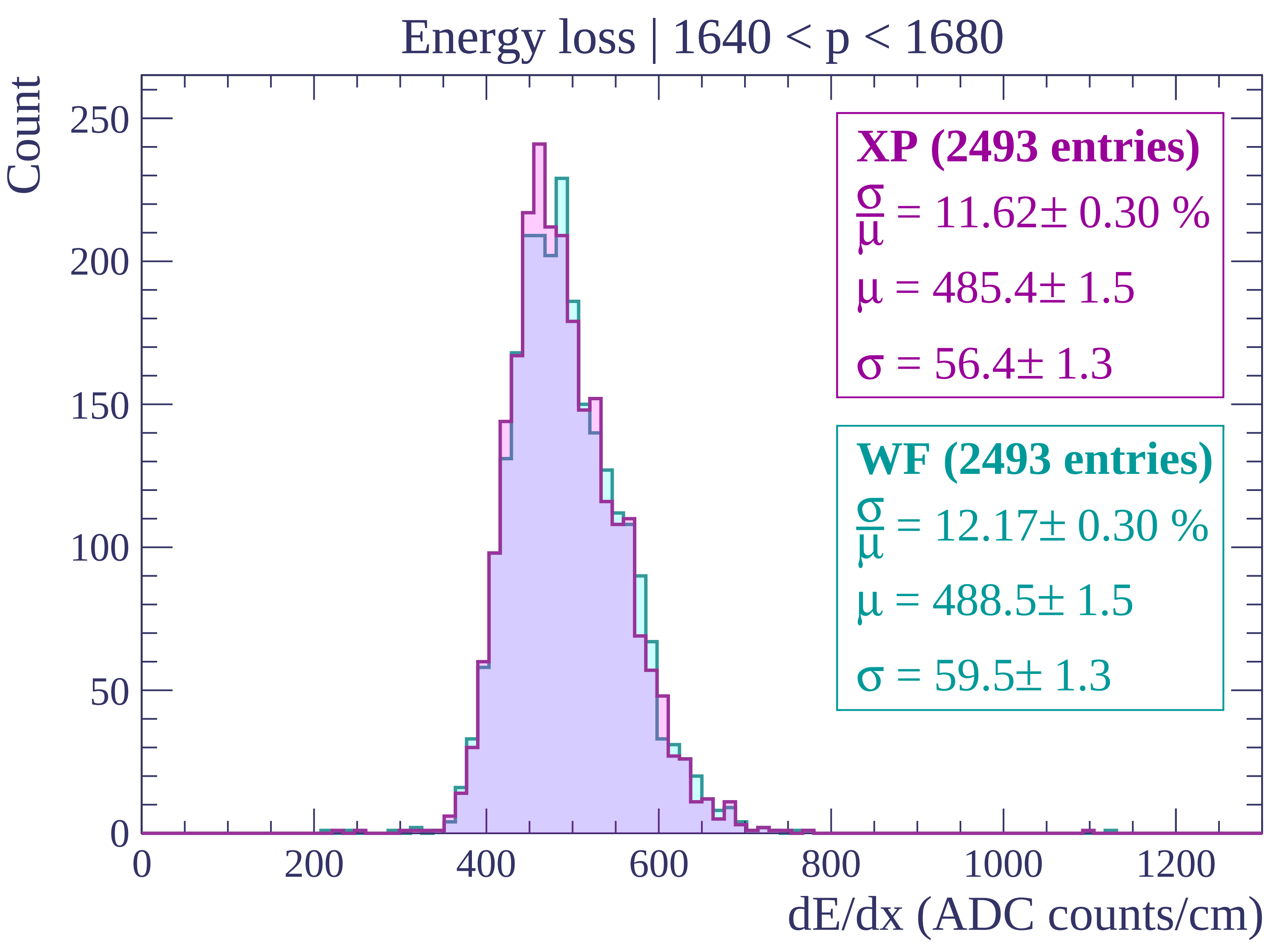

Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2632 entries)
σ/μ = 11.10± 0.29 %
μ = 474.2± 1.3
σ = 52.6± 1.2
WF (2632 entries)
σ/μ = 11.25± 0.27 %
μ = 478.1± 1.3
σ = 53.8± 1.1

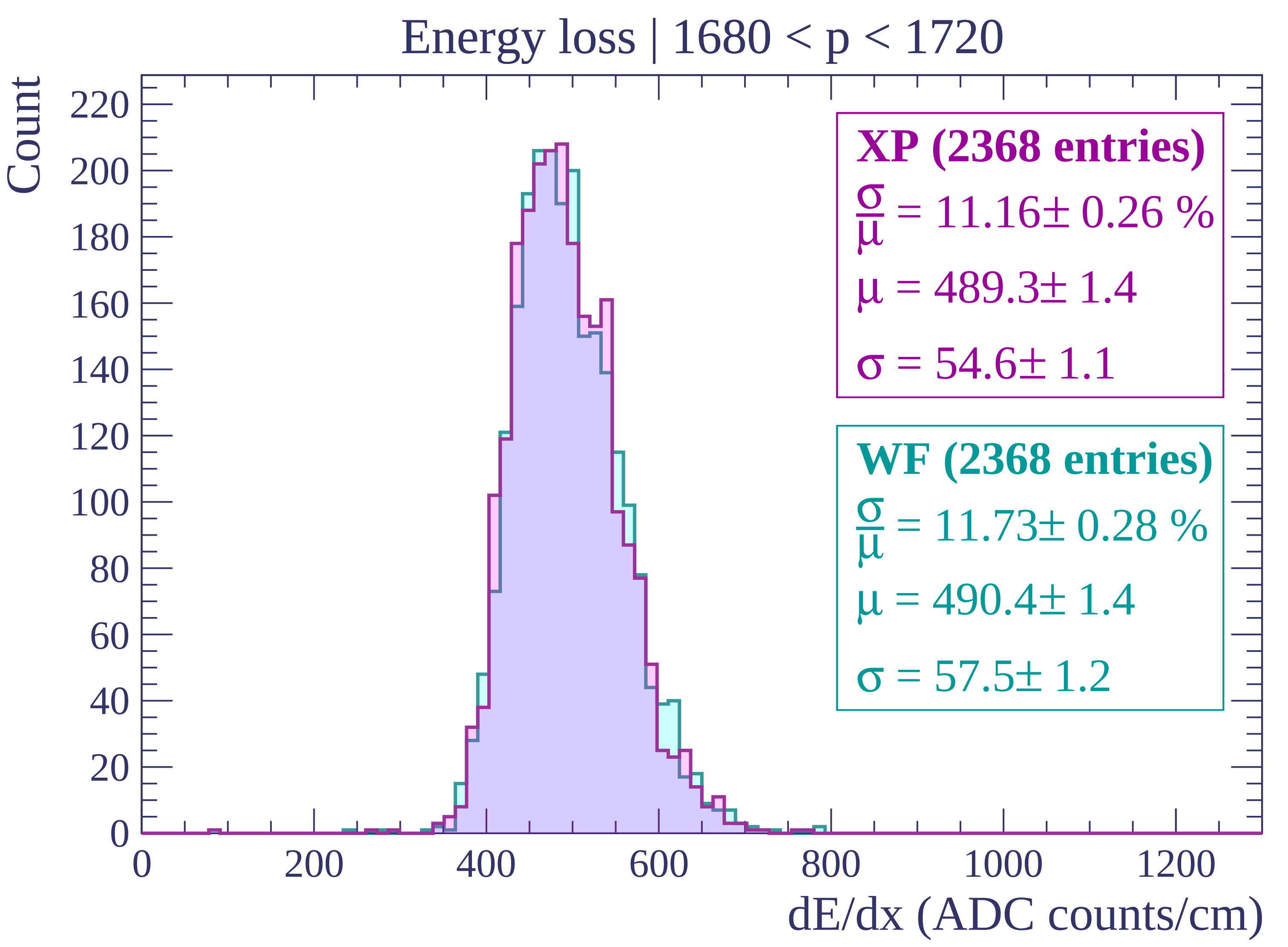

Energy loss | 1520 < p < 1560
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2668 entries)
σ/μ = 11.68± 0.30 %
μ = 479.6± 1.4
σ = 56.0± 1.3
WF (2668 entries)
σ/μ = 11.34± 0.28 %
μ = 481.4± 1.4
σ = 54.6± 1.2


Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2456 entries)
$\frac{\sigma}{\mu}$ = 12.10± 0.32 %
μ = 480.9± 1.4
σ = 58.2± 1.4
WF (2456 entries)
$\frac{\sigma}{\mu}$ = 11.96± 0.32 %
μ = 483.0± 1.4
σ = 57.8± 1.4

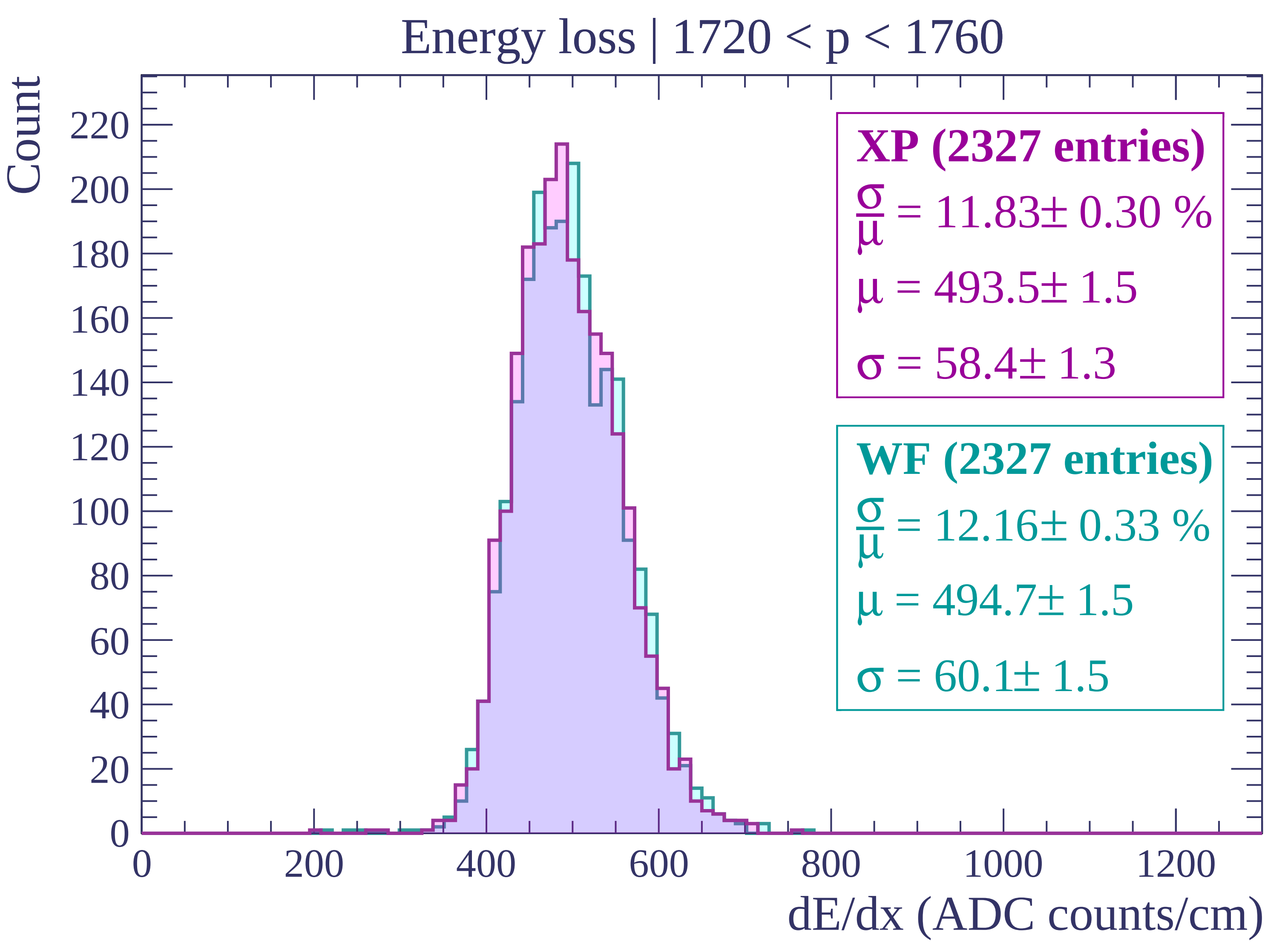


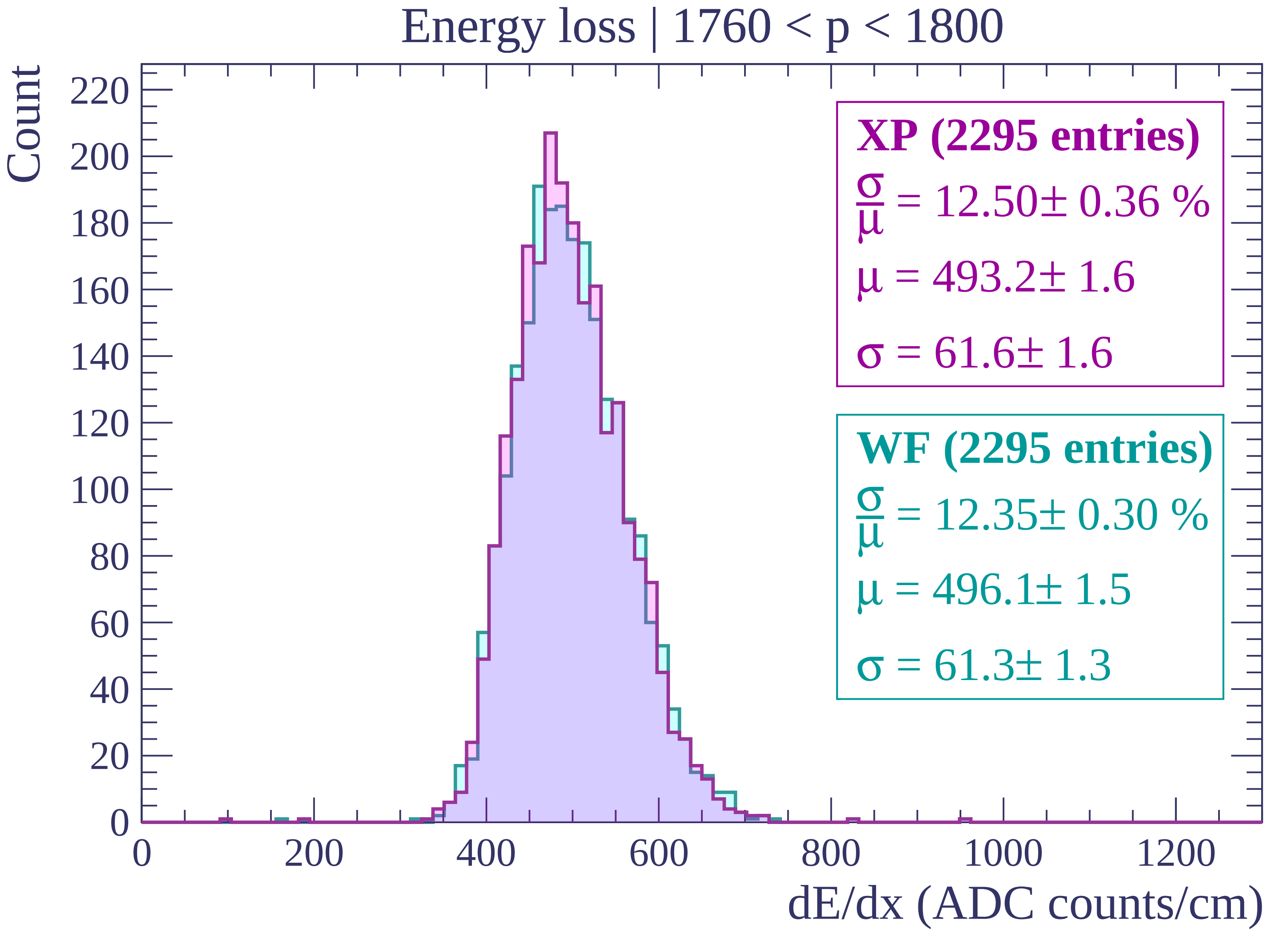
Energy loss | 1600 < p < 1640
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2498 entries)
σ/μ = 11.20± 0.28 %
μ = 480.7± 1.4
σ = 53.8± 1.2
WF (2498 entries)
σ/μ = 11.38± 0.28 %
μ = 484.0± 1.3
σ = 55.1± 1.2

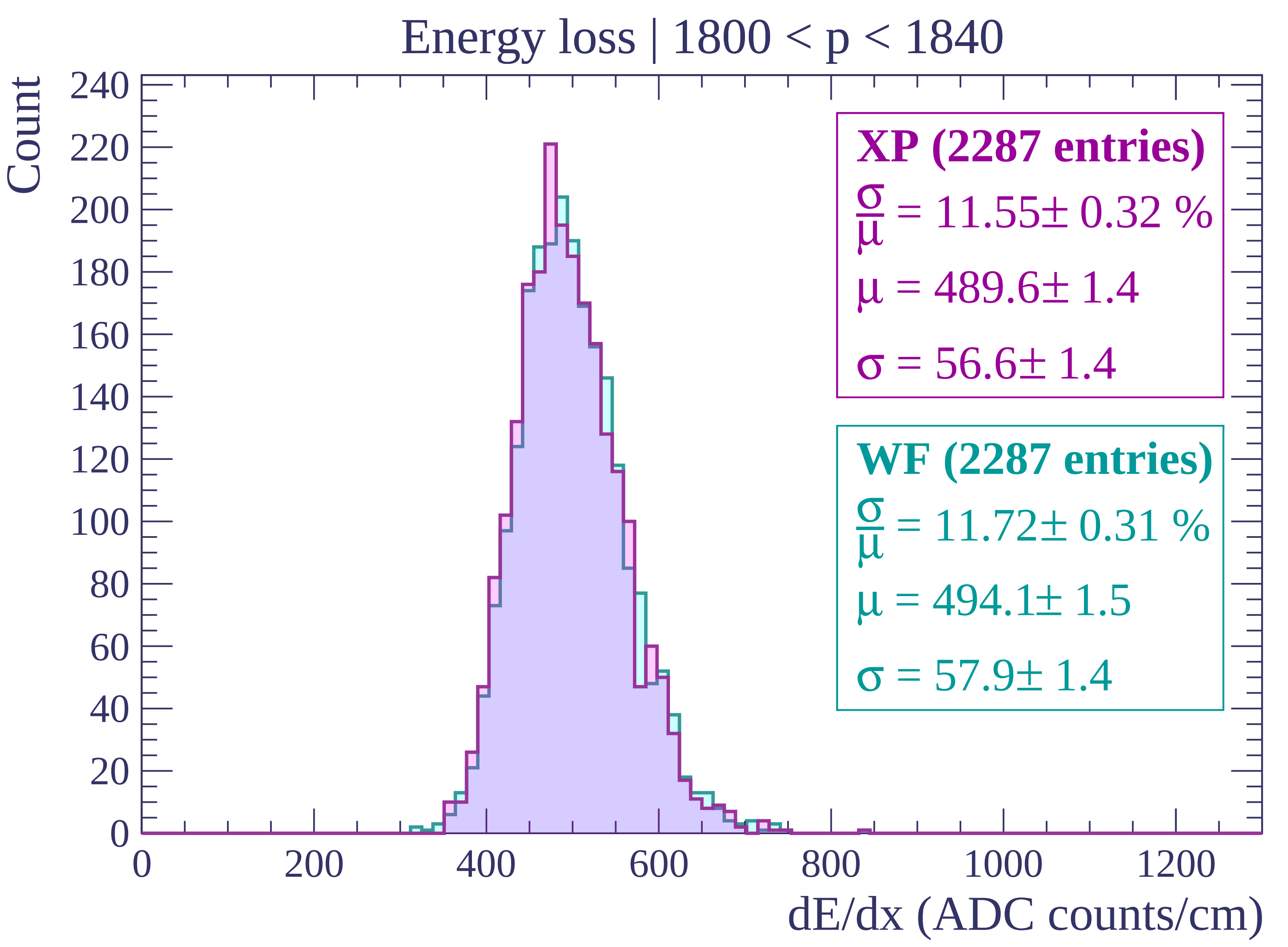

Energy loss | 1640 < p < 1680
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2493 entries)
σ/μ = 11.62± 0.30 %
μ = 485.4± 1.5
σ = 56.4± 1.3
WF (2493 entries)
σ/μ = 12.17± 0.30 %
μ = 488.5± 1.5
σ = 59.5± 1.3

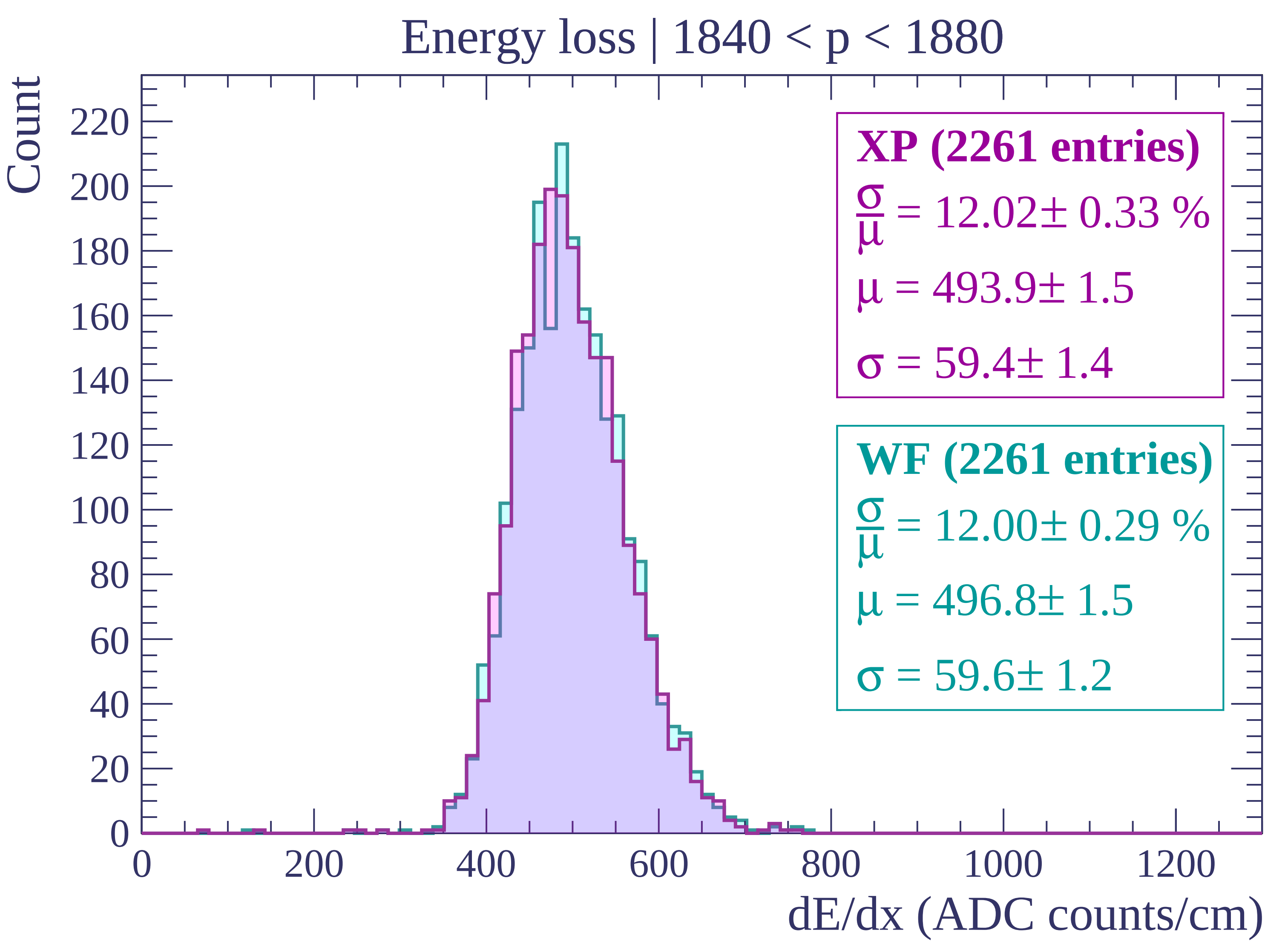


Energy loss | 1680 < p < 1720
Count
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2368 entries)
$\frac{\sigma}{\mu}$ = 11.16± 0.26 %
μ = 489.3± 1.4
σ = 54.6± 1.1
WF (2368 entries)
$\frac{\sigma}{\mu}$ = 11.73± 0.28 %
μ = 490.4± 1.4
σ = 57.5± 1.2

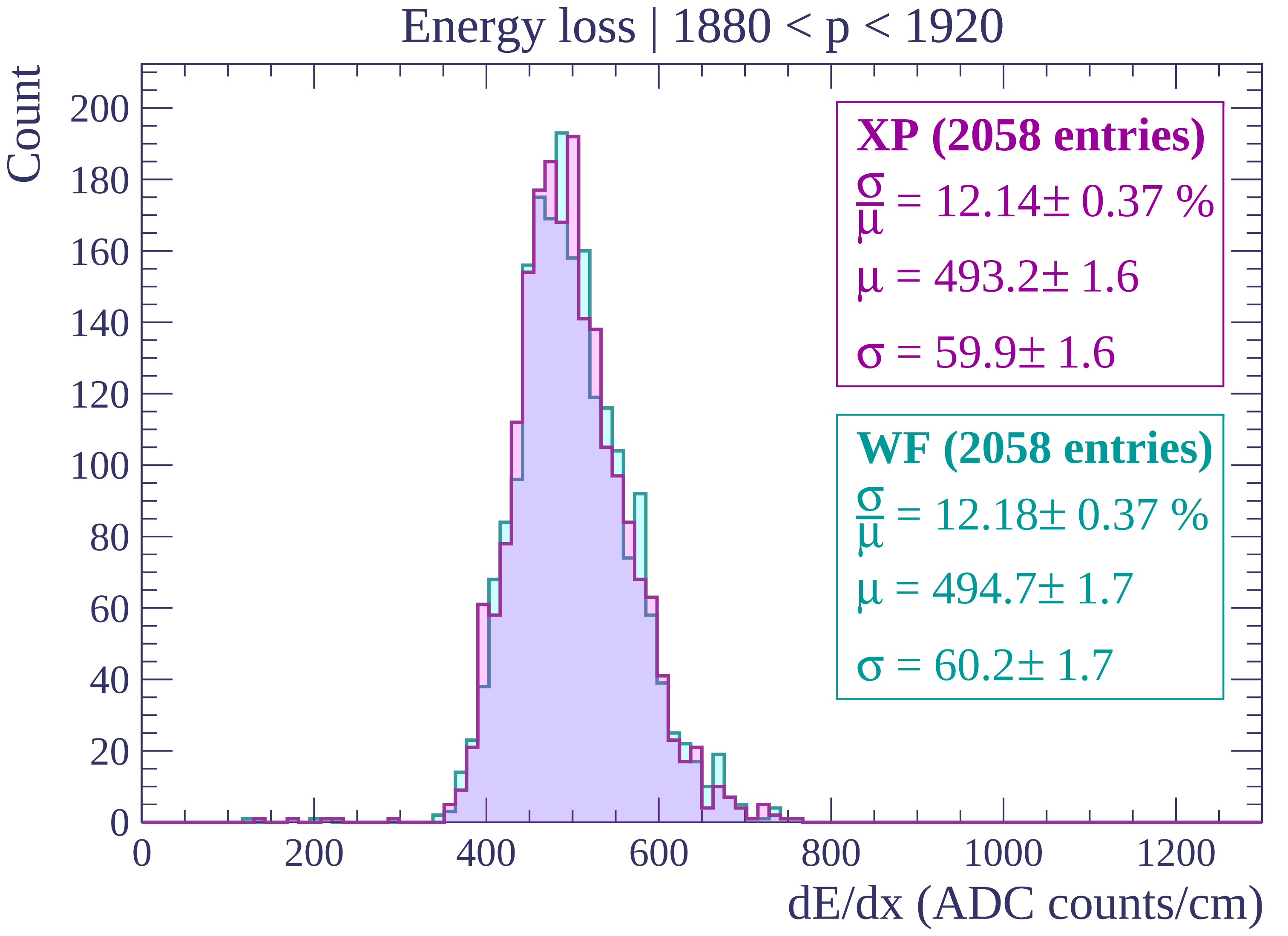

Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (2327 entries)
σ/μ = 11.83± 0.30 %
μ = 493.5± 1.5
σ = 58.4± 1.3
WF (2327 entries)
σ/μ = 12.16± 0.33 %
μ = 494.7± 1.5
σ = 60.1± 1.5


Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP (2295 entries)
σ/μ = 12.50± 0.36 %
μ = 493.2± 1.6
σ = 61.6± 1.6
WF (2295 entries)
σ/μ = 12.35± 0.30 %
μ = 496.1± 1.5
σ = 61.3± 1.3

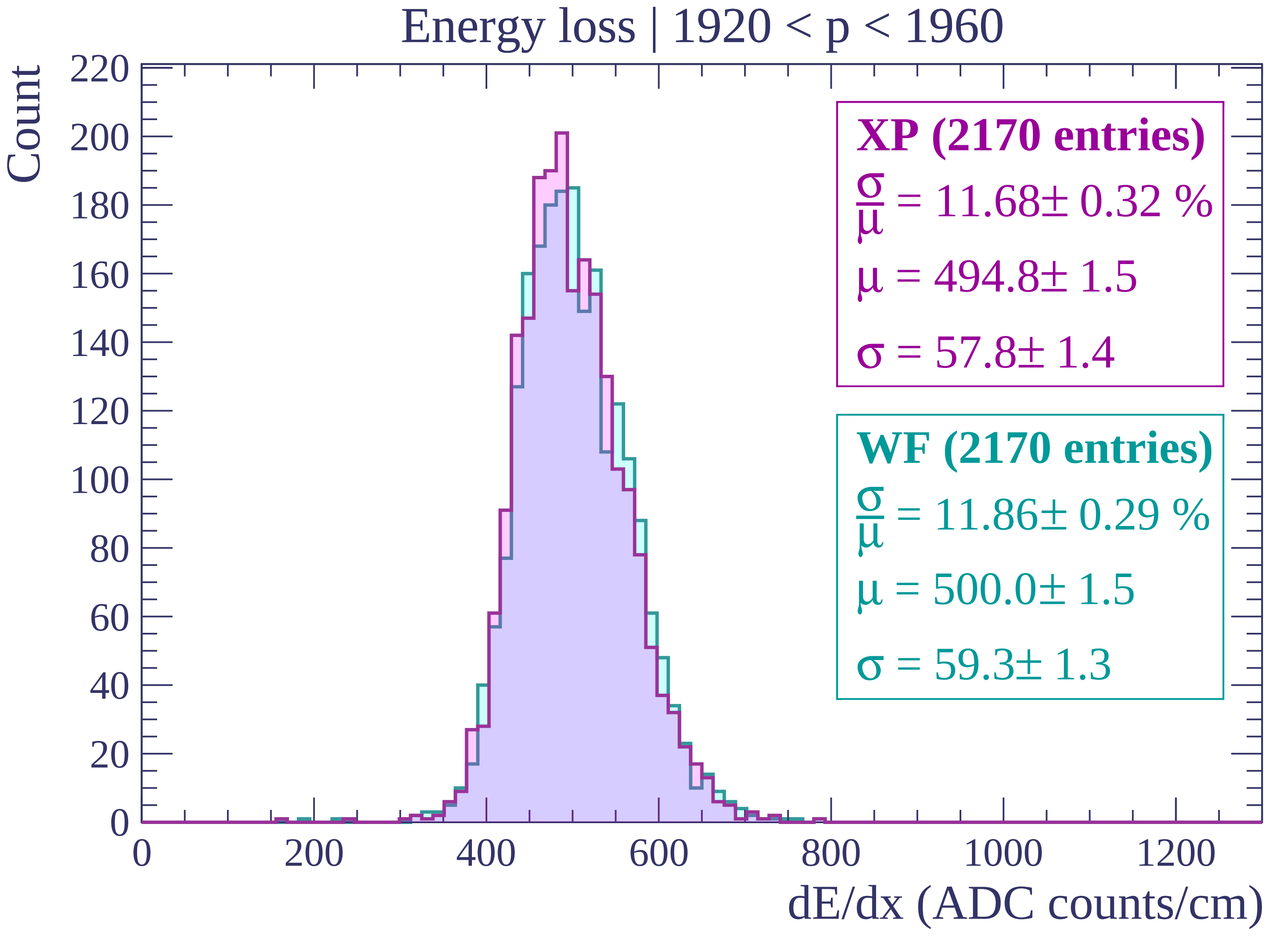

Energy loss | 1800 < p < 1840
Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2287 entries)
σ/μ = 11.55± 0.32 %
μ = 489.6± 1.4
σ = 56.6± 1.4
WF (2287 entries)
σ/μ = 11.72± 0.31 %
μ = 494.1± 1.5
σ = 57.9± 1.4



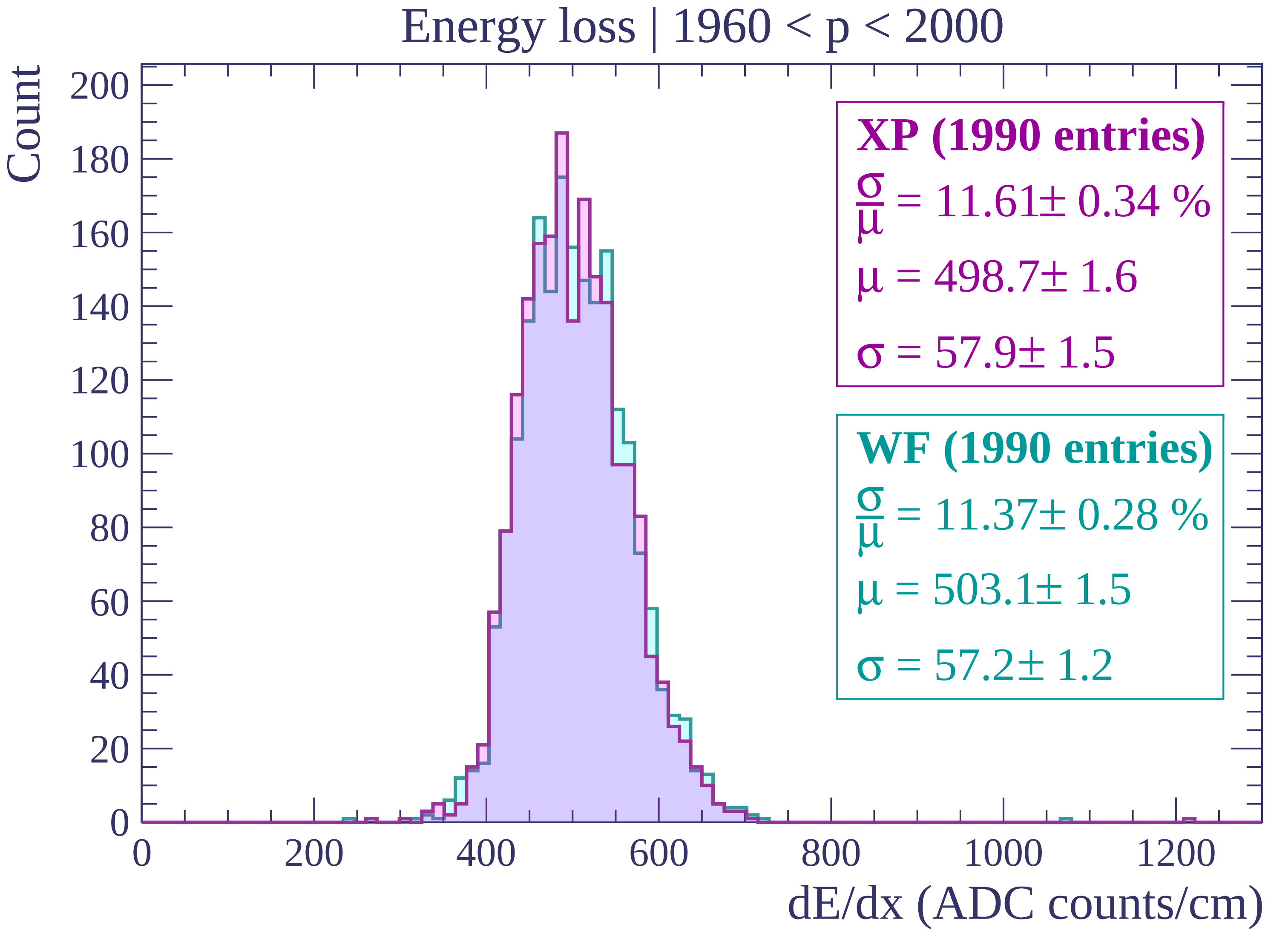
Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (2261 entries)
$\frac{\sigma}{\mu}$ = 12.02± 0.33 %
μ = 493.9± 1.5
σ = 59.4± 1.4
WF (2261 entries)
$\frac{\sigma}{\mu}$ = 12.00± 0.29 %
μ = 496.8± 1.5
σ = 59.6± 1.2

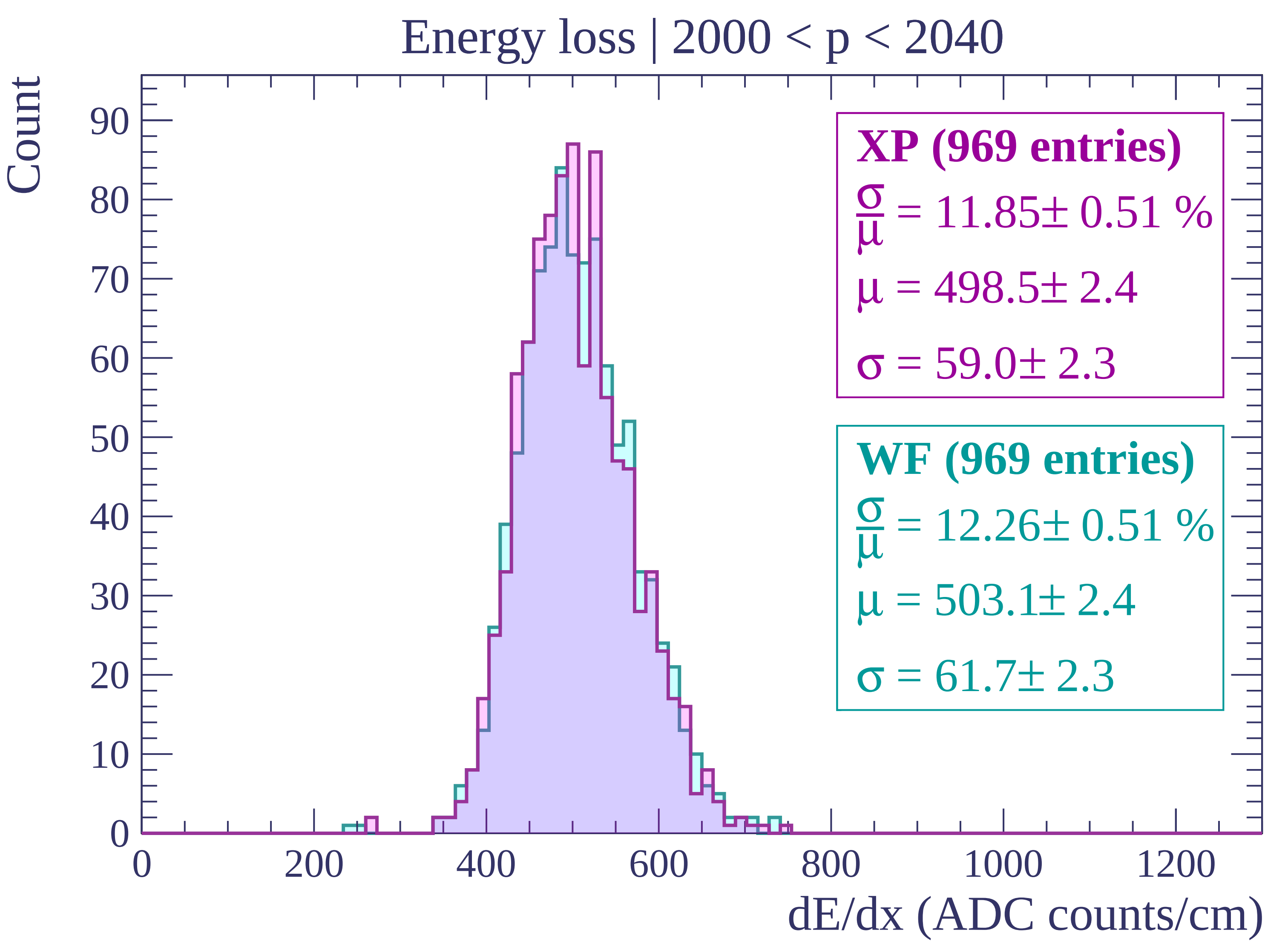


Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
XP (2058 entries)
σ/μ = 12.14± 0.37 %
μ = 493.2± 1.6
σ = 59.9± 1.6
WF (2058 entries)
σ/μ = 12.18± 0.37 %
μ = 494.7± 1.7
σ = 60.2± 1.7

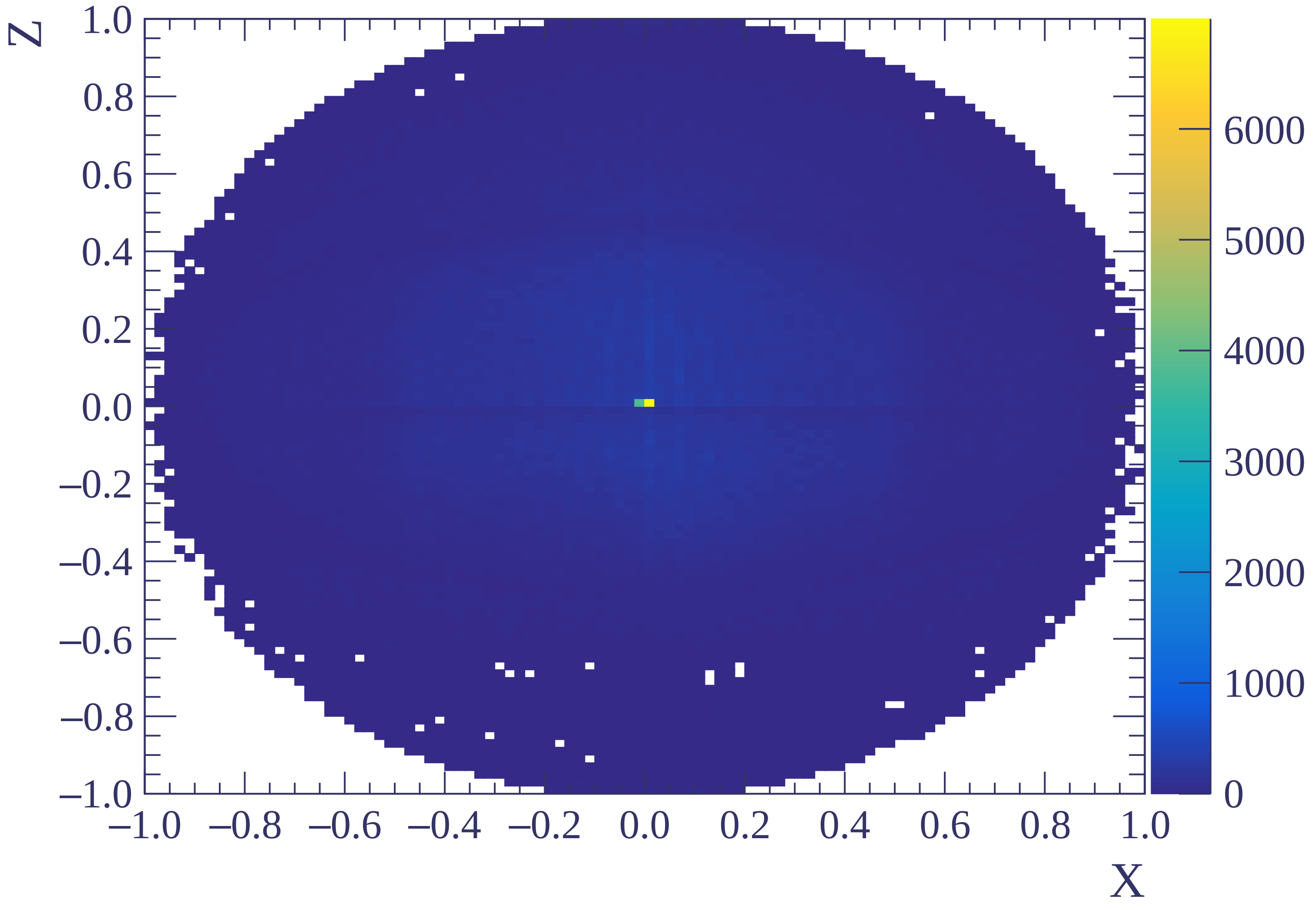


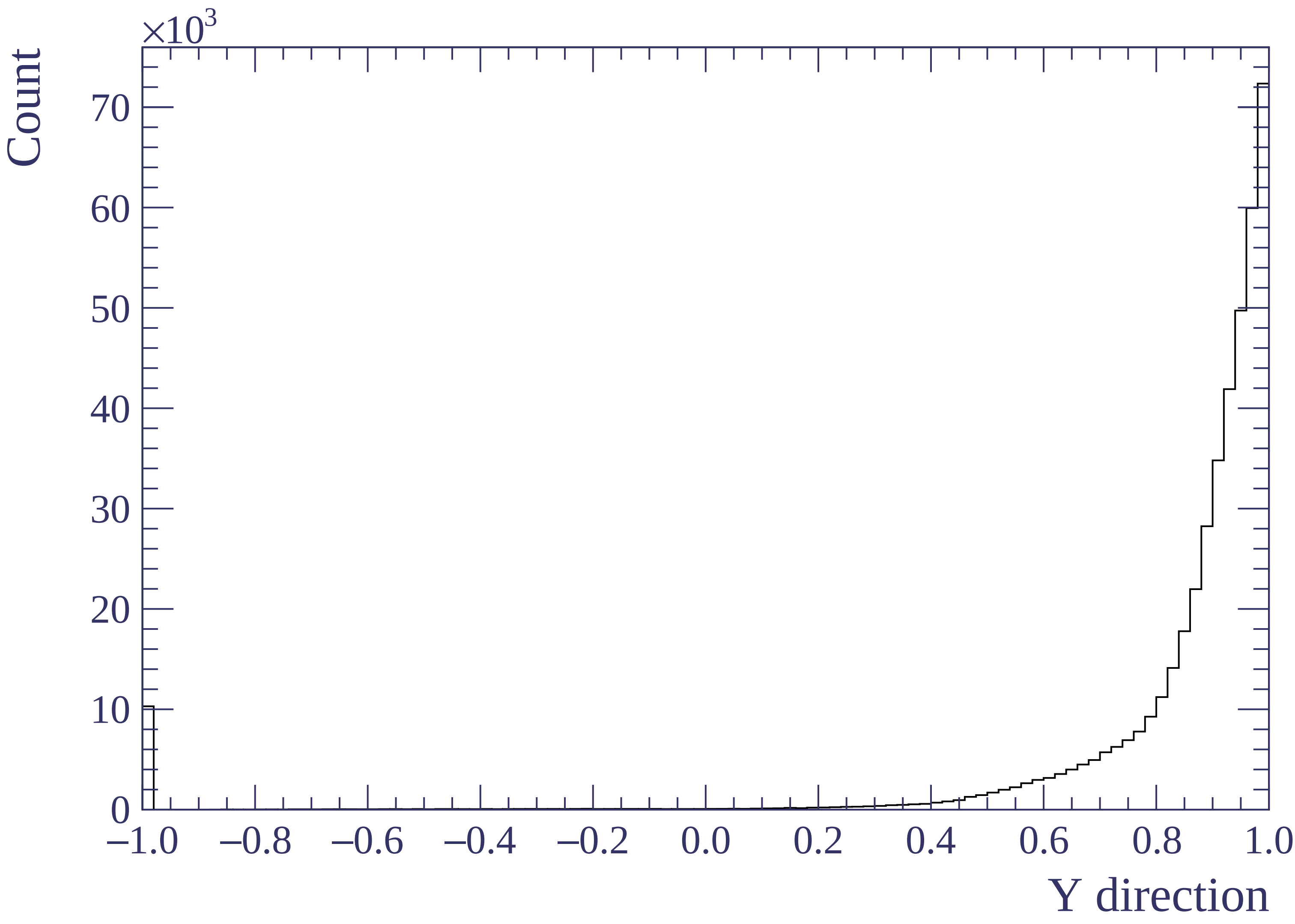
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (2170 entries)
$\frac{\sigma}{\mu}$ = 11.68± 0.32 %
μ = 494.8± 1.5
σ = 57.8± 1.4
WF (2170 entries)
$\frac{\sigma}{\mu}$ = 11.86± 0.29 %
μ = 500.0± 1.5
σ = 59.3± 1.3

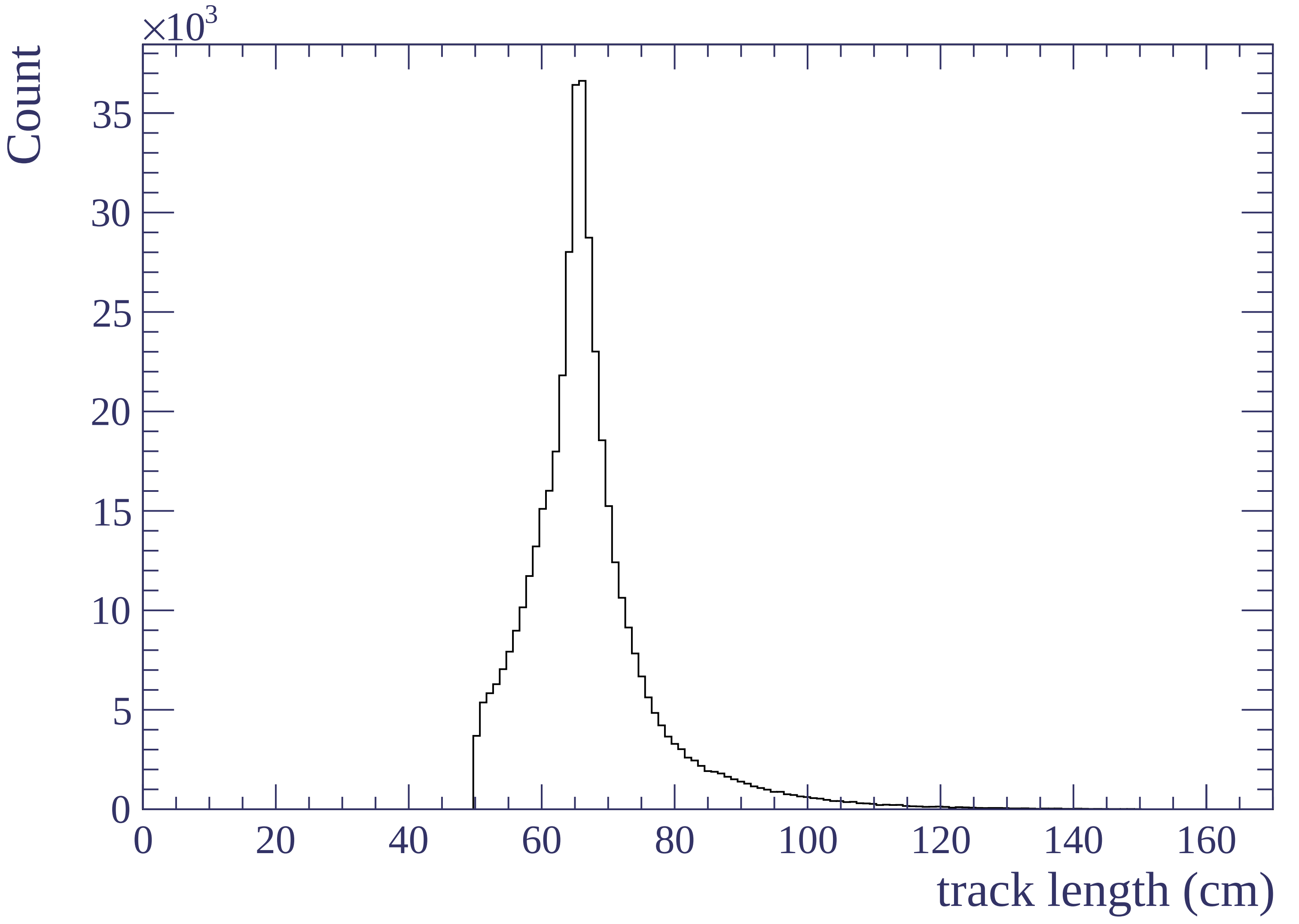

Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
XP (1990 entries)
σ/μ = 11.61± 0.34 %
μ = 498.7± 1.6
σ = 57.9± 1.5
WF (1990 entries)
σ/μ = 11.37± 0.28 %
μ = 503.1± 1.5
σ = 57.2± 1.2
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200



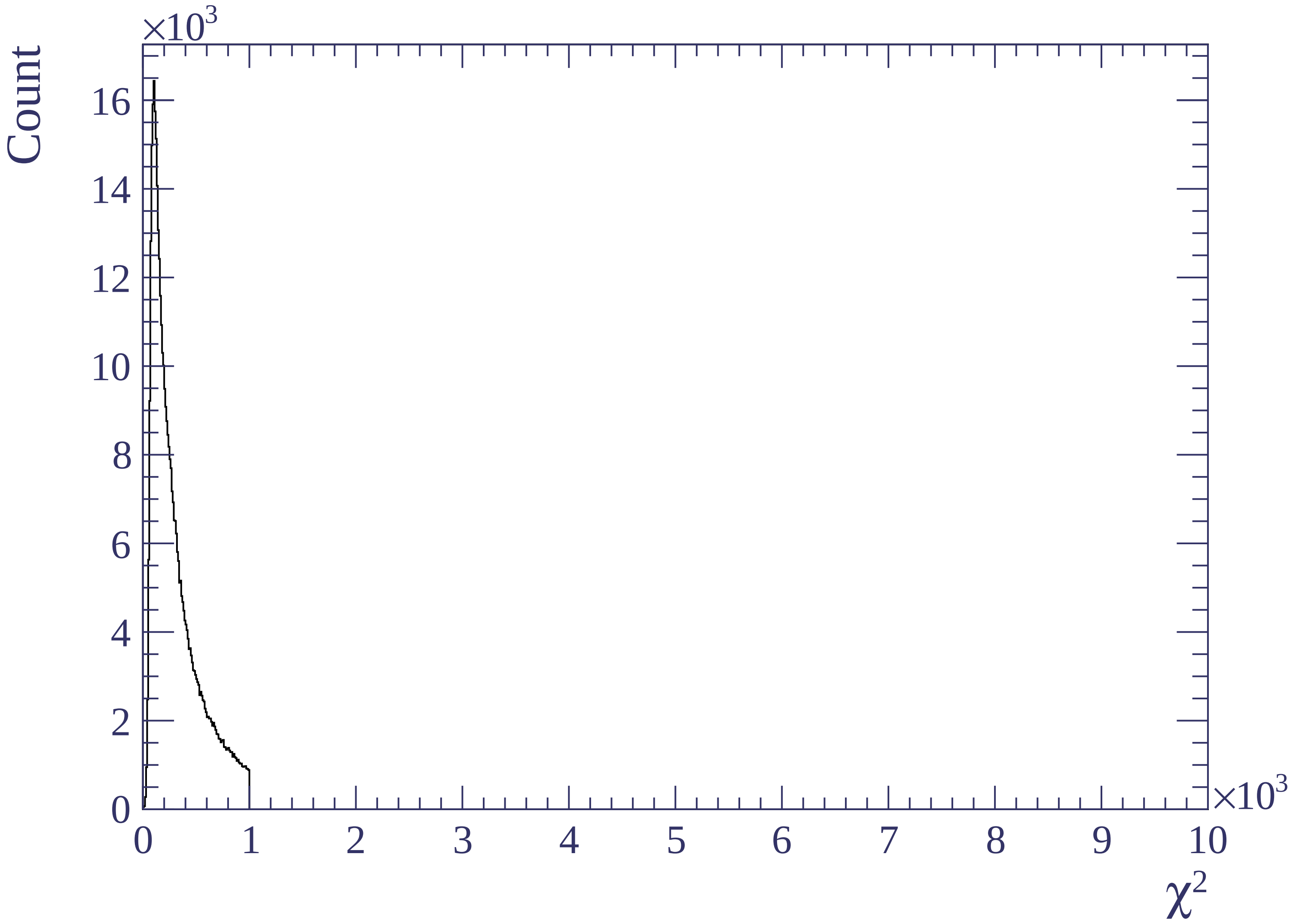
Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP (969 entries)
σ/μ = 11.85± 0.51 %
μ = 498.5± 2.4
σ = 59.0± 2.3
WF (969 entries)
σ/μ = 12.26± 0.51 %
μ = 503.1± 2.4
σ = 61.7± 2.3

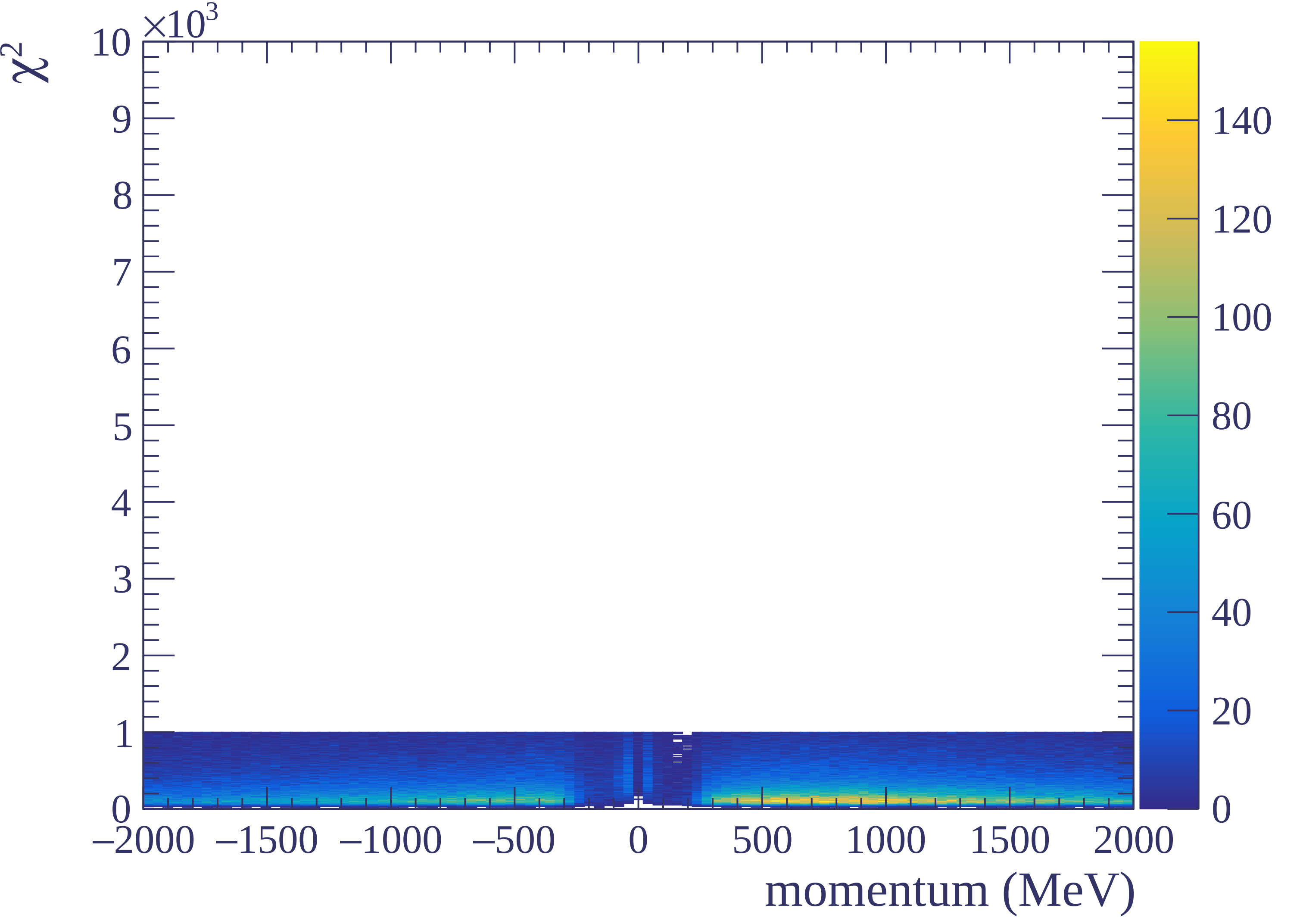

Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X
6000
5000
4000
3000
2000
1000
0

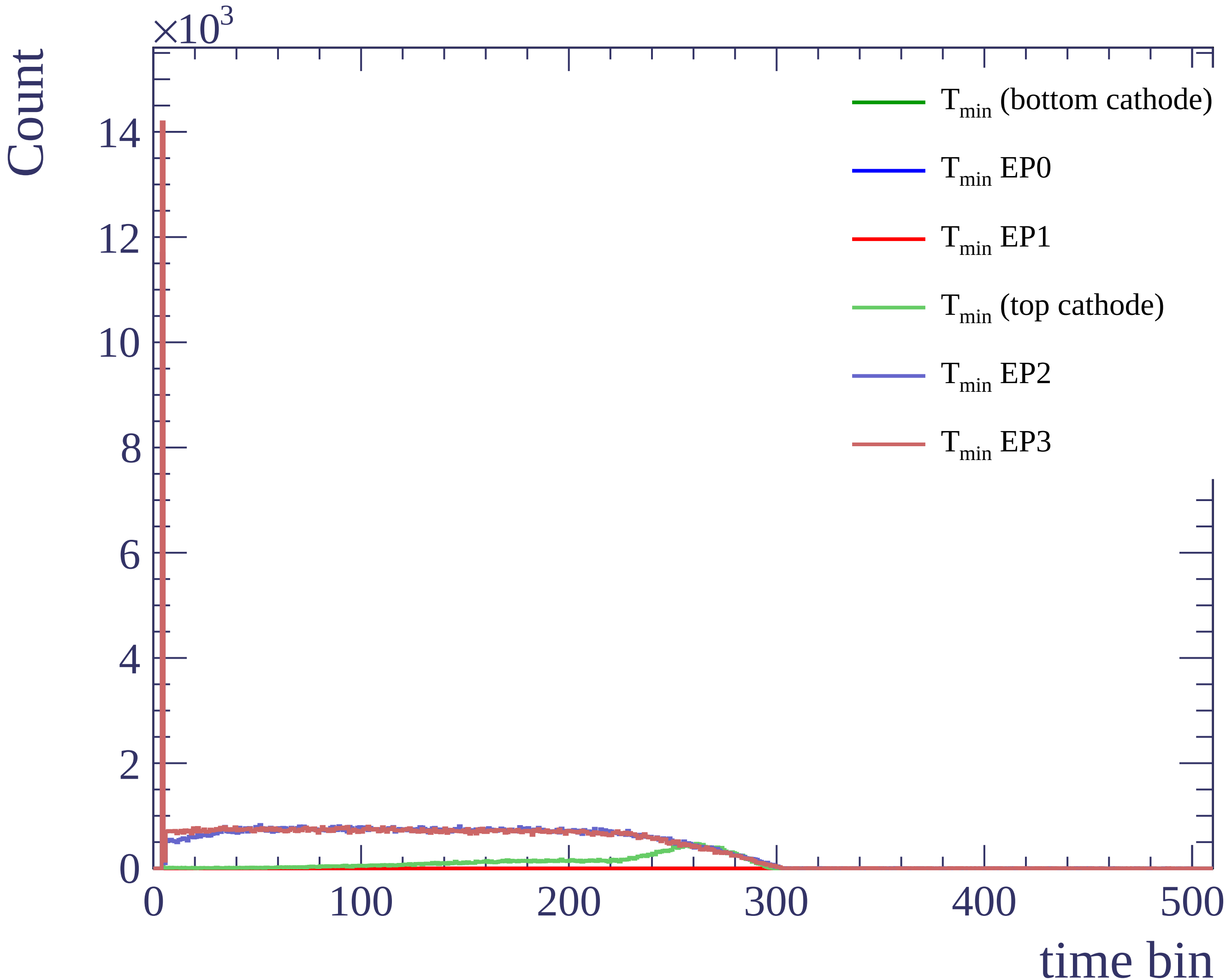

×10³
Count
70
60
50
40
30
20
10
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



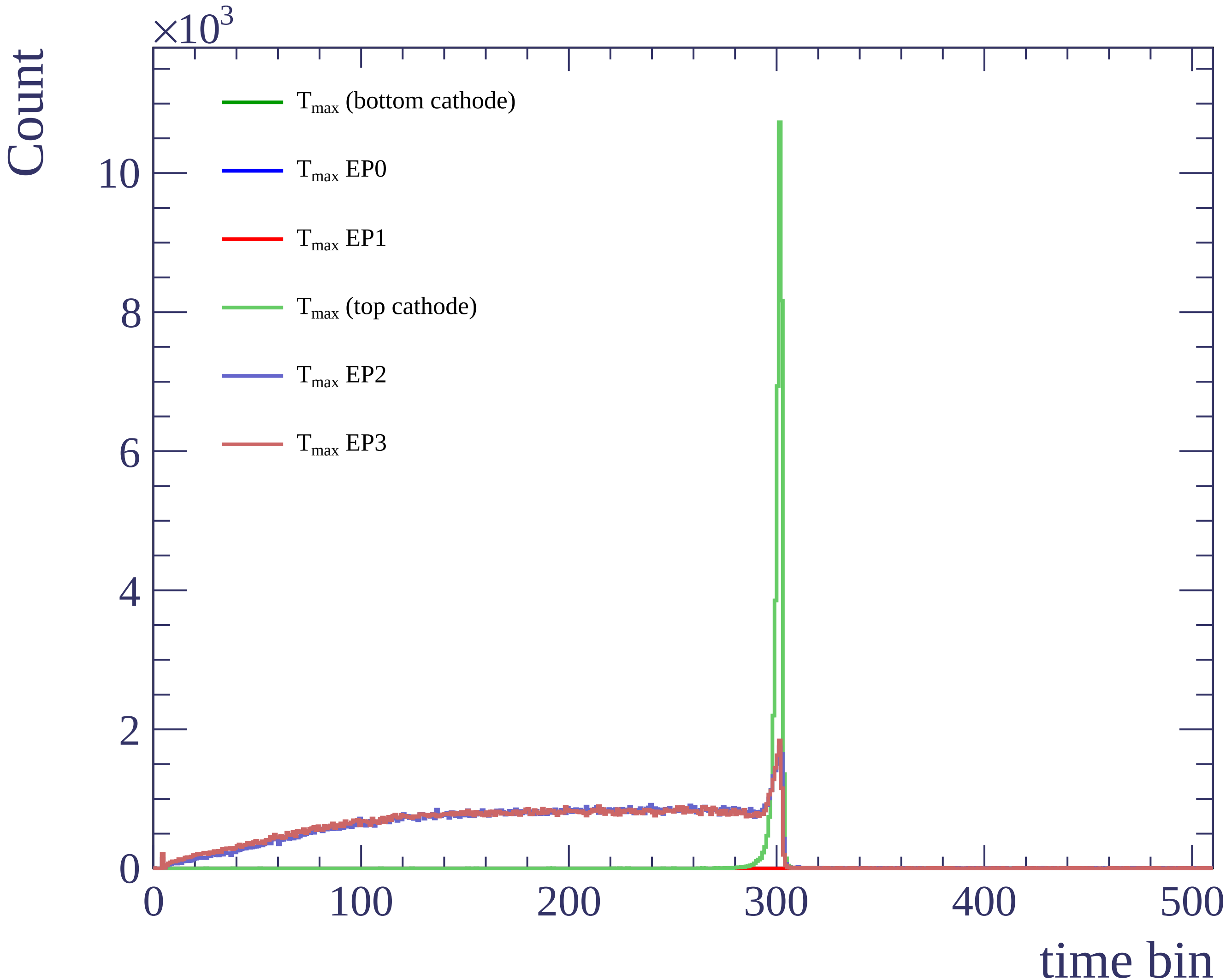
×10^3
Count
0
5
10
15
20
25
30
35
0
20
40
60
80
100
120
140
160
track length (cm)


Count
×10^3
16
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2


χ^2
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
140
120
100
80
60
40
20
0



$\times 10^3$
Count
14
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



×10^3
Count
T_max (bottom cathode)
T_max EP0
T_max EP1
T_max (top cathode)
T_max EP2
T_max EP3
0
2
4
6
8
10
0
100
200
300
400
500
time bin